

Mandarin Verb Doubling and Its Implications for the Syntactic Theory

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Doctor of Philosophy

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Abstract

This thesis is concerned with Mandarin verb doubling and its implications for the syntactic theory. I propose a phrasal movement analysis of verb doubling, namely that the doubling configuration in question is derived from verbal phrasal movement out of an articulated verbal structure. After the overall introduction in chapter 1, chapter 2 lays out the core properties of the verbal phrasal movement involved in verb doubling, that is, the fronted verbal phrase can target two landing sites in the clausal spine. I argue that this two-landing-site analysis has both theoretical and empirical advantages over other hypothetical alternatives. Chapter 3 discusses the role of postverbal adverbials in verb doubling constructions. I argue that this group of elements in fact comprises two sub-groups (i.e., freely-adjoined adjuncts and selected arguments), and propose that the way they are introduced to verbal projections differs from one to the other. Chapter 4 puts forwards the idea that the fronted verbal phrases that engage in the formation of verb doubling can be larger than simple VPs. This proposal not only offers a direct account of distinct patterns that different ditransitive verb doubling configurations exhibit, but also makes correct predictions about the systematic distribution of multiple postverbal adverbials within verb doubling. Chapter 5 develops an analysis of realising verb doubling effects at PF, consisting of a series of operations that take place post-syntactically. I propose that various doubling effects, thus, can be attributed to different combinations of these PF operations. Chapter 6 examines the locality profile of verb doubling in bi-clausal contexts. I show that long-distance verb doubling behaves differently in three embedding contexts, and propose that the varying locality profiles of verb doubling can be properly accounted for by the Williams Cycle. Chapter 7 concludes the thesis.

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If I were asked almost ten years ago what I would be doing in my early thirties, doing a Ph.D. in linguistics definitely would not be the answer. But, as we all know, life is full of uncertainties: now I am already standing at the end of this wonderful journey, awaiting the beginning of the next one. I have been really fortunate to meet many people in this journey over the past four years, and I think there is no better place than here to express my heartfelt gratitude to them.

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for my years in London
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Abbreviations

1 = first person

2 = second person

3 = third person

CL = classifier

CLITIC = clitic

DE = postverbal modification marker

DE_{nom} = pre-nominal modification marker

DOU = distributive operator *dōu*

EXP = experiential aspect

GEN = genitive

INF = infinitive

LOC = locative

NEG = negation

PFV = perfective aspect

POSS = possessive marker

PL = plural

PROG = progressive

SFP = sentence-final particle

SG = singular

TOP = topic marker

Chapter 1

Introduction

This thesis is concerned with verb doubling in Mandarin Chinese, a structural phenomenon that has been long observed and discussed in literature on Mandarin syntax (Li and Thompson 1981; Huang 1982; Tai 1999; Cheng 2007, 2017; Fang and Sells 2007; H.-L. Huang 2018; Bartos 2019; Lai 2021, 2024; Meadows and Yan 2023, 2025; Yan and Meadows 2025, among many). Two relevant examples are given as below.

- (1) a. Piet huà nà-gè jùfǎ shù huà de hěn zǐxì.
Piet draw that-CL syntax tree draw DE very carefully
‘Piet drew the syntax tree very carefully.’
- b. Piet huà nà-gè jùfǎ shù huà-le sān biàn.
Piet draw that-CL syntax tree draw-PFV three time
‘Pite drew the syntax tree three times.’

In both sentences, the same lexical verb *huà* ‘draw’ occurs twice, thus being termed ‘verb doubling’. Similar doubling effects are also found across a variety of languages (see Harbour 1999; Abels 2001; Cable 2004; Landau 2006; Vicente 2009; Hein 2017, 2018, 2020; Bleaman 2022, among others), but there are several structural properties which are seemingly unique to Mandarin Chinese. First, the first occurrence of the doubled verb and its following object, together as a unit, are able to occur in a clause-medial, post-subject position, as illustrated above, which is not commonly found cross-linguistically. Second, there is always an adverbial showing up after the second occurrence of the doubled verb, a position which is generally referred to as the postverbal field. The occurrence of the postverbal adverbial is required in every verb doubling configuration in question, which, again, has not been observed in other languages as far as I am aware. Third, between the second verb and the postverbal field occurs a particle, which varies from one doubling configuration to another, depending on the type of the postverbal adverbial. It

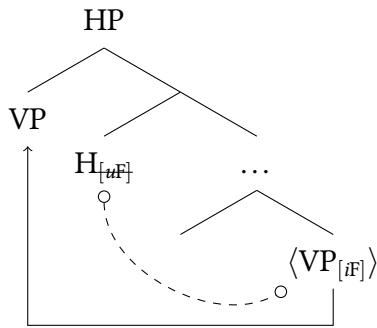
is these intriguing properties that motivate me to conduct this research into Mandarin verb doubling, think about how this configuration can be accounted for by the current syntactic theory, and ask what implications the former can have for the latter.

1.1 Main proposal

The central goal of this thesis is to offer an in-depth analysis of how to derive Mandarin verb doubling, thereby obtaining better understanding of the three language-specific properties listed above. In what follows I sketch out several core ideas that my main proposal comprises.

First, I propose that derivation of verb doubling uniformly involves verbal phrasal movement, which minimally consists of a lexical verb and its internal argument (i.e., a VP), out of an articulated verbal structure. The examples in (1) only show that the fronted VP lands in a clause-medial position that follows the subject. We will find that the same VP can also land in a clause-initial position which precedes the subject. To capture the distribution of the fronted VPs, I propose that there are (at least) two landing sites for VP-fronting in the clausal spine, one in the specifier of a specific functional projection FP (which occurs between TP and *v*P), and the other in the left periphery. As schematised in (2), I posit that verbal phrasal movement to either position (i.e., Spec,HP) is feature-triggered, and that the triggering feature (i.e., the F-feature on H) varies from one position to the other.

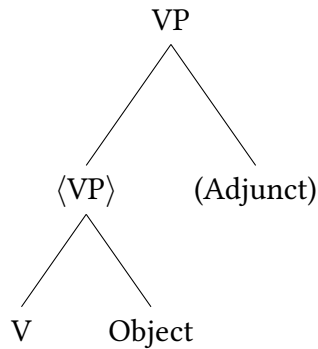
(2) *Schema for verbal phrasal movement involved in verb doubling*



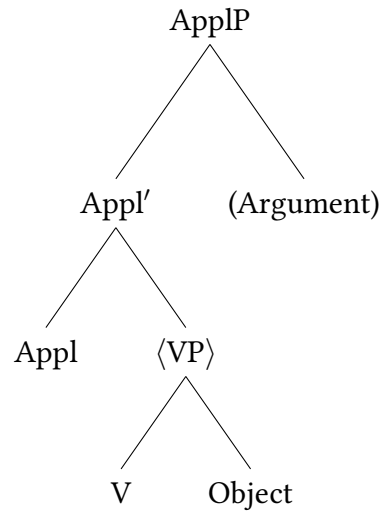
Second, I argue that there is an adjunct/argument dichotomy among postverbal adverbials. That is, adverbials that show up in the postverbal field of verb doubling do not share the same structural role: some of them behave like adjuncts that can freely show up in any configuration, while others behave more like arguments that are subject to the selection of the verbal head. To capture this dichotomy, I propose that these adverbials are introduced to verbal projections in two different ways. For adverbial adjuncts, my proposal is that they function as V-projection adjuncts which can be freely adjoined to either side of the verbal phrase (e.g., VP or ApplP). For argument-like

adjuncts, I propose that they are selected by some higher verbal head (i.e., the applicative head) and occupy either a leftward or a rightward specifier position. For the sake of clarity, only the rightward version of these two verbal projections are illustrated in (3) and (4) below.

(3) *Adverbial as a non-selected adjunct*

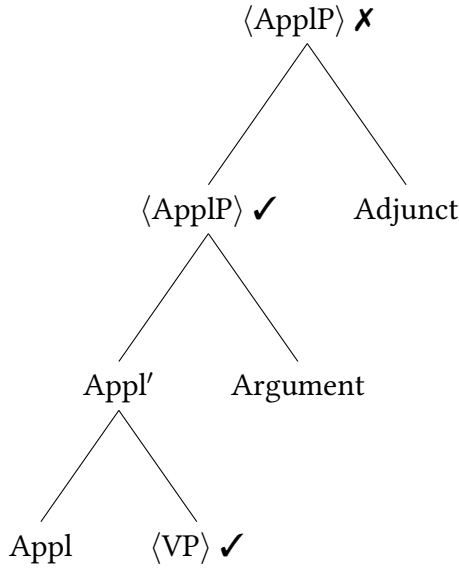


(4) *Adverbial as a selected argument*



Third, I propose that verbal phrasal movement involved in verb doubling is not confined to simple VPs only; instead, bigger verbal phrases like ApplPs are also able to engage in constructing verb doubling, e.g., the second-highest ApplP as in (5). However, although the size of the fronted verbal phrase displays flexibility to some degree, I will show that it is impossible for the entire verbal phrase, e.g., the topmost ApplP as given below, to undergo the same movement. Thus, there is in fact some constraint on how big the fronted verbal phrase can be, which leads to the result that the postverbal adverbial is always stranded in the verb doubling configuration.

- (5) *Flexible size of the fronted verbal phrase (but with some constraint)*



Last but not least, to account for how verb doubling effects are realised at PF, I propose that there are a series of operations that take place post-syntactically, as schematised in (6). Depending on the specific particle involved, different verb doubling configurations make use of their own toolkit of post-syntactic operations. That is to say, realising verb doubling effects at PF is not a uniform procedure, though these constructions do not significantly differ from each other on the surface.

- (6) *Post-syntactic operations involved in verb doubling*

... >> Head Movement >> Lowering >> Chain Reduction >> Local Dislocation >> ...

What I have presented so far is just a condensed description of the major part of my proposal. As the thesis proceeds, more details of my proposed analysis of verb doubling will be uncovered in order, starting from chapter 2 to chapter 5. Furthermore, there is a remaining part of my proposal concerning the long-distance locality profile of VP-movement I have not touched upon yet, which will be the main topic of chapter 6. In order to understand the puzzle discussed there, it is necessary for us to first have a comprehensive understanding of the key characteristics of Mandarin verb doubling, which is exactly what the next four chapters are designed for.

1.2 Comparison with prior literature

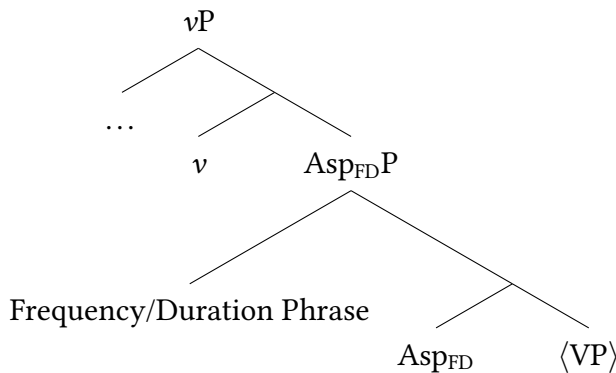
As mentioned at the very beginning, there is vast literature on verb doubling, not only in Mandarin Chinese alone (see Cheng 2017 for an overview), but also across languages (see Hein 2018, 2020 for details). Nevertheless, the phrasal movement analysis proposed in this thesis is not as

widely entertained in prior work as one might expect, except in Lai (2021, 2024) and my own one (Meadows and Yan 2023, 2025). Here I would like to place a particular focus on Lai’s work, and make a brief comparison between his proposal and mine.

As with my proposed analysis sketched in the previous section, Lai proposes that verbal phrasal movement is involved in deriving verb doubling, and that there is a clause-medial landing site that can host the fronted VP (in addition to the one in the left periphery). He identifies this post-subject position with the specifier of the aspectual phrase (i.e., Spec,AspP), a functional projection that occurs between TP and v P. This clause-medial domain is exactly where my proposed FP is located in the clausal spine. In this respect, it is fair to say that both Lai’s work and mine share the same intuition regarding the underlying mechanism for constructing verb doubling.

The main differences between our analyses are reflected in the following three aspects. The first one is concerned with how adverbials are introduced to verb doubling configurations in general. Lai focuses on cases where frequency and duration phrases show up postverbally, and posits that these adverbials are base-generated in the specifier of some low functional projection in the Cinquean Hierarchy (Cinque 1999) called $Asp_{FD}P$, as schematised in (7). Since his analysis does not take into account other types of verb doubling (such as manner-type, locative-type, etc.; see chapter 3 for in-depth discussion), Lai does not recognise the adjunct/argument dichotomy that I argue for, which results in further division between the ways different adverbials are introduced to the verbal projection (recall (3) and (4)).

(7) *Frequency/duration phrase introduced by $Asp_{FD}P$ in the Cinquean Hierarchy*



The second difference between Lai’s proposed analysis and mine lies in the size of the fronted verbal phrase. In Lai’s proposal, only VPs can undergo movement and contribute to the formation of verb doubling. By contrast, my proposal is that verbal phrases which are larger than simple VPs (i.e., a verb-object unit) can also be fronted and become part of the doubling configurations (recall (5)). This crucial analytical difference leads us to make totally opposite predictions as follows: (8) demonstrates three non-canonical strings of verb doubling; in each scenario, what undergoes

fronting is clearly larger than a simple VP, which falls under my proposed analysis. But according to Lai's proposal, neither of these strings is predicted to be licit, contrary to the fact that I will show in chapter 4.

(8) *Non-canonical strings of verb doubling*

- a. Subj [[V₁ Obj] Argument] [[V₂-Part] Adjunct]
- b. Subj [[V₁ Obj] Adjunct] [[V₂-Part] Adjunct]
- c. Subj [[[V₁ Obj] Argument] Adjunct] [[V₂-Part] Adjunct]

The third difference between our proposals is related to how verb doubling effects are realised at PF. In Lai's work, he adopts a theory of pronunciation proposed by Saab (2017), and directly applies it to verb doubling in Mandarin. Putting aside its details, this algorithm offers a plain account of how VP-fronting interacts with copy deletion and realisation at PF, especially in cases where tense and aspect markers (e.g., *le* in (1b)) are involved in the derivation of verb doubling. In contrast, though I do not provide a detailed PF algorithm as Lai does, my proposal recognises in the first place that particles involved in verb doubling can never be treated in the same way (e.g., the particle *de* in (1a) vs. the aspectual marker *le* just mentioned). Therefore, it makes full use of various post-syntactic operations (recall 6)), and accounts for a much wider range of verb doubling effects than its counterpart (see more details in chapter 5).

The comparison made above is sufficient to show how my proposal is different from Lai's. However, it is worth emphasising that this does not necessarily mean that our proposed analyses are incompatible with each other. In fact, I borrow a lot of insights from Lai's work and incorporate them into my own one, as will be clearer soon. A lot of our differences can eventually be reduced to a couple of distinct core assumptions that we respectively adopt and begin with to investigate verb doubling (e.g., the syntax of adverbials). I view these divergences as a promising point of departure for further discussion about relevant topics.

1.3 Overview of the thesis

The thesis is organised as follows. In chapter 2, I focus on the distribution of the first verb and the object that follows it. The displacement of this verb-object unit motivates me to propose that verb doubling involves verbal phrasal movement which targets two separate landing sites in the clausal spine. I show evidence to support the phrasal movement analysis as well as the two-landing-site approach to the distribution of the fronted verbal phrases.

In chapter 3, I investigate the role of adverbials in verb doubling configurations. I argue that what are labelled as 'postverbal adverbials' can actually be further classified into two sub-groups:

while most of them behave like canonical adjuncts that can be freely-adjoined to the verbal projection, some of them act more like arguments that are selected by some verbal head. In accordance with this adjunct/argument dichotomy, I propose that the way these two sub-groups of adverbials are introduced to verbal projections is not identical. I show that this division makes correct predictions about co-occurrence of multiple adverbials in the postverbal field.

In chapter 4, I pay close attention to the distribution of verbal dependents across verb doubling configurations. By discussing verb doubling with ditransitives in detail, I propose that verbal phrasal movement involved in verb doubling can actually be larger than simple VPs, such as ApplPs. I demonstrate that this proposal makes a series of precise predictions about articulated verbal structures that contains multiple adverbials, and that postverbal adverbials can be distributed within verb doubling configurations in a systematic way. After that, I discuss two crucial analytic problems, one related to the adverbials being stranded in the postverbal field, the other related to a specific group of simple verb-object units that are generally defined as verbal compounds.

In chapter 5, I develop a way of realising verb doubling effects at PF. I draw detailed comparisons between particles involved in verb doubling in the first place, and then make use of their similarities and differences to motivate my post-syntactic account. I propose that there are different operations that take place post-syntactically, and that the way they interact with each other will lead to distinct effects on the realisation of both verb doubling and other closely-related configurations.

In chapter 6, I turn to the locality profile of cross-clausal verb doubling. I show that long-distance verb doubling does not behave identically in three embedding contexts: while the higher verbal phrasal movement is always permitted, the lower one is more restricted and only possible out of one embedding context. I argue that these three embedding contexts differ in clause size. Based on this idea, I propose that the varying locality profiles of long-distance verb doubling falls under the Williams Cycle, and adopt a Level Embedding approach to accounting for them. I extend the same approach to investigating the locality profile of long-distance object shift, the result of which offers further support for the Williams Cycle. I close the chapter with two problems related to interaction between verb doubling and other movement-related phenomena. I offer a brief conclusion in chapter 7.

Chapter 2

Positioning Verb Doubling

One of the most striking characteristics of verb doubling, as indicated by the term itself, is that a lexical verb occurs twice in one sentence. For instance, the lexical verb *kàn* ‘read’ has two separate occurrences in both sentences in (9). This chapter focuses on the first occurrence of the doubled verb, and provides a preliminary account of its distribution in the clausal spine.

- (9) a. Piet **kàn** nà-(yī)-běn xiǎoshuō **kàn**-le sān-biàn.
Piet read that-one-CL novel read-PFV three-time
‘Piet read that novel three times.’
- b. **Kàn** nà-(yī)-běn xiǎoshuō Piet **kàn**-le sān-biàn.
read that-one-CL novel Piet read-PFV three-time
‘As for reading that novel, Piet read it three times.’

The string of canonical verb doubling like (9a) is illustrated in (10). The two occurrences of the same verb are respectively followed a verbal dependent: the first one (V_1) is followed by the object, while the second one (V_2) precedes a particle and an adverbial. Considering the position of the V_1 plus its following object relative to the subject, this V_1 -object unit, i.e., a simple verb phrase associated with the V_1 , can be characterised as occupying a clause-medial, post-subject position.

- (10) *Scenario 1 of verb doubling: V_1 -object unit in a clause-medial position*
Subj [V_1 Obj] [[V_2 -Part] Adverbial] (e.g., (9a))

This is not the only position that can host this verb phrase though. A lot of previous work has documented that the V_1 -object unit can also occur in the clausal left periphery (e.g., Cheng 2007; Lai 2021, 2024; Meadows and Yan 2023, 2025; Yan and Meadows 2025, among others), as in (9b). The corresponding string of this kind of verb doubling is illustrated in (11). It is clear that,

compared to the string in (10), nothing happens to the V_2 and its surroundings here; the only difference is that the verb phrase associated with the V_1 shows up in a clause-initial, pre-subject position with the object.

- (11) *Scenario 2 of verb doubling: V_1 -object unit in a clause-initial position*
 [V_1 Obj] Subj [[V_2 -Part] Adverbial] (e.g., (9b))

Based on the relative position between the V_1 -object unit and the subject, I refer to the two configurations in (10) and (11) respectively as *low* and *high verb doubling* in what follows. To account for this distributional difference, in section 2.1, I postulate that verb doubling involves verbal phrasal movement. I propose that (i) there are *two* landing sites of verbal phrasal movement in the clausal spine, and that (ii) verbal phrasal movement to each landing site is triggered by a different feature. After presenting diagnostics for phrasal movement in section 2.2, I argue against an alternative, potential account of the distributional difference just seen above in section 2.3. That is, there is only *one* available position in the clausal left periphery that is able to host the fronted verbal phrase, and the *prima facie* two landing sites relative to the subject is actually derived by some further topicalisation of the subject. I demonstrate that this *fronting-plus-topicalisation* derivation encounters several issues, but my proposed *two-landing-site* analysis does not, which provides valuable support for the latter. Finally in section 2.4, I show that this two-landing-site account of low and high verb doubling is compatible with the widely-accepted clausal architecture in Mandarin Chinese, which is perceived to contain a clause-medial domain below TP that parallels the left periphery (à la Belletti 2004).

2.1 Verb doubling as phrasal movement to two landing sites

The majority of verb doubling constructions that we can find in the relevant literature are what I refer to as low verb doubling (e.g., Li and Thompson 1981; Tai 1999; Cheng 2017, a.o.). Regardless of the meaning of the adverbial in the clause-final position (which I will discuss in the subsequent section 3), all sorts of low verb doubling share one commonality: the first occurrence of the doubled verb plus the object takes up a clause-medial, post-subject position. Some of the examples are presented in (12) (all of which are adapted from Meadows and Yan 2025:(1)&(21)).

- (12) a. Piet $kàn_1$ $yáyī$ $kàn_2$ -guò sān cì.
 Piet see dentist see-EXP three time
 ‘Piet has gone to the dentist three times.’

- b. Piet **kàn₁ diànyǐng** kàn₂-le sān tiān.
Piet watch movie watch-PFV three day
'Piet watched movies for three days.'
- c. Piet **kàn₁ xiǎoshuō** kàn₂ de hěn kuài.
Piet read novel read DE very quickly
'Piet read novels very quickly.'
- d. Piet **kàn₁ zìmù** kàn₂ de hěn tòngkǔ.
Piet read subtitles read DE very painful
'Piet read subtitles and felt painful (as a result).'

In each case above, the V_1 -object unit is the first element that shows up after the subject *Piet*, viz., *kàn yáyī* 'go to the dentist' in (12a), *kàn diànyǐng* 'watch movies' in (12b), *kàn xiǎoshuō* 'read novels' in (12c), and *kàn zìmù* 'read subtitles' in (12d).

The V_1 -object unit is not confined to this clause-medial, post-subject position only. Every example in (12) has a counterpart that I refer to as high verb doubling: all else being equal, the same V_1 -object unit occurs differently in a clause-initial position and precedes the subject. These high verb doubling configurations are given in (13).

- (13) a. **Kàn₁ yáyī** Piet kàn₂-guò sān cì.
see dentist Piet see-EXP three time
'As for going to the dentist, Piet has gone there three times.' (cf. (12a))
- b. **Kàn₁ diànyǐng** Piet kàn₂-le sān tiān.
watch movie Piet watch-PFV three day
'As for watching movies, Piet watched them for three days.' (cf. (12b))
- c. **Kàn₁ xiǎoshuō** Piet kàn₂ de hěn kuài.
read novel Piet read DE very quickly
'As for reading novels, Piet read them very quickly.' (cf. (12c))
- d. **Kàn₁ zìmù** Piet kàn₂ de hěn tòngkǔ.
read subtitles Piet read DE very painful
'As for reading subtitles, Piet read them and felt painful (as a result).' (cf. (12d))

Between low and high verb doubling, there are one similarity and one difference that are worth highlighting here. First of all, the similarity is concerned with the V_1 -object unit. Consider (14) first, which shows two canonical, non-doubling configurations with two postverbal elements in Mandarin Chinese. In both examples, the single lexical verb *kàn* is followed by the object and the adverbial at the same time: in (14a), the object *yáyī* 'dentist' follows the frequency adverbial *sān cì*

‘three times’ linearly; whereas in (14b), the object *xiǎoshuō* ‘novel’ precedes the manner adverbial *hěn kuài* ‘very quickly’ in linear order.

- (14) a. Piet **kàn**-guò sān cì yáyī.
 Piet see-EXP three time dentist
 ‘Piet has gone to the dentist three times.’
 b. Piet **kàn** xiǎoshuō hěn kuài.
 Piet read novel very quickly
 ‘Piet read novels very quickly.’

We do not need to bother ourselves with the relative word order between the two postverbal elements in canonical structures at the moment. What we need to keep in mind is that the so-called postverbal field in Mandarin Chinese does not follow the *Phrase Structure Condition* proposed by Huang (1982, 1984), as given in (15) and (16): according to this structural constraint, there is no way of hosting more than one element postverbally in Mandarin, which is clearly against what (14) demonstrates. In later sections, we will find more cases in favour of the idea the Phrase Structure Condition is not a correct generalisation (see Lai 2021, 2024 for more detailed discussion on this).

(15) *The $\bar{X}$ structure of Chinese is of the form:*

- a. $[_{X^n} X^{n-1} YP^*]$ iff $n=1$ and $X \neq N$
 b. $[_{X^n} YP^* X^{n-1}]$ otherwise (Huang 1982:41, (20))

(16) *The Phrase Structure Condition*

Within a given sentence in Chinese, the head (the verb of VP) may branch to the left only once, and only on the lowest level of expansion. (Huang 1984:(1))

Now let us have a look at low and high verb doubling in (12) and (13) again: in all these given examples, the first occurrence of the doubled verb is always followed by the object. Is it possible for the V_1 to show up alone, without being followed by the object? As shown in (17), such a configuration is impossible for low verb doubling, no matter where the object shows up in the post- V_2 field.

- (17) a. *Piet **kàn₁** **kàn₂**-guò sān cì yáyī.
 Piet see see-EXP three time dentist
 Int.: ‘Piet has gone to the dentist three times.’ (cf. (12a))

- b. *Piet **kàn₁** **kàn₂**-le sān tiān **diànyǐng**.
 Piet watch watch-PFV three day movie
 Int.: ‘Piet watched movies for three days.’ (cf. (12b))
- c. *Piet **kàn₁** **kàn₂** **xiǎoshuō** de hěn kuài.
 Piet read read novel DE very quickly
 Int.: ‘Piet read novels very quickly.’ (cf. (12c))
- d. *Piet **kàn₁** **kàn₂** **zìmù** de hěn tòngkǔ.
 Piet read read subtitles DE very painful
 Int.: ‘Piet read subtitles and felt painful (as a result).’ (cf. (12d))

(18) demonstrates that it is also impossible for the single V_1 to occupy the pre-subject position on its own in high verb doubling configurations.

- (18) a. ***Kàn₁** Piet **kàn₂**-guò sān cì **yáyī**.
 see Piet see-EXP three time dentist
 Int.: ‘As for going to the dentist, Piet has gone there three times.’ (cf. (13a))
- b. ***Kàn₁** Piet **kàn₂**-le sān tiān **diànyǐng**.
 watch Piet watch-PFV three day movie
 ‘As for watching movies, Piet watched them for three days.’ (cf. (13b))
- c. ***Kàn₁** Piet **kàn₂** **xiǎoshuō** de hěn kuài.
 read Piet read novel DE very quickly
 Int.: ‘As for reading novels, Piet read them very quickly.’ (cf. (13c))
- d. ***Kàn₁** Piet **kàn₂** **zìmù** de hěn tòngkǔ.
 read Piet read subtitles DE very painful
 Int.: ‘As for reading subtitles, Piet read them and felt painful (as a result).’ (cf. (13d))

The ungrammatical examples in (17) and (18), as schematised in (19), suggest that it is obligatory that the V_1 and the object have to co-occur as a unit when they engage in the formation of verb doubling (no matter it is low or high). This observation is crucial to motivate a derivation of verb doubling which uniformly involves verbal phrasal movement (i.e. the V_1 -object unit as a VP that undergoes movement), which I will sketch out soon.

(19) *Illicit string of verb doubling: No V_1 -object unit*

- a. *Subj [V_1] [[V_2 (**Obj**) Part (**Obj**)] Adverbial]
 b. * [V_1] Subj [[V_2 (**Obj**) Part (**Obj**)] Adverbial]

Turning to the difference between low and high verb doubling, we find that the interpretation of the two doubling constructions is not the same. The translation of low verb doubling in (12) and high verb doubling (13) has already indicated this interpretational difference: though the former looks informational-structurally neutral, the latter seems to involve either topicalisation or focalisation. This contrast is better illustrated in the following scenario (20) (which itself is informational-structurally neutral as there is no preceding discourse):

(20) *Scenario: One day Pablo walked into the classroom and saw Piet reading Three-Body Problem very quickly. Later Pablo mentioned this to Wassily ...*

- a. (Wǒ fāxiàn) Piet **kàn** **Sāntǐ** tèbié kuài.
 1SG find Piet read TBP particularly quickly
 ‘(I found that) Piet was reading the *Three-Body Problem* very quickly.’
 (no verb doubling: informational-structurally neutral)
- b. (Wǒ fāxiàn) Piet **kàn₁** **Sāntǐ** **kàn₂** de tèbié kuài.
 1SG find Piet read TBP read DE particularly quickly
 ‘(I found that) Piet was reading the *Three-Body Problem* very quickly.’
 (low verb doubling: informational-structurally neutral)
- c. # (Wǒ fāxiàn) **Kàn₁** **Sāntǐ** Piet **kàn₂** de tèbié kuài.
 1SG find read TBP Piet read DE particularly quickly
 ‘(I found that) As for reading the *Three-Body Problem*, Piet was reading it very quickly.’
 (high verb doubling: informational-structurally marked)

In this given scenario, where Piet was seen reading the *Three-Body Problem* very quickly by Pablo, it is totally acceptable to utter both (20a), which is a canonical structure without verb doubling, and (20b), which is a low verb doubling configuration, suggesting that they are both neutral in terms of information structure, and can be used interchangeably in most situations. In contrast, (20c), as an instance of high verb doubling, is informational-structurally marked when uttered in this scenario, which favours the idea that the V_1 -object unit is construed as either a topic or a focus in the clausal left periphery.

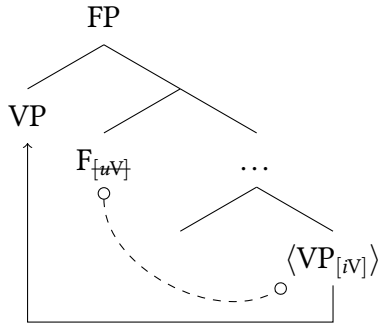
So far we have seen that verb doubling can show up in two different positions in the clausal spine. Low verb doubling occurs in a clause-medial position following the subject, whose interpretation is informational-structurally neutral. High verb doubling, by comparison, shows up in a clause-initial position and precedes the subject as a topic or a focus. Although their distribution and interpretation are not identical, they are subject to the same well-formedness condition that the first occurrence of the doubled verb must form a unit with the object.

To account for their differences and similarities at the same time, I postulate that the deriva-

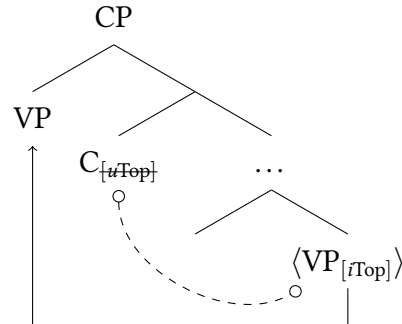
tion of verb doubling involves verbal phrasal movement, namely verbal predicate fronting of the lowest thematic domain which minimally contains the direct object. For simplicity, I refer to this movement as VP-fronting in the following discussion.¹ The displacement of the V_1 -object unit relative to the subject provides a first indication that VP-fronting might underlie the formation of verb doubling. This standpoint has already been argued for independently by Lai (2021, 2024) and Yan and Meadows (2025) (also see Meadows and Yan 2023, 2025 for some precursory discussion), and it will be adopted throughout this work. This account is supported by conventional phrasal movement diagnostics, such as island and reconstruction effects, which I will present soon in the upcoming section 2.2.

Following the idea that the surface subject in Mandarin Chinese typically occupies Spec,TP (e.g., Lin 2011; Paul 2015, among many others), I propose that there are two landing sites around the subject in the clausal spine that can be respectively targeted by VP-fronting. The low position is the specifier of a designated functional projection FP in the clause-medial domain below the TP, as illustrated in (21). The high one is the specifier of the CP, which is undoubtedly above the TP, as illustrated in (22).²

(21) *Post-subject, low verb doubling*



(22) *Pre-subject, high verb doubling*



Given that low verb doubling is informational-structurally neutral compared to high verb doubling (recall the contrast in (20)), I propose that verbal phrasal movement involved in the two verb doubling constructions is triggered by different features. For low verb doubling, I follow Massam and Smallwood (1997), Coon (2010), Collins (2017), Travis and Massam (2021), and van Urk (2024), and propose that verbal predicate fronting is triggered by the categorial feature of the verb $[uV]$ located in the medial clausal spine for producing one kind of unmarked word order. As to high verb doubling, I propose that the corresponding predicate fronting is driven by the feature

¹ As we proceed, we will find cases where the internal structure of the fronted VP is not as simple as that of the transitive verb doubling configurations we see here. For example, verb doubling can be built upon ditransitives, whose internal structure consists of more than one object. Furthermore, in some extremely complex cases, verbal dependents that we label as adverbials can also be contained in the fronted verbal phrase. These more complicated configurations will be discussed in detail in section 4.

² Throughout this thesis, I adopt the assumption that vP is a phase, and that movement out of the vP phase has to go through its edge (see van Urk 2020 for an overview). This requirement also applies to verbal phrasal movement here.

[*u*Top] in the left periphery for the sake of information structure. There is no evidence to show that these two positions are correlated in terms of movement: specifically, if the head C triggers VP-fronting to its specifier, the verbal phrase does not need to move to the clause-medial Spec,FP first to land into the final position. In other words, VP-fronting to either position is triggered independently of the other.

2.2 Diagnostics for phrasal movement

Evidence in favour of a phrasal movement analysis includes a language-specific diagnostic, i.e., *lexical identity effects*, and the conventional movement tests, namely, *island effects* and *reconstruction effects*. Let me start off with the lexical identity effects in what follows. In Mandarin Chinese, it is possible for two words to be interchangeable in many occasions, if they share the same lexical meaning but not the same form.³ As exemplified in (23), while *kàn* ‘read’ is used in the question, the answer employs *dù*, a different word form with (nearly) the same meaning ‘read’, and is still considered a licit response.

(23) *Interchangeable lexical items*

- Q: Nǐ zài **kàn** shénme ya?
 2SG PROG read what SFP
 ‘What are you reading?’
 A: Wǒ zài **dù** Sāntǐ ne.
 1SG PROG read TBP SFP
 ‘I am reading the *Three-Body Problem*.’

However, verb doubling is one of those occasions in which the interchange of words with same meaning is prohibited, as given in (24). In either low (24a) or high verb doubling (24b), *kàn* and *dù* are unable to form a grammatical doubling construction. Instead, doubled verbs have to be identical (that is, *kàn* occurs twice in the structure), leading to the lexical identity effects.

(24) *Lexical identity effects in verb doubling*

- a. Piet **kàn** nà-běn xiǎoshuō **kàn**/***dù** de hěn kuài.
 Piet read that-CL novel read DE very quickly
 ‘Piet reads the novel quickly.’

³ Faruk Akkuş (pers.comm.) wonders whether there exists a subset-superset relation among the lexically-identical words, and whether such a relation would affect verb doubling. As far as we can tell, at least with respect to the examples concerned here, it is not clear if there exists such a relationship between *kàn* and *dù*, but swapping the positions of these two words would not change the results given in (23) and (24).

- b. **Kàn** nà-běn xiǎoshuō Piet **kàn**/***dú** de hěn kuài.
 read that-CL novel Piet read DE very quickly
 ‘As for reading the novel, Piet reads it quickly.’

Under the copy theory of movement (e.g., Chomsky, 1993, 1995; Nunes, 1995), such effects can receive a direct explanation, namely, both verbs are actually VP-copies, and one of them undergoes phrasal movement with the direct object to some higher position in the clausal spine. In other words, the higher VP-copy gets spelled out completely. I will come back to the spell-out of the lower VP-copy in chapter 5.

Further evidence for a VP-movement analysis for verb doubling comes from island and reconstruction effects. In (25a) and (26a), I construct two canonical structures without verb doubling, and put the VP *kàn xiǎoshuō* ‘read novels’ in a *wh*-island and a complex NP-island respectively. As exemplified in (25b) and (26b), low verb doubling out of islands is impossible. In both (25c) and (26c), high verb doubling out of islands also results in ungrammatical sentences. The results shown here are expected if verb doubling involves VP-movement.⁴

(25) *Wh-islands for high verb doubling*

- a. Wassily xiǎng zhīdào [wèishéme Piet **kàn xiǎoshuō** hěn kuài].
 Wassily want know why Piet read novel very quickly
 ‘Wassily wonders why Piet reads novels quickly.’
- b. *Wassily **kàn xiǎoshuō** xiǎng zhīdào [wèishéme Piet kàn de hěn kuài].
 Wassily read novel want know why Piet read DE very quickly
 Int.: ‘Wassily wonders why Piet reads novels quickly.’
- c. ??**Kàn xiǎoshuō** Wassily xiǎng zhīdào [wèishéme Piet kàn de hěn kuài].
 read novel Wassily want know why Piet read DE very quickly
 Int.: ‘As for reading novels, Wassily wonders why Piet reads them quickly.’

(26) *Complex NP-islands for high verb doubling*

- a. Wassily jùjué [Piet **kàn xiǎoshuō** hěn kuài de shuōfǎ].
 Wassily reject Piet read novel very quickly DE_{nom} claim.
 ‘Wassily rejects the claim that Piet reads novels quickly.’

⁴ As will be seen in chapter 6, long-distance VP-movement to the matrix clause-medial position is generally restricted. It is likely that both ungrammatical examples of low verb doubling (25b) and (26b) can be ruled out independently of moving across islands (see a similar line of reasoning in Lai 2021, 2024). However, if that was the case, we would miss a uniform account of all the ungrammatical configurations presented here, among which those high verb doubling ones are clearly affected by islandhood. Therefore, I do not entertain the alternative account here.

- b. *Wassily **kàn xiǎoshuō** jùjué [Piet kàn de hěn kuài de *shuōfǎ*].
 Wassily read novel reject Piet read DE very quickly DE_{nom} claim.
 Int.: ‘Wassily rejects the claim that Piet reads novels quickly.’
- c. ***Kàn xiǎoshuō** Wassily jùjué [Piet kàn de hěn kuài de *shuōfǎ*].
 read novel Wassily reject Piet read DE very quickly DE_{nom} claim.
 Int.: ‘As for reading novels, Wassily rejects the claim that Piet reads them quickly.’

Moving on to reconstruction effects, the examples of high verb doubling in (27-29) all support the VP-movement analysis. In (27b), the VP *huà tāzìjǐ-de-xiàoxiàng* ‘draw the portrait of himself_i’ has to reconstruct to the lower thematic domain so that the reflexive can be bound by *méi-rén_i* ‘nobody_i’.⁵ Similarly in (28b), obligatory VP-reconstruction means that an instance of *Piet_i* is c-commanded by the pronoun *tā_i*, thereby violating Principle C. The example in (29b) shows that ungrammaticality only results if the pronoun c-commands the lower VP-copy. Further embedding of the pronoun removes the Principle C violation.

(27) *Principle A reconstruction effects in verb doubling*

- a. Méi-rén_i **huà tāzìjǐ-de-xiàoxiàng** huà de hěn kuài.
 NEG-people draw oneself-POSS-portrait draw DE very quickly
 ‘No one_i draws the portrait of himself_i quickly.’
- b. **Huà tāzìjǐ-de-xiàoxiàng** méi-rén_i huà de hěn kuài.
 draw oneself-POSS-portrait NEG-people draw DE very quickly
 ‘No one_i draws the portrait of himself_i quickly.’

(28) *Principle C reconstruction effects in verb doubling*

- a. *Tā_i **huà Piet_i-de-xiàoxiàng** huà de hěn kuài.
 3SG draw Piet-POSS-portrait draw DE very quickly
 *‘He_i draws the portrait of Piet_i quickly.’
- b. ***Huà Piet_i-de-xiàoxiàng** tā_i huà de hěn kuài.
 draw Piet-POSS-portrait 3SG draw DE very quickly
 *‘He_i draws the portrait of Piet_i quickly.’

(29) *Removal of Principle C violation in verb doubling*

- a. Tā_i-de-māmā **huà Piet_i-de-xiàoxiàng** huà de hěn kuài.
 3SG-POSS-mother draw Piet-POSS-portrait draw DE very quickly
 ‘His_i mother draws the portrait of Piet_i quickly.’

⁵ Here I ignore the problem of whether *méi-rén* is really a constituent that can function as a subject in Mandarin Chinese. See footnote 8 for relevant discussion.

- b. **Huà Piet_i-de-xiàoxiàng** tā_i-de-māmā huà de hěn kuài.
 draw Piet-POSS-portrait 3SG-POSS-mother draw DE very quickly
 ‘His_i mother draws the portrait of Piet_i quickly.’

All the low verb doubling constructions in (27a), (28a) and (29a) show the same results as their counterparts. Nonetheless, since the doubled VP does not move across its binding antecedent, the reconstruction diagnostics are not directly supportive to suggest a phrasal movement approach to low verb doubling.⁶

The hypothesis I pursue is that both low and high verb doubling are constructed by the same mechanism of moving the VP to some higher positions. Though the conventional movement diagnostics given above do not provide direct arguments for the low case, the overall distribution of verb doubling shows that both constructions are subject to identical well-formedness conditions. One of them has already been shown in (17) and (18) in the previous section, that is, the V₁-object unit must undergo fronting as a complete verbal phrase. Lai (2021, 2024) further demonstrates that, without the postverbal adverbial, neither low nor high verb doubling can be constructed, as illustrated in (30).

- (30) a. Piet kàn nà-běn xiǎoshuō kàn *(de hěn kuài).
 Piet read that-CL novel read DE very quickly
 Int.: ‘Piet reads the novel quickly.’
 b. Kàn nà-běn xiǎoshuō Piet kàn *(de hěn kuài).
 read that-CL novel Piet read DE very quickly
 Int.: ‘As for reading the novel, Piet reads it.’

This parallel suggests that both kinds of verb doubling do not differ from each other with respect to their syntactic derivations. Considering that there exists compelling evidence in favour of the movement approach to high verb doubling, I conclude that low verb doubling displays no fundamental difference from its high counterpart in this respect.

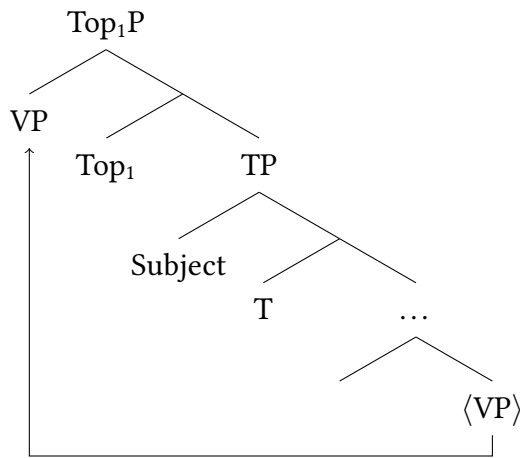
2.3 Alternative account of deriving low/high verb doubling

In the previous two sections, I have outlined my approach to deriving low and high verb doubling, which relies on the idea that there are two landing sites in the clausal spine available for hosting VP-fronting. For the sake of exposition, I refer to this line of analysis as the two-landing-site ap-

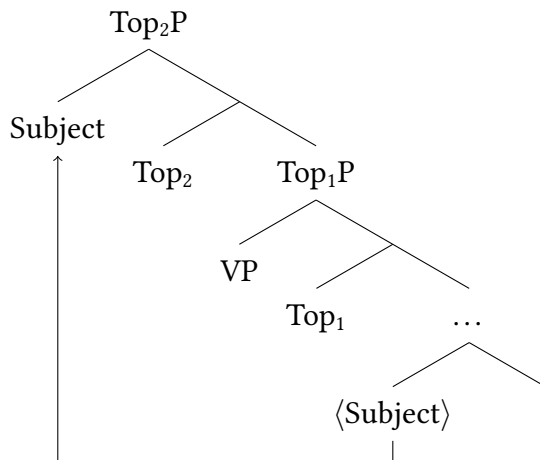
⁶ The fact that low verb doubling does not receive enough direct support from movement diagnostics (compared to its high counterpart) leads to a few non-movement analyses. See Huang (1982, 1992); Tang (1990); Paul (2002a); Fang and Sells (2007); Hsu (2008, 2013), among others.

proach (see also Lai 2021, 2024). In this section, I would like to argue against a different, potential account of deriving the two positions of verb doubling, before proceeding to investigation into their other properties. Suppose that the clausal left periphery in Mandarin Chinese contains multiple topics, as proposed by Rizzi (1997). An alternative way of capturing the distinction between high and low verb doubling is to assume that there is only one topic position in the left periphery which is able to host VP-fronting and lead to high verb doubling, and that the *prima facie* ‘low verb doubling’ is in effect derived by some further topicalisation of the subject across the fronted VP. Such a derivation is illustrated sequentially in (31a) and (31b): first, the VP moves across the subject which is already in Spec,TP, and lands in the specifier of a relatively lower topic phrase, i.e., Spec,Top₁P in (31a); then, the subject undergoes further movement from Spec,TP to another topic phrase which is higher than where the fronted VP is hosted, i.e., Spec,Top₂P in (31b). I refer to this hypothetical derivation as the fronting-plus-topicalisation approach.

- (31) a. *Step 1: VP-fronting to the left periphery across the subject*



- b. *Step 2: Further topicalisation of the subject across the fronted VP*



There are at least two reasons to believe that this fronting-plus-topicalisation analysis is seemingly plausible as well as feasible. The first one is concerned with verb doubling across many other languages. It has been reported in a lot of languages that verb doubling exists (which is more commonly termed *predicate fronting* in the relevant literature), including Haitian (Harbour 1999), Classical Hebrew (Harbour 1999), Modern Hebrew (Landau 2006), Russian (Abels 2001), Spanish (Vicente 2009), Yiddish (Cable 2004; Bleaman 2022), and so on (see Hein 2017, 2018, 2020 for a comprehensive typological survey of this phenomenon). Two examples are provided as follows, respectively from Spanish (32a) and Yiddish (32b).

- (32) a. **Leer** el libro, Juan lo ha **leído**.
 read-INF the book Juan CLITIC have.3SG read
 ‘As for reading the book, Juan has indeed read it.’
 (Vicente 2009:167, (10a); Spanish; emphasis mine; glossing adjusted)
- b. **red-n** mame-loshn **red** ikh.
 speak-INF mama-language speak.1SG I
 ‘As for speaking Yiddish, I speak it.’ (Bleaman 2022:394, (2); Yiddish; emphasis mine)

The first occurrence of the doubled verb, with its following object, occurs in the clause-initial position in both examples, namely, *leer el libro* ‘reading the book’ in (32a) and *red-n mame-loshn* ‘speaking Yiddish’ in (32b). Interpretation-wise, the fronted verbal phrase is topicalised in either sentence, as suggested by their translation. These predicate cleft constructions are clearly what I define as high verb doubling. As far as I am aware, so far no studies have reported that these verb doubling languages also have constructions that I analyse as low verb doubling. If verb doubling is indeed a widespread syntactic phenomenon across languages, it is a bit surprising that low and high verb doubling are not equally distributed in every language. Since only the topic position in the left periphery seems cross-linguistically available, this implies that the post-subject position of the so-called ‘low verb doubling’ is just an illusion, which has to be derived by some other mechanism.

The second reason for pursuing the fronting-plus-topicalisation analysis is related to one of the specific idiosyncrasies that Mandarin Chinese has, which can provide a direct solution to the ‘low position illusion’ just mentioned above. Mandarin has been long taken as a representative of topic-prominent languages (Xu 2015:393). Subject in this language, for sure, can undergo movement to a topic position, which opens up the possibility that the fronted verbal phrase appears to be following the subject in a clause-medial position because of the further movement of subject to a higher topic. One relevant example which displays subject’s movement to a topic is given below.

If so, according to the fronting-plus-topicalisation approach (31), one may predict that the fronted VP would never occur in the post-subject position if the subject was the *wh*-word *shéi*, because the *wh*-subject would be unable to undergo topicalisation across the fronted VP. However, this is contrary to what we see in (35a), where a post-*wh*-subject verb doubling construction is perfectly grammatical. (35b) further demonstrates that, once the *wh*-subject in this verb doubling precedes the overt topic marker *ne*, the ungrammaticality arises. These results strongly support the idea that the post-subject verb doubling is derived from VP-fronting that directly targets a position lower than the subject, rather than further topicalisation of the subject.

- (35) a. Shéi rènshí zhè-ge rén rènshí-le wǔ nián?
 who know this-CL person know-PFV five year
 ‘Who has known this person for five years?’ (Lai 2021:59, (119))
- b. *Shéi ne, rènshí zhè-ge rén rènshí-le wǔ nián?
 who TOP know this-CL person know-PFV five year

Lai (2021) also uses examples like (36) and (37) that contain *méiyǒurén* ‘nobody’ to support the conclusion just sketched above. As with Ko (2005:886), Lai regards *méiyǒurén* as a single constituent that can function as the subject (cf. Soh 2005). The reasoning behind (36) and (37) is identical to what I have provided for (34) and (35): (36) shows that *méiyǒurén*, analysed as a subject, resists topicalisation; however, we can still find examples where *méiyǒurén* is followed by a fronted verbal phrase, such as (37), which, once again, argues against the fronting-plus-topicalisation derivation sketched in (31).⁸

- (36) a. Méiyǒurén kàn-le nà-běn shū.
 nobody read-PFV that-CL book
 ‘Nobody read that book.’ (Lai 2021:58, (116a))
- b. *Méiyǒurén ne, kàn-le nà-běn shū.
 nobody TOP read-PFV that-CL book
 (Lai 2021:58, (116b))
- (37) a. Méiyǒurén kàn nà-běn shū kàn-le liǎng cì.
 nobody read that-CL book read-PFV two time
 ‘Nobody read that book twice.’ (Lai 2021:59, (118))

⁸ Note that it is actually controversial to assume that *méiyǒurén* is a nominal projection equivalent to its English counterpart *nobody*. Paul (2021b:281) argues that *méiyǒurén* (together with many other similar nominal-like expressions) is essentially a full-fledged proposition “involving the existential verb *yǒu* ‘have, exist’ preceded by negation ... whose unique internal argument is merged *vP*-internally.” There are good reasons to believe that Paul’s analysis is on the right track (e.g., *méiyǒurén* cannot function as an object in any case, which is unexpected if it is indeed a nominal constituent on its own). See Paul (2021b) for more details.

- b. *Méiyǒurén ne, kàn nà-běn shū kàn-le liǎng cì.
nobody TOP read that-CL book read-PFV two time

2.3.2 Long-distance locality profile of verb doubling

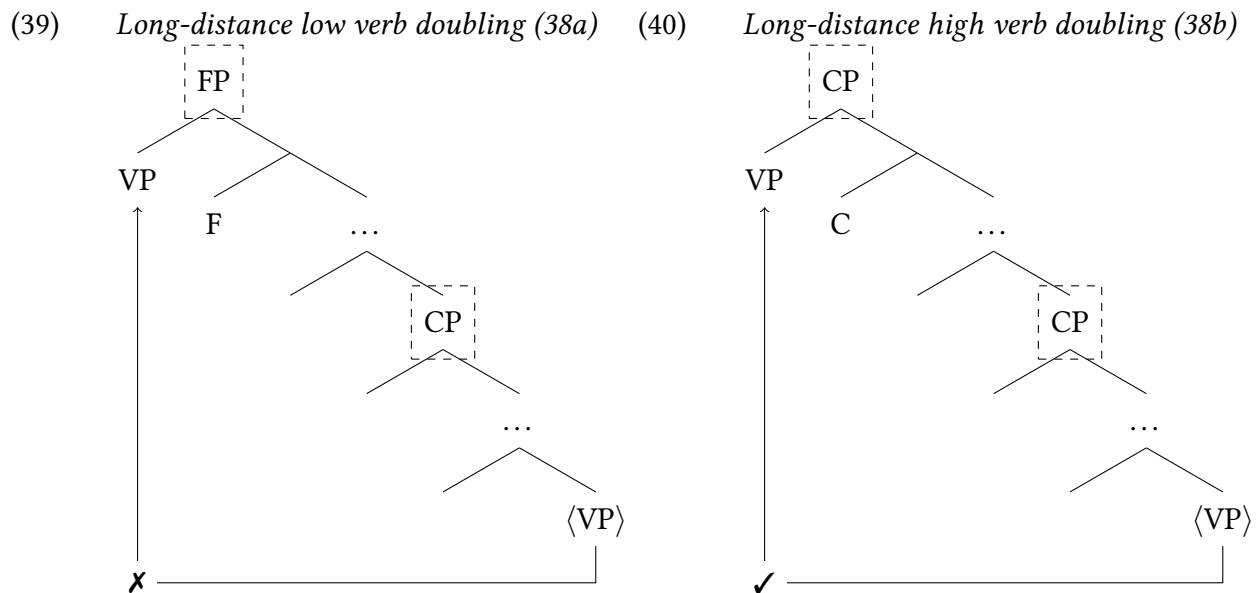
The second argument against the further-topicalisation approach concerns the different locality behaviour that these two types of verb doubling constructions exhibit. Consider (38a) first: this is a grammatical configuration where the verb in the fronted VP *qí nèi-pǐ mǎ* ‘ride the horse’ shows the doubling effect with the embedded predicate *qí* ‘ride’ instead of the matrix one *shuō* ‘say’. Regardless of the approach one takes, it must be recognised by both of them that the verbal phrase *qí nèi-pǐ mǎ* undergoes long-distance movement from the embedded clause to the left periphery of the matrix clause in this scenario. An interesting contrast, then, arises in (38b): unlike the previous example in which the fronted VP precedes the subject, here the VP occurs postverbally in the matrix clause; as a result, the whole sentence becomes ungrammatical.

- (38) a. *Qí nèi-pǐ mǎ, Zhāngsān shuō Lǐsì qí-le liǎng-gè xiǎoshí.*
ride that-sc cl horse Zhangsan say Lisi ride-PFV two-CL hour
‘Zhangsan said that Lisi rode the horse for two hours.’ (Lai 2024:690, (63))
- b. **Zhāngsān qí nèi-pǐ mǎ shuō Lǐsì qí-le liǎng-gè xiǎoshí.*
Zhangsan ride that-sc cl horse say Lisi ride-PFV two-CL hour
Int.: ‘Zhangsan said that Lisi rode the horse for two hours.’ (Lai 2024:690, (64))

As discussed by Lai (2024:690), the above contrast would be mysterious if the fronted VPs in both examples took up the same position, and it was in fact the matrix subject that had undergone further movement in (38b), namely under the fronting-plus-topicalisation approach. Let me make this point a bit more explicit here: since both VP-fronting and topicalisation are seen to be able to undergo long-distance movement, it is fair to assume that both of them are some sort of $\bar{A}$ -movement. If one would like to argue for the fronting-plus-topicalisation approach, then they might need to attribute the ungrammaticality of (38b) to, e.g., some intervention effect induced by the fronted VP, given that the topicalisation of the subject is hypothesised to happen after VP-fronting (recall (31)). However, this kind of analysis cannot hold under any circumstances, because it would lead to a more severe problem that why further topicalisation of the subject across the fronted VP is even possible in the canonical mono-clausal context in the first place. This result leaves us with no choice but to abandon the fronting-plus-topicalisation approach to deriving the high/low distinction among verb doubling constructions.

How does the other approach account for the locality contrast in (38), then? The result above clearly has nothing to say in favour of the two-landing-site approach. In Yan and Meadows (2025),

we characterise the ungrammatical (38b) as an instance of illicit VP-movement and term it *improper verb doubling*. We argue there that long-distance VP-fronting, as with many other long-distance movements, is subject to a specific locality constraint called the *Williams Cycle* (Williams 1974, 2003, 2011, 2013). Informally speaking, the Williams Cycle prohibits a long-distance moved constituent from landing in a position lower than the size of the clause it moves across (in the sense of clausal spine). According to this constraint, the reason why (38b) is ungrammatical is that the landing site for this long-distance low verb doubling, i.e., Spec,FP, is lower than the size of the embedded clause, which is argued to be a full-fledged CP, as schematised in (39). By contrast, the same ungrammaticality does not show up in (38a) because the landing site of this fronted VP is arguably as high as the CP, as schematised in (40).



The contrast in (38) is just part of the full paradigm of improper verb doubling, which is the topic of chapter 6. I will present more detailed analyses then. What I want to emphasise here is that there is a fully-developed account of the long-distance locality contrast seen above, which is compatible with the two-landing-site approach.

2.3.3 Difference in information structure

Last but not least, the third argument is related to the information structure of these two types of verb doubling constructions, which we have already seen in (20), and has been used as a motivation for proposing the two-landing-site approach. The scenario that displays the information-structural difference between low and high verb doubling is repeated in (41).

(41) *Scenario: One day Pablo walked into the classroom and saw Piet reading Three-Body Problem very quickly. Later Pablo mentioned this to Wassily ...*

- a. (Wǒ fāxiàn) Piet kàn Sāntǐ kàn de tèbié kuài.
 1SG find Piet read TBP read DE particularly quickly
 ‘(I found that) Piet was reading the *Three-Body Problem* very quickly.’
 (low verb doubling: informationally neutral)
- b. #(Wǒ fāxiàn) Kàn Sāntǐ Piet kàn de tèbié kuài.
 1SG find read TBP Piet read DE particularly quickly
 ‘(I found that) As for reading the *Three-Body Problem*, Piet was reading it very quickly.’
 (high verb doubling: informationally marked)

We find that (41b) is informationally marked, where the VP *kàn Sāntǐ* ‘read the *Three-Body Problem*’ occurs before the subject Piet. Since this fronted VP is assumed to move to the left periphery under both approaches, the markedness that (41b) displays does not go beyond our expectation. By comparison, the fact that it is totally acceptable to utter (41a) within the same contexts indicates that this kind of post-subject verb doubling is informationally neutral.

The information structure of (41a) constitutes a problem to the fronting-plus-topicalisation approach. This is because, in principle, the topicalisation of the subject should not change the information structure that has already been established by the first-applied VP-fronting. That is to say, (41a) is predicted to be at least as informationally marked as (41b). Needless to say one would expect to see that the subject is interpreted as a topic under this approach, which is clearly not the case according to the translation above. The two-landing-site approach, however, offers a direct account for the difference seen in (41): since these two types of verb doubling target different positions in the clausal spine, it is reasonable to assume that the fronted VP in high verb doubling enters a domain where it can become informationally marked (i.e., the left periphery), whereas low verb doubling does not involve the same kind of VP-fronting as its high counterpart does, thereby remaining neutral in terms of its information structure.

2.3.4 Interim summary

In this section I have presented three arguments to argue against the alternative approach to the distinction between low and high verb doubling which consists of VP-fronting to the left periphery and further topicalisation of the subject across the fronted VP. More importantly, I have used the same arguments to argue for my proposed analysis that verb doubling derives from verbal phrasal movement that lands in two different positions in the clausal spine. The conclusion that I have come to here will lay the foundation for further discussion of the locality of VP-fronting in chapter 6.

2.4 Broader picture of the clausal architecture in Mandarin

The previous section has shown clear evidence that there is indeed a post-subject position in the clausal spine in Mandarin Chinese that can host verbal predicate fronting. This idea offers a direct account for the fact that, unlike many other doubling languages which only exhibit high verb doubling (or predicate fronting) (recall (32)), Mandarin exhibits not only high verb doubling but also the low one. In this section I provide further evidence in favour of the existence of this clause-medial position. I will show that the distribution of low and high verb doubling is actually compatible with the widely-accepted clausal architecture in Mandarin Chinese: that is, there is a clause-medial domain below TP that parallels the left periphery (à la Belletti 2004), where a variety of phrasal movements are able to move to.

It has been documented in a lot of work on word order in Mandarin Chinese that nominal constituents like objects can be displaced and occur in positions either after or before the subject. Let us have a brief look at three prominent cases. The first one is termed *object shift* (see Ernst and Wang 1995; Shyu 1995; Paul 2002b, 2005; Chen 2023, 2025, among many others), as exemplified in (42). In both sentences, the object *nà-(yī)-běn xiǎoshuō* ‘that novel’ does not show up in its canonical postverbal position; instead, it is shifted to some preverbal position. In (42a), the object is positioned between the subject *Piet* and the verb *kàn-le* ‘read’. In (42b), this nominal is shifted to a position even before the subject.

(42) *Object shift in Mandarin Chinese*

- a. Piet **nà-(yī)-běn xiǎoshuō** *kàn-le* sān biàn.
Piet that-one-CL novel read-PFV three time
‘Piet read THAT NOVEL three times.’
- b. **Nà-(yī)-běn xiǎoshuō** Piet *kàn-le* sān biàn.
that-one-CL novel Piet read-PFV three time
‘As for THAT NOVEL, Piet read it three times.’

The second displacement phenomenon is a language-specific configuration termed *lián ... dōu construction* (see Shyu 1995; Paul 2002b, 2005; Chen 2023, 2025; Cao 2025, among many others), as shown in (43). As with object shift just seen above, here the object *nà-(yī)-běn xiǎoshuō* is also displaced to two preverbal positions, either following or preceding the subject as respectively shown in (43a) and (43b). What is different between these two kinds of displacement is that the displaced object in (43) is accompanied by this *lián ... dōu* construction, whose interpretation is equivalent to ‘even’ in English. In both examples, the occurrence of *lián* is optional, and only shows up before the displaced object. By contrast, the position of *dōu* is obligatorily fixed to

one between the subject and the verbal predicate *kàn-le*. Therefore, even though the *lián ... dōu* construction with the object contained inside looks like a continuous string of words when all these components co-occur in the post-subject position (43a), this configuration can actually be discontinuous when the displaced object occurs before the subject (43b).

(43) *Lián ... dōu construction in Mandarin Chinese*

- a. Piet (**lián**) **nà-(yī)-běn xiǎoshuō dōu** kàn-le sān biàn.
Piet LIAN that-one-CL novel DOU read-PFV three time
‘Piet even read THAT NOVEL three times.’
- b. (**Lián**) **Nà-(yī)-běn xiǎoshuō** Piet **dōu** kàn-le sān biàn.
LIAN that-one-CL novel Piet DOU read-PFV three time
‘As for THAT NOVEL, Piet even read it three times.’

(44) demonstrates the third case of object displacement in Mandarin Chinese, which I refer to as *overt wh-fronting*. It has been well known since Huang (1982) that Mandarin Chinese is a *wh*-in-situ language. However, as noticed by Pan (2011, 2014) and Ren and Yan (to appear), among many, it is not the same case for every *wh*-word in Mandarin that it is unable to undergo overt movement. Unlike bare *wh*-words like *shénme* ‘what’ that are rarely possible to move out of their base position, it is never a problem for more complex *which*-phrases to display overt fronting, as exemplified in (44). In (44a), the complex *which*-phrase *nǎ-(yī)-běn xiǎoshuō* ‘which novel’ stays in its canonical object position and introduces a question. In (44b) and 44c, we can clearly find that the same *which*-phrase is displaced, and once again, shows up in those two familiar positions in the clausal spine, namely, either below or above the subject. The same displacement is found in complex *what*-phrases as well, such as *shénme xiǎoshuō* ‘what novel’. Relevant examples are given in (45).

(44) *Overt wh-fronting in Mandarin Chinese (involving complex which-phrases)*

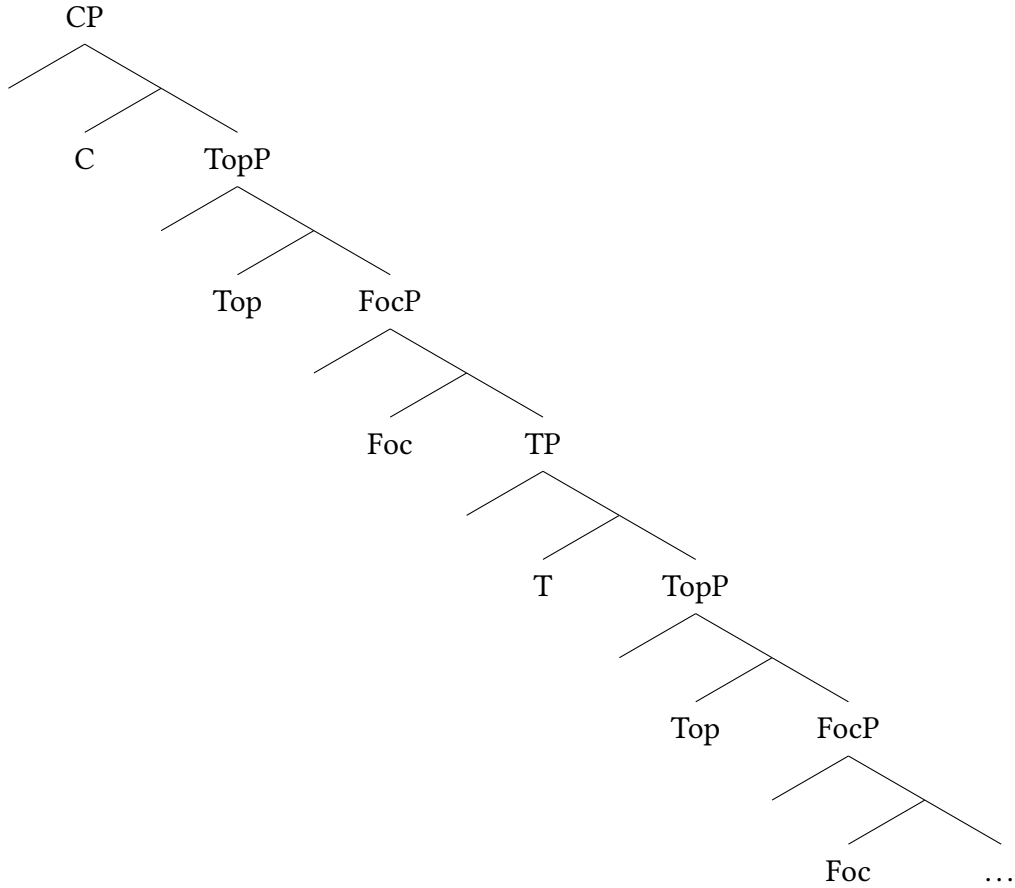
- a. Piet kàn-le **nǎ-(yī)-běn xiǎoshuō** sān biàn?
Piet read-PFV which-one-CL novel three time
‘Which novel did Piet read three times?’
- b. Piet **nǎ-(yī)-běn xiǎoshuō** kàn-le sān biàn?
Piet which-one-CL novel read-PFV three time
‘Which novel did Piet read three times?’
- c. **Nǎ-(yī)-běn xiǎoshuō** Piet kàn-le sān biàn?
that-one-CL novel Piet read-PFV three time
‘Which novel did Piet read three times?’

(45) *Overt wh-fronting in Mandarin Chinese (involving complex what-phrases)*

- a. Piet kàn-le **shénme xiǎoshuō** sān biàn?
Piet read-PFV what novel three time
'What novel did Piet read three times?'
- b. Piet **shénme xiǎoshuō** kàn-le sān biàn?
Piet what novel read-PFV three time
'What novel did Piet read three times?'
- c. **Shénme xiǎoshuō** Piet kàn-le sān biàn?
what novel Piet read-PFV three time
'What novel did Piet read three times?'

The overall distribution of the three displacement phenomena seen above suggests, if the object displacement constantly involves movement, nominal movements in Mandarin can target not only positions in the left periphery, but also those in the clause-medial domain. This is exactly the conclusion that previous literature arrives at: the TP-internal, clause-medial domain in Mandarin Chinese parallels the left periphery in the sense that it contains at least a topic position and a focus one (à la Belletti 2004). Every instance of object displacement in the clause-medial position, thus, is analysed as resulting from phrasal movement triggered by topicalisation (e.g., object shift in (42)), focalisation (e.g., *lián ... dōu* construction in (43)), or either of them (e.g., overt *wh*-fronting in (44)) (see more detailed discussions in references cited above). The corresponding clausal architecture in Mandarin Chinese is schematised in (46) (cf. Paul 2005).

(46) *Clause-initial and medial domains in Mandarin Chinese*



We have seen that, in addition to verb doubling, there are other three kinds of phrasal movement of which the distribution exhibits the low/high distinction, as summarised in (47). This means that the clause-medial, low position is not unique to verb doubling, and that the distribution of verb doubling fits in well with the general clausal architecture in Mandarin Chinese. One remaining question is whether what I propose as the FP (i.e., the clause-medial position used to host low verb doubling) is identical to the TP-internal TopP or FocP. If not, what is the relative hierarchical order among these clause-medial positions in the clausal spine?

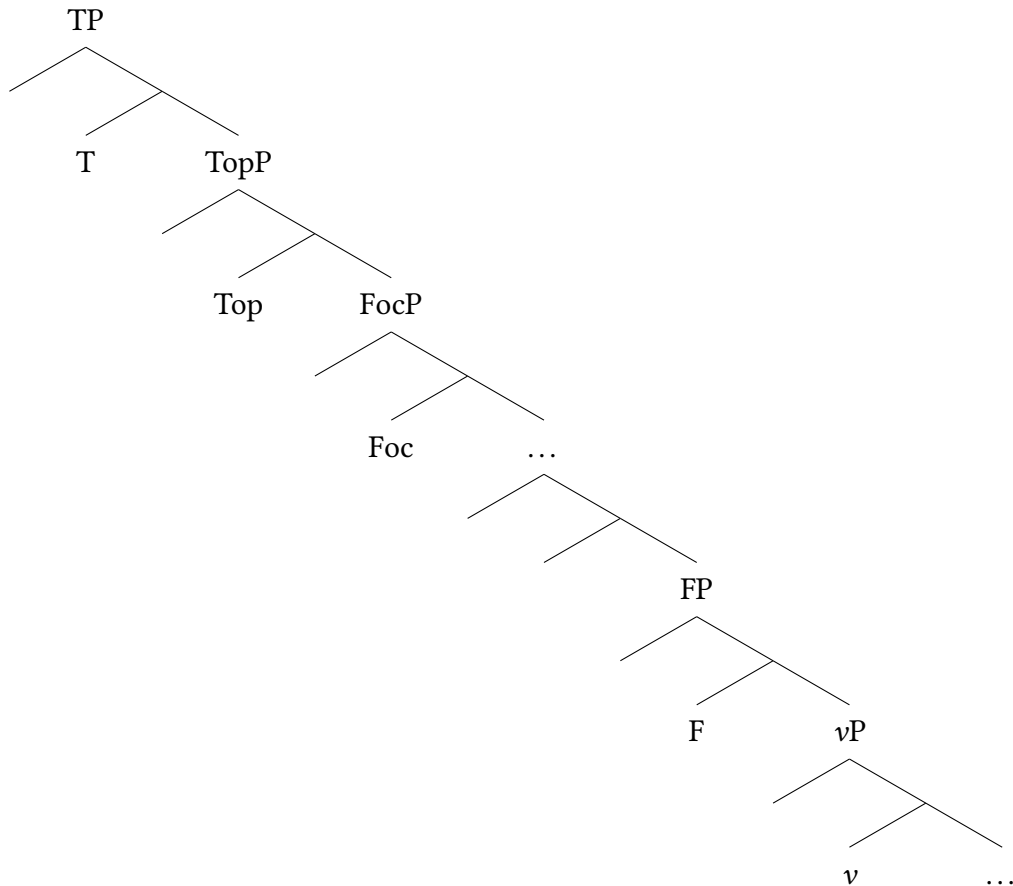
(47) *Landing sites for a variety of phrasal movements in Mandarin Chinese*

Landing sites	Object shift	<i>Lián ... dōu</i>	<i>Wh</i> -fronting	Verb doubling
Clause-medial, low positions	✓	✓	✓	✓
Clause-initial, high positions	✓	✓	✓	✓

For the first question, we can immediately reply that the answer is ‘no’. Recall that one characteristic of low verb doubling is that it is informationally neutral (as demonstrated in (20)), which indicates that the landing site (i.e., Spec,FP) the verbal phrase moves to has nothing to do with

the information structure. Therefore, it is plausible to postulate that FP is not any position related to information structure below the TP. As to the second question, I will leave this discussion till chapter 6, where I will present evidence to support the idea that FP is lower than both TopP and FocP below the TP, as schematised in (48).

(48) *A split clause-medial domain in Mandarin Chinese*



To sum up, in this section I have provided empirical support for the existence of the clause-medial, low landing site where low verb doubling can move to. This further favours the two-landing-site approach that I put forward to accounting for the low and high distinction among verb doubling. In the next chapter I will turn my attention from the clause-initial and medial positions to the field after the second occurrence of the doubled verb, with a particular focus on the role of postverbal adverbials in verb doubling.

Chapter 3

The Role of Adverbials in Verb Doubling

In the previous discussion, we have seen a variety of verb doubling constructions, but not discussed how many types of verb doubling there are in Mandarin Chinese, and what the criterion for differentiating verb doubling is. The general method of classifying verb doubling adopted in most literature is to distinguish verb doubling on the basis of the meaning of adverbial expressions that follow the second occurrence of the doubled verb (viz., the postverbal field) (e.g., Li and Thompson 1981; Tai 1999; Cheng 2017, among many others). I refer to this kind of classification as the *meaning-based taxonomy* of verb doubling, simply illustrated in (49) (with the determining element highlighted in bold, i.e., the postverbal adverbial).¹

- (49) *Meaning-based taxonomy of verb doubling*
Subj [V₁ Obj] [[V₂-Part] **Adverbial**]

The purpose of this section is to discuss the role of adverbials in various verb doubling constructions. In section 3.1 and 3.2, I show that, among the six verb doubling constructions identified by the meaning-based taxonomy, there is a subtle but crucial distinction concerning the configurational relationship between the postverbal adverbial and the verbal projection (in the sense of the conventional X-bar template). In section 3.3, I argue that what is labelled as an adverbial in the postverbal field can actually be analysed as either a freely-adjoined adjunct of the verbal phrase (e.g., VP or ApplP), or an argument selected by some higher verbal head (i.e., the applicative head). In section 3.4, I demonstrate that the dichotomy between adjuncts and arguments makes several correct predictions about co-occurrence of multiple postverbal adverbials, but also brings into a problem that there is a specific ban on multiple adjuncts in the postverbal field. To address the latter, I propose that the nature of this ban lies at the side of PF instead of in narrow syntax.

¹ Although I only use a string of low verb doubling to illustrate the meaning-based taxonomy here, this method of classification also applies to high verb doubling.

3.1 Meaning-based taxonomy of verb doubling

According to the meaning of the postverbal adverbials, traditional literature identifies six types of verb doubling constructions (e.g., Li and Thompson 1981; Tai 1999; Cheng 2017, among many), as listed in (50) (for the sake of convenience, I only present examples of low verb doubling in the following discussion).

- (50) a. Piet dú běnkē dú-le liǎng huí.
Piet study undergraduate study-PFV two time
'Piet studied for an undergraduate degree twice.' (frequency-type)
- b. Piet dú běnkē dú-le wǔ nián.
Piet study undergraduate study-PFV five year
'Piet studied for an undergraduate degree for five years.' (duration-type)
- c. Piet kàn mànhuà kàn de hěn màn.
Piet read comic read DE very slowly
'Piet read comics very slowly.' (manner-type)
- d. Piet kàn mànhuà kàn de hěn yūn.
Piet read comic read DE very dizzy
'Piet read comics and felt dizzy (as a result).' (resultative-type)
- e. Piet xiě-zì xiě-zài-le qiáng-shàng.
Piet write-word write-LOC-PFV wall-up
'Piet wrote on the wall.' (locative-type)
- f. Piet sàn-bù sàn-qù-le Wéiduōliǎ Gōngyuán.
Piet disperse-step disperse-to-PFV Victoria park
'Piet went for a walk to Victoria Park.' (directional-type)

Both (50a) and (50b) contain a temporal expression as their modifying adverbial: in the former case the postverbal position is occupied by a frequency phrase *liǎng huí* 'two times', whereas in the latter case it becomes a duration phrase *wǔ nián* 'five years' that takes up the same position. (50c) shows a different situation in which a manner expression *hěn màn* 'very slowly' shows up to the right of the second occurrence of the verb *kàn* 'read', specifying the way that Piet read comics. A similar configuration is shown in (50d), but here the postverbal expression *hěn yūn* 'very dizzy' is interpreted as a result of Piet reading comics.² (50e) and (50f) respectively demonstrate that

² One crucial difference between manner-type and resultative-type verb doubling constructions is that the adverbials in the latter are not confined to simple adjectival phrases: in the relevant literature (e.g., Cheng 2007; Lai 2021), there are many cases where the resultative particle *de* is followed by clausal constituents. I will not discuss this difference in detail in this work.

the postverbal field can even host a locative expression like *zài qiáng-shàng* ‘on the wall’ or a directional expression like *qù Wéiduōlìyǎ Gōngyuán* ‘to Victoria Park’.³

At first sight there seems to be no significant distinction among these six types of verb doubling except the different meanings of adverbials (abstracting away from some trivial dissimilarities related to particles, which I will return to in chapter 5). However, further investigation will reveal that this is not the case: the configurational relationship between the postverbal adverbial and the verbal projection varies from one to another. To better illustrate this, I will pay particular attention to the locative type (e.g., (50e)) and the directional type (e.g., (50f)) in the following discussion, since neither of them has been seriously taken into consideration in the previous literature, as far as I am aware.

3.2 Postverbal adverbials as adjuncts or arguments

The configurational relationship between the postverbal adverbial and the verbal projection is not an uncontroversial topic in the relevant literature: generally, adverbials are analysed as either adjuncts that can be freely adjoined to any projection (e.g., Ernst 2002), or specifiers of a sequence of dedicated functional projections (in the sense of Cinque 1999, 2004a, 2006; cf. Alexiadou 1997). Correspondingly, the postverbal status of adverbials can result from their being right-adjoined to the modified projection, or from upward movement of constituents that are originally below the projection that hosts the adverbials (see Ernst 2020 for an overview of these two approaches to the syntax of adverbials). It has also been argued by, e.g., Larson (2004, 2014) that postverbal adverbials can be perceived as complements of the lowest V head within the Larsonian VP-shell.

The same divergence can still be found even we just focus on the Mandarin literature. For instance, with respect to the position of frequency and duration phrases, Lai (2021, 2024) argues that they are introduced in the specifier of a functional head that might optionally occur between *v* and *V* in the Cinquean Hierarchy (cf. Gougnet 2006), whereas many other linguists analyse these adverbials as V-projection adjuncts, including Tang (1990, 2001), Soh (1998), Huang et al. (2009), Ernst (2014), H.-L. Huang (2018), Bartos (2019), etc. In addition, the idea of treating frequency and duration phrases as complements to *V* can also be found, e.g., in Huang (1994) and Paul (2017) (though this is not stated explicitly in the latter). As for postverbal manner expressions, they are treated either as right adjoined adjuncts (e.g., Ernst 2014; Meadows and Yan 2023, 2025), or as

³ Here I treat both *zài* and *qù* originally as part of the locative and the directional phrases, which are probably prepositional phrases. The way I analyse them is not uncontroversial, especially for *qù*. For instance, Z. Chen (2023) regards the same word as a directional verb which can form serial verb constructions with other verbal heads (à la Ramchand 2008’s first-phase model). It seems difficult to extend this analysis from *qù* to *zài*, since the latter rarely demonstrates verbal properties. In addition, this analysis will miss the parallelism between these two words, which will be clearer as we proceed. Therefore, I stick to treating both of them as prepositions in the following discussion. At the moment, I remain agnostic towards how they occur in the position between the lexical verb and the aspect marker.

complements taken by the verbal head (e.g., Tang 1990, 2001; Huang et al. 2009; Larson 2018).

For the time being, I follow Ernst (2002, 2014) and assume that in Mandarin Chinese adverbials are generally adjuncts. Evidence in favour of this assumption comes from the distribution of frequency, duration, manner, and resultative expressions. According to Ernst (2014:50), these four are the only adverbials that can occur postverbally in Mandarin (contra Travis 1984 and Paul 2017, who claim that Mandarin adverbs exclusively occur preverbally). Since they are treated as adjuncts, they can co-occur freely with any verbs. The first illustration of this property of adjuncts is those examples shown in (50a)-(50d), where both *dú* ‘study’ and *kàn* ‘read’ can be viewed as *learn* verbs (in the sense of Levin 1993). A further illustration is given in (51) and (52).

- (51) a. Piet *sàn-bù* *sàn-le* ***hǎoduō huí***.
Piet disperse-step disperse-PFV many time
‘Piet went for many walks.’ (frequency-type)
- b. Piet *sàn-bù* *sàn-le* ***bàn-gè-duō zhōngtóu***.
Piet disperse-step disperse-PFV half-CL-more hour
‘Piet went for a walk for more than half an hour.’ (duration-type)

In (51) the doubled verb is *sàn* ‘disperse’, which is previously used in the directional-type of verb doubling (50f). According to Levin (1993)’s taxonomy, this one can be regarded as a *motion* verb. It is clear that both the frequency expression *hǎoduō huí* ‘many times’ in (51a) and the duration expression *bàn-gè-duō zhōngtóu* ‘more than half an hour’ in (51b) have no difficulty in occurring to the right of the second occurrence of *sàn*. This fact is compatible with the idea that both postverbal expressions are freely-adjoined adverbial adjuncts. In other words, they are not selected by the motion verb used here.

- (52) a. Piet *xiě-zì* *xiě* *de* ***hěn kuài***.
Piet write-word write DE very quickly
‘Piet wrote very quickly.’ (manner-type)
- b. Piet *xiě-zì* *xiě* *de* ***hěn lèi***.
Piet write-word write DE very tired
‘Piet wrote and felt tired (as a result).’ (resultative-type)

In (52) the verb that demonstrate the doubling effect is *xiě* ‘write’, a *creation* verb (again, in the sense of Levin 1993) taken from the locative-type (50e). Here we can make the same conclusion about manner and resultative adverbials that they are adjuncts, as *hěn kuài* ‘very quickly’ in (52a) and *hěn lèi* ‘very tired’ in (52b) are both capable of showing up in the postverbal position. Combining (51) with (52), what these examples show together is that there is no selection of these

postverbal adverbials by verbs (no matter what types they are), an expected property of adjuncts.

Now what about locative and directional expressions that occur in the postverbal field? If we maintain the assumption that adverbials are in general adjuncts, we would predict that both kinds of expression should also have no problem with any verbs, just like what frequency, duration, manner, and resultative adverbials have demonstrated in (51) and (52). Nonetheless, this is contrary to the fact. Consider (53) first, where the locative expression *zài chuáng-shàng* ‘on the bed’ is used in a verb doubling construction grounded on *kàn* ‘read’. Neither (53a) nor (53b) is ungrammatical, unlike (50c) and (50d) (both repeated in (54)), where the same verb *kàn* is compatible with both manner and resultative adverbials.

- (53) a. *Piet *kàn* màn huà *kàn-zài-le* **chuáng-shàng**.
 Piet read comic read-LOC-PFV bed-up
 Int.: ‘Piet read comics on the bed.’ (*locative-type)
- b. *Piet *kàn* màn huà *kàn-qù-le* **túshūguǎn**.
 Piet read comic read-to-PFV library
 Int.: ‘Piet went to the library to read comics.’ (*directional-type)
- (54) a. Piet *kàn* màn huà *kàn de* **hěn màn**.
 Piet read comic read DE very slowly
 ‘Piet read comics very slowly.’ (= (50c), manner-type)
- b. Piet *kàn* màn huà *kàn de* **hěn yūn**.
 Piet read comic read DE very dizzy
 ‘Piet read comics and felt dizzy (as a result).’ (= (50d), resultative-type)

One may imagine that the contrast between (53a)/(53b) and (54a)/(54b) could be due to the reason that there might be some general semantic incompatibility between verbs like *kàn* (i.e., *learn* verbs) and adverbials like locatives and directionals that prevents constructions in (53) from being into existence. However, there are counterpart examples (55) which can refute this idea. Here in both constructions (55a) and (55b), the *learn* verb *kàn* is not doubled, and it is fine with locative and directional adverbials that occur preverbally. Considering that the main difference between examples in (53) and in (55) is that adverbials occur postverbally in the former while preverbally in the latter, it is reasonable to narrow down the range of questions, and at least rule out the possibility that it is some purely semantic constraint that stops locative and directional adverbials from being licensed in certain postverbal position.

- (55) a. Piet jīngcháng **zài chuáng-shàng** kàn mànhuà.
 Piet often LOC bed-up read comic
 ‘Piet often reads comics on the bed.’ (preverbal locative adverbial)
- b. Piet jīngcháng **qù túshūguǎn** kàn mànhuà.
 Piet often to library read comic
 ‘Piet often goes to the library to read comics.’ (preverbal directional adverbial)

Similar ungrammatical verb doubling constructions built on locative/directional adverbials are given in (56), where a different *learn* verb *dú* ‘study’ is used. Again, we find that *dú* is unable to form a doubling construction with either the locative expression *zài Lúndūn* ‘in London’ (56a) or the directional expression *qù Yìdàlì* ‘to Italy’ (56b) (though its doubled counterpart does not face the issue of co-occurring with the frequency adverbial (50a) and the duration one (50b), as shown in the repeated (57)).

- (56) a. *Piet *dú* běnkē *dú-zài-le* **Lúndūn**.
 Piet study undergraduate study-LOC-PFV London
 Int.: ‘Piet studied for an undergraduate degree in London.’ (*locative-type)
- b. *Piet *dú* běnkē *dú-qù-le* **Yìdàlì**.
 Piet study undergraduate study-to-PFV Italy
 Int.: ‘Piet went to Italy to study for an undergraduate degree.’ (*directional-type)
- (57) a. Piet *dú* běnkē *dú-le* **liǎng huí**.
 Piet study undergraduate study-PFV two time
 ‘Piet studied for an undergraduate degree twice.’ (= (50a), frequency-type)
- b. Piet *dú* běnkē *dú-le* **wǔ nián**.
 Piet study undergraduate study-PFV five year
 ‘Piet studied for an undergraduate degree for five years.’ (= (50b), duration-type)

Examples in (58), together with (56) provide another piece of evidence for the conclusion that the contrast between (53) and (55) contributes to: as evidenced by both (58a) and (58b), there is no independent, meaning-related constraint on the co-occurrence between *learn* verbs like *dú* and preverbal locative/directional adverbials.

- (58) a. Piet **zài Lúndūn** *dú-le* běnkē.
 Piet LOC London study-PFV undergraduate
 ‘Piet studied for an undergraduate degree in London.’ (preverbal locative adverbial)

- b. Piet **qù Yìdàlì** *dú-le* běnkē.
Piet to Italy study-PFV undergraduate
'Piet went to Italy to study for an undergraduate degree.'
(preverbal directional adverbial)

What underlies the ungrammaticality in both (53) and (56), then, should be attributed to some structural (rather than semantic) requirement of verbs: recall that in (50e) and (50f), the locative and directional adverbials are licensed in verb doubling constructions respectively built upon a *creation* verb that requires a location (i.e., *xiě*) and a *motion* verb (i.e., *sàn*). That is to say, as far as locative and directional expressions are concerned, they, as postverbal adverbials, are only subject to structural-building requirement of certain types of verbs but not any kinds. This consequence is totally different from what we have seen above, that is, the other four adverbials do not behave identically in terms of showing up postverbally.

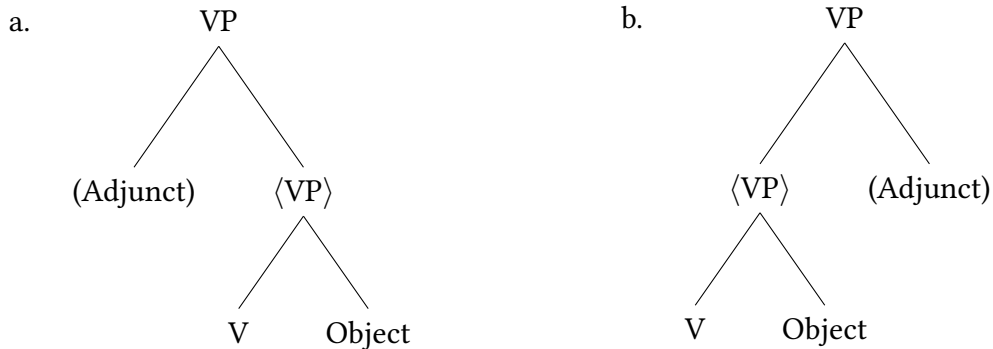
We have seen that there appears a further division among the six types of postverbal adverbials identified by the meaning-based taxonomy of verb doubling: frequency, duration, manner, and resultative expressions, as one group, can freely occur in the postverbal field, whereas locative and directional expressions, as another group, are constrained to some extent when they are present postverbally. I have already mentioned that the unconstrained occurrence of the first four adverbials suggests that these four can be understood as adverbial adjuncts, and that they are not selected by the doubled verbs (cf. Ernst 2014). Along this line of reasoning, I argue that the constrained locative and directional adverbials in verb doubling constructions, in effect, are arguments selected by verbs. To put it differently, I agree with the literature that adverbials that *prima facie* occur in the same postverbal field should not be analysed indiscriminatingly, but I disagree that the right cut should be made between specifiers, complements, and adjuncts. The distinction between arguments and adjuncts, instead, provides a better and more direct account for all the contrasts seen above: in (53) and (56), neither of the *learn* verbs *kàn* nor *dú* is able to select postverbal locative and directional expressions as complements, thereby rendering these examples ungrammatical. In contrast, the same constituents in (55) and (58) function as adjuncts when they occur preverbally; thus there is no selection between the preverbal adjuncts and verbs, and no ungrammaticality arises.⁴ In section 3.4, I will further show how this argument/adjunct dichotomy draws support from co-occurrence of multiple postverbal adverbials.

⁴ One may wonder how constituents switch between arguments and adjuncts on the basis of their position relative to the verb. This is just one property of Mandarin syntax that I take for granted: as noted by Ernst (2014:50), “even though Mandarin is head-initial in terms of its arguments ... it acts more like a head-final language with respect to adjuncts”.

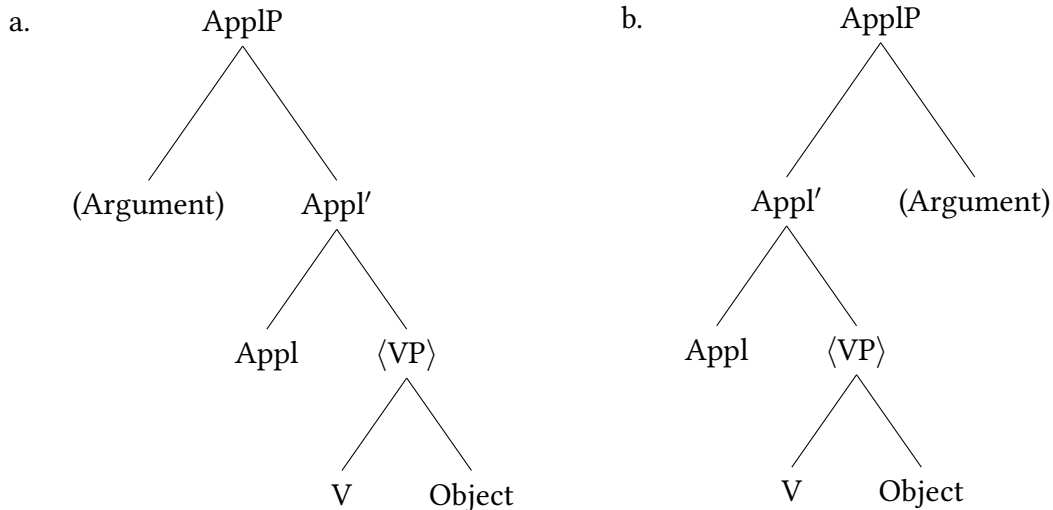
3.3 The adjunct/argument dichotomy and verbal projections

We now need two configurations which can reflect the different relations between these two groups of adverbials and the verbal projection, on one hand, and capture the dichotomy between adjuncts and arguments, on the other hand. Following Janke and Neeleman (2012) and Abels (2024) (also see Abels and Neeleman 2012), I propose that verbal projections are not constantly rightward descending in Mandarin Chinese: not only can verbal projections be rightward ascending (in the case of adjunction), but also can introduce arguments via a rightward specifier. The two kinds of verbal projections that I propose are respectively illustrated in (59) and (60).

(59) *Adverbial as a non-selected adjunct*



(60) *Adverbial as a selected argument*

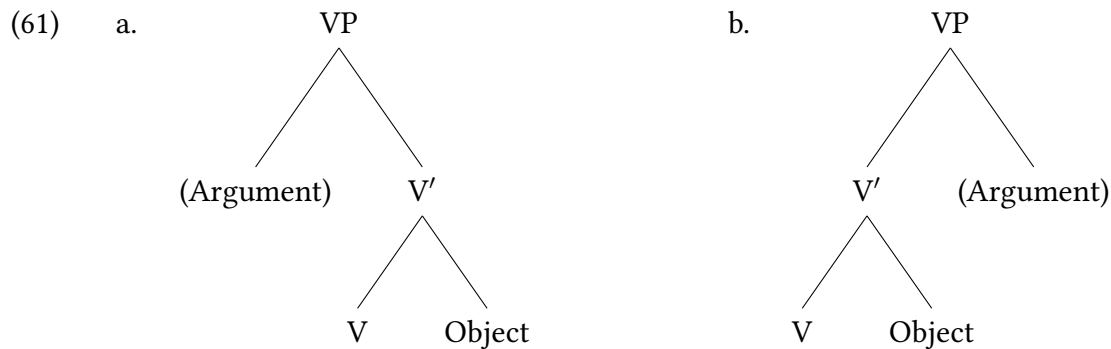


(59) shows a situation in which adverbials function as V-projection adjuncts. (60) represents a different scenario where adverbials are selected as arguments by the applicative head. In each verbal projection, there are two positions for introducing adjuncts and arguments. As seen in (59), adverbial adjuncts can be freely adjoined to either side of the VP; while in (60), arguments are introduced into the verbal projection by either a leftward or a rightward specifier. Here I

remain agnostic towards the precise positions of the adverbial adjuncts and arguments discussed above, as eventually it is just a matter concerned with empirical facts. What really matters is that there is a selectional relation between the argument and the verb in (60), but not between the adjunct and the verb in (59). Following Adger (2003:105-115), I posit that in the case of introducing arguments (60), the Appl head bears one and only one argument-related feature, which can get checked (or deleted) by the merger of the argument in the specifier position. This results in a configuration where the node of the argument is the sister of the Appl'. In contrast, there is no similar feature trigger in the case of adjunction (59), and the sister of the adjunct is always the maximal verbal projection. This explains why adverbial adjuncts seem to be less constrained than their argument counterparts.⁵

I further propose that the fronted verbal phrase involved in verb doubling is initially part of the multi-layered verbal projection proposed here, which gets triggered by some feature to move as the derivation proceeds. In each configuration illustrated above (and below), the lowest VP is contained in angle brackets, which indicates that it is the one that undergoes predicate fronting and ultimately gets spelled out as the V_1 in verb doubling. I will talk more about the relationship between the fronted verbal phrase and the proposed verbal projections here in chapter 4.

Before moving to the next section, I would like to make some brief remarks on my proposed applicative structure (60). Even though this structure can correctly reflect the selectional relationship between the argument and the verb, one may wonder why the high applicative projection (in the sense of Pylkkänen 2002, 2008) is needed in the first place. An alternative structure like (61) can also maintain the selectional relation without introducing a higher verbal projection: that is, the argument is introduced within the VP, and is the sister of the V'.



I have two reasons to adopt the applicative structure in (60) instead of the alternative one just sketched above. The first reason is concerned with the literature on applicatives. As sum-

⁵ Exploring any elaborate implementation of adjunction will be beyond the scope of this work, therefore I do not commit myself to any of them. I refer interested readers to various endeavours carried out at different stages of the Minimalist Program, e.g., the early pair-merge idea in Rubin (2003) and Chomsky (2004), a separate basic operation Adjoin (different from Merge) in Adger (2003), or the latest multi-dimensional approach in Pan and Du (2024).

marised by McGinnis (2008:1225-1126), “(a)n applicative is a syntactic element adding an extra, ‘non-core’ object to a clause”, whose thematic categories “include benefactive, malefactive, recipients, sources, locatives, and instrumentals, among others” (see also McGinnis 2001 and Pylkkänen 2002, 2008). Recall that among the six types of postverbal adverbials, I only analyse locative and directional phrases as arguments. Relevant examples are repeated in (62) below.

- (62) a. Piet xiě-zì xiě-zài-le **qiáng-shàng**.
 Piet write-word write-LOC-PFV wall-up
 ‘Piet wrote on the wall.’ (= (50e), locative-type)
- b. Piet sǎn-bù sǎn-qù-le **Wéiduōlìyǎ Gōngyuán**.
 Piet disperse-step disperse-to-PFV Victoria park
 ‘Piet went for a walk to Victoria Park.’ (= (50f), directional-type)

Since both kinds of phrases can be understood as ‘non-core’ elements to some extent, and one of them (i.e., locatives) has been used as a thematic category of applicatives in other languages (e.g., in the Bantu language Kinyarwanda; see relevant examples in McGinnis 2001, 2004), it is plausible to assume that in Mandarin Chinese, the locative and directional phrases can be introduced into the verbal projection as applied arguments as well, thereby requiring the presence of the applicative projection.

The second reason is more theoretically-driven: in section 2.1, I proposed that forming verb doubling involves verbal phrasal movement to two landing sites in the clausal spine. What I mean by ‘verbal phrasal movement’ is the maximal projection of the verbal head, such as VP or ApplP, rather than the intermediate projection, such as V’ or Appl’. If this idea is adopted, then only the applicative structure in (60) can correctly result in the distribution of the locative and directional phrases in (62): the VP that contains the object is fronted to some higher position, while the argument inside the ApplP still stays in the postverbal field.

By contrast, the alternative structure in (61) cannot achieve the same result, since the arguments would be pied-piped by the VP-fronting, as schematised in (63). This movement *only* leads to ungrammatical sentences as shown in (64) (examples of high verb doubling are omitted).⁶

- (63) *Illicit string of verb doubling: Argument (=Adverbial) pied-piped by the V-object unit*
- a. *Subj [[V₁ Obj] **Adverbial**] [V₂ Part]
- b. *[[V₁ Obj] **Adverbial**] Subj [V₂ Part]

⁶ Note that, even if we adopt the applicative structure in (60), (64) still constitutes a problem: since the ApplP is also a maximal verbal projection, this can undergo verbal phrasal movement as well under my proposed analysis, which gives rise to the same ungrammatical result. The reason why only the VP instead of the ApplP can be triggered to move here will become clearer in section 4.3.

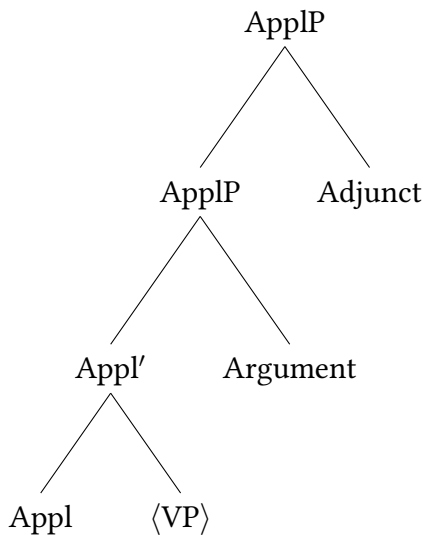
- (64) a. *Piet xiě-zì zài qiáng-shàng xiě-le.
 Piet write-word LOC wall-up write-PFV
 Int.: 'Piet wrote on the wall.' (cf. (62a))
- b. *Piet sàn-bù qù Wéiduōliǎ Gōngyuán sàn-le.
 Piet disperse-step to Victoria park disperse-to-PFV
 Int.: 'Piet went for a walk to Victoria Park.' (cf. (62b))

It is obvious that my selection of the structure here relies heavily on this idea that only the maximal projection of the verbal head can be the target for movement. At this point there is nothing that prevents us from assuming that the verbal phrasal movement can also target the intermediate projection of the VP, thereby adopting the alternative structure (61). In section 4.1, I will present examples of verb doubling built upon ditransitives, which will provide further support for my proposed analysis and verbal structures.

3.4 Co-occurrence of multiple postverbal adverbials

Distinguishing adjuncts from arguments leads us to make several correct predictions about co-occurrence of multiple adverbials in the postverbal field. The first one is that an argument and an adjunct are able to co-occur postverbally, as illustrated in (65) (for the sake of simplicity, I only use rightward ascending verbal projections to illustrate the predicted configurations throughout this section).

- (65) *Possible co-occurrence of an argument and an adjunct*



As exemplified in (66a), the postverbal filed of this verb doubling construction is filled by the directional argument *qù Wéiduōliǎ Gōngyuán* and the frequency adjunct *liǎng huí* at the same time, and the sentence turns out to be grammatical as expected. In (66b), it becomes the duration adjunct *bàn-gè-duō zhōngtóu* that co-occurs with the same directional argument. The judgement here is a bit degraded compared to (66a), but still seemingly acceptable.⁷

- (66) a. Piet *sàn-bù* *sàn-qù-le* **Wéiduōliǎ Gōngyuán liǎng huí.**
 Piet disperse-step disperse-to-PFV Victoria park two time
 ‘Piet went for a walk to Victoria Park twice.’
 (directional argument+frequency adjunct)
- b. %Piet *sàn-bù* *sàn-qù-le* **Wéiduōliǎ Gōngyuán bàn-gè-duō zhōngtóu.**
 Piet disperse-step disperse-to-PFV Victoria park half-CL-more
 hour
 ‘Piet went for a walk to Victoria Park for more than half an hour.’
 (%directional argument+duration adjunct)

Another example is given in (67): the locative argument *zài qiáng-shàng* is combined with the frequency adjunct *liǎng huí* in (67a), and with the duration adjunct *bàn-gè-duō zhōngtóu* in (67b). The judgement of the former is pretty close to that of (66b), which I still consider acceptable, but the latter is certainly ungrammatical.⁸ A plausible account of the contrast here might be relevant to the event structure of the verbal phrase. It seems that the locative argument puts an endpoint to the event of writing: once the word is written onto the wall, the whole event can be considered a completed one. There is nothing that prevents this event from being done again and again, thus it is possible to be modified by the frequency adjunct as in (67a). By contrast, the same completed event cannot be further modified by the duration adjunct as in (67b), because it conflicts with interpretation of an event which contains an endpoint.

- (67) a. %Piet *xiě-zì* *xiě-zài-le* **qiáng-shàng liǎng huí.**
 Piet write-word write-LOC-PFV wall-up two time
 ‘Piet wrote on the wall twice.’ (locative argument+frequency adjunct)
- b. *Piet *xiě-zì* *xiě-zài-le* **qiáng-shàng bàn-gè-duō zhōngtóu.**
 Piet write-word write-LOC-PFV wall-up half-CL-more hour
 (*locative argument+duration adjunct)

⁷ The judgements in (66) are provided by one speaker I consult, which I personally share as well. Another native speaker also accepts (66a) but rejects (66b).

⁸ The judgements in (67) are shared by me and one of my consultants. Another consultant rejects both examples here.

Other postverbal argument-adjunct combinations are not as good as what one may expect, but they seem to be subject to some independent factors. For instance, in (68) the directional argument *qù Wéiduōlìyǎ Gōngyuán* is unable to co-occur with either the manner adjunct *hěn kuài* (68a) or the resultative adjunct *hěn yūn* (68b).

- (68) a. *Piet sǎn-bù sǎn-qù-le Wéiduōlìyǎ Gōngyuán hěn kuài.
 Piet disperse-step disperse-to-PFV Victoria park very quickly
 (*directional argument+manner adjunct)
- b. *Piet sǎn-bù sǎn-qù-le Wéiduōlìyǎ Gōngyuán hěn yūn.
 Piet disperse-step disperse-to-PFV Victoria park very dizzy
 (*directional argument+resultative adjunct)

Similar ungrammaticality arises in (69) as well. (69a) demonstrates that the locative argument *zài qiáng-shàng* cannot show up together with the manner adjunct in the postverbal field. (69b) shows that the same locative argument is incompatible with the resultative adjunct either.

- (69) a. *Piet xiě-zì xiě-zài-le qiáng-shàng hěn kuài.
 Piet write-word write-LOC-PFV wall-up very quickly
 (*locative argument+manner adjunct)
- b. *Piet xiě-zì xiě-zài-le qiáng-shàng hěn yūn.
 Piet write-word write-LOC-PFV wall-up very dizzy
 (*locative argument+resultative adjunct)

For these cases, I attribute their ungrammaticality to a specific licensing condition for manner and resultative adverbials. Recall that in all the manner/resultative-type verb doubling constructions (e.g., (50c)/(50d)) seen above, these two kinds postverbal adverbials are introduced by a same particle *de*. Adopting from Huang (1988), Cheng (2007), and Meadows and Yan (2023, 2025), among many, I assume that the particle *de* is closer to the adverbial than the verb in terms of constituency, and that when it is spelled out, it must be linearly right-adjacent to a lexical verb or some verbal element. This linearisation requirement of *de* is summarised in (70).

(70) *Linearisation requirement of de*

The particle *de* must be right-adjacent to a verbal element when sent to linearisation.

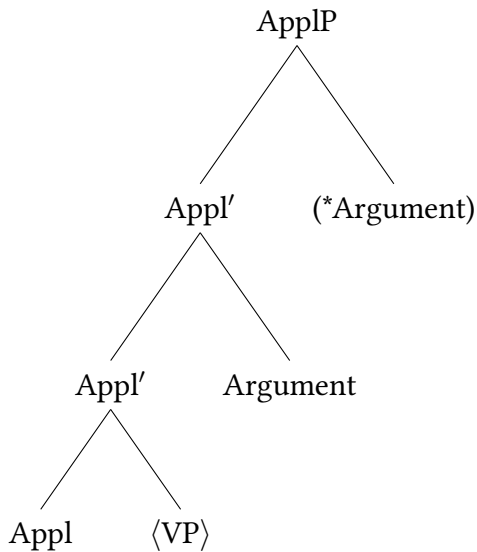
The linearisation requirement given above provides a direct account of why the directional argument cannot co-occur with either the manner or the resultative adjunct: not only is its existence necessary in the postverbal field (which is not the case in (68)), but also it has to be right-adjacent

to the second occurrence of the doubled verb (which is also impossible as in (71) and (72), since the directional argument intervenes between the verb *sàn* and the particle *de*). I will talk about this unique property of *de* in more detail in chapter 5.

- (71) a. *Piet *sàn-bù* *sàn-qù-le* **Wéiduōlìyǎ Gōngyuán** *de* **hěn kuài**.
Piet disperse-step disperse-to-PFV Victoria park DE very quickly
(*directional argument+manner adjunct)
- b. *Piet *sàn-bù* *sàn-qù-le* **Wéiduōlìyǎ Gōngyuán** *de* **hěn yūn**.
Piet disperse-step disperse-to-PFV Victoria park DE very dizzy
(*directional argument+resultative adjunct)
- (72) a. *Piet *xiě-zì* *xiě-zài-le* **qiáng-shàng** *de* **hěn kuài**.
Piet write-word write-LOC-PFV wall-up DE very quickly
(*locative argument+manner adjunct)
- b. *Piet *xiě-zì* *xiě-zài-le* **qiáng-shàng** *de* **hěn yūn**.
Piet write-word write-LOC-PFV wall-up DE very dizzy
(*locative argument+resultative adjunct)

The second correct prediction which results from the adjunct/argument dichotomy is that, it is impossible for two arguments to show up postverbally at the same time, as illustrated in (73), since the Appl head is supposed to select one and only one argument.

- (73) *Impossible co-occurrence of two arguments*

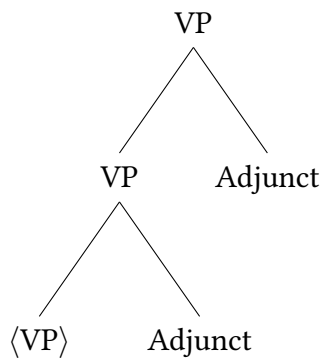


As shown in (74a), the directional argument *qù Wéiduōliǎ Gōngyuán* cannot precede the locative one *zài Lúndūn* in the postverbal field. Swapping their positions still renders the sentence ungrammatical, as in (74b). The fact that two arguments are unable to co-occur provides empirical support for the idea that there is only one applied argument position available within the ApplP.

- (74) a. *Piet sà-n-bù sà-n-qù-le Wéiduōliǎ Gōngyuán zài Lúndūn.
 Piet disperse-step disperse-to-PFV Victoria park LOC London
 (*directional argument+locative argument)
- b. *Piet sà-n-bù sà-n-zài-le Lúndūn qù Wéiduōliǎ Gōngyuán.
 Piet disperse-step disperse-LOC-PFV London to Victoria park
 (*locative argument+directional argument)

Although I have illustrated two correct predictions above, it is necessary to point out that the currently-adopted approach to postverbal adverbials also make some wrong predictions, which seemingly poses a serious challenge to the approach itself. For instance, Paul (2017) claims that adverbial adjuncts in Mandarin Chinese exclusively occur in preverbal positions (cf. Travis 1984), and correspondingly, that frequency, duration, and manner expressions are not adverbial adjuncts at all (resultative expressions are not included in her discussion).⁹ She argues that, if there is right adjunction of adverbials (à la Ernst 2002, 2014), we would expect to see, e.g., the co-occurrence of manner and frequency adverbials in the postverbal field. The corresponding verbal projection is schematised in (75).

- (75) *Predicted co-occurrence of two adjuncts*



However, this predicted co-occurrence is contrary to what we find in (76). As shown in (76a), the manner adverbial *hěn kuài* is unable to linearly precede the frequency one *liǎng huí*. Swapping their positions does not help change the ungrammaticality, as shown in (76b).

⁹ Paul (2017) uses the terms *adverb*, *adjunct*, and *adverbials* interchangeably. I stick to my own terms here to ensure that I will not complicate the following discussion.

- (76) a. *Piet sà-n-bù sà-n-de **hěn kuài liǎng huí.**
 Piet disperse-step disperse-DE very quickly two time
 (*manner adjunct+frequency adjunct)
- b. *Piet sà-n-bù sà-n-le **liǎng huí hěn kuài.**
 Piet disperse-step disperse-PFV two time very quickly
 (*frequency adjunct+manner adjunct)

A similar problem is noted by Ernst (2014:66) himself: if duration and frequency expressions are freely adjoined adjuncts, they are supposed to be able to co-occur postverbally. Again, this is not borne out by (77): neither the duration phrase *bàn gè duō zhōngtóu* can occur before the frequency one *liǎng huí* as in (77a), nor can the other way around as in (77b).

- (77) a. *Piet sà-n-bù sà-n-le **bàn-gè-duō zhōngtóu liǎng huí.**
 Piet disperse-step disperse-DE half-CL-more hour two time
 (*duration adjunct+frequency adjunct)
- b. *Piet sà-n-bù sà-n-le **liǎng huí bàn-gè-duō zhōngtóu.**
 Piet disperse-step disperse-PFV two time half-CL-more hour
 (*frequency adjunct+duration adjunct)

To complete the empirical picture, I give examples that contain manner and duration adjuncts in the postverbal field in (78). Once again, neither of the configurations is grammatical: as seen in (78a), the manner phrase *hěn kuài* cannot be followed by the duration one *bàn-gè-duō zhōngtóu*; and there is no point of changing their relative order, as shown in (78b).

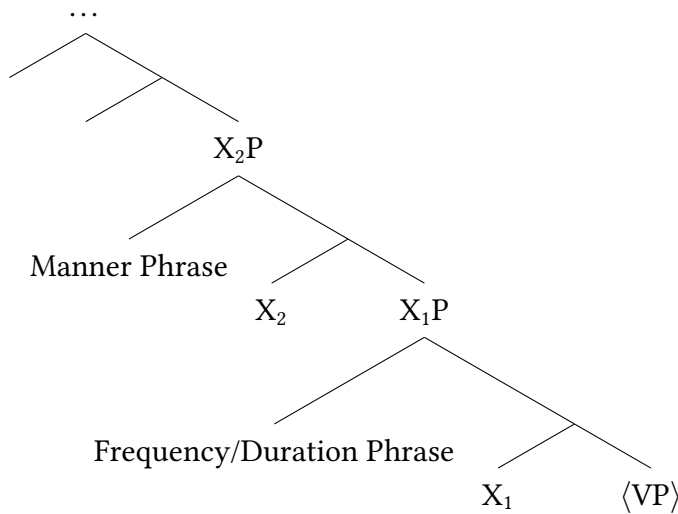
- (78) a. *Piet sà-n-bù sà-n-de **hěn kuài bàn-gè-duō zhōngtóu.**
 Piet disperse-step disperse-DE very quickly half-CL-more hour
 (*manner adjunct+duration adjunct)
- b. *Piet sà-n-bù sà-n-le **bàn-gè-duō zhōngtóu hěn kuài.**
 Piet disperse-step disperse-PFV half-CL-more hour very quickly
 (*duration adjunct+manner adjunct)

How do we account for this co-occurrence issue under the adjunct approach? And does this mean that we have to give up the adjunct treatment of postverbal adverbials? In what follows, I would like to have some discussion on the second question first, before sketching out my solution to the ungrammatical co-occurrence of multiple adjuncts.

My answer to the second question is ‘no’. This is mainly because the co-occurrence issue actually constitutes a challenge to other non-adjunct approaches to postverbal adverbials as well. Let

me take Cinque (1999)'s specifier approach as an instance. Lai (2021, 2024) assumes that frequency and duration phrases belong to the same class of adverbials, and that they are introduced by the same functional head in the Cinquean Hierarchy. Due to their complementary distribution, the ungrammatical co-occurrence of frequency and duration adverbials in (77) gets explained right away. But this line of thinking cannot provide an account of those examples in (76) and (78), as it is reasonable to assume that frequency/duration and manner adverbials respectively belong to different functional categories, and thus are introduced by different functional heads. This hypothetical scenario is illustrated as below (cf. Lai 2021, 2024).¹⁰

(79) *Co-occurrence issue: A hypothetical Cinquean Hierarchy*



Given that different adverbials are generally possible to co-occur in the Cinquean Hierarchy (which is actually the way how their relative positions are determined), it would become mysterious why frequency or duration adverbials are banned from co-occurring with manner ones, regardless of their relative orders. One can imagine that the same difficulty would still recur, even if we adopt other approaches to analysing postverbal adverbials. Considering that Ernst's and Cinque's ideas are currently the two most developed theories regarding the syntax of adverbials, we should not blame the unavailability of multiple postverbal adverbials on the adjunct approach only. The co-occurrence issue seen above should be considered as a general challenge to every approach to postverbal adverbials.

With the discussion above in mind, now we can turn to the first question. Since the syntax of adverbials cannot offer a direct solution to the co-occurrence issue, it is not unreasonable to conjecture that what prevents the co-occurrence of multiple adjuncts does not lie in narrow syntax,

¹⁰ It does not matter which type of adverbials goes to which specifier position in this hypothetical Cinquean Hierarchy, as suggested by all the ungrammatical examples in (76)-(78).

but in other components of grammar instead, e.g., at PF. Actually, the ban we are looking at here is reminiscent of a well-formedness condition on the linearisation statements called *distinctness* that Richards (2010) proposes.¹¹ The formulation of distinctness is given as below.

(80) *Distinctness*

If a linearisation statement $\langle \alpha, \alpha \rangle$ is generated, the derivation crashes.

(Richards 2010:5, (5))

According to this condition, two elements with the same category are banned from being sent to the same Spell-Out domain, because they are not distinguishable enough from each other; sending two similar items to the same Spell-Out domain would only generate a statement that PF cannot linearise. Richards uses multiple sluicing in English (81) as an example to explain the effect of distinctness. (81a) and (81b) differ from (81c) and (81d) in the category of the sluicing remnants: in the former two cases, the remnants are respectively DPs and PPs; while in the latter two examples, the sluicing remnants are all DPs.

- (81) a. I know everyone danced with someone, but I don't know [_{DP} who] [_{PP} with whom].
 b. I know every man danced with a woman, but I don't know [_{DP} which man] [_{PP} with which woman].
 c. *I know everyone insulted someone, but I don't know [_{DP} who] [_{DP} whom].
 d. *I know every man insulted a woman, but I don't know [_{DP} which man] [_{DP} which woman].

(adapted from Richards 2010:3, (1)-(2))

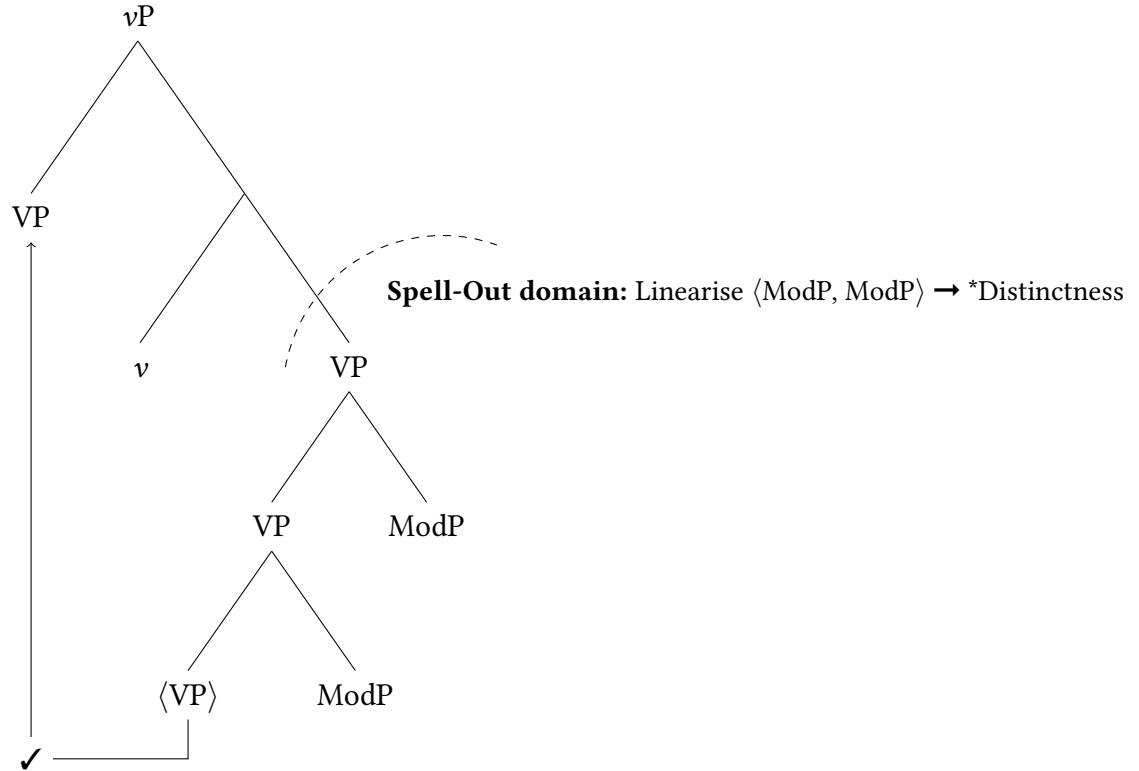
Richards argues that, due to the categorial difference noted above, when (81c) and (81d) are built at the ν P level, which he assumes is a strong phase, the material that is sent to the Spell-Out domain contains two DPs, which would result in a linearisation statement $\langle \text{DP}, \text{DP} \rangle$. However, PF cannot linearise this, as it is a violation of distinctness (80). Thus, the corresponding derivation crashes. The same problem does not arise in (81a) and (81b), because the PP remnant there constitutes a strong phase on its own; therefore, the DP contained inside the PP gets spelled out at a domain different from that outside the PP, which does not constitute a violation at PF.

How does distinctness help account for the co-occurrence issue of multiple adjuncts? We have known that the violation of distinctness is subject to two conditions: first, two items are similar to each other in terms of their category; and second, these two items are sent to the same Spell-Out domain. Although the postverbal adjuncts do not look like a uniform category at first sight, there are theories which assume that is the case. For example, Rubin (2003) proposes that there exists

¹¹ I thank Coppe van Urk for suggesting distinctness as a solution to the co-occurrence problem here.

a functional head called Mod(ification) which forms an extended projection with all adjuncts. If this idea is adopted, every adjunct in the verbal projection is then a ModP, as illustrated in (82).

(82) *Co-occurrence issue: A violation of distinctness*



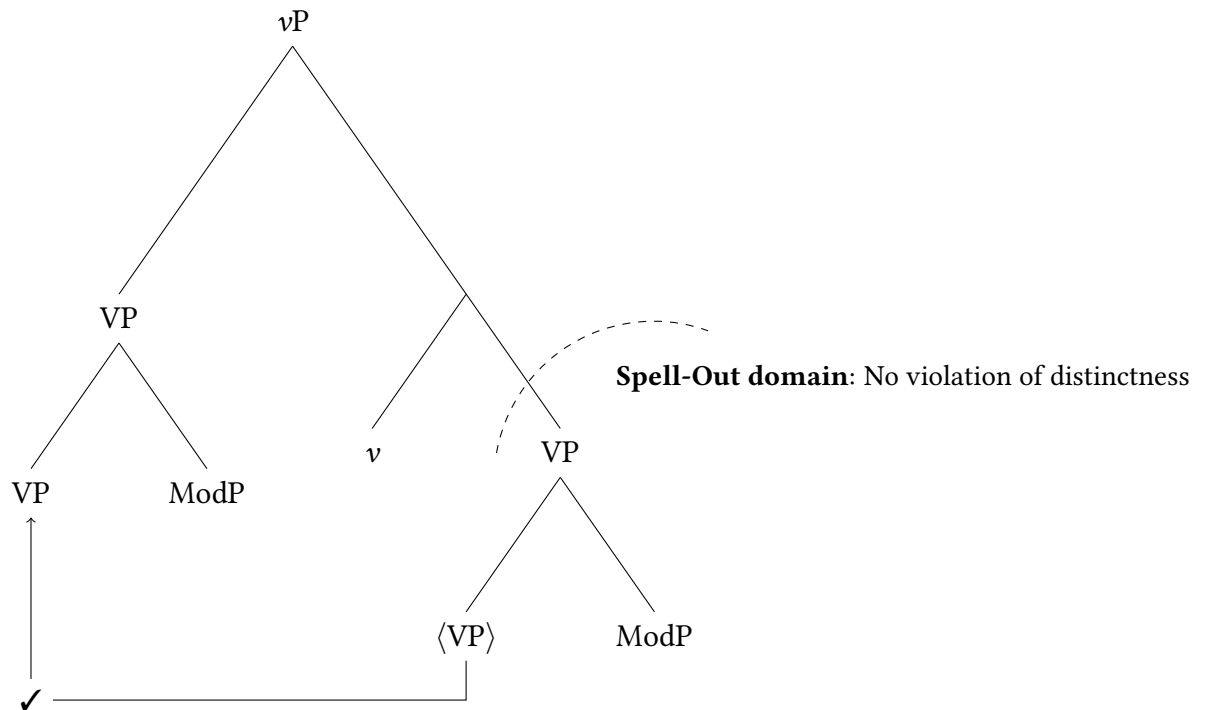
In this configuration, the lowest VP is right-adjoined by two ModPs. Following Richards (2010), I assume that as soon as the vP (i.e., a strong phase) is constructed, its complement VP will be sent to PF via Spell-Out. It is clear that within this Spell-Out domain, since the two adjuncts are both ModPs, they would generate a linearisation statement $\langle \text{ModP}, \text{ModP} \rangle$ that violates distinctness. This thus explains the ban on the co-occurrence of multiple adjuncts in the postverbal field that we have seen above: the nature of this ban is a violation of a well-formedness condition at PF rather than some principle in narrow syntax.¹²

This distinctness-based account of the co-occurrence issue leaves us with a prediction that is closely-related to VP-fronting. Richards (2010:54 *et seq.*) proposes that languages use different methods for avoiding distinctness violations, one of which is movement. If one of the two identical items can be moved out of the Spell-Out domain, then there will be no linearisation statement

¹² It is worth noting that the co-occurrence of multiple adjuncts is not prohibited cross-linguistically. For example, in English it is grammatical to say *Piet read the book slowly twice*. It seems that distinctness, as a well-formedness constraint at PF, might be too strong to apply to all the cross-linguistic cases. Therefore, we have to assume that languages are parameterised in terms of various respects, including but not limited to the size of phases, the category of postverbal adverbials, and the site of adjunction. I thank David Adger for pointing this out to me.

that violates distinctness. In the case of (82), the most straightforward method of rescuing this otherwise ungrammatical configuration is to separate the lower ModP from the higher one. Given that VP-fronting is involved in the derivation, we may predict the following configuration (83): the lower ModP is pied-piped by the VP-fronting to the phase edge, and only the higher ModP still stays inside the Spell-Out domain. Since distinctness will not be violated in this scenario, this configuration is predicted to be acceptable.

(83) *Co-occurrence issue: No violation of distinctness via VP-fronting*



In section 4.2, we will find that our prediction here is on the right track. This configuration will not only provide strong evidence that the fronted verbal phrase involved in verb doubling originates from the same verbal projection where the postverbal adverbial is introduced, but also further confirm our observation here that the constraint on co-occurrence of adjuncts is specifically being disallowed in a linear sequence to the right of the verb. As we proceed, we will see more and more cases where the verbal phrases that undergo fronting are in fact larger than a simple VP that is only made up of a lexical verb and an object.

To summarise, in this chapter I have explored an adjunct/argument dichotomy among postverbal adverbials, the core element that determines the classification of verb doubling. Even though they all occur in the same linear position within verb doubling, some of them behave more like arguments that are selected by the verbs, while others do not display the same selectional relation with verbs and behave more like adjuncts. In order to capture this distinction, I have proposed two

different verbal projections: for those argument-like adverbials, they are selected by the higher Appl head and introduced to the only specifier position of the ApplP; by contrast, for those adverbial adjuncts, they can be freely-adjoined to any maximal verbal projection. I have correctly predicted that in the postverbal field, an argument and an adjunct can co-occur, and that the same situation cannot take place between two postverbal arguments. Moreover, I have provided a distinctness-based account of the co-occurrence issue of multiple adjuncts. The discussion here opens up the possibility that the verbal predicate that undergoes fronting in verb doubling can be a maximal projection larger than a simple verb-object unit (i.e., a VP). This will be corroborated in the upcoming chapter.

Chapter 4

On the Distribution of Verbal Dependents

So far, to derive the two strings of verb doubling in (84), I have respectively proposed that (i) the displacement of the V_1 -object unit results from verbal phrasal movement, and that (ii) the postverbal adverbial is either a freely-adjoined adjunct of some maximal verbal projection, or an argument selected by the applicative head.

(84) *Two strings of verb doubling*

- a. Subj [V_1 **Obj**] [[V_2 -Part] **Adverbial**] (= (10), low verb doubling)
- b. [V_1 **Obj**] Subj [[V_2 -Part] **Adverbial**] (= (11), high verb doubling)

I have also proposed that the fronted verbal phrase and the postverbal adverbial are initially contained within the same verbal projection. Since the verb-object unit is more embedded than the adverbial, it follows that the object and the adverbial are predicted to be never able to swap their positions inside a verb doubling configuration, as schematised in (85).

(85) *Illicit string of verb doubling: No swapping between the object and the adverbial*

- a. *Subj [V_1 **Adverbial**] [[V_2 -Part] **Obj**]
- b. *[V_1 **Adverbial**] Subj [[V_2 -Part] **Obj**]

This prediction is borne out by the contrast as below. In (86), the object *nà-bù jìlùpiàn* ‘the documentary’ linearly precedes the frequency adverbial *shí biàn* ‘ten times’ in either low or high verb doubling. By comparison, in (87) these two verbal dependents swap their positions with each other: now it becomes the frequency adverbial that precedes the object in linear order, which renders both sentences ungrammatical.

- (86) a. Piet kàn **nà-bù jìlùpiàn** kàn-le **shí biàn**.
 Piet watch that-CL documentary watch-PFV ten time
 ‘Piet watched the documentary ten times.’
 (low verb doubling: object ...adverbial)
- b. Kàn **nà-bù jìlùpiàn** Piet kàn-le **shí biàn**.
 watch that-CL documentary Piet watch-PFV ten time
 ‘As for watching the documentary, Piet watched it ten times.’
 (high verb doubling: object ...adverbial)
- (87) a. *Piet kàn **shí biàn** kàn-le **nà-bù jìlùpiàn**.
 Piet watch ten time watch-PFV that-CL documentary
 Int.: ‘Piet watched the documentary ten times.’
 (low verb doubling: *adverbial ...object)
- b. *Kàn **shí biàn** Piet kàn-le **nà-bù jìlùpiàn**.
 watch ten time Piet watch-PFV that-CL documentary
 Int.: ‘As for watching the documentary, Piet watched it ten times.’
 (high verb doubling: *adverbial ...object)

The illustrations given above might convey an impression that, across verb doubling, objects are only permitted to follow the V_1 , and adverbials are only allowed to occur in the postverbal field. This is probably because up to now we have only entertained canonical cases of verb doubling which are built upon transitive verbs plus one adverbial. The primary purpose of this chapter is to present a much bigger, and certainly more complex, picture of the distribution of the verbal dependents. First in section 4.1, I demonstrate how ditransitive verbs engage in verb doubling, and focus on differences in the distribution of objects between double object constructions and prepositional dative ones. I propose that the differences can be easily accounted for once we adopt the idea that ditransitives have asymmetric verbal structures (i.e., either an ApplP or a VP), and that verbal phrasal movement can target verbal projections that are larger than VPs (contra Lai 2021, 2024). Then in section 4.2, I show that my proposal of verbal phrasal movement makes a specific prediction about articulated verbal structures with multiple adverbials, and that there is a systematic way of distributing adverbials within verb doubling constructions. In section 4.3, I focus on an analytic problem caused by the current proposal, which I refer to as the *stranding problem* of verb doubling. I propose a solution based on Abels (2003)’ version of anti-locality, applying this constraint as a derivational filter in syntax. Finally, I have some brief discussion on a specific group of simple verb-object units which are conventionally defined as verbal compounds in section 4.4. I argue that the unique characteristics of these verbal compounds come from the properties of the dummy objects (in the sense of Cheng and Sybesma 1998) taken by the verbs,

and that they are still transitive VPs in nature.

4.1 Ditransitive verb doubling

So far we have not seen any verb doubling constructions that are built upon ditransitive verbs yet. As with many languages (e.g., English), ditransitive verbs in Mandarin can form either double object constructions (with the linear order V-IO-DO) or prepositional dative constructions (with the linear order V-DO-P-IO).¹ In what follows I will start off by presenting ditransitive verb doubling in these two constructions in turn, and then provide an account of their differences by adopting asymmetrical verbal structures. I will show that whether a string of ditransitive verb doubling is licit or not is closely-related to its verb structure, and that remnant movement of verbal phrase is generally not available in Mandarin Chinese.

4.1.1 Ditransitive verb doubling in double object constructions

Let me begin with ditransitive verb doubling in double object constructions. Consider (88) first: here what displays doubling effects is a ditransitive verb *jiào* ‘teach’, which respectively takes *Wassily* as its indirect object and *fēntílùn jùfǎxué* ‘mereological syntax’ as its direct object. Since the indirect object *Wassily* precedes the direct object *fēntílùn jùfǎxué* in linear order, this indicates that the ditransitive verb doubling in question here is a double object construction. Just like the transitive verb doubling examples that we have seen before, this ditransitive verb doubling configuration is compatible with various types of postverbal adverbials.

(88) *Doubling with ditransitive verbs in double object constructions*

- a. Piet jiào **Wassily fēntílùn jùfǎxué** jiào-le shí biàn.
Piet teach Wassily mereology syntax teach-PFV ten time
‘Piet taught Wassily mereological syntax ten times.’ (frequency-type)
- b. Piet jiào **Wassily fēntílùn jùfǎxué** jiào-le sān-gè yuè.
Piet teach Wassily mereology syntax teach-PFV three-CL month
‘Piet taught Wassily mereological syntax for three months.’ (duration-type)

¹ There is arguably a third ditransitive structure in Mandarin Chinese called V-*gěi* double object construction (Paul and Whitman 2010). The only difference between this construction and the canonical double object one is that it contains a preposition-like morpheme *gěi* before the IO-DO string, which leads to a slightly different linear order V-*gěi*-IO-DO. Interestingly, this *gěi* morpheme is usually found as the spell-out of the prepositional head in the dative construction, which suggests that there might be some derivational relationship between the prepositional dative construction and the V-*gěi* double object construction. But between the two kinds of double object constructions, it is unclear to me at the moment what their relationship is. I do not include the V-*gěi* double object construction here, but refer interested readers to Paul and Whitman (2010) and Abels (2024) for more discussion.

- c. Piet jiào **Wassily fēntílùn jùfǎxué** jiào de hěn rènzhēn.
 Piet teach Wassily mereology syntax teach DE very earnestly
 ‘Piet taught Wassily mereological syntax with dedication.’ (manner-type)
- d. Piet jiào **Wassily fēntílùn jùfǎxué** jiào de hěn tòngkǔ.
 Piet teach Wassily mereology syntax teach DE very painful
 ‘Piet_i taught Wassily mereological syntax and he_i became very painful (as a result).’
 (resultative-type)

In the examples above, both the indirect and the direct objects follow the first occurrence of *jiào*. However, it is not always the case that both objects have to show up right after the *V*₁. In certain situations, the first occurrence of the doubled ditransitive can be followed by the direct object alone. This is what we find in the first two examples in (89).

- (89) a. Piet jiào **fēntílùn jùfǎxué** jiào-le **Wassily** shí biàn.
 Piet teach mereology syntax teach-PFV Wassily ten time
 ‘Piet taught Wassily mereological syntax ten times.’ (frequency-type)
- b. Piet jiào **fēntílùn jùfǎxué** jiào-le **Wassily** sān-gè yuè.
 Piet teach mereology syntax teach-PFV Wassily three-CL month
 ‘Piet taught Wassily mereological syntax for three months.’ (duration-type)
- c. *Piet jiào **fēntílùn jùfǎxué** jiào **Wassily** de hěn rènzhēn.
 Piet teach mereology syntax teach Wassily DE very earnestly
 Int.: ‘Piet taught Wassily mereological syntax with dedication.’ (manner-type)
- d. *Piet jiào **fēntílùn jùfǎxué** jiào **Wassily** de hěn tòngkǔ.
 Piet teach mereology syntax teach Wassily DE very painful
 Int.: ‘Piet taught Wassily mereological syntax and he_i became very painful (as a result).’
 (resultative-type)

What both (89a) and (89b) demonstrate is a grammatical ditransitive verb doubling configuration in which only the direct object *fēntílùn jùfǎxué* follows the first *jiào*. The indirect object *Wassily* does not follow the same occurrence of *jiào* as the direct one does; instead, it occurs in the postverbal field, and intervenes between the second occurrence of *jiào* and the frequency/duration adverbial. By contrast, neither (89c) nor (89d) is grammatical, even though the distribution of the double objects here is identical to that in (89a) and (89b): that is, the direct object and the indirect object respectively follow the *V*₁ and the *V*₂.

Where does this contrast between (89a)/(89b) and (89c)/(89d) come from? The only difference among these four examples lies in the postverbal field: in the first two cases, we have known that the indirect object *Wassily* is surrounded by the second occurrence of *jiào*, the aspect parti-

cle *le*, and the frequency/duration adverbial; while in the second two cases, the postverbal field surrounding the indirect object changes to the manner/resultative adverbial plus the particle *de*. Recall that in section 3.4, I briefly mentioned that the particle *de* has some unique requirement in terms of linearisation, repeated in (90) below.

(90) *Linearisation requirement of de*

The particle *de* must be right-adjacent to a verbal element when sent to linearisation.

(=(70))

Both (89c) and (89d) are clearly violations of this requirement: since *Wassily* stays between the second *jiào* and the particle *de*, the latter cannot be immediately right-adjacent to the former, which is why they are ruled out.

Now let us come back to grammatical cases like (89a) and (89b). Since these two have already shown that a ditransitive verb doubling with the linear word order V_1 -DO- V_2 -IO is possible, it seems reasonable to ask whether changing the positions of the two objects still makes ditransitive verb doubling acceptable. As shown in (91), this string turns out to be ungrammatical.

- (91) a. *Piet jiào **Wassily** jiào-le **fēntílùn jùfǎxué** shí biàn.
 Piet teach Wassily teach-PFV mereology syntax ten time
 Int.: ‘Piet taught Wassily mereological syntax ten times.’ (frequency-type)
- b. *Piet jiào **Wassily** jiào-le **fēntílùn jùfǎxué** sān-gè yuè.
 Piet teach Wassily teach-PFV mereology syntax three-CL month
 Int.: ‘Piet taught Wassily mereological syntax for three months.’ (duration-type)
- c. *Piet jiào **Wassily** jiào **fēntílùn jùfǎxué** de hěn rènzhēn.
 Piet teach Wassily teach mereology syntax *DE* very earnestly
 Int.: ‘Piet taught Wassily mereological syntax with dedication.’ (manner-type)
- d. *Piet jiào **Wassily** jiào **fēntílùn jùfǎxué** de hěn tòngkǔ.
 Piet teach Wassily teach mereology syntax *DE* very painful
 Int.: ‘Piet taught Wassily mereological syntax and he_i became very painful (as a result).’ (resultative-type)

In both (91a) and (91b), the first occurrence of the ditransitive *jiào* is followed by the indirect object *Wassily*, and the second occurrence of the same verb by the direct object *fēntílùn jùfǎxué*. The result that neither of the two sentences is acceptable clearly indicates that the linear order $*V_1$ -IO- V_2 -DO is not a licit string of verb doubling for double object constructions. For the sake of completeness, I also construct the same linear order for the manner-type verb doubling in (91c) and the resultative-type in (91d). Since both examples still violate the linearisation requirement

of *de* (90) (as the examples in (89c) and (89d) do), they are ruled out as expected.

The three strings of ditransitive verb doubling in double object constructions are summarised in (92). If we interpret these strings via the lens of verbal phrasal movement, we can conclude that the ditransitive verbs form verbal phrases with both indirect and direct objects, or with the direct object only. However, they are unable to form verbal phrases just with the indirect object. These properties will become crucial later when we think about the verbal structure of ditransitives in double object constructions.

(92) *Strings of verb doubling with ditransitives in double object constructions*

- a. Subj [V₁ **IO DO**] [[V₂-Part] Adverbial]
- b. Subj [V₁ **DO**] [[V₂-Part **IO**] Adverbial]
- c. *Subj [V₁ **IO**] [[V₂-Part **DO**] Adverbial]

4.1.2 Ditransitive verb doubling in prepositional dative constructions

Now we can move on to ditransitive verb doubling in prepositional dative constructions, as exemplified in (93). Here the doubled ditransitive verb becomes *jièshào* ‘introduce’, which takes *tā-de péngyǒu* ‘his friend’ as its direct object. As to the indirect object *Wassily*, it is not directly taken by the verb, but rather introduced by the preposition *gěi* ‘to’. The linear order V₁-DO-P-IO confirms that these examples are all prepositional dative constructions. As with their double object counterparts in (88), ditransitives in prepositional dative constructions here are able to construct verb doubling with different meanings.

(93) *Doubling with ditransitive verbs (in prepositional dative constructions)*

- a. Piet_i jièshào **tā_i-de péngyǒu gěi Wassily** jièshào-le hěnduō biàn.
Piet introduce 3SG-POSS friend to Wassily introduce-PFV many time
‘Piet introduced his friend to Wassily many times.’ (frequency-type)
- b. Piet_i jièshào **tā_i-de péngyǒu gěi Wassily** jièshào-le bàn-gè-duō
Piet introduce 3SG-POSS friend to Wassily introduce-PFV half-CL-more
zhōngtóu.
hour
‘Piet introduced his friend to Wassily for more than half an hour.’ (duration-type)
- c. Piet_i jièshào **tā_i-de péngyǒu gěi Wassily** jièshào de hěn dàshēng.
Piet introduce 3SG-POSS friend to Wassily introduce DE very loudly
‘Piet introduced his friend to Wassily very loudly.’ (manner-type)

- d. Piet_i jièshào **tā_i-de péngyǒu** gěi **Wassily** jièshào de tèbié jīdòng.
 Piet introduce 3SG-POSS friend to Wassily introduce DE particularly excited
 ‘Piet introduced his friend to Wassily and became very excited (as a result).’
 (resultative-type)

However, there is no way of separating the two objects and letting them respectively follow one of the double verbs in prepositional dative constructions, unlike what we have seen in (89a) and (89b) above. Let us see the hypothetical linear order V₁-DO-V₂-P-IO in (94) first:

- (94) a. *Piet_i jièshào **tā_i-de péngyǒu** jièshào-gěi-le **Wassily** hěnduō biàn.
 Piet introduce 3SG-POSS friend introduce-to-PFV Wassily many time
 Int.: ‘Piet introduced his friend to Wassily many times.’ (frequency-type)
- b. *Piet_i jièshào **tā_i-de péngyǒu** jièshào-gěi-le **Wassily** bàn-gè-duō
 Piet introduce 3SG-POSS friend introduce-to-PFV Wassily half-CL-more
 zhōngtóu.
 hour
 Int.: ‘Piet introduced his friend to Wassily for more than half an hour.’
 (duration-type)
- c. *Piet_i jièshào **tā_i-de péngyǒu** jièshào gěi **Wassily** de hěn dàshēng.
 Piet introduce 3SG-POSS friend introduce to Wassily DE very loudly
 Int.: ‘Piet introduced his friend to Wassily very loudly.’ (manner-type)
- d. *Piet_i jièshào **tā_i-de péngyǒu** jièshào gěi **Wassily** de tèbié jīdòng.
 Piet introduce 3SG-POSS friend introduce to Wassily DE particularly excited
 Int.: ‘Piet introduced his friend to Wassily and became very excited.’
 (resultative-type)

None of the examples given above is grammatical. In each sentence, the first occurrence of the ditransitive *jiào* is only followed by the direct object *tā-de péngyǒu*. The indirect object *Wassily* (plus the preposition *gěi*) occurs after the second *jiào* instead.² The result here suggests that, unlike ditransitive verb doubling in double object constructions, both objects have to follow the V₁ at the same time in prepositional dative constructions.

This idea is further supported by (95), which demonstrates the other hypothetical linear order V₁-P-IO-V₂-DO: even though the two objects change the occurrence of the doubled verb they follow, all of the examples are still ungrammatical. Once again, this suggests that it is impossible

² Note that in (94a) and (94b) I insert the preposition *gěi* into the position between the V₂ and the aspect marker *le*. This is because this is the canonical position of *gěi* in prepositional dative constructions without verb doubling. Placing this preposition after the aspect marker will not change the ungrammatical status.

to break the string V₁-DO-P-IO in prepositional dative constructions.

- (95) a. *Piet_i jièshào **gěi Wassily** jièshào-le **tā_i-de péngyǒu** hěnduō biàn.
 Piet introduce to Wassily introduce-PFV 3SG-POSS friend many time
 Int.: ‘Piet introduced his friend to Wassily many times.’ (frequency-type)
- b. *Piet_i jièshào **gěi Wassily** jièshào-le **tā_i-de péngyǒu** bàn-gè-duō
 Piet introduce to Wassily introduce-PFV 3SG-POSS friend half-CL-more
 zhōngtóu.
 hour
 Int.: ‘Piet introduced his friend to Wassily for more than half an hour.’
 (duration-type)
- c. *Piet_i jièshào **gěi Wassily** jièshào **tā_i-de péngyǒu** de hěn dàshēng.
 Piet introduce to Wassily introduce 3SG-POSS friend DE very loudly
 Int.: ‘Piet introduced his friend to Wassily very loudly.’ (manner-type)
- d. *Piet_i jièshào **gěi Wassily** jièshào **tā_i-de péngyǒu** de tèbié jīdòng.
 Piet introduce to Wassily introduce 3SG-POSS friend DE particularly excited
 Int.: ‘Piet introduced his friend to Wassily and became very excited.’
 (resultative-type)

(96) summarises the three strings of ditransitive verb doubling in prepositional dative constructions we have seen above. It is obvious that in this kind of ditransitive structure, the lexical verb forms a verbal phrase with both the direct object and the preposition-indirect object unit, which explains why (96a) is the only grammatical verb doubling configuration. The difference that we found between (92) and here offers an important clue that the underlying structure of ditransitives in these two constructions is not the same.

(96) *Strings of verb doubling with ditransitives in prepositional dative constructions*

- a. Subj [V₁ **DO** [**P IO**]] [[V₂-Part] Adverbial]
 b. *Subj [V₁ **DO**] [[V₂-**P**-Part **IO**] Adverbial]
 c. *Subj [V₁ [**P IO**]] [[V₂-Part **DO**] Adverbial]

4.1.3 Asymmetrical verbal structures of ditransitives

In the previous two sections, I have shown that ditransitive verb doubling in double object constructions and in prepositional dative constructions are different in terms of the distribution of objects. If we only focus on the first occurrence of the doubled verb and its following dependent(s), and adopt the idea that their occurrence is derived from verbal phrasal movement, the patterns

of ditransitive verb doubling can be re-generalised as follows.

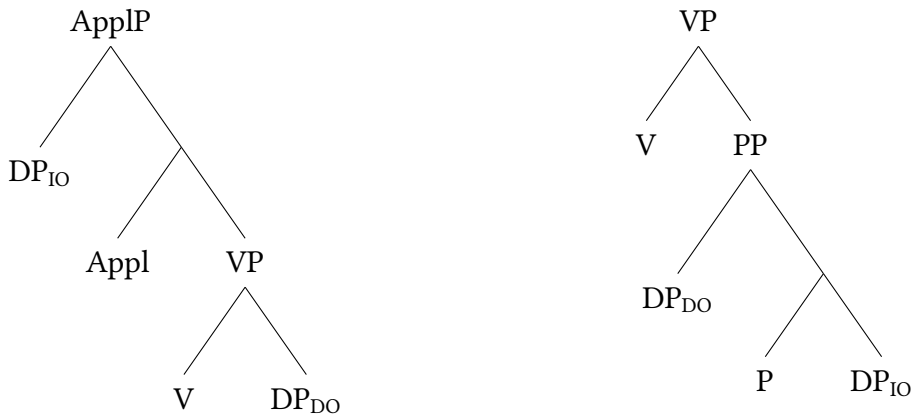
(97) *Fronted verbal phrases involved in ditransitive verb doubling*

	V ₁ -IO-DO	V ₁ -DO-P-IO	V ₁ -DO	V ₁ -(P)-IO
Double object constructions	✓	NA	✓	✗
Prepositional object constructions	NA	✓	✗	✗

From the table (97), we know that both kinds of ditransitive verb doubling can involve verbal predicate fronting that contains both the direct and the indirect objects. In addition to that, a verb-direct object unit (i.e., a simple VP) can also engage in forming verb doubling in double object constructions, but not in prepositional dative constructions. Last but not least, the verb and the indirect object are unable to form a verbal phrase under any circumstances.

To account for all these patterns related to ditransitive verb doubling, I propose that the verbal structures underlying the two ditransitive constructions are asymmetrical, in line with Marantz (1993), Bruening (2001), Abels (2024), van Urk (2024), and so on. There are a lot of proposed ditransitive structures in the literature, e.g., Larson (1988), Marantz (1993), Pesetsky (1995), Harley (2002), Pykkänen (2002, 2008), Janke and Neeleman (2012), Abels (2024), among many others. For concreteness, I adopt a high applicative structure (à la Marantz 1993) for the double object construction, as in (98) (cf. Pykkänen 2002, 2008), and a Pesetskyan (1995) cascade structure for the prepositional dative construction, as in (99) (cf. Harley 2002). These two verbal structures are asymmetrical in the sense that the former is an ApplP with a VP inside, which is essentially a verbal extended projection (in the sense of Grimshaw 1991, 2000, 2005), while the latter is an VP that takes a PP as its complement, which is non-verbal at all.

(98) *Double object construction as an ApplP* (99) *Prepositional dative construction as a VP*



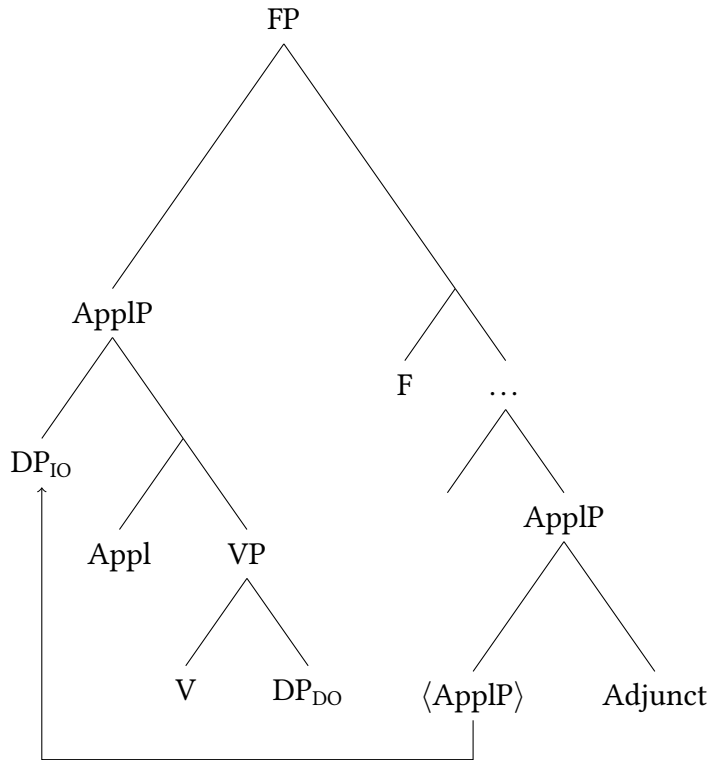
Let us now have a look at how these two asymmetrical verbal structures can derive ditransitive verb doubling case by case. I will begin with verb doubling in double object constructions and the high applicative structure (98). We have seen that the fronted verbal phrase in this kind of

ditransitive verb doubling can alter between one string (i.e., V₁-IO-DO) and the other (i.e., V₁-DO). The relevant examples are repeated as below.

- (100) a. Piet jiào **Wassily fēntìlùn jùfǎxué** jiào-le shí biàn.
 Piet teach Wassily mereology syntax teach-PFV ten time
 ‘Piet taught Wassily mereological syntax ten times.’ (= (88a), frequency-type)
- b. Piet jiào **fēntìlùn jùfǎxué** jiào-le **Wassily** shí biàn.
 Piet teach mereology syntax teach-PFV Wassily ten time
 ‘Piet taught Wassily mereological syntax ten times.’ (= (89a), frequency-type)

In (100a), the fronted verbal phrase has to contain both the indirect and the direct objects, which is exactly the size of the ApplP. Therefore, I propose that, in cases like (100a), what undergoes verbal predicate fronting is the entire ApplP, as illustrated in (101). This ApplP-fronting is compatible with my proposed analysis of deriving verb doubling: as mentioned above, the ApplP is in natural a verbal extended projection, so nothing prevents it from being the target of verbal phrasal movement.

- (101) *Ditransitive verb doubling in double object constructions involving ApplP-fronting*



There are two things I would like to draw our attention to. The first one is related to features that trigger movement in this scenario. Recall that in section 2.1, I proposed that low and high verb doubling is triggered by different features. The one which is responsible for the low movement is the categorial feature of the verbal phrase, and it is usually the feature [*uV*]. The one that triggers the high movement is some information-structure-related feature like [*uTop*]. Since what undergoes fronting here is the whole ApplP, when it targets the low position in the clausal spine, I propose that it is triggered by its corresponding categorial feature [*uAppl*] (instead of [*uV*]). When the ApplP is moved to the left periphery, I suppose that this high movement is still triggered by topicalisation.

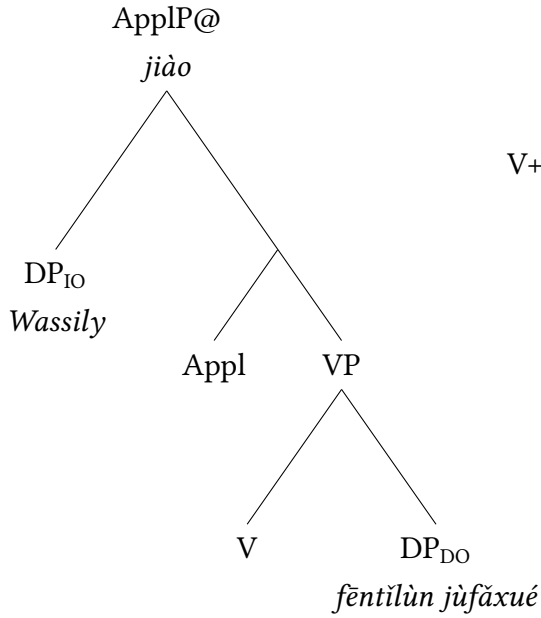
Another point that is worthwhile mentioning here is concerned with the word order internal to the fronted ApplP. One reason Lai (2021:110-112) used to argue against the high applicative structure (98) for double object constructions is that this structure would create wrong word order. Suppose that in (101), the V head undergoes movement to the Appl head - no matter in narrow syntax (Travis 1984) or post-syntactically (Chomsky 2001) - inside the fronted ApplP. Since the indirect object occupies the Spec,ApplP, which asymmetrically c-commands the Appl head, the lexical verb can never be spelled out before the indirect object, according to Kayne (1994)'s Linear Correspondence Axiom. The relevant example is given in (102).

- (102) *Piet **Wassily** jiào **fēntílùn jùfǎxué** jiào-le shí biàn.
Piet Wassily teach mereology syntax teach-PFV ten time
Int.: 'Piet taught Wassily mereological syntax ten times.'

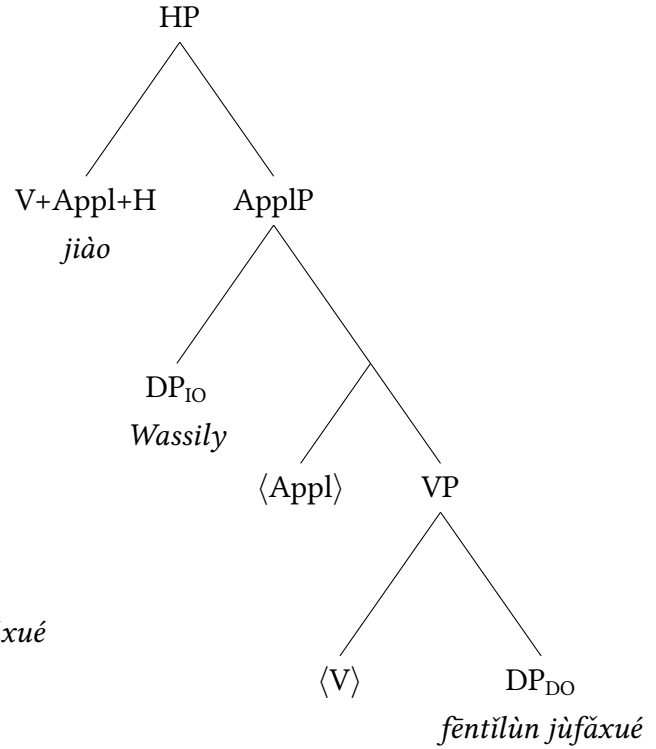
There are at least two methods of addressing this word order issue. The first one is to adopt the idea that points of exponence are not borne on the heads but on the maximal projections instead. This is the core idea that Brody (2000) proposes in his Mirror Theory: the traditional head-movement-effect is replaced by a diacritic whose position in the clausal spine determines the point of phonologically realising a morphological word.³ As illustrated in (103), we can posit that there is a spell-out diacritic borne on the maximal Appl projection instead of the Appl head. The presence of this diacritic indicates that the ditransitive verb *jiào* would be spelled out at the top of the ApplP, and that the top node would be the first one that gets realised phonologically when the entire phrase is sent to linearisation. This ensures that the verb is linearised before both objects within the fronted ApplP (also see Bye and Svenonius 2012).

³ My interpretation of Brody (2000)'s Mirror Theory is actually over-simplified: he also proposes a telescoped structure in which only heads are required to form what he calls the syntactic complementation line. The distinction among projections is no longer important within Brody's framework, therefore only heads are maintained. For more detailed discussions, I refer interested readers to Abels (2003), Adger et al. (2009), Adger (2013), and Adger (2025).

(103)



(104)



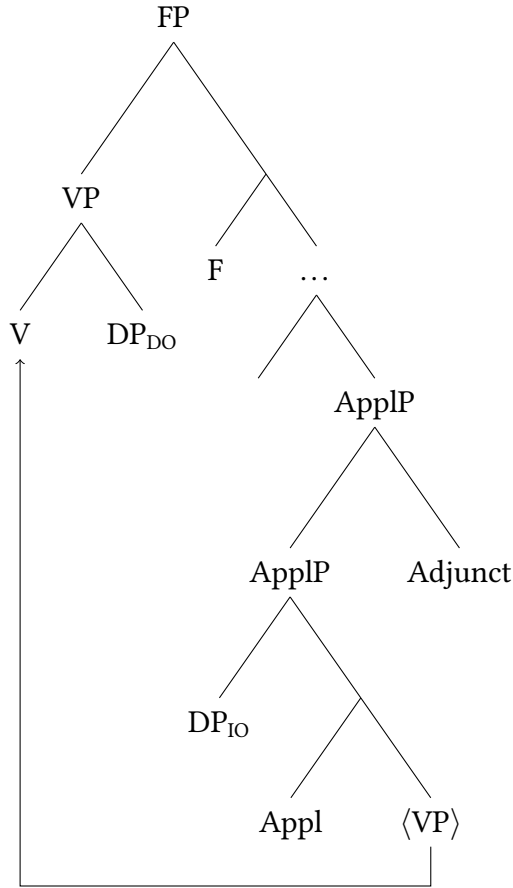
The alternative solution to the word order issue is to posit that the size of the fronted verbal phrase is even larger than *ApplP*, as illustrated in (104). Here I label this hypothetical verbal projection as *HP*, which I assume is still smaller than *vP*. In this structure, we can still maintain the head movement analysis, and postulate that *jiào* is spelled out at the highest head. This method can easily avoid the problem of linearisation within the applicative structure. However, this approach does not come for free: when the *HP* is targeted for movement to *Spec,FP*, the feature trigger, correspondingly, has to change to [*uH*].

There is nothing at this point that can help decide between the two methods sketched above. However, for the sake of exposition, I adopt the diacritic approach throughout the discussion here, mainly because this idea is compatible with all the cases of verb doubling investigated in this work: regardless of the size of the fronted verbal phrase, the verb needs to be linearised first.

We can now return to the alternative verb doubling configuration (100b) that occurs in double object constructions. This demonstrates a different scenario from (100a) in which the fronted verbal phrase only contains the direct object. This basic verb-direct object unit is just the size of a simple *VP*, which is also part of the applicative structure (98). Thus, I propose that, in cases like (100b), it is still the lowest *VP* inside the entire verbal extended projection that undergoes fronting, as illustrated in (105). This derivation has nothing different from other canonical verb doubling configurations built on transitive verbs: the *VP* movement can be triggered by either the categorial feature [*uV*] to the clause-medial position *Spec,FP*, or the information-structure-related

feature [*u*Top] to the clause-initial position Spec,CP.

- (105) *Ditransitive verb doubling in double object constructions involving VP-fronting*

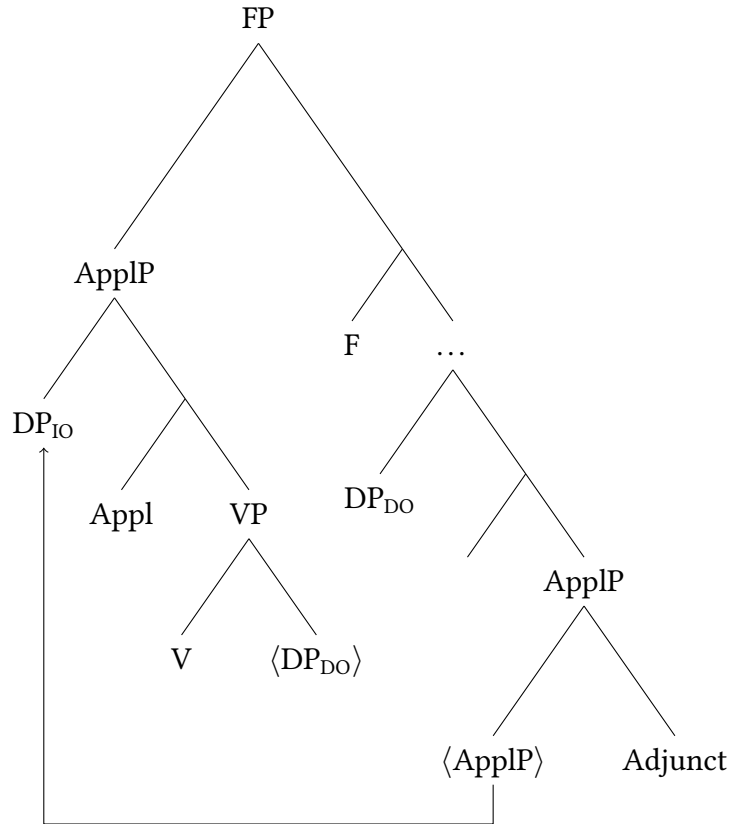


The two cases discussed above have clearly shown how the high applicative structure (98) accounts for the alternation of verb doubling in double object constructions. It is equally important to ask whether the same structure can also explain the ungrammaticality of the string *V₁-IO in double object constructions, as given in (106).

- (106) *Piet jiào **Wassily** jiào-le **fēntilùn jùfǎxué** shí biàn.
 Piet teach Wassily teach-PFV mereology syntax ten time
 Int.: 'Piet taught Wassily mereological syntax ten times.' (= (91a), frequency-type)

Let us first consider the derivation of this configuration if it is grammatical: the simplest method of deriving (106) seems to involve scrambling plus remnant movement. (107) illustrates this hypothetical derivation: first, the direct object undergoes scrambling to some position outside ApplP but still below vP; then, the remnant ApplP, in which the DP is already evacuated, undergoes the same predicate fronting to Spec,FP that we have seen earlier.

(107)



The hypothetical derivation outlined above displays no conflict with my current proposal, but the example is still ungrammatical. This implies that the ungrammaticality of the illicit string $*V_1$ -IO is not related to the structure itself, but to the two derivational steps. It has been proposed by Soh (1998) that Mandarin Chinese does have object scrambling in the postverbal field, but, as far as I am aware, the remnant movement of verbal phrase has not been widely acknowledged in the Mandarin literature (however, see N. Li 2025 for a recent proposal). Therefore, I attribute the illicitness of the string $*V_1$ -IO in double object constructions to the idea that remnant movement is generally unavailable in Mandarin Chinese.⁴

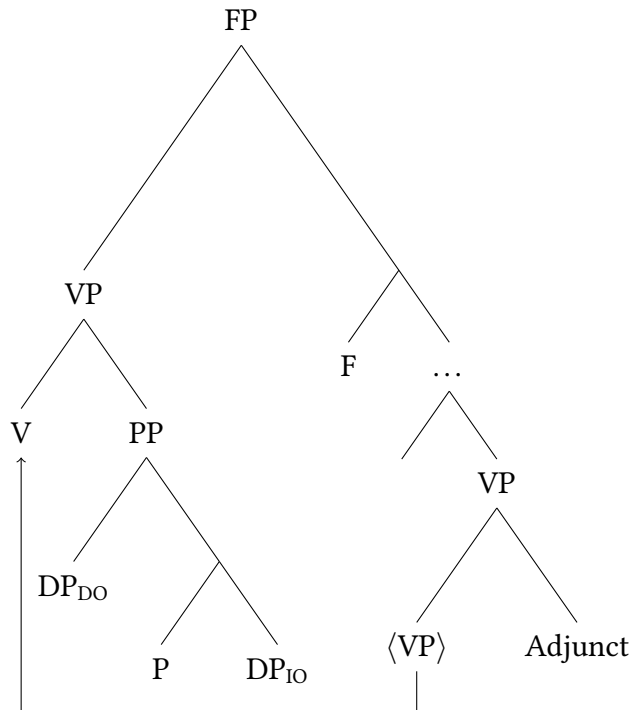
Next, I am going to explain the licit and the illicit strings of verb doubling in prepositional dative constructions, with the cascade VP structure (99). The only valid verb doubling in this configuration is repeated in (108), where the first occurrence of the doubled verb is followed in turn by the direct object and the preposition-indirect object unit (i.e., V_1 -DO-P-IO).

⁴ Note that in the derivation (107), the scrambling of the direct object might be blocked by the indirect object due to the intervention effects caused by relativised minimality. This can be another way of accounting for the ungrammaticality of (106). However, we will find soon that the lack of remnant movement can also provide a proper account of other ungrammatical cases in prepositional dative constructions, which the intervention effects cannot. Therefore, I do not consider this possible account in the rest of the discussion. I thank David Adger for bringing my attention to this.

- (108) Piet_i jièshào **tā_i-de péngyǒu** gěi Wassily jièshào-le hěnduō biàn.
 Piet introduce 3SG-POSS friend to Wassily introduce-PFV many time
 ‘Piet introduced his friend to Wassily many times.’ (= (93a), frequency-type)

The grammaticality of (108) results directly from VP-fronting to Spec,FP, as illustrated in (109). Here the cascade VP structure is the only candidate for verbal predicate fronting. Thus, when the functional head F starts looking for some verbal phrase, only this VP can be the goal and triggered to move. Along with the fronted VP, the PP inside also undergoes fronting to a higher position, in which the direct object has already preceded the prepositional head and the direct object, which eventually leading to the string V₁-DO-P-IO.

- (109) *Ditransitive verb doubling in prepositional dative construction involving VP-fronting*

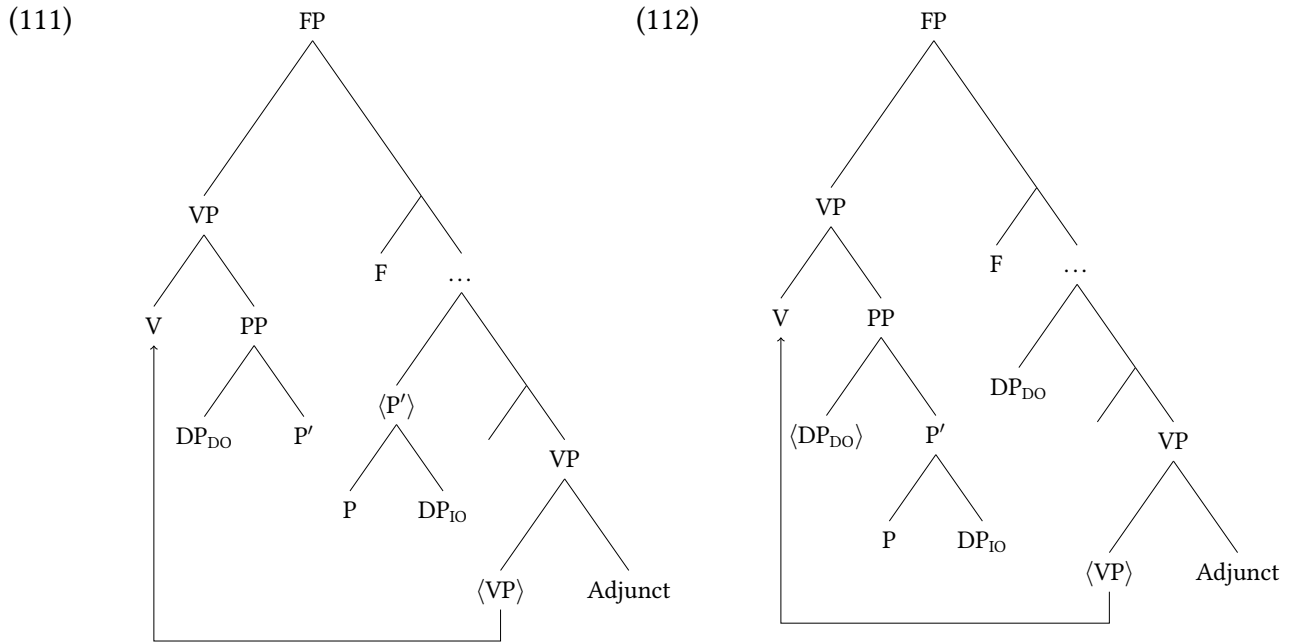


Once the V₁ is followed by either the direct object or the preposition-indirect object unit alone, the string of verb doubling becomes ungrammatical, as respectively exemplified in (110a) and (110b).

- (110) a. *Piet_i jièshào **tā_i-de péngyǒu** jièshào-gěi-le Wassily hěnduō biàn.
 Piet introduce 3SG-POSS friend introduce-to-PFV Wassily many time
 Int.: ‘Piet introduced his friend to Wassily many times.’ (= (94a), frequency-type)

- b. *Piet_i jièshào gěi Wassily jièshào-le tā_i-de péngyǒu hěnduō biàn.
 Piet introduce to Wassily introduce-PFV 3SG-POSS friend many time
 Int.: ‘Piet introduced his friend to Wassily many times.’ (= (95a), frequency-type)

As with the hypothetical remnant ApplP-fronting illustrated in (107) above, the simplest way of deriving these two configurations is seemingly to rely on remnant VP-fronting: (111) shows the hypothetical derivation of the string *V₁-DO in (110a), where the intermediate projection P’ undergoes scrambling first, followed by the remnant VP-fronting; whereas (112) displays the conjectured derivation of the string *V₁-P-IO in (110b), where it becomes the direct object that moves out of the VP first, then followed by the remnant VP movement. No matter whether the first scrambling is licit or not in either derivation, as long as we hold the assumption that remnant movement of verbal phrase is not available in Mandarin, these two derivations can be ruled out directly, which explains straightforwardly why neither of the strings illustrated above is grammatical in prepositional dative constructions.



To summarise, in this section I have shown that verb doubling with ditransitives in double object and prepositional dative constructions are different in terms of the distribution of objects. I have adopted asymmetrical ditransitive verbal structures, and proposed that all the licit strings of ditransitive verb doubling are direct consequences of verbal phrasal movement: under this proposal, verbal predicate fronting is not confined to the VP; larger verbal projections like ApplP can be the target of movement as well (contra Lai 2021, 2024). For those illicit strings, I have attributed their illicitness to the general unavailability of verbal remnant movement in Mandarin Chinese. The existence of ditransitive verb doubling has clearly indicated that verbal predicate fronting can be

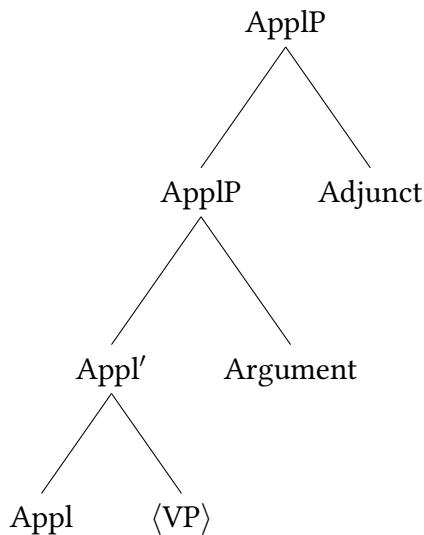
carried out by verbal projections that are larger than simple verb-object units. In the next section I will show more cases that favour this idea.

4.2 Verb doubling with distributed adverbials

The current proposal of verbal phrasal movement makes a specific prediction about articulated verbal structures with multiple postverbal adverbials: we expect to find verb doubling configurations in which each occurrence of the doubled verb is followed by at least one adverbial. In this section, I will show that this prediction is on the right track, and that there exists a systematic way of distributing multiple adverbials within verb doubling configurations.

Recall that in section 3.3, I proposed a number of articulated verbal structures which contain both the fronted VP and the postverbal adverbial (no matter it is an adjunct or an argument). Later in section 3.4, I offered several predictions about the co-occurrence of multiple adverbials in the postverbal field. The adjunct/argument dichotomy led us to predict that, in an applicative structure where the argument is selected by the Appl head, the adjunct can be freely adjoined to the topmost ApplP, as illustrated in (113).

- (113) *Co-occurrence of an argument and an adjunct in the same postverbal field*



The co-occurrence of the argument and the adjunct in this applicative structure is borne out empirically. As shown in (114a) and (114b), the directional argument *qù Niújīn* ‘to Oxford’ can be followed by either the duration adjunct *liù-gè-duō liù-gè-duō* ‘more than six hours’ or the frequency adjunct *wǔ huí* ‘five times’. However, the co-occurrence does not happen between the same directional argument and the manner adjunct *tèbié màn* ‘very slowly’ as in (114c), which I attributed to some independent requirement of the particle *de* (see the original discussion in

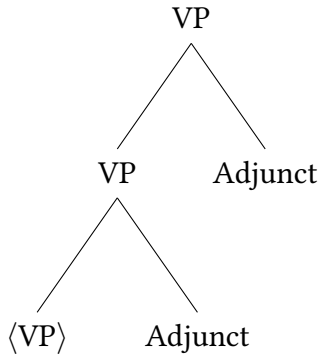
section 3.4).

(114) *String of verb doubling*: Subj [V₁ Obj] [[[V₂-Part] Argument] Adjunct]

- a. Piet (cóng Lúndūn) qí-chē qí-qù-le **Niújīn liù-gè-duō zhōngtóu.**
 Piet from London ride-bike ride-to-PFV Oxford six-CL-more hour
 ‘Piet cycled to Oxford (from London) for more than six hours.’ (direction+duration)
- b. Piet (cóng Lúndūn) qí-chē qí-qù-guò **Niújīn wǔ huí.**
 Piet from London ride-bike ride-to-EXP Oxford five time
 ‘Piet has cycled to Oxford (from London) five times.’ (direction+frequency)
- c. *Piet (cóng Lúndūn) qí-chē qí-qù **Niújīn tèbié màn.**
 Piet from London ride-bike ride-to Oxford particularly slowly
 Int.: ‘Piet cycled to Oxford (from London) very slowly.’ (direction+manner)

I also predicted that multiple adjuncts are able to co-occur postverbally, as schematised in (115).

(115) *Co-occurrence of two adjuncts in the same postverbal field*



But it turns out that there seems to be no way of hosting two adjuncts within the same postverbal field, as shown in (116): regardless of the relative order between the co-occurring adjuncts, none of the combinations listed here can generate a grammatical string of verb doubling. To account for this co-occurrence issue, I proposed that the co-occurrence of multiple adjuncts constitutes a violation of distinctness (in the sense of Richards 2010), thereby attributing the ungrammaticality of (116) to the side of PF (also see section 3.4 for the original discussion).

(116) *String of verb doubling*: *Subj [V₁ Obj] [[[V₂-Part] Adjunct] Adjunct]

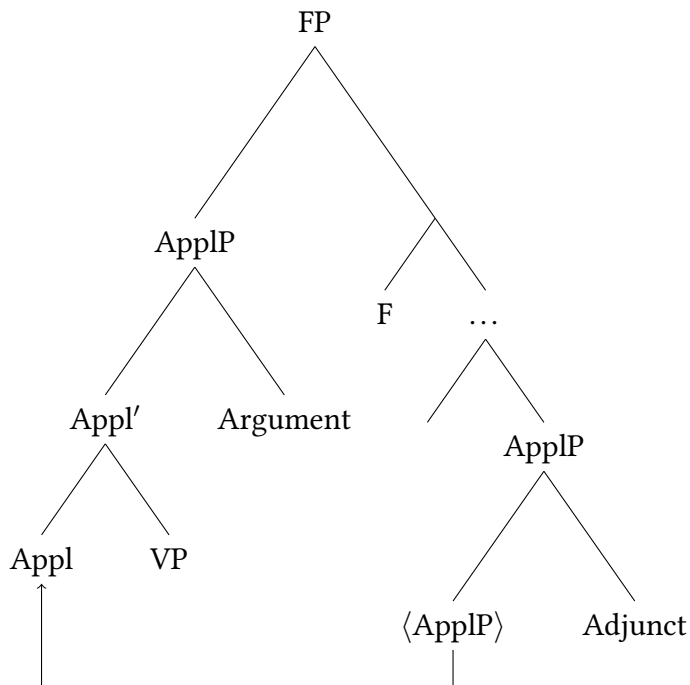
- a. *Piet qí-chē qí-le **wǔ huí liù-gè-duō zhōngtóu.**
 Piet ride-bike ride-PFV five time six-CL-more hour
 (*frequency+duration)

- b. *Piet qí-chē qí-le liù-gè-duō zhōngtóu tèbié màn.
 Piet ride-bike ride-PFV six-CL-more hour particularly slowly
 (*duration+manner)
- c. *Piet qí-chē qí-le wǔ huí tèbié màn.
 Piet ride-bike ride-PFV five time particularly slowly
 (*frequency+manner)

With the two articulated verbal structures (113) and (115) in mind, now let us have a closer look at how they can be related to verbal phrasal movement. In both configurations illustrated above, what undergoes predicate fronting is the lowest VP. Nevertheless, according to my proposal in section 4.1, verbal predicate fronting is not confined to simple VPs only; verbal projections larger than VPs are also targets for verbal phrasal movement, which certainly include those maximal verbal projections that contain an adverbial inside. Correspondingly, this leads us to predict that in certain verb doubling configurations, multiple adverbials can still co-occur without being in the same postverbal field, since one of them is pied-piped by the fronted verbal phrase.

Consider (117) first: comparing with its counterpart (113), here the argument and the adjunct are distributed to two different positions. The former is contained within the ApplP that undergoes fronting to Spec,FP, while the latter still stays in the postverbal field. When the entire verb doubling configuration is sent to linearisation, these two adverbials are predicted to respectively follow the first and the second occurrence of the doubled verb in linear order.

(117) *Distributed co-occurrence of an argument and an adjunct*



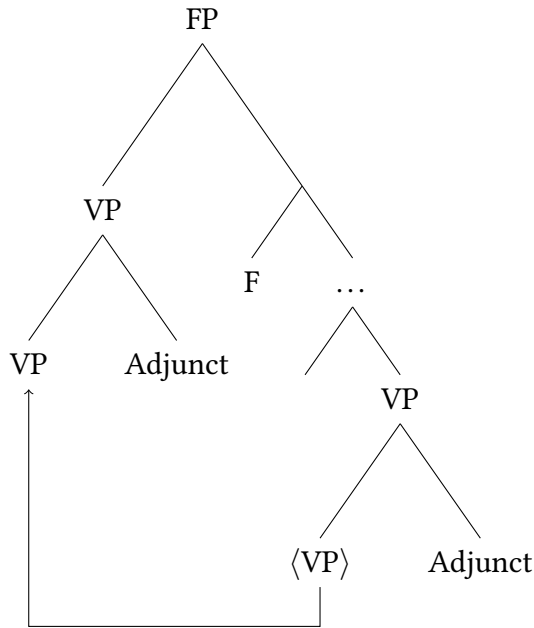
The predicted separation of the argument and the adjunct in (117) is borne out by (118). In each of the following examples, the directional argument *qù Niújīn* always follows the V₁-object unit *qí-chē* ‘cycle’, and only the adjunct alone shows up postverbally. It is worth pointing out that in (118c), the particle *de* is inserted into a position after the V₂. Since there is nothing that intervenes between them, the linearisation requirement of *de* is satisfied, which is why the ungrammaticality seen in (114c) does not arise in this case.

(118) *String of verb doubling*: Subj [[V₁ Obj] Argument] [[V₂-Part] Adjunct]

- a. Piet (cóng Lúndūn) qí-chē **qù Niújīn** qí-le **liù-gè-duō zhōngtóu**.
Piet from London ride-bike to Oxford ride-PFV six-CL-more hour
‘Piet cycled to Oxford (from London) for more than six hours.’
(direction...duration)
- b. Piet (cóng Lúndūn) qí-chē **qù Niújīn** qí-guò **wǔ huí**.
Piet from London ride-bike to Oxford ride-EXP five time
‘Piet has cycled to Oxford (from London) five times.’ (direction...frequency)
- c. Piet (cóng Lúndūn) qí-chē **qù Niújīn** qí de **tèbié màn**.
Piet from London ride-bike to Oxford ride DE particularly slowly
‘Piet cycled to Oxford (from London) very slowly.’ (direction...manner)

Another predicted verb doubling configuration with distributed adverbials is illustrated in (119): here the co-occurring adjuncts are also distributed to two different positions within the same verb doubling construction. One of them is pied-piped by the VP that undergoes movement; the other is stranded in the topmost layer of the articulated verbal structure.

(119) *Distributed co-occurrence of an adjunct and an adjunct*



Note that this is actually not the first time we see the configuration above. At the end of section 3.4, a similar configuration was given to illustrate a crucial prediction related to distinctness: since the two adjuncts are no longer in the same Spell-Out domain (i.e., the complement to the ν P, which is not represented here) after verbal predicate fronting, we predict that there will be no violation of distinctness any more in this scenario, and that the both adjuncts can be phonologically realised without causing the co-occurrence issue again. This prediction is corroborated by (120).

(120) *String of verb doubling: Subj [[V₁ Obj] Adjunct] [[V₂-Part] Adjunct]*

- a. Piet qí-chē **wǔ huí** (zǒnggòng) qí-le **liù-gè-duō zhōngtóu**.
 Piet ride-bike five time in.total ride-PFV six-CL-more hour
 ‘Piet cycled five times, and he cycled for more than six hours (in total).’
 (frequency... zǒnggòng ... duration)
- b. Piet qí-chē **liù-gè-duō zhōngtóu** qí de (dōu) **tèbié màn**.
 Piet ride-bike six-CL-more hour ride DE DOU particularly slowly
 ‘Piet cycled for more than six hours, and he cycled very slowly (every hour).’
 (duration... dōu ... manner)
- c. Piet qí-chē **wǔ huí** qí de (dōu) **tèbié màn**.
 Piet ride-bike five time ride DE DOU particularly slowly
 ‘Piet cycled five times, and he cycled very slowly (every time).’
 (frequency... dōu ... manner)

Compared to those in (116), these examples demonstrate that multiple adjuncts are able to co-occur in verb doubling, when they do not show up together in the same postverbal field. In (120a), the frequency adjunct *wǔ huí* gets spelled out within the fronted VP, while the duration adjunct *liù-gè-duō zhōngtóu* inside the stranded verbal structure. In (120b) and (120c), the position after the V_2 is occupied by the same manner adjunct *tèbié màn* in both scenarios, whereas that after the V_1 is respectively taken up by the duration adjunct and the frequency one.

There is one point that is worth noting here: to render these configurations with distributed adjuncts sound more natural and acceptable, it would be better to contain an adverb of quantification like *zǒnggòng* ‘in total’ in (120a) and a distributive marker *dōu* (J.-W. Lin 1998; cf. Cheng 1995) in (120b) and (120c). The main difference between these two quantificational markers is that their direction of scope is not the same, as indicated by the translation above. In (120a), it is the duration adjunct *liù-gè-duō zhōngtóu* that takes scope over the frequency adjunct *wǔ huí* following the fronted VP. In (120b) and (120c), what takes the wide scope becomes the the duration and the frequency phrase that precedes the manner adjunct *tèbié màn* in linear order.

The scopal direction of each quantificational marker is seemingly compatible with the relative scope between adverbials. According to Cinque (1990) and Ernst (2002), the scope between frequency and duration adverbials is pretty variable; whereas the scope of manner adverbials is generally lower than that of frequency and duration ones. The relative scope among adverbials allows us to predict that, it will be possible to replace the quantificational adverb *zǒnggòng* with the distributive marker *dōu* in (120a) (and receive a distributive reading as a result), but impossible to do this the other way around in (120b) and (120c). This is borne out by (121).

- (121) a. Piet qí-chē **wǔ huí** (*dōu*) qí-le **liù-gè-duō zhōngtóu**.
 Piet ride-bike five time DOU ride-PFV six-CL-more hour
 ‘Piet cycled five times, and he cycled for more than six hours (every time).’
 (frequency... *dōu* ...duration; cf. (120a))
- b. *Piet qí-chē **liù-gè-duō zhōngtóu** qí de (*zǒnggòng*) **tèbié** **màn**.
 Piet ride-bike six-CL-more hour ride DE IN.TOTAL particularly slowly
 (duration... *zǒnggòng* ...manner; cf. (120b))
- c. *Piet qí-chē **wǔ huí** qí de (*zǒnggòng*) **tèbié** **màn**.
 Piet ride-bike five time ride DE IN.TOTAL particularly slowly
 (frequency... *zǒnggòng* ...manner; cf. (120c))

In (121a), due to the distributive marker *dōu*, it now becomes the frequency adjunct *wǔ huí* that scopes over the duration adjunct *liù-gè-duō zhōngtóu* in the postverbal field. Together with (120a), this supports the idea that there is no fixed scope between frequency and duration adverbials. By

contrast, neither (121b) nor (121c) is grammatical, since the quantificational adverb *zǒnggòng* requires the manner adverb *tèbié màn* to take a wider scope over the duration/frequency phrase that linearly precedes it, which is impossible due to its own, relatively low scope.

With the examples in (120) and (121) in mind, we can now have a look at those given in (122). Compared to (120), the positions of the two co-occurring adjuncts here are swapped. What deserves attention is that, in (122a) where the duration adjunct precedes the frequency one in linear order, only the adverb of quantification *zǒnggòng* is able to show up in this configuration rather than the distributive marker *dōu*. In contrast, no matter which quantificational marker is adopted, the manner adverb can never linearly precede either the duration phrase or the frequency one in a verb doubling configuration with distributed adjuncts, as shown in (122b) and (122c).

- (122) a. Piet qí-chē **liù-gè-duō zhōngtóu** (%*zǒnggòng*/**dōu*) qí-le **wǔ huí**.
 Piet ride-bike six-CL-more hour in.total DOU ride-PFV five time
 ‘Piet cycled for more than six hours, and he cycled five times (%in total/*every hour).’
 (duration...%*zǒnggòng*/**dōu* ...frequency)
- b. *Piet qí-chē **tèbié màn** (*zǒnggòng*/*dōu*) qí-le **liù-gè-duō zhōngtóu**.
 Piet ride-bike particularly slowly in.total DOU ride-PFV six-CL-more
 hour
 (*manner...*zǒnggòng*/*dōu* ...duration)
- c. *Piet qí-chē **tèbié màn** (*zǒnggòng*/*dōu*) qí-le **wǔ huí**.
 Piet ride-bike particularly slowly in.total DOU ride-PFV five time
 (*manner...*zǒnggòng*/*dōu* ...frequency)

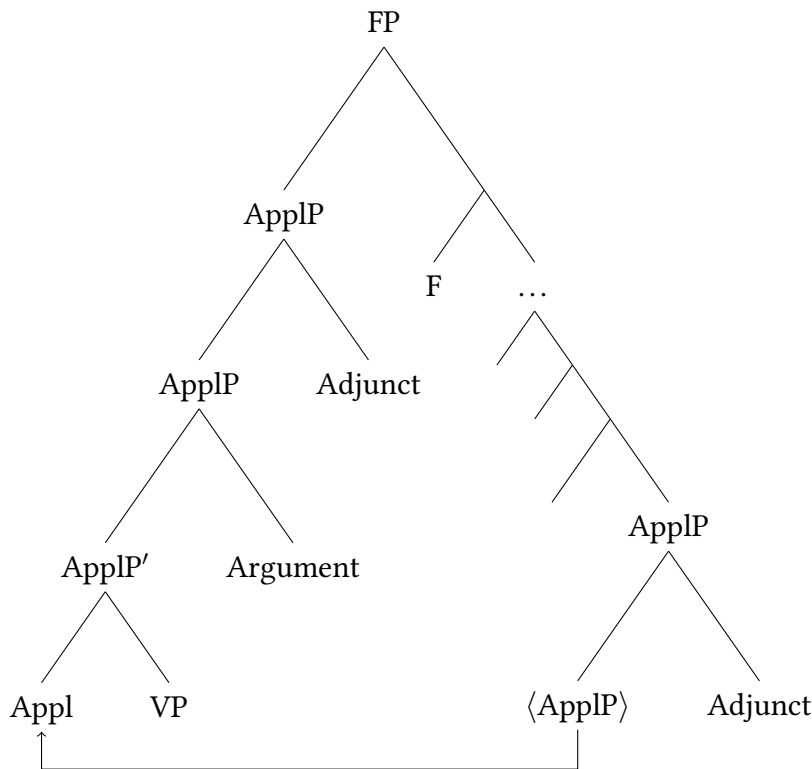
How do we account for the impossible orders of co-occurring adjuncts seen above? Let us consider them one by one. In (122a), that *zǒnggòng* is able to show up in the configuration does not go beyond the expectation, since, as mentioned above, there is no rigid relative scope between duration and frequency adverbials. Due to the scopal direction of this quantificational adverb, here it is the frequency phrase *wǔ huí* that takes scope over the duration one *liù-gè-duō zhōngtóu*. As to the impossible occurrence of *dōu* in the same configuration, I attribute this to the fact that the distributive reading of the duration phrase in this scenario does not make sense.

In (122b) and (122c), it is expected that *dōu* cannot appear in both strings of verb doubling. This is because *dōu* requires the adverbial on its left side to scope over the one on its right side, but the scope of the manner adverb itself is lower than that of its duration/frequency counterpart, thereby leading to a scopal contradiction. Along the same line of reasoning, the appearance of the quantificational adverb *zǒnggòng* should be permitted, since its scopal direction would not contradict the relative scope between adverbials. However, this is contrary to the fact. Here I

propose that the impossible verb doubling in (122b) and (122c) can all be attributed to the same reason that the string $[[V_1 \text{ Obj}] \text{ Adjunct}_{\text{Manner}}]$ actually involves a string vacuous movement: even though this string looks similar to an articulated verbal structure, it in fact involves VP movement inside already which renders it too big to be a target of verbal phrasal movement. To put it differently, this string is identical to a manner-type low verb doubling in terms of involving verbal phrasal movement, but differs in whether the V_2 gets spelled out. I will present more detailed discussion in chapter 5.

Last but not least, there is one more predicted verb doubling configuration with distributed adverbials I would like to show, which is illustrated in (123). This configuration is actually a mixture of the possible co-occurrences of multiple adverbials that I have shown above: here what is targeted for verbal predicate fronting is part of an articulated applicative structure in which an adjunct co-occurs with an argument; the rest of the applicative structure, which is stranded in the postverbal field, only contains another adjunct.

(123)



Although this verb doubling configuration looks pretty complex, it is still borne out empirically, as shown in (124). Note that in all these examples, the scope between adverbials still results from the scopal direction of the quantificational markers *zǒnggòng* and *dōu* that I outlined above.

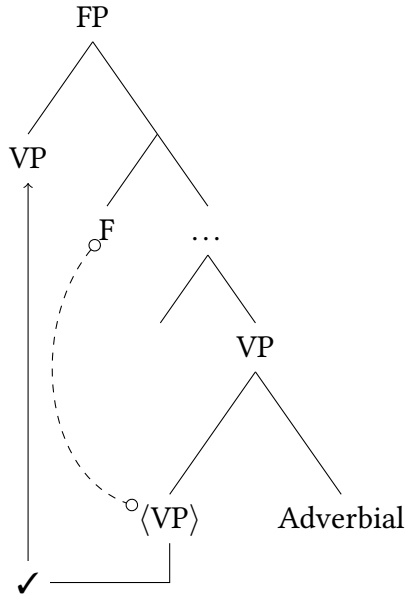
- (124) *String of verb doubling*: Subj [[[V₁ Obj] Argument] Adjunct] [[V₂-Part] Adjunct]
- a. Piet (cóng Lúndūn) qí-chē **qù Niújīn wǔ huí** (zǒnggòng) qí-le
 Piet from London ride-bike to Oxford five times in.total ride-PFV
liù-gè-duō zhōngtóu.
 six-CL-more hour
 ‘Piet cycled to Oxford (from London) five times, and he cycled for more than six hours (in total).’ (direction+frequency... zǒnggòng ... duration)
- b. Piet (cóng Lúndūn) qí-chē **qù Niújīn wǔ huí** (dōu) qí-le **liù-gè-duō zhōngtóu.**
 Piet from London ride-bike to Oxford five times DOU ride-PFV six-CL-more hour
 ‘Piet cycled to Oxford (from London) five times, and he cycled for more than six hours (every time).’ (direction+frequency... dōu ... duration)
- c. Piet (cóng Lúndūn) qí-chē **qù Niújīn liù-gè-duō zhōngtóu** (zǒnggòng) qí-le **wǔ huí.**
 Piet from London ride-bike to Oxford six-CL-more hour in.total ride-PFV five times
 ‘Piet cycled to Oxford (from London) for more than six hours, and he cycled five times (in total).’ (direction+duration... zǒnggòng ... frequency)
- d. Piet (cóng Lúndūn) qí-chē **qù Niújīn liù-gè-duō zhōngtóu** qí de (dōu) **tèbié màn.**
 Piet from London ride-bike to Oxford six-CL-more hour ride DE DOU particularly slowly
 ‘Piet cycled to Oxford (from London) for more than six hours, and he cycled very slowly (every hour).’ (direction+duration... dōu ... manner)
- e. Piet (cóng Lúndūn) qí-chē **qù Niújīn wǔ huí** qí de (dōu) **tèbié màn.**
 Piet from London ride-bike to Oxford five times ride DE DOU extremely slowly
 ‘Piet cycled to Oxford (from London) five times, and he cycled very slowly (every time).’ (direction+frequency... dōu ... manner)

To sum up, in this section I have presented three verb doubling configurations predicted by the proposal that verbal projections larger than simple VPs can also be target for verbal predicate fronting. The existence of these configurations not only provides further support for the articulated verbal structures that I proposed in section 3.3, but also reveals that co-occurring adverbials can be systematically arranged within verb doubling constructions.

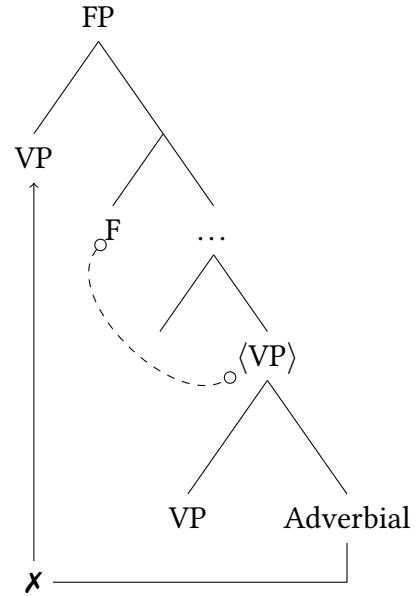
4.3 The stranding problem of the postverbal adverbials

There is one analytic problem that I would like to discuss in this section, which is concerned with the postverbal adverbials. From the previous two sections, we have known that verbal predicate fronting is not confined to simple VPs (e.g., ApplPs), and that part of an articulated verbal structure (e.g., verbal phrases with multiple adverbials) can also be targeted for verbal phrasal movement. What is interesting is that, the topmost adverbial of the entire verbal structure is stranded in all the grammatical verb doubling configurations that we have seen so far, as illustrated in (125a). We have never seen a single case where the entire verbal phrase undergoes fronting and no adverbial is stranded in the postverbal field, as illustrated in (125b).

(125) a. *Fronting without the adverbial*



b. *Fronting with the adverbial*



Empirical evidence confirms that the second scenario (125b) is never possible. For instance, (126) contains two cases of canonical transitive verb doubling with only one adverbial. No matter the adverbial is a freely-adjoined adjunct (as in (126a)) or a selected argument (as in (126b)), neither of them can leave the postverbal field and be pied-piped by the fronted verbal phrase.

(126) a. *Piet kàn nà-bù jìlùpiàn **shí biàn** kàn-le.

Piet watch that-CL documentary ten time watch-PFV

Int.: 'Piet watched the documentary ten times.'

(cf. (86a))

b. *Piet sǎn-bù qù Wéiduōliǎ Gōngyuán sǎn-le.

Piet disperse-step to Victoria park disperse-to-PFV

Int.: 'Piet went for a walk to Victoria Park.'

(=(64b))

The same situation is also found in ditransitive verb doubling in (127). (127a) displays a double object construction, while (127b) a prepositional dative one. In both configurations, verbal predicate fronting cannot target the entire verbal phrase within which the adverbial is contained.

- (127) a. *Piet jiào Wassily fēntǐlùn jùfǎxué **shí biàn** jiào-le.
 Piet teach Wassily mereology syntax ten time teach-PFV
 Int.: ‘Piet taught Wassily mereological syntax ten times.’ (cf. (88a))
- b. *Piet_i jièshào tā_i-de péngyǒu gěi Wassily **hěnduō biàn** jièshào-le.
 Piet introduce 3SG-POSS friend to Wassily many time introduce-PFV
 Int.: ‘Piet introduced his friend to Wassily many times.’ (cf. (93a))

In verb doubling configurations with multiple adverbials (128), we find that these examples are not different from what we have seen above: as long as the postverbal field is not filled by a stranded adverbial, the sentence becomes ungrammatical.

- (128) a. *Piet (cóng Lúndūn) qí-chē qù Niújīn **liù-gè-duō zhōngtóu** (zǒnggòng) qí-le.
 Piet from London ride-bike to Oxford six-CL-more hour in.total
 ride-PFV
 Int.: ‘Piet cycled to Oxford (from London) for more than six hours (in total).’ (cf. (118a))
- b. *Piet qí-chē wǔ huí **liù-gè-duō zhōngtóu** (zǒnggòng) qí-le.
 Piet ride-bike five time six-CL-more hour in.total ride-PFV
 Int.: ‘Piet cycled five times, and he cycled for more than six hours (in total).’ (cf. (120a))

Based on the examples above, I generalise this problem as in (129), and refer to it as the *stranding problem* of verb doubling.

- (129) *The stranding problem of verb doubling*
 Why does the topmost adverbial of the entire verbal phrase have to be stranded postverbally in verb doubling?

An ideal solution to the stranding problem should be some constraint which can be found elsewhere in the grammar and take the same effect in other similar situations. The solution I would like to suggest here is to appeal to Abels (2003)’ anti-locality constraint, an idea that I adopt from Lai (2021, 2024).

In Lai (2021:68 *et seq.*)’s own work on verb doubling, he notices another puzzle which is similar but not totally identical to the stranding problem, which I refer to as the *obligatoriness problem* of verb doubling. The problem is as follows: in canonical non-doubling configurations, the appearance of postverbal adverbials like frequency or duration phrases is optional, as shown in (130a). Given the non-selected status of the frequency phrase *shí biàn* ‘ten times’, its optionality is actually what we expect. However, the appearance of the same adverbial becomes obligatory in verb doubling: (130b) and (130c) are respectively low and high verb doubling of the same type; in either example, the doubling configuration becomes ungrammatical without the frequency adjunct.

(130) *Examples of the obligatoriness puzzle*

- a. Piet kàn-le nà-bù jìlùpiàn (shí biàn).
Piet watch-PFV that-CL documentary ten time
‘Piet watched the documentary (ten times).’
- b. Piet kàn nà-bù jìlùpiàn kàn-le *(shí biàn).
Piet watch that-CL documentary watch-PFV ten time
‘Piet watched the documentary ten times.’ (=86a))
- c. Kàn nà-bù jìlùpiàn Piet kàn-le *(shí biàn).
watch that-CL documentary Piet watch-PFV ten time
‘As for watching the documentary, Piet watched it ten times.’ (=86b))

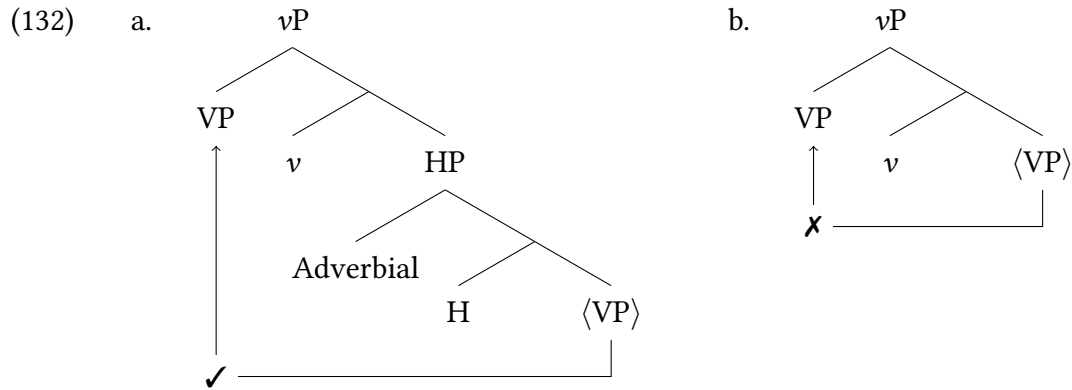
Lai relates the obligatoriness problem to the complement-to-specifier version of anti-locality proposed by Abels (2003, 2012), as given in (131) (also see Newman 2020 for a more generalised version). According to this constraint, complement-to-specifier movement cannot take place within the same projection because that movement distance is not sufficiently long enough.

(131) *Comp-to-Spec Anti-locality Constraint*

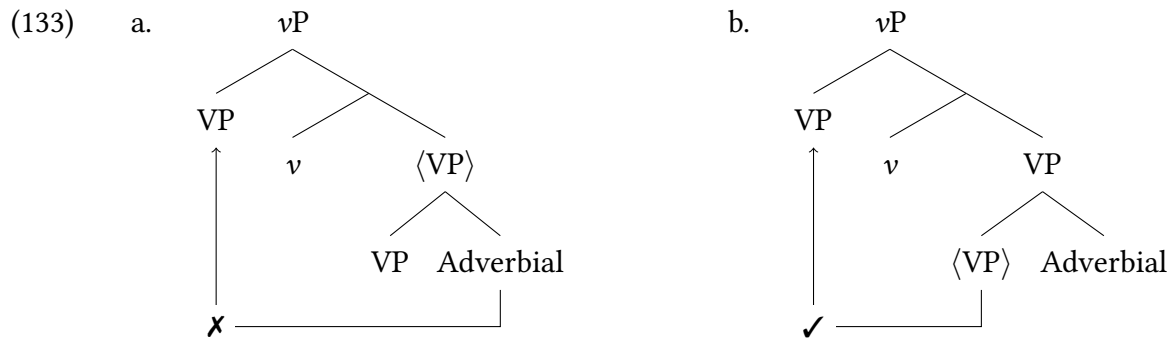
Movement of the complement of a head X must cross a head other than X.

(Abels 2003:12-14, *et seq.*)

Lai argues that the obligatory presence of the postverbal adverbial in verb doubling is to make sure that VP-fronting would not violate the anti-locality constraint. Unlike the current proposal, he analyses postverbal adverbials as specifiers of some functional projection between VP and vP, as illustrated in (132a). In this scenario, VP-movement to the phase edge (i.e., Spec,vP) does not violate the anti-locality constraint, because this movement starts from the complement of the functional head H and eventually stops at the specifier of a projection higher than HP. By contrast, (132b) demonstrates a clear violation of anti-locality due to the disappearance of HP.



Borrowing insights from Lai's treatment of the adverbials, we can find that Abels' anti-locality constraint takes similar effect in the stranding problem. Let us consider (133a) first and assume that vP is a phase:⁵ in this derivation, what is moved to the phase edge is the entire VP with the adverbial included, which, in principle, is permitted under the current proposal. However, if this movement takes place, it will violate the anti-locality constraint given above, because this is a local movement from the complement of vP to its own specifier. Therefore, the entire VP cannot be involved in verb doubling.



The result of ruling out the first derivation is that the probe (i.e., the phase head in this case) needs to continue to look for a matching goal in its c-command domain. As illustrated in (133b), the next target for verbal predicate fronting is the second-highest VP within the entire verbal structure, which does not contain the adverbial. Unlike the previous scenario, this new target VP will not face the risk of violating anti-locality when it moves to the phase edge, because, due to the 'salvage' of the adverbial, it is NOT really the complement of vP : together with the adverbial,

⁵ For Abels (2003)'s version of anti-locality, this assumption is obligatory, since it has further connection with distribution of features in his framework (see relevant discussion right below). To address the stranding problem here, whether this assumption is accepted or not does not make huge difference, because anti-locality will only be applied as a filter in derivation. But for my analysis elsewhere, e.g., to account for the constraint on co-occurrence of multiple postverbal adjuncts in section 3.4, that vP is a phase is independently required to make the distinctness analysis work.

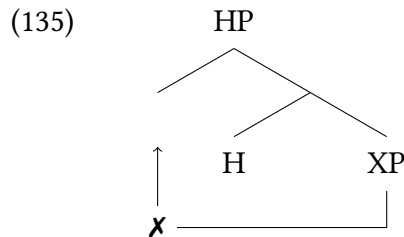
it is just part of the genuine VP complement.⁶ In other words, this verbal phrasal movement is actually sub-extraction out of an articulated VP. Since the adverbial is not contained within the target VP in the first place, this explains why the topmost adverbial of the entire verbal structure is always stranded in the postverbal field.

It appears that the anti-locality constraint offers a direct solution to the stranding problem. However, there is one crucial question about how this constraint is implemented. In Abels (2003)' original proposal, the anti-locality constraint works tightly with another condition termed *last resort*, as given in (134).

(134) *Last resort*

A constituent α may only be merged, i.e., base-merged or re-merged, if that leads to the immediate satisfaction of a previously unsatisfiable feature. (Abels 2003:92, (92))

In a scenario like (135), the H head first merges with the XP complement. Assuming that merge (either external or internal) is feature-triggered, that means that there is a feature satisfaction relation established between H and XP. If XP is then moved to the specifier position of HP, this movement will not lead to a new feature satisfaction relation, which violates the last resort condition. In this sense, we can conclude that the anti-locality constraint is in effect blocked by the last resort condition.



If the last resort condition is also taken into account in my proposed anti-locality solution to the stranding problem, we may encounter an issue related to feature satisfaction in (133b). Recall that in section 3.3, I proposed that adverbial adjuncts are freely-adjoined to the maximal verbal projection. This leads to the result that there is no difference between the topmost verbal projection and the second-highest one in terms of feature specification (e.g., they are both VPs). In the derivation (133b), the first feature satisfaction relation is established between the v head and the VP complement. Next, when the second-highest VP is extracted out of the VP complement and

⁶ Just to clarify, I define 'complement' as the constituent that mutually c-commands with a head. In (133), the topmost VP and the v head mutually c-command each other, therefore the former is the complement of the latter. But the second-highest VP, which is the sister of the adverbial, is not the complement of v due to the absence of the mutual c-commanding relation.

moved to Spec,vP, that will create another feature satisfaction relation between *v* and VP, which does not count as an “immediate satisfaction of a previously unsatisfiable feature”, a violation of the last resort condition. If so, this derivation (133b) should be ruled out, not because of the anti-locality constraint, but because of the last resort condition instead. The same issue will not arise in scenarios where the postverbal adverbial is an argument, because there the topmost category of the entire verbal structure is an ApplP instead of a VP, which ensures that the two feature satisfaction relations that the *v* head has with its complement and specifier are different.

The problem sketched above requires us to think more about the implementation of the anti-locality constraint in general (see Ershova 2024 for some discussion). At the moment, I can only apply anti-locality as a derivational filter effective in narrow syntax.

There is another potential solution to the stranding problem that deserves attention, which shifts the burden of explanation from narrow syntax to the syntax-phonology interface. The basic idea is as follows: since the second occurrence of the doubled verb cannot stand on its own and has to be followed by (at least) an adverbial, these properties can be re-interpreted as some constraints on the prosodic structure of verb doubling. That is, the V_2 in verb doubling disprefers forming a prosodic phrase in isolation from other prosodic words; instead, it prefers a derivation in which it is followed by some other element (e.g., the adverbial) and able to form a bi-branching, prosodic structure.

I will not flesh out this alternative solution in the rest of the section due to two reasons. On the one hand, this requires us to be more familiar with the prosody of Mandarin Chinese in the first place. On the other hand, it would be better to implement this idea with other Optimality-Theoretic constraints, which is beyond the scope here (see the Appendix to this chapter for an initial attempt). What matters to us is that, nothing stops the possibility that the stranding problem could be prosodic in nature: recall that in section 3.4, I have already proposed a distinctness-based solution to the co-occurrence issue of multiple adjuncts; so it is plausible that the solution to the stranding problem also lies at the syntax-phonology interface.

4.4 Simple verb-object units as verbal phrases or compounds

Before wrapping up the whole chapter, I would like to have some brief discussion on simple verb-object units involved in verb doubling constructions. For the purpose of clarity, I use the examples (50) introduced at the beginning of section 3.1 to illustrate these units, repeated in (136) below.

- (136) a. Piet **dú** **běnkē** dú-le liǎng huí.
 Piet study undergraduate study-PFV two time
 ‘Piet studied for an undergraduate degree twice.’ (frequency-type)

- b. Piet **dú** **běnkē** dú-le wǔ nián.
Piet study undergraduate study-PFV five year
'Piet studied for an undergraduate degree for five years.' (duration-type)
- c. Piet **kàn** **mànhuà** kàn de hěn màn.
Piet read comic read DE very slowly
'Piet read comics very slowly.' (manner-type)
- d. Piet **kàn** **mànhuà** kàn de hěn yūn.
Piet read comic read DE very dizzy
'Piet read comics and felt dizzy (as a result).' (resultative-type)
- e. Piet **xiě-zì** xiě-zài-le qiáng-shàng.
Piet write-word write-LOC-PFV wall-up
'Piet wrote on the wall.' (locative-type)
- f. Piet **sàn-bù** sàn-qù-le Wéiduōlìyǎ Gōngyuán.
Piet disperse-step disperse-to-PFV Victoria park
'Piet went for a walk to Victoria Park.' (directional-type)

Li and Thompson (1981:442) notice that in a verb doubling configuration, “what is labelled direct object can be either an *actual direct object* or the object component of a *verb-object compound*” (emphasis mine). They offer three defining properties of verb-object compounds, as listed below:

- (137) 1. One or both of the constituents being bound morphemes
2. Idiomaticity of the meaning of the entire unit
3. Inseparability or limited separability of the constituents

(Li and Thompson (1981:73))

If we apply these criteria to all the verb-object units in (136), we will find that some of them do behave like verbal compounds rather than verbal phrases. In what follows, I will use the last two criteria to show their differences.⁷

Let me begin with the second criterion, which is concerned with the meaning of the verb-object unit. As suggested by the glossing and the translation, neither *dú běnkē* ‘study for an undergraduate degree’ in (136a)/(136b) nor *kàn mànhuà* ‘read comics’ in (136c)/(136d) falls into the group of verb-object compounds; instead, these two are standard verbal phrases since they both convey a non-idiomatic, compositional meaning. The non-compound nature of these two phrases is further supported by the fact that both verbal elements *dú* and *kàn* are very productive

⁷ Given the fact that hat Mandarin Chinese does not have a productive morphological system, which renders the recognition of bound morphemes a bit tricky, I leave the first criterion aside in the discussion here.

verbs in Mandarin which can combine with numerous nominals to form verbal phrases, as shown in (138a) and (138b). Similarly, both *běnkē* and *mànhuà* can be taken by many other verbs as direct complements, as in (138c) and (138d). As what we expect, the meaning of all the verbal phrases in (138) is compositional.

- (138) a. **dú** yánjiūshēng / bóshì / dàxué
study postgraduate doctorate university
'study for a postgraduate degree / study for a doctorate / study at university'
- b. **kàn** bàozhǐ / zázhì / xiǎoshuō
read newspaper magazine novel
'read newspapers / magazines / novels'
- c. zhuǎnrù / jìnrù **běnkē**
transfer enter undergraduate
'transferred to / enter undergraduate'
- d. chuàngzuò / xiūgǎi **mànhuà**
create revise comic
'create / revise comics'

By contrast, this meaning-related property of verb-object compounds can more or like be found on *xiě-zì* 'write' in (136e) and 'sàn-bù' *go for a walk* in (136f). Consider *xiě-zì* first: though both components have their own meanings and are often used with other verbal/nominal elements, their combined meaning is idiomatic to some extent (which only means 'write'), since the individual meaning of *zì* (i.e., 'word') is lost, which suggests that *xiě-zì* is a verb-object compound. The compound status of *sàn-bù* is much easier to notice: its meaning is clearly idiomatic, since it is not compositionally derivable from the meaning of each component (i.e., *sàn* 'disperse', and *bù* 'step').

The dichotomy between verb-object phrases and verb-object compounds is further corroborated by the third property in (137): if an object-like element is an object component of a verbal compound rather than an actual object of a verbal phrase, we will not see total separation between the verb and the object. As exemplified in (139) below, the object is introduced by a demonstrative-classifier cluster and rendered definite in each case, namely, *nà-gè běnkē* 'that undergraduate degree' in (139a)/(139b), and *nà-běn mànhuà* 'that comic' in (139c)/(139d). Since definite expressions are certainly separable constituents and ungrammaticality does not arise in any of these examples, these confirm that the objects *běnkē* and *mànhuà* do not form compounds with the verbs *dú* and *kàn*.

- (139) a. Piet **dú** *nà-gè běnkē* dú-le liǎng huí.
 Piet study that-CL undergraduate study-PFV two time
 ‘Piet studied for that undergraduate degree twice.’ (frequency-type)
- b. Piet **dú** *nà-gè běnkē* dú-le wǔ nián.
 Piet study that-CL undergraduate study-PFV five year
 ‘Piet studied for that undergraduate degree for five years.’ (duration-type)
- c. Piet **kàn** *nà-běn mànhuà kàn* de hěn màn.
 Piet read that-CL comic read DE very slowly
 ‘Piet read that comic very slowly.’ (manner-type)
- d. Piet **kàn** *nà-běn mànhuà kàn* de hěn yūn.
 Piet read that-CL comic read DE very dizzy
 ‘Piet read that comic and felt dizzy (as a result).’ (resultative-type)

Now consider (140a) and (140b): compared to (136e) and (136f), the only change to both verb doubling constructions is that the original object components *zì* ‘word’ and *bù* ‘step’ are replaced by definite expressions, introduced by the same demonstrative-classifier cluster as above. However, unlike the grammatical results seen in (139), neither of the examples in (140) is totally acceptable,⁸ which suggests that the object component in these two configurations is hardly separable from the preceding verbal component. This result provides further confirmation that these objects-like elements constitute compounds with verbs.

- (140) a. ?Piet **xiě** *nà-gè zì* xiě-zài-le qiáng-shàng.
 Piet write that-CL word write-on-PFV wall-up
 (locative-type)
- b. *Piet **sàn** *nà-chǎng bù* sàn-qù-le Wéiduōliǎ Gōngyuán.
 Piet disperse that-CL step disperse-to-PFV Victoria park
 (directional-type)

We have seen that among simple verb-object units in Mandarin, there are two subtypes: one of them is full verbal phrases, and the other is verbal compounds. What matters to the current proposal is as follows: so far I have only presented doubling configurations that are built upon either

⁸ I find the judgement of (140a) a bit trickier than that of (140b), therefore I give a question mark to the former but an asterisk to the latter. One possible reason could be that, as mentioned earlier, the compound status of *xiě-zì* is not as steady as that of *sàn-bù*: if enough context is provided, *nà-gè zì* ‘that word’ in (140a) can refer to a specific, salient entity in the context. In that sense this element is understood as a single constituent rather than part of a verb-object compound. However, a similar situation can never happen to (140b). Such a difference between compounds conforms to Li and Thompson (1981:80)’s generalisation that “the behaviour of verb-object compounds forms a continuum, with any specific compound falling at some point on the continuum”.

transitive verbs or ditransitive ones, each of which contains one internal argument at least. The reason why I do not mention intransitive verbs is that doubling configurations built upon them are usually judged degraded or totally unacceptable, as shown in (141), where the intransitive *kū* ‘cry’ is doubled.

(141) *Doubling with intransitive verbs*

- a. ??Piet **kū kū** de hěn dàshēng.
Piet cry cry DE very loudly
‘Piet cried very loudly.’ (manner-type)
- b. ??Piet **kū kū** de hěn nánshòu.
Piet cry cry DE very uncomfortable
‘Piet cried and became very uncomfortable.’ (resultative-type)
- c. ??Piet **kū kū-le** yī-bǎi huí.
Piet cry cry-PFV one-hundred time
‘Piet cried a hundred times.’ (frequency-type)
- d. ??Piet **kū kū-le** sān-tiān-sān-yè.
Piet cry cry-PFV three-day-three-night
‘Piet cried for three whole days.’ (duration-type)

The properties of verbal compounds we have seen above might leave one wondering whether they are intransitive or transitive in nature. The fact that verbal compounds, as with full verbal phrases, can be the target for predicate fronting involved in verb doubling (e.g., in (136)) suggests that verbal compounds are not really intransitive (otherwise doubling would not be possible in the first place). If this is the case, then we need to ask ourselves the following two questions. First, what is independent evidence supporting the idea that verbal compounds are transitive VPs in nature? Second, what makes verbal compounds so different from other verbal phrases?

I argue that the answers to these two questions are both concerned with the property of the object component in verbal compounds. To be specific, these object components show the behaviour of what Cheng and Sybesma (1998) define as *dummy objects* (also see Paul 1998 and Badan 2013): in a verb-object unit, a dummy object does not make much contribution to the meaning of the combination as a whole, though it is syntactically active. As already mentioned above, semantically, the object components *zì* and *bù* do behave like dummy elements, since their meanings are lost (or invisible) in verbal compounds. Syntactically, there are two arguments that favour the idea that these object components still behave as active objects. First, they are in complementary distribution with other objects, as shown in (142) and (143)

- (142) a. xiě-zì / xiě xiǎoshuō
 write- word write novel
 ‘write / write novels’
- b. *xiě-zì xiǎoshuō / *xiě xiǎoshuō zì
 write-word novel write novel word
- (143) a. sàn-bù / sàn-xīn
 disperse-step disperse-heart
 ‘go for a walk / relax’
- b. *sàn-bù-xīn / *sàn-xīn-bù
 disperse-step-heart disperse-heart-step

In (142), we can find that, although *zì* and *xiǎoshuō* can respectively form a verbal compound and a verbal phrase with *xiě* (as in (142a)), they are unable to co-occur after the same verb, regardless of the relative order (as in (142b)). The same situation is found in (143): either *bù* or *xīn* ‘heart’ can individually form a verbal compound with *sàn* (as in (143a)); however, these two object components cannot appear after the same verb at the same time (as in (143b)). Suppose that both *xiě* and *sàn* can license one and only one internal argument, it follows naturally that the object components in (142b) and (143b) are competing for the same object position, thereby showing complementary distribution.

Second, the object components in verbal compounds can be modified by time expressions, as exemplified in (144).

- (144) a. Piet xiě-le yī xiàwǔ de zì.
 Piet write-PFV one afternoon DE_{nom} word
 ‘Piet wrote all afternoon.’
- b. Piet sàn-le bàn-gè-duō zhōngtóu de bù.
 Piet disperse-PFV half-CL-more hour DE_{nom} step
 ‘Piet went for a walk for more than half an hour.’

In both examples, the object component is preceded by a time expression plus a particle *de*, which is generally considered as a modification marker: in (144a), *zì* is modified by *yī xiàwǔ de* ‘all afternoon’; and in (144b), *bù* is modified by *bàn-gè-duō zhōngtóu de* ‘more than half an hour’. Since this kind of modification is also found in (145), which shows verbal phrases consisting of active objects, what (142) demonstrates suggests that the object components in verbal compounds should be treated equally.

- (145) a. Piet dú-le sì nián de běnkē.
 Piet study-PFV four year DE_{nom} undergraduate
 ‘Piet studied for an undergraduate degree for four years.’
- b. Piet kàn-le yī wǎn de mànhuà.
 Piet read-PFV one night DE_{nom} comic
 ‘Piet read comics all night.’

Based on the discussion above, I conclude that there are reasons to regard the dummy object components in verbal compounds as active objects, which further supports the idea that the verbal compounds are essentially transitive VPs. The remaining problem is to account for the differences between verbal phrases and compounds. To put it differently, we need explanations for the two properties of verbal compounds seen earlier, i.e., the idiomatic interpretation, and the inseparability or limited separability of the constituents in (140).

For the former property, I attribute it to the lexical meaning of the verbal component alone. In certain contexts, the verbal component itself is sufficient to contribute to the entire idiomatic meaning, as shown in (146).

- (146) a. Piet hái zài **xiě** ne.
 Piet still PROG write SFP
 ‘Piet is still writing.’ (cf. (136e))
- b. Piet gāng qù **sàn**-le yīhuǐ’er.
 Piet just to disperse-PFV a.while
 ‘Piet just went for a walk.’ (cf. (136f))

Considering that the object component of the verbal compound is a dummy element with little meaning, it will not alter the meaning conveyed by the verbal component when these two are combined. Therefore, the idiomatic meaning of the verbal compound is in fact totally decided by that of the verbal component used as a single verb phrase, which is certainly different from a verbal phrase whose meaning is compositional.

As to the latter property, I argue that ‘(in)separability’ is not defining enough as a property. In (144) and (145), we have seen that both the object component of the verbal compound and the direct object of the verbal phrase can be modified by time expressions and be separated from the verbal component. This fact is against the third defining properties proposed by Li and Thompson (1981), and furthermore, suggests that (in)separability might not a proper criterion for distinguishing verbal compounds from verbal phrases. If this is the case, then we need to provide a different account of why the definite expressions are ungrammatical in verbal compounds (140). I attribute this to the fact that dummy object components are also characterised as non-referential objects

(Cheng and Sybesma 1998), which means that they are essentially incompatible with definite markers. Why dummy objects have to be non-referential requires further investigation, which I will leave it aside for future study.

4.5 Chapter summary

To summarise, in this chapter I have shown a series of verb doubling configurations which consist of more than one direct object and one adverbial as their verbal dependents. By first investigating ditransitive verb doubling in double object and prepositional dative constructions, I have proposed that verb doubling can target either an ApplP or a VP for verbal phrasal movement. Based on this proposal, I have correctly predicted that verb doubling can contain multiple co-occurring adverbials, as long as they are distributed across the string of verb doubling and do not show up in the same postverbal field. I have also discussed the stranding problem of verb doubling, and argued for the transitive VP status of verbal compounds. In the next chapter, I will focus on the question of how the same verb gets pronounced twice, and propose a uniform PF-account of all types of verb doubling.

Appendix

This section presents some initial attempt to address the stranding problem via the syntax-phonology interface.⁹ It has been long recognised that there are correspondences and mismatches found between the syntactic structure and the prosodic structure of an utterance (Selkirk 1984, among many others). One of the general tendencies spotted in a prosodic structure but not in a syntactic one is *binarity effects*, that is, two prosodic words tend to be grouped together into one prosodic phrase (see Elfner 2012 for relevant discussion). For instance, (147) shows a nominal phrase in Japanese, in which *wain-no* ‘wine’ and *e* ‘painting’ do not directly form a constituent, leading to the interpretation ‘painting of awfully tasty wine’. However, prosodically, these two words do form a single prosodic phrase φ , as illustrated in (148).

- (147) $[_{NP} [_{NP} [_{AP} \text{yakeni umai}] \text{wain-no}] \text{e}]$
 awfully tasty wine-GEN painting
 ‘painting of awfully tasty wine’ (adapted from Ishihara and Kalivoda 2022:(10))
- (148) $[_{NP} [_{NP} [_{AP} \text{yakeni umai}] \text{wain-no}] \text{e}] \rightarrow ({}_{\varphi} ({}_{\varphi} \text{yakeni umai}) ({}_{\varphi} \text{wain-no e}))$

⁹ I thank Coppe van Urk again for inspiring me on the following idea.

Under Match Theory (Selkirk 2011; also see Ishihara and Kalivoda 2022 for an overview), divergences between syntactic and prosodic structures, like the binarity effects given above, can be implemented by various markedness constraints. The two constraints in (149) are adopted on purpose, as they have been applied in analysing the binarity effects in Japanese (e.g., Selkirk 2011; Ishihara 2014).

(149) *Binarity constraints*

- a. $\text{BINMIN}(\varphi, \omega)$
Assign a violation for every node φ that dominates fewer than 2 nodes ω .
- b. $\text{BINMAX}(\varphi, \omega)$
Assign a violation for every node φ that dominates more than 2 nodes ω .

Due to $\text{BINMAX}(\varphi, \omega)$, *wain-no* is prevented from forming a φ with its preceding modifier *yakeni umani* ‘awfully tasty’ (which itself is already a φ) in (147), since that will create a phrase consisting of more than two ω . Meanwhile, due to $\text{BINMIN}(\varphi, \omega)$, *wain-no* and *e* are prevented from being phrased individually, leading to the result that they two form a single φ as in (148).

Match Theory offers us an opportunity to re-examine the stranding problem through the lens of binarity effects. Following the basic syntax-to-prosody mapping constraint $\text{MATCH}(\text{XP}, \varphi)$ in (150), I suppose that in Mandarin, syntactic VPs correspond to prosodic phrases φ .

(150) $\text{MATCH}(\text{XP}, \varphi)$

The left and right edges of a constituent of type XP must correspond to the left and right edges of a constituent of type φ .

Based on this idea, the prosodic structure of the verb doubling string (151) is illustrated in (152). I propose that the binarity constraints given in (149) also take effect in the prosodic structure here.

- (151) $[\text{FP } [\text{VP } \text{kàn } \text{nà-bù } \text{jìlùpiàn }] [\text{VP } \text{kàn-le } *(\text{shí biàn})]]$
 watch that-CL documentary watch-PFV ten time
 ‘watched the documentary ten times’ (=(130b))

- (152) $[\text{FP } [\text{VP } \text{kàn } \text{nà-bù } \text{jìlùpiàn }] [\text{VP } \text{kàn-le } *(\text{shí biàn})]]$
 $\rightarrow (\varphi (\varphi (\omega \text{kàn}) (\omega \text{nà-bù jìlùpiàn})) (\varphi (\omega \text{kàn-le}) *(\omega \text{shí biàn})))$

Due to $\text{BINMAX}(\varphi, \omega)$, the frequency adjunct *shí biàn* is prevented from being pied-piped by the VP *kàn nà-bù jìlùpiàn* ‘watch the documentary’, because moving the entire verbal structure

will create a φ consisting of three ω . At the same time, $\text{BINMIN}(\varphi, \omega)$ requires that the adjunct has to stay in the postverbal field; otherwise, the second VP will form a φ with only one ω .

The way I implement the binarity constraints (149) here is quite simplified. For example, I do not put these two constraints with other ones and rank all of them in the conventional Optimality-Theoretic tableau. Furthermore, there are a lot of related questions that need to be investigated, such as the prosodic structure of verb doubling with multiple and/or distributed adverbials, as given in section 4.2. In addition, how adjunction can be properly mapped from the syntactic structure to the prosodic one is still a debatable question in the relevant literature (e.g., Truckenbrodt 1999; Bellik and Kalivoda 2016; Brodtkin 2025, among many). I will have to leave all these questions for future research.

Chapter 5

PF Algorithm of Verb Doubling

My discussion so far has only been concerned with the internal syntax of verb doubling, and has not touched upon the question on how the same verb gets pronounced twice at PF. The purpose of this chapter is to show how verb doubling effects are realised at PF.

My point of departure in this chapter is those particles that appear on the second occurrence of the doubled verb. In the previous discussion, we have seen two kinds of particles associated with adverbial adjuncts in verb doubling. One of them is commonly known as ‘tense and aspect markers’ (henceforth TAMs), such as *le* and *guò* in (153). Both of them denote perfective aspect, and function as verbal suffixes to express the event as terminated and bounded by a reference time. However, *guò* differs from *le* in further implying discontinuity: that is, *guò* implies that the final state of an event no longer holds in the current time, whereas *le* offers no information beyond the final point of the event (M. Liu 2015).

(153) *Verb doubling with TAMs*

- a. Piet kàn nà-běn xiǎoshuō kàn-**le/guò** shí biàn.
Piet read that-CL novel read-PFV/EXP ten time
‘Piet read/has read the novel ten times.’
- b. Kàn nà-běn xiǎoshuō Piet kàn-**le/guò** shí biàn.
read that-CL novel Piet read-PFV/EXP ten time
‘As for reading the novel, Piet read/has read it ten times.’

The other kind of particles is *de*, as given in (154). The function of *de* is to introduce what is traditionally labelled as ‘descriptive complements’ (e.g., Tang 1990; Li 1990; Huang et al. 2009), which leads to either a manner or a resultative interpretation (see Huang 1988; Cheng 2007; Meadows and Yan 2023, 2025, among many). To maintain consistency with my proposal in chapter 3, here I label the manner and resultative expressions as ‘descriptive adjuncts’ instead, and corre-

spondingly, refer to *de* as a marker of descriptive adjunct (henceforth DAM).

(154) *Verb doubling with DAMs*

- a. Piet kàn nà-běn xiǎoshuō kàn **de** tèbié zǐxì/nánshòu.
Piet read that-CL novel read DE particularly carefully/uncomfortable
‘Piet read the novel very carefully/...and became uncomfortable.’
- b. Kàn nà-běn xiǎoshuō Piet kàn **de** tèbié zǐxì/nánshòu.
read that-CL novel Piet read DE particularly carefully/uncomfortable
‘As for reading the novel, Piet read it very carefully/...and became uncomfortable.’

In section 5.1, I first compare these two groups of particles, and show that they have more differences than similarities they share. I argue that these particles cannot originate from the same source in the clausal spine, and that their fundamental differences can be attributed to their different base positions. Then in section 5.2, I develop a PF-based analysis of realising verb doubling effects. I propose that there are a series of post-syntactic operations that take place at PF, and that their relative order to each other varies in different verb doubling configurations. In section 5.3, I extend my proposed analysis to two other configurations that are closely-related to canonical cases of verb doubling, and show how they can be derived by the same post-syntactic mechanism.

5.1 A tale of two sources of particles

In this section, I will use *le* and manner *de* as representatives of the two groups of particles, and focus on comparisons between these two. I will first show similarities between *le* and manner *de*, which suggests that they are manipulated by the same mechanism at PF. I will further show that there are more differences than similarities between these two particles, which indicates that particles involved in multiple spell-outs of verbs are actually from two different sources.

5.1.1 Similarities among particles

I will begin with presenting two similarities shared by *le* and manner *de*. The first similarity is that in verb doubling configurations, both particles must appear on the V_2 instead of the V_1 (Li and Thompson 1981; Huang 1982; Paul 2002a; Lai 2021, 2024), as shown in (153) and (154) above. (155) demonstrates a configuration where *le* is attached to the first occurrence of *kàn* ‘read’ rather than the second one. It turns out that TAMs cannot be attached to the V_1 when the verb is doubled. This fact is compatible with the proposal I put forward in chapter 4, namely that the size of the

fronted verbal phrases is either a VP or an ApplP: either of them is too small to include TAMs.

- (155) a. *Piet *kàn-le* nà-běn xiǎoshuō *kàn* shí biàn.
 Piet read-PFV that-CL novel read ten time
 Int.: ‘Piet read the novel ten times.’
- b. **Kàn-le* nà-běn xiǎoshuō Piet *kàn* shí biàn.
 read-PFV that-CL novel Piet read ten time
 Int.: ‘As for reading the novel, Piet read it ten times.’

A similar configuration is given in (156), where manner *de* appears on the V_1 , unlike its canonical appearance on the V_2 . As with what (155) shows, neither of the examples here is grammatical, which means that DAMs are also prohibited from appearing on the first occurrence of the doubled verb.

- (156) a. *Piet *kàn de* nà-běn xiǎoshuō *kàn* tèbié zǐxì.
 Piet read DE that-CL novel read particularly carefully
 Int.: ‘Piet read the novel very carefully.’
- b. **Kàn de* nà-běn xiǎoshuō Piet *kàn* tèbié zǐxì.
 read DE that-CL novel Piet read particularly carefully
 Int.: ‘As for reading the novel, Piet read it very carefully.’

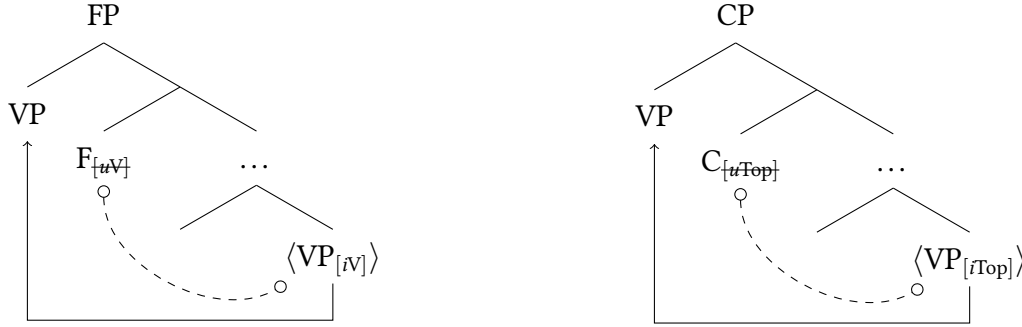
A crucial question that arises here is why particles have to appear on the V_2 . I attribute this requirement of particles to the fact that only lexical verbs that are part of the clausal spine can be the target for the attachment of particles.¹ Consider (157) first, where both examples are canonical configurations without verb doubling and contain a TAM. It is clear that in these constructions, the lexical verb *kàn* and the TAM *le/guò* are in the same clausal spine, no matter how the latter appears on the former (I will specify their method of displacement in section 5.2.1).

- (157) a. Piet *kàn-le* nà-běn xiǎoshuō shí biàn.
 Piet read-PFV that-CL novel ten time
 ‘Piet read the novel ten times.’
- b. Piet *kàn-guò* nà-běn xiǎoshuō shí biàn.
 Piet read-EXP that-CL novel ten time
 ‘Piet has read the novel ten times.’

¹ Just to clarify, what I refer to as the ‘clausal spine’ here is the verbal extended projection (in the sense of Grimshaw 1991, 2000, 2005) that in general constitutes a clause. See more relevant discussion below.

With (157) in mind, let us reconsider the ungrammatical configurations in (155) and (156). Previously in chapter 2, I proposed that verb doubling involves verbal phrasal movement, which can respectively target a low position Spec,FP or a high one Spec,CP in the clausal spine. The illustrations of these two landing sites for the fronted VPs are repeated as follows.

- (158) *Low verb doubling in Spec,FP* (= (21)) (159) *High verb doubling in Spec,CP* (= (22))



Since the fronted verbal phrases occupies the specifier position of some functional projection, it can be said that the higher verb copy forms a verbal extended projection on its own, and does not constitute part of the clausal spine. By contrast, the lower verb copy in the base position counts as part of the clausal spine, as it forms a verbal extended projection with FP and CP. This distributional difference between the two copies of the same verb offers a direct explanation to the question above: the particles have to appear on the V_2 instead of the V_1 because only the V_2 forms part of the clausal spine.

Let us move on to the second similarity between *le* and *de*, that is, they are unable to co-occur with components other than the lexical verb within the lower verbal copy. Under the copy theory of movement (Chomsky 1993, 1995), when, e.g., VP-fronting takes place in the derivation, it leaves an entire VP copy in the base position, which consists of a lexical verb and its internal argument. Both (160) and (161) demonstrate a configuration where the lower copy of the internal argument *nà-běn xiǎoshuō* ‘that novel’ is spelled out and shows up between the lower verb *kàn* and the particle *le/de*. Since the particles are blocked from appearing on the V_2 due to the intervening internal argument, the result that all these sentences are ungrammatical is expected, based on the first similarity discussed above.

- (160) a. *Piet *kàn* *nà-běn xiǎoshuō* *kàn* *nà-běn xiǎoshuō* **le** *shí biàn*.
 Piet read that-CL novel read that-CL novel PFV ten time
 Int.: ‘Piet read the novel ten times.’
- b. **Kàn* *nà-běn xiǎoshuō* Piet *kàn* *nà-běn xiǎoshuō* **le** *shí biàn*.
 read that-CL novel Piet read that-CL novel PFV ten time
 Int.: ‘As for reading the novel, Piet read it ten times.’

- (161) a. *Piet kàn nà-běn xiǎoshuō kàn *nà-běn xiǎoshuō* **de** tèbié zǐxì.
 Piet read that-CL novel read that-CL novel DE particularly carefully
 Int.: ‘Piet read the novel very carefully.’
- b. *Kàn nà-běn xiǎoshuō Piet kàn *nà-běn xiǎoshuō* **de** tèbié zǐxì.
 read that-CL novel Piet read that-CL novel DE particularly carefully
 Int.: ‘As for reading the novel, Piet read it very carefully.’

Even if the particle and the lower copy of the internal argument swap their positions, these examples do not exhibit any improvement in terms of grammaticality. As given in (162) and (163), here particles are all right-adjacent to the lower verb without any intervention. The lower copy of the internal argument, instead, follows the V_2 -particle unit and precedes the adverbial. The result is that these configurations are still unacceptable.

- (162) a. *Piet kàn nà-běn xiǎoshuō kàn-**le** *nà-běn xiǎoshuō* shí biàn.
 Piet read that-CL novel read-PFV that-CL novel ten time
 Int.: ‘Piet read the novel ten times.’
- b. *Kàn nà-běn xiǎoshuō Piet kàn-**le** *nà-běn xiǎoshuō* shí biàn.
 read that-CL novel Piet readPFV that-CL novel ten time
 Int.: ‘As for reading the novel, Piet read it ten times.’
- (163) a. *Piet kàn nà-běn xiǎoshuō kàn **de** *nà-běn xiǎoshuō* tèbié zǐxì.
 Piet read that-CL novel read DE that-CL novel particularly carefully
 Int.: ‘Piet read the novel very carefully.’
- b. *Kàn nà-běn xiǎoshuō Piet kàn **de** *nà-běn xiǎoshuō* tèbié zǐxì.
 read that-CL novel Piet read DE that-CL novel particularly carefully
 Int.: ‘As for reading the novel, Piet read it very carefully.’

What the examples in (160)-(163) show is that particles are just unable to co-occur with the lower copy of the internal argument, regardless of the position of the latter. Combined with the first similarity discussed above, we can also conclude that particles’ attachment to the lower verb copy is subject to the spell-out of the internal argument’s copy. If we interpret these similarities via the lens of PF, what we have found here suggests that the displacement of all these particles is subject to some identical mechanism at PF. I will elaborate on this idea in section 5.2.1.

5.1.2 Differences among particles

In what follows I will show that there are more differences between TAMs and DAMs, compared to the similarities illustrated in the previous section. In section 3.4 and section 4.1, we have found in several occasions that *le* and *de* do not behave totally identical to each other. Let me start off by reviewing these differences first (for the purpose of clarity, the relevant examples are re-arranged together with some new ones).

The first difference is concerned with multiple adverbials in the postverbal field. (164) consists of two examples in which the frequency adjunct *liǎng huí* ‘two times’ co-occurs with the locative argument *qù Wéiduōliǎ Gōngyuán* ‘to Victoria Park’. (164a) shows that, when *le* is right-adjacent to the lower verb copy and attached to it (ignoring the existence of the intervening preposition *qù*), the sentence is acceptable. In contrast, (164b) shows that the same particle is unable to be separated from the lower verb copy by the intervening locative argument.

- (164) a. Piet sà-n-bù sà-n-qù-**le** Wéiduōliǎ Gōngyuán liǎng huí.
Piet disperse-step disperse-to-PFV Victoria park two time
'Piet went for a walk to Victoria Park twice.'
(= (66a))
- b. *Piet sà-n-bù sà-n-qù Wéiduōliǎ Gōngyuán **le** liǎng huí.
Piet disperse-step disperse-to Victoria park PFV two time
Int.: 'Piet went for a walk to Victoria Park twice.'

The ungrammatical result in (164b) is pretty similar to what we have seen in (160) above, though the source of the intervenor is not the same: in the latter scenario, the intervenor between the lower verb copy and the TAM *le* is the lower copy of the internal argument; while in the former scenario, the intervenor becomes the locative argument, which itself is not a copy left by movement. Due to this difference, I do not attribute the ungrammaticality of these two scenarios to the same reason, that is, why (164b) is unacceptable has nothing to do with the co-occurrence of *le* and the locative argument. In fact, if the co-occurrence was the reason, (164a) should be unacceptable as well (cf. (162)), which is contrary to the fact. To account for the contrast seen in (164), I postulate that TAMs in general resist separation from the lexical verbs they are supposed to appear on. Thus, I attribute the ungrammaticality of (164b) to the intervening locative argument that occurs between the V_2 and the particle.

If we adopt the assumption that TAMs and DAMs are from the same source, we might expect to find the same contrast that *le* displays in (164) in verb doubling configurations containing *de*. However, this is not what (165) shows us. In (165a), the particle *de* appears on the lower verb copy and is followed by the locative argument. Unlike (164a), this configuration is ungrammatical. In (165b), since the locative argument intervenes between *de* and the lower verb copy, this sentence

is, once again, ungrammatical, as with (164b).

- (165) a. *Piet sàn-bù sàn-qù-(le)-**de** Wéiduōliǎ Gōngyuán hěn kuài.
 Piet disperse-step disperse-to-PFV-DE Victoria park very quickly
- b. *Piet sàn-bù sàn-qù-le Wéiduōliǎ Gōngyuán **de** hěn kuài.
 Piet disperse-step disperse-to-PFV Victoria park DE very quickly
 (= (71a))

How do we account for the ungrammaticality of both examples above? The solution to (165b) is straightforward: following Huang (1988), Cheng (2007), and Meadows and Yan (2023, 2025), I posit that DAMs must be right-adjacent to the lexical verbal element. In other words, as with TAMs (164b), DAMs resist separation from the lower verb copy as well. In (165b), the locative argument *qù Wéiduōliǎ Gōngyuán* prevents *de* from being adjacent to the lower verb copy, thereby giving rise to ungrammaticality. Note that the proposal concerning TAMs and DAMs here follows the first similarity just discussed in section 5.1.1, that is, all particles have to appear on the V₂.

As to (165a), I propose that the intervention effect found in (165b) also takes place in this configuration. That is, even though the locative argument does not intervene between manner *de* and the lower verb copy, it shows up between the particle and the manner adjunct and still causes the intervention effect. By comparison, there is no similar intervention effect found in (164a) where the same locative argument shows up between *le* and the frequency adjunct. This provides the first argument that TAMs and DAMs are distinct in nature: *de* has some unique property of being subject to rightward intervenors, while *le* does not.

The second difference we have already seen previously is related to verb doubling with ditransitives, which presents another instance of the intervention effect. (166) shows a ditransitive verb doubling in a double object construction, in which the fronted VP only contains the direct object *fēntílùn jùfǎxué* ‘mereological syntax’, and the indirect object *Wassily* is stranded in the postverbal field.

- (166) a. Piet jiào fēntílùn jùfǎxué jiào-**le** Wassily shí biàn.
 Piet teach mereology syntax teach-PFV Wassily ten time
 ‘Piet taught Wassily mereological syntax ten times.’
 (= (89a))
- b. *Piet jiào fēntílùn jùfǎxué jiào Wassily **le** shí biàn.
 Piet teach mereology syntax teach Wassily PFV ten time
 ‘Piet taught Wassily mereological syntax ten times.’

In (166a), the TAM *le* is attached to the second occurrence of the doubled verb *jiào* ‘teach’

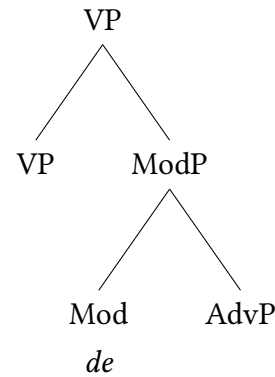
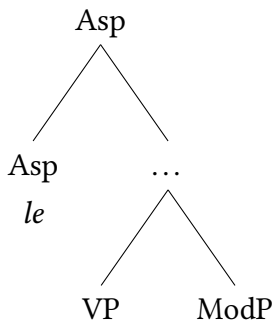
without any intervention. It is clear that the existence of the indirect object that follows *le* does not lead to any ungrammaticality. In (166b), the configuration is different as here the indirect object becomes the intervenor that shows up between the lower verb and *le*. According to the requirement that TAMs cannot be separated from the lexical verb, this example is ungrammatical as expected.

By comparison, (167) exhibits a slightly different ditransitive verb doubling configuration which contains the DAM *de* and the manner adjunct *hěn rènzhēn* ‘very earnestly’. Since manner *de*, as summarised above, is sensitive to intervenors that show up on its either side, neither of the examples below is acceptable: in (167a), the indirect object *Wassily* functions as the intervenor between *de* and the manner adjunct; while in (167b), the same intervenor occurs between the particle and the lower verb.

- (167) a. *Piet jiào fēntílùn jùfǎxué jiào **de** Wassily hěn rènzhēn.
 Piet teach mereology syntax teach DE Wassily very earnestly
 Int.: ‘Piet taught Wassily mereological syntax with dedication.’
 b. *Piet jiào fēntílùn jùfǎxué jiào Wassily **de** hěn rènzhēn.
 Piet teach mereology syntax teach Wassily DE very earnestly
 Int.: ‘Piet taught Wassily mereological syntax with dedication.’ (=89c)

The contrast between (166) and (167) provides another piece of evidence in favour of the idea that TAMs and DAMs are fundamentally different from each other. Given that *le*, as a representative of TAMs, is canonically perceived as the head of AspP (e.g., Lai 2021, 2024), it is reasonable to postulate that *de*, as a representative of DAMs, originates from a different source in the clausal spine. Following Meadows and Yan (2023, 2025), I posit that DAMs like *de* are base-generated in the head of ModP (Rubin 2003), which takes adverbial adjuncts as its complements. The base positions of *le* and *de* are respectively illustrated in (168) and (169).

- (168) *Le* base-generated in the head of AspP (169) *De* base-generated in the head of ModP



As shown in (168), the TAM *le* is generated as the head of AspP, which itself is outside the ModP. By contrast, the DAM *de* is generated within the ModP, as shown in (169). Combined with the contrast between *le* and *de* in terms of the intervention effect, the difference here implies that whether the particle is sensitive to the intervenor on its right side is related to its base position. I will return to this point in section 5.2.1.

There are other three differences which further argue for the idea that TAMs and DAMs are originally from different sources in the clausal spine. The third difference is concerned with whether particles are sensitive to tense and aspect. In (170) and (171), all the verb doubling configurations are constructed in the past tense, as indicated by the temporal expression *zuówǎn* ‘last night’ that occurs in the sentence-initial position. In this situation, neither *le* nor *de* are able to be omitted in the postverbal position. In other words, the existence of these particles is obligatory in this scenario.

- (170) a. *Zuówǎn* Piet kàn nà-běn xiǎoshuō kàn*(-**le**) shí biàn.
yesterday.night Piet read that-CL novel read-PFV ten time
‘Last night Piet read the novel ten times.’
- b. *Zuówǎn* kàn nà-běn xiǎoshuō Piet kàn*(-**le**) shí biàn.
yesterday.night read that-CL novel Piet read-PFV ten time
‘As for reading the novel, last night Piet read it ten times.’
- (171) a. *Zuówǎn* Piet kàn nà-běn xiǎoshuō kàn *(**de**) tèbié zǐxì.
yesterday.night Piet read that-CL novel read DE particularly carefully
‘Last night Piet read the novel very carefully.’
- b. *Zuówǎn* kàn nà-běn xiǎoshuō Piet kàn *(**de**) tèbié zǐxì.
yesterday.night read that-CL novel Piet read DE particularly carefully
‘As for reading the novel, last night Piet read it very carefully.’

However, when verb doubling configurations are constructed in the present tense, we find a difference that arises between these two particles. As shown in (172), the present tense of the entire sentence is marked by the temporal expression *měi tiān* ‘every day’ that occurs in the post-subject position. Here it is impossible for *le* to appear on the lower verb copy. In contrast, (173) shows that *de* is still obligatory to occur in this present-tense context.

- (172) a. Piet *měi tiān* kàn nà-běn xiǎoshuō kàn(*-**le**) shí biàn.
Piet every day read that-CL novel read-PFV ten time
‘Piet reads the novel ten times every day.’

- b. Kàn nà-běn xiǎoshuō Piet *měi tiān* kàn(*-le) shí biàn.
 read that-CL novel Piet every day read-PFV ten time
 ‘As for reading the novel, Piet reads it ten times every day.’
- (173) a. Piet *měi tiān* kàn nà-běn xiǎoshuō kàn *(de) dōu tèbié zǐxì.
 Piet every day read that-CL novel read DE DOU particularly carefully
 ‘Piet reads the novel very carefully every day.’
- b. Kàn nà-běn xiǎoshuō Piet *měi tiān* kàn *(de) dōu tèbié zǐxì.
 read that-CL novel Piet every day read DE DOU particularly carefully
 ‘As for reading the novel, Piet reads it very carefully every day.’

The contrast seen above suggests that, while *le* is sensitive to tense and aspect, *de* shows the opposite. This result is compatible with the current proposal that these two particles are base-generated in different positions in the clausal spine. Since *le*, as a perfective marker, is base-generated in the head of AspP, it is plausible to assume that its occurrence is subject to the entire clause’s tense and aspect. By comparison, the same constraint does not take effect on *de*, because it is not base-generated in a position directly related to tense and aspect.

The fourth difference between *le* and *de* shows a similar situation as what we just have seen in (172) and (173), that is, these two particles differ in whether their occurrence is sensitive to modals or not. As shown in (174), when the post-subject position here is occupied by, e.g., a deontic modal *bìxū* ‘must’, the TAM *le* is unable to occur postverbally. However, the presence of the DAM *de* is still obligatory when the same deontic modal occurs in manner-type verb doubling configurations, as given in (175).

- (174) a. Piet *bìxū* kàn nà-běn xiǎoshuō kàn(*-le) shí biàn.
 Piet must read that-CL novel read-PFV ten time
 ‘Piet must read the novel ten times.’
- b. Kàn nà-běn xiǎoshuō Piet *bìxū* kàn(*-le) shí biàn.
 read that-CL novel Piet must read-PFV ten time
 ‘As for reading the novel, Piet must read it ten times.’
- (175) a. Piet *bìxū* kàn nà-běn xiǎoshuō kàn *(de) gèng zǐxì yīdiǎn.
 Piet must read that-CL novel read DE more carefully a.bit
 ‘Piet must read the novel a bit more carefully.’
- b. Kàn nà-běn xiǎoshuō Piet *bìxū* kàn *(de) gèng zǐxì yīdiǎn.
 read that-CL novel Piet must read DE more carefully a.bit
 ‘As for reading the novel, Piet must read it a bit more carefully.’

The account of this difference is not fundamentally different from that of the third one. The impossible co-occurrence of the deontic modal and the TAM *le* is expected, since cross-linguistically, modals in general are only compatible with the infinitive form of verbs, which rules out *le*'s attachment to the verb in (174). But because *de* is not a marker of tense and aspect, its attachment to the verb in (175) is not prohibited.²

Last but not least, there is a fifth difference between *le* and *de* relevant to coordination, which is worthwhile mentioning here. (176) shows two examples of coordinating verb doubling constructions of the same meaning-type. In (176a), both conjuncts are of duration-type verb doubling, and in (176b), it is two manner-type verb doubling constructions that engage in coordination. Here each conjunct contains its own particle, depending on the type of verb doubling.³

(176) *Verb doubling in coordination*

- a. Piet [**pǎo**-bù **pǎo**-le sān-gè zhōngtóu], (bìngqiě) [**dú**-shū yě **dú**-le
Piet run-step run-PFV three-CL hour and read-book also read-PFV
sān-gè zhōngtóu].
three-CL hour
‘Piet ran for three hours, and also read for three hours.’ (duration + duration)
- b. Piet [**pǎo**-bù **pǎo** de hěn kuài], (bìngqiě) [**dú**-shū yě **dú** de hěn kuài].
Piet run-step run DE very quickly and read-book also read DE very quickly
- ‘Piet ran very quickly, and also read very quickly.’ (manner + manner)

Now let us have a look at (177), where verb doubling in coordination is connected with right node raising (RNR).⁴ Consider (177b) first: compared to (176b), the coordination here also involves two manner-type verb doubling conjuncts. However, the two verb doubling conjuncts are more reduced here, as their common manner adverbial *hěn kuài* ‘very quickly’ occurs at the end of the entire string of words, forming a right node raising configuration. Note that here both conjuncts still contain the particle *de* within, even though the particle forms a constituent with the manner adverbial in the first place, as illustrated in (169) above.⁵

² One may wonder whether the contrast between (174) and (175) is concerned with the meaning of modals. As far as I am aware, this contrast still holds when the deontic *bìxū* is replaced by other modals, such as the epistemic *yīnggāi* ‘should’, the future *huì* ‘will’, the deontic *děi* ‘have to’, etc. This fact rules out the possibility that the ill-formedness of these examples stems from semantics.

³ One thing to note here is that the coordinator *bìngqiě* ‘and’ is in fact optional. This might be related to the obligatory existence of *yě* ‘also’ in the second conjunct which has already taken some effect of coordination.

⁴ I thank Zhouyi Sun for sharing the data in (177) with me.

⁵ There are another two trivial properties of (177b) that cannot be found in (176b): on the one hand, the coordinator changes to *hé* instead of *bìngqiě*, whose existence is obligatory in this coordination; on the other hand, the second reduced conjunct is followed by a plural-related marker *dōu* instead of co-occurring with *yě*.

(177) *Coordinated verb doubling with RNR*

- a. *Piet [**pǎo**-bù **pǎo**-le] hé [**dú**-shū **dú**-le] dōu sān-gè zhōngtóu.
Piet run-step run-PFV and read-book read-PFV DOU three-CL hour
Int.: ‘Piet ran for three hours, and read for three hours as well.’
- b. Piet [**pǎo**-bù **pǎo** de] hé [**dú**-shū **dú** de] dōu hěn kuài.
Piet run-step run DE and read-book read DE DOU very quickly
‘Piet ran and read very quickly.’

The difference arises when both verb doubling conjuncts become duration-type and contain the TAM *le*, as given in (177a). Here what undergoes right node raising is a duration adverbial *sān-gè zhōngtóu* ‘three hours’. Unlike its manner counterpart in (177b), this coordination turns out to be ungrammatical. Given that both duration and manner expressions are adverbial adjuncts (recall section (3.2)), it seems reasonable to still attribute the contrast here to the difference between *le* and *de*. However, it is unclear to me at this point how this fifth difference can be linked to the different base positions of these two particles. Therefore, I will have to leave this for future study.

To summarise, in this section I have shown five differences between *le* and *de*, and proposed that these two (groups of) particles have distinct base positions. Although they are from different sources in the clausal spine, both particles need to appear on the lower verb, which implies that they might rely on different methods of being attached to the same target. This is the idea I will flesh out in the next section.

5.2 Deriving verb doubling effects at PF

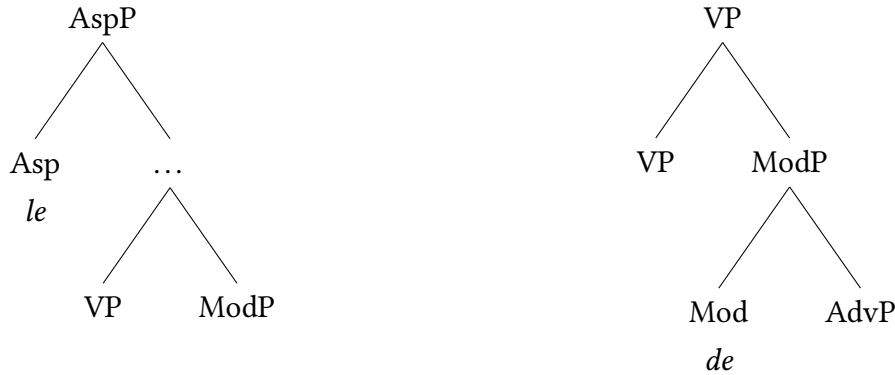
In this section I will develop an analysis of realising verb doubling effects at PF. I will pursue the idea that verb doubling results from multiple spell-outs of the same verb in the PF component of the grammar, in line with Nunes (2004), Landau (2006), Cheng (2007), Hein (2017), Lai (2021, 2024), and Meadows and Yan (2023, 2025), among many others. I will first lay out a series of post-syntactic operations at the PF-branch which my proposed analysis will rely on. Then I will show in turn how these operations can be applied to realise various verb doubling effects.

5.2.1 Post-syntactic operations and their relative order at PF

It has been proposed in a lot of literature that, in the PF component of the grammar, there are a series of post-syntactic operations that can alter the output of narrow syntax (e.g., Embick and Noyer 2001; Embick 2007; Hein 2017, 2018, 2020, among others). There are (at least) four of them that matter to my proposed analysis of realising verb doubling at the PF-branch of the grammar.

The first two are concerned with the post-syntactic displacement of particles. In section 5.1.1, I have compared the TAM *le* and the DAM *de*, and proposed that they are base-generated in two different positions in the clausal spine, repeated in (178) and (179) below.

- (178) *Le* base-generated in the head of AspP (179) *De* base-generated in the head of ModP



Due to their different base positions, it is reasonable to posit that they do not appear on the lexical verb in a uniform way. Following Embick and Noyer (2001) and Embick (2007), I propose that there are two methods of implementing displacement of particles post-syntactically, namely, *Lowering* and *Local Dislocation*. According to Embick and Noyer, a crucial difference between these two post-syntactic movement operations lies in whether they apply on linearised or unlinearised structures. As schematised in (180), Lowering applies on hierarchical, unlinearised structures. This operation achieves the canonical head movement effects (in the sense of Travis 1984) by adjoining a higher head to a lower head, and these heads do not have to be linearly adjacent. This indicates that Lowering has a non-local or non-adjacent character (Embick and Noyer 2001:562)

- (180) *Lowering of X to Y*

$$[_{XP} X \dots [_{YP} \dots Y \dots]] \mapsto [_{XP} \dots [_{YP} \dots [_{Y} Y + X] \dots]]$$

(adopted from Embick and Noyer 2001:562, (6))

In contrast, Local Dislocation can only apply on linearised structures, and is subject to linear precedence and adjacency rather than hierarchy. The proper environment where this post-syntactic operation can be implemented is schematised in (181). Here the notation ' $X * Z$ ' denotes that X have to linearly precede Z, and these two elements must be adjacent to each other.

- (181) *Local Dislocation of X to Z*

$$[X * [Z * Y]] \mapsto [[_Z Z + X] * Y]$$

(adopted from Embick 2007:562-563, (10)-(11))

The structural difference between (180) and (181) comes from the idea Lowering and Local Dislocation happen at different points at PF relative to *Vocabulary Insertion/Linearisation*, another post-syntactic operation which maps a hierarchical structure where vocabulary items are inserted onto a linear string in accordance with, e.g., Kayne (1994)'s Linear Correspondence Axiom (LCA).⁶ Since the hierarchical structure that Lowering applies on can be converted to the linearised one that Local Dislocation makes use of (but not vice versa), this suggests that Lowering takes place before Local Dislocation at PF, as schematised in (182).⁷

(182) *Order of post-syntactic operations (to be continued)*
 ... >> Lowering >> ... >> Local Dislocation >> ...

Combining with the different base positions of particles illustrated above, I posit that TAMs like *le* appear on the verbs via Lowering, and that the displacement of DAMs like *de* is realised by Local Dislocation. These two post-syntactic movement operations are respectively illustrated in (183) and (184).⁸

⁶ In fact, Embick and Noyer (2001) propose that Vocabulary Insertion and Linearisation are two different operations, and the latter takes place either at or right after the former, according to the *Late Linearisation Hypothesis*:

(i) *The Late Linearisation Hypothesis*

The elements of a phrase marker are linearised at Vocabulary Insertion.

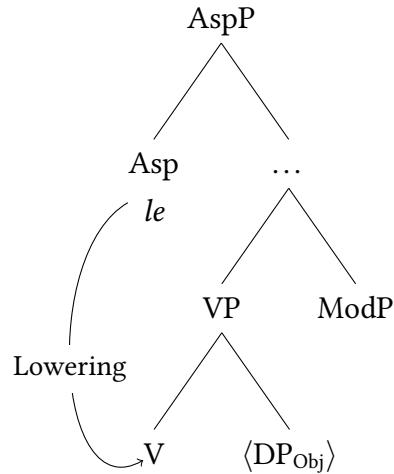
(Embick and Noyer 2001:562, (8))

As will be clear soon, it does not matter to my proposed analysis whether these two operations are the same or not. Besides, Kayne's LCA is not compatible with the right-adjunction structures adopted in this work, a problem I have to leave for future research at this point. Therefore I will maintain my simplification in what follows.

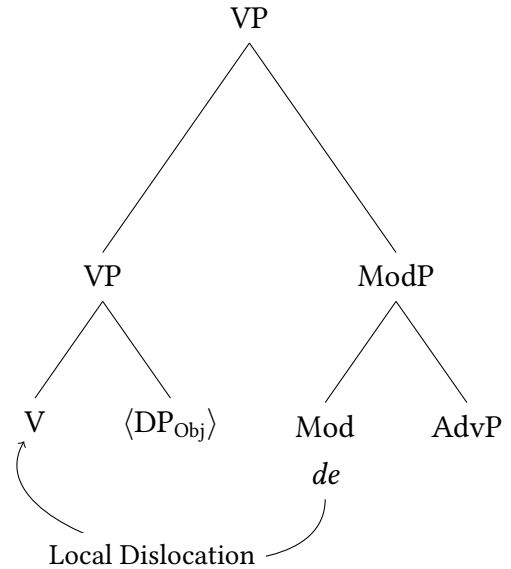
⁷ Though I will continue to bring up Vocabulary Insertion and/or Linearisation in the following discussion, they will not be illustrated in the order of post-syntactic operations like (182) for the purpose of clarity.

⁸ Note that in (184), the direct object is already deleted at this stage of applying Local Dislocation, which ensures that the V head and the Mod head (i.e., *de*) are adjacent to each other in linear order.

(183) Lowering of *le* to V



(184) Local Dislocation of *de* to V



The distinction between these two post-syntactic operations offers a plausible account of why *le* and *de* differ in rightward intervention effects found in section 5.1.2 (recall (164a)/(166a) vs. (165a)/(167a)): when the ModP requires to be locally dislocated, it is already vocabulary-inserted and spelled out as *de*. Therefore, the rightward intervention effects can be directly considered as a unique property of *de*, that is, it cannot be separated from the manner adverb by any intervenors on its right side. By contrast, when the AspP is lowered to the verbal head at PF, *le* is not spelled out yet, which explains why *le* behaves differently in terms of the rightward intervention effects.

Let us move on to the third post-syntactic operation, namely, *Chain Reduction* as proposed by Nunes (2004). Under the copy theory of movement (Chomsky 1993, 1995), whenever a constituent moves in narrow syntax, it is assumed to leave a copy at its launching site and creates a chain of movement. One problem of the copy theory is that, in standard cases (like phrasal movement), the higher copy (or the head of the movement chain) is generally the one that gets spelled out, whereas the lower copy (or the tail of the chain) is not pronounced. Considering that the two copies of the same chain are (almost) identical, one needs to explain (i) why it is the higher copy that receives the pronunciation, and (ii) how the lower copy does not get spelled out. The solution to these two problems that Nunes (2004) proposes is Chain Reduction, which deletes the lower copy c-commanded by the higher one at PF.

To realise the doubling effects, I posit that Chain Reduction also plays the same role of deleting the lower verbal copy in verb doubling configurations. Interestingly, unlike standard cases where the entire lower copy gets deleted, verb doubling seems to involve partial deletion (cf. Fanselow 1990): that is, everything inside the lower verbal copy gets deleted except the verb head. Given

that in most cases the lower verb head is spelled out together with the particle, suppose that the role of these particles is to help the verb prevent Chain Reduction, an idea which has been proposed in a lot of relevant literature (e.g., Landau 2006, Cheng 2007, and Meadows and Yan 2023, 2025, among many).⁹ This idea offers a direct account of why particles are unable to co-occur with the copy of the internal argument of the verb, a similarity among particles that we have seen in section 5.1.1 (recall (160)-(163)): the internal argument's copy, unlike the lower verb head, cannot be rescued by the particle and escape from Chain Reduction, thus it has to be totally deleted at PF.

What matters to my proposed analysis here is how Chain Reduction interacts with other post-syntactic movement operations, namely, Lowering and Local Dislocation. The idea that particles can help the lower verbal head escape from deletion prevents Chain Reduction from happening before any post-syntactic movement operations in the case of verb doubling, because if this was the case, the lower verbal copy could not be spelled out in the first place, leading to a situation where none of the particles would find a verbal host to be attached to in the clausal spine.¹⁰ Thus, Chain Reduction has to take place either between the two post-syntactic movement operations, as schematised in (185a), or after both of them, as schematised in (185b).

(185) *Hypothetical order of post-syntactic operations*

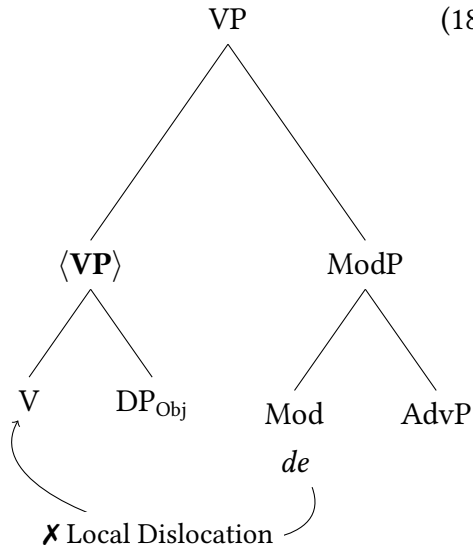
- a. ... >> Lowering >> Chain Reduction >> Local Dislocation >> ...
- b. ... >> Lowering >> ... >> Local Dislocation >> Chain Reduction >> ...

In both scenarios, Chain Reduction poses no problem to TAMs that will undergo Lowering: Lowering allows TAMs to appear on the lower verbal head before Chain Reduction applies, leading to the result of partial deletion. However, it is not easy for Chain Reduction to deal with DAMs in either situation. Consider (185a) first: as illustrated in (186), Chain Reduction deletes the entire lower verbal copy before Local Dislocation applies; when *de* is spelled out and looks for a hosting verb in the clausal spine, there will be no such element then. Consider (185b) next: as illustrated in (187), here when the Mod head is spelled out as *de*, it cannot be locally dislocated to the V head, as the direct object is not deleted yet and blocks the adjacency between these two elements.

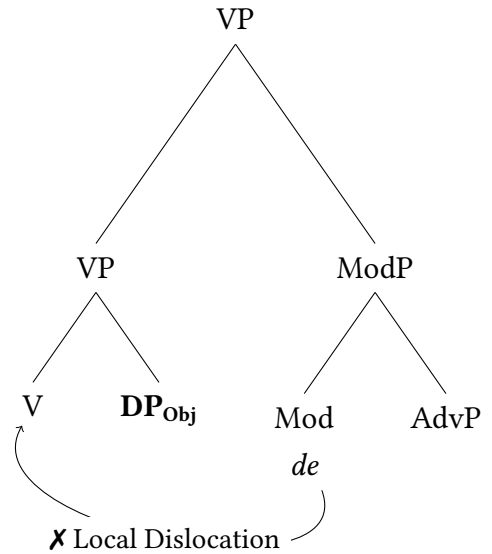
⁹ As will become clearer soon, the relation between the lower verb head and the particle in my proposed analysis is actually different from what is stated here: it is the existence of the lower verb head that provides an opportunity for the particle to be hosted; the lower verb head can escape from Chain Reduction on its own by Head Movement.

¹⁰ Note that, even though Chain Reduction cannot precede either Lowering or Local Dislocation here, there are configurations which can be derived by this order of post-syntactic operations. I will return to this idea in section 5.3.

(186)



(187)



It turns out that neither of the hypothetical orders presented in (185) can create a proper environment for combining Local Dislocation with Chain Reduction. To address this issue, I have to appeal to a fourth post-syntactic operation, namely, *Head Movement*. In the relevant literature (e.g., Tsai 1994; Huang 2015, among many other), Mandarin verbs are usually assumed to undergo head movement to the *v* head in narrow syntax. Here I adopt this idea that V-to-*v* movement exists in Mandarin Chinese, but differ in positing that Head Movement takes place post-syntactically, in line with Chomsky (2001) (also see Boeckx and Stjepanovic 2001; Sauerland and Elbourne 2002; Hein 2017, among many). With respect to its order relative to other operations at PF, I propose that Head Movement always happens before other movement operations. For concreteness, I further propose that the four post-syntactic operations discussed here are ordered as given in (188).¹¹

(188) *Order of post-syntactic operations*

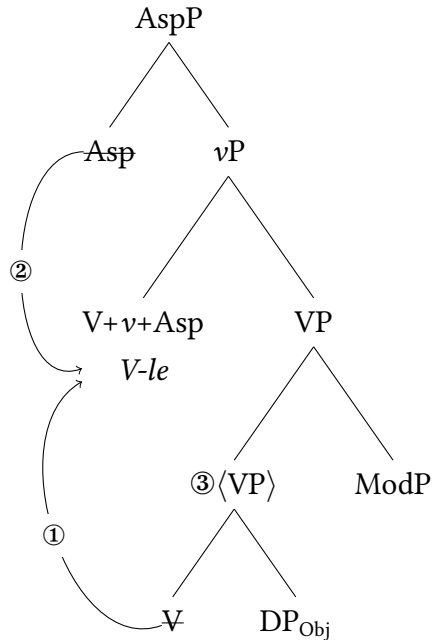
... » Head Movement » Lowering » Chain Reduction » Local Dislocation » ...

(189) and (190) respectively illustrate how these post-syntactic operations interact and apply to TAMs and DAMs. For expository purposes, the four operations in (188) are numbered in turn. As shown in (189), the first post-syntactic operation here is Head Movement (①), which triggers the V-to-*v* movement. Following Boeckx and Stjepanovic (2001), Sauerland and Elbourne (2002), and Hein (2017), I assume that Head Movement at PF leaves no copy or trace at its base position, thus indicated by the strikethrough. Once the first operation is done, Lowering (②) follows to happen at PF (which I assume also leaves no trace at all), and renders the Asp head attached to

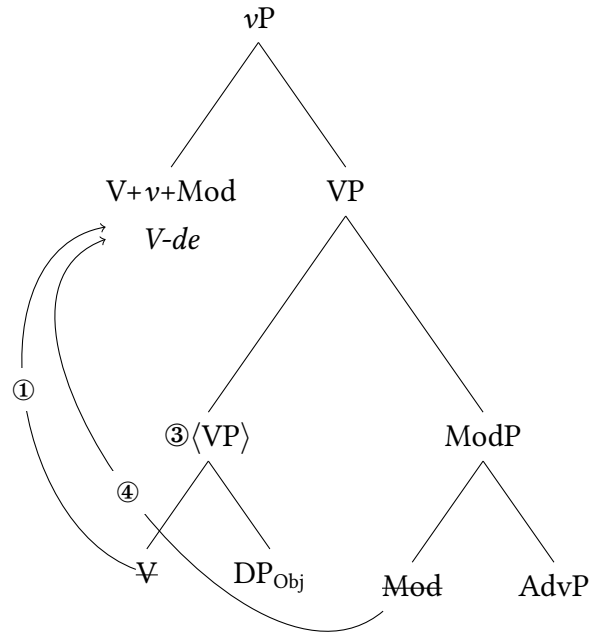
¹¹ Though I do not include Vocabulary Insertion/Linearisation in (188), I posit that it happens between Chain Reduction and Local Dislocation.

the V+v complex. After the second step, Chain Reduction (③) takes place and deletes the lower VP copy including the object (which is assumed to be fronted already in narrow syntax). Eventually, the entire V+v+Asp complex gets vocabulary-inserted and spelled out as a verb attached by *le*.

(189) *Lowering of le to V+v*



(190) *Local Dislocation of de to V+v*



The post-syntactic scenario in (190) is slightly different from what we just have seen in (189). Here the first operation is still Head Movement, which permits the lower V head to be adjoined to the *v* head. Since the DAMs are not base-generated in the Asp head, there is no Lowering happening here, leading to a situation where Chain Reduction happens right after Head Movement. Once the lower VP copy is deleted, Vocabulary Insertion/Linearisation takes place, and the ModP is spelled out as *de*. As a final step, this DAM undergoes Local Dislocation (④), and appears on the V+v complex.¹²

¹² It might be of interest to someone whether Head Movement has to take place post-syntactically. For the two cases illustrated here, it actually does not make any difference if one regards Head Movement as an operation taking place in narrow syntax, since the PF component of the grammar always comes after narrow syntax under the current minimalist assumption (Chomsky 1995). However, as will be seen in section 5.3, there are other configurations closely-related to verb doubling that seem to show no Head Movement effect at all, which I take as a tough challenge to the syntactic treatment of Head Movement (that is, what happens to syntactic Head Movement there?). By contrast, those cases will receive a more direct account from the post-syntactic treatment of Head Movement proposed here, as will be clearer soon.

5.2.2 Implementation

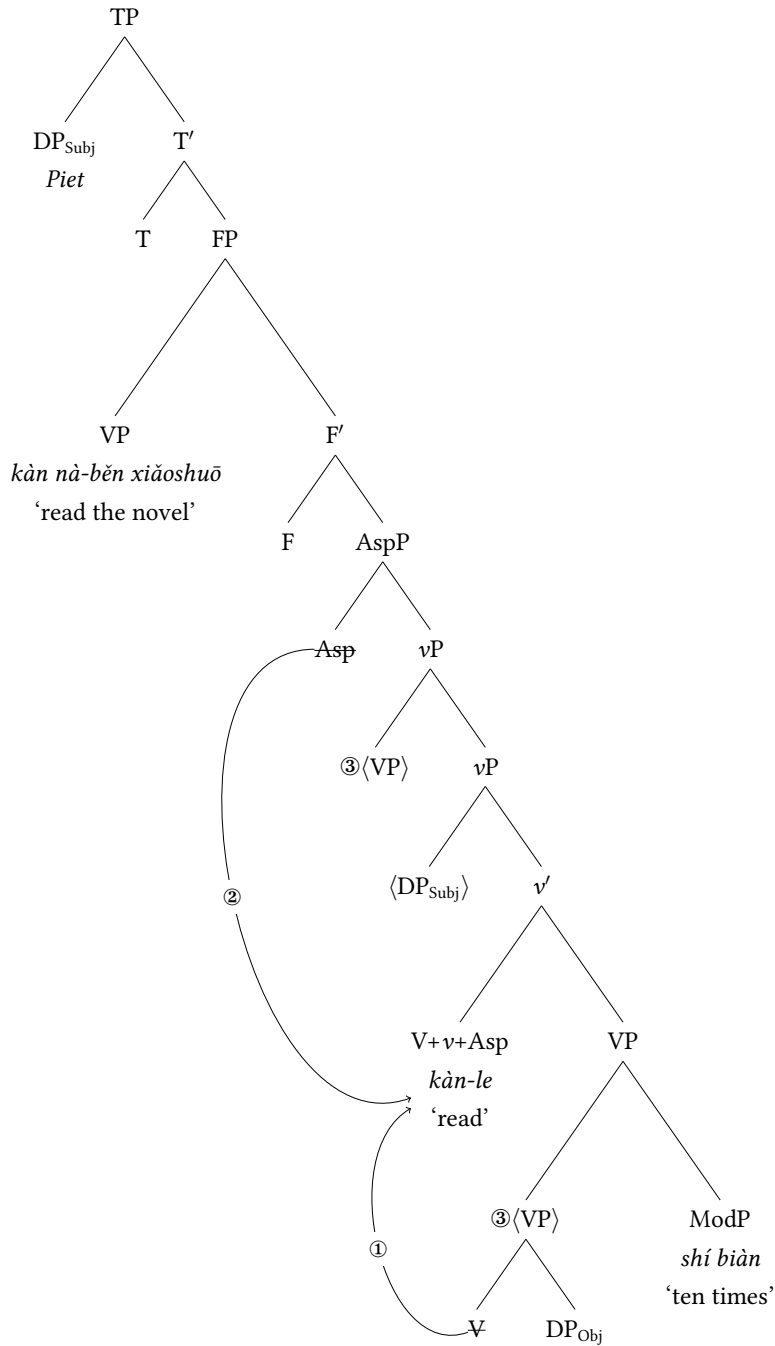
In what follows I will demonstrate how the four post-syntactic operations I just proposed can be implemented to realise various verb doubling effects. I will focus on four configurations here, namely, low and high verb doubling which contain either the TAM *le* or the DAM *de*. The first example is given in (191), which is a low verb doubling construction with *le* appearing on the V_2 .

- (191) Piet kàn nà-běn xiǎoshuō kàn-**le** shí biàn.
Piet read that-CL novel read-PFV ten time
'Piet read the novel ten times.' (=153a))

The post-syntactic structure of (191) is illustrated in (192). As with what we have seen above, the first operation at PF is Head Movement (①) which allows the V -to- v movement to happen, and the second operation is the Lowering (②) of the Asp head to the $V+v$ complex. What is worth attention here is concerned with the third operation Chain Reduction (③): in narrow syntax, the fronted VP moves across the edge of vP due to the requirement of phasehood (recall section 3.4 and 4.3), leading to a result that a VP copy is left in Spec, vP . Here I adopt the assumption that Spell-Out only happens once at PF (contra Uriagereka 1999): when Chain Reduction applies to both the lowest and the intermediate VP copies, they get deleted at the same time, which ensures that only the highest VP in Spec,FP gets spelled out.¹³

¹³ As far as I can tell, this Single Spell-Out idea is still compatible with, e.g., the distinctness condition that I adopted in section 3.4, which requires the complement of vP to form a Spell-Out domain at PF (cf. Fox and Pesetsky 2005).

(192)

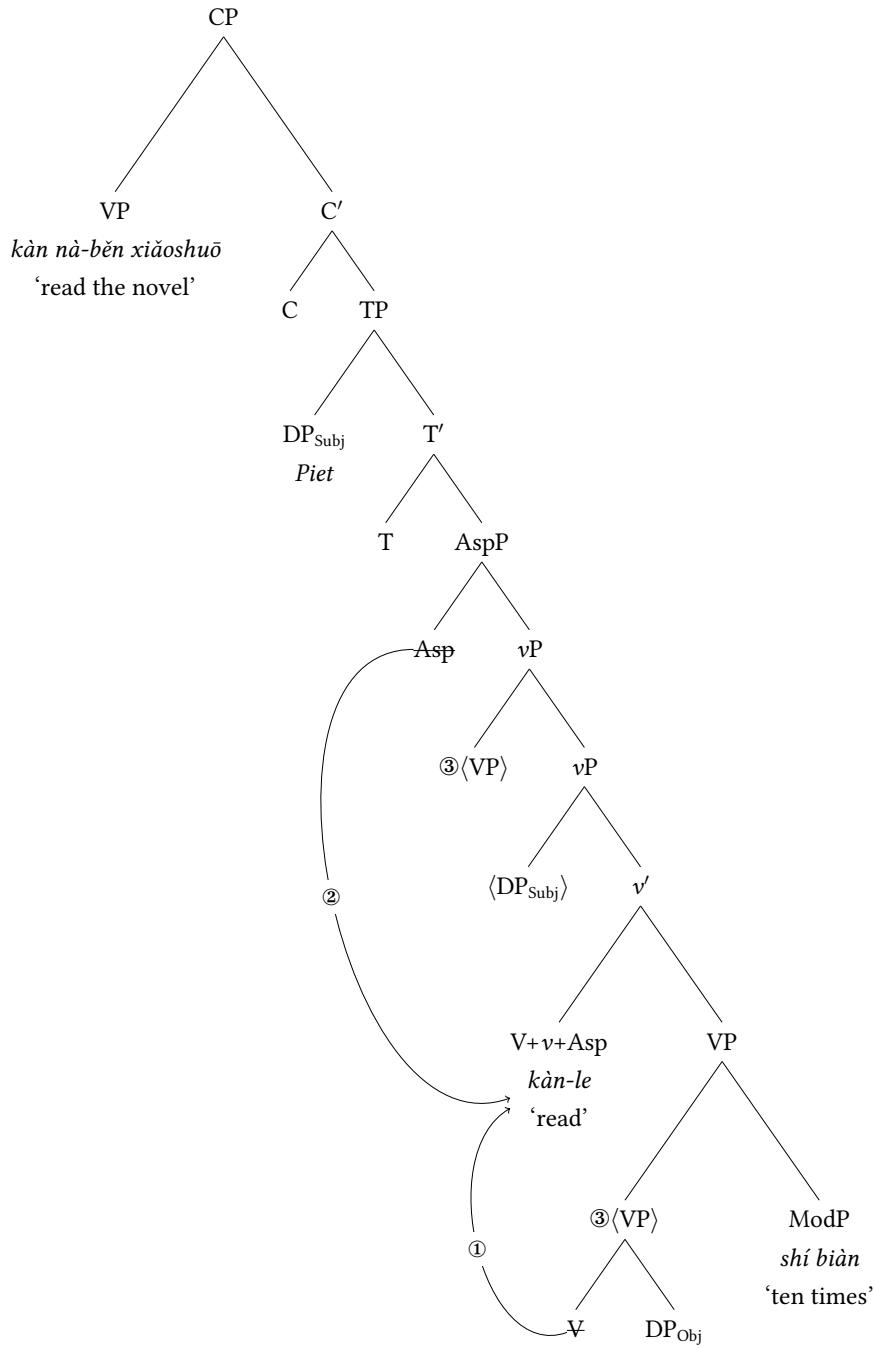


The second example (193) is the high counterpart of (191), whose post-syntactic structure is presented in (194). The only difference between (194) and (192) lies in the position where the fronted VP gets spelled out: in the former, it is in Spec,CP; whereas in the latter, that happens in Spec,FP.

- (193) Kàn nà-běn xiǎoshuō Piet kàn-**le** shí biàn.
 read that-CL novel Piet read-PFV ten time
 'As for reading the novel, Piet read it ten times.'

(=(153b))

(194)



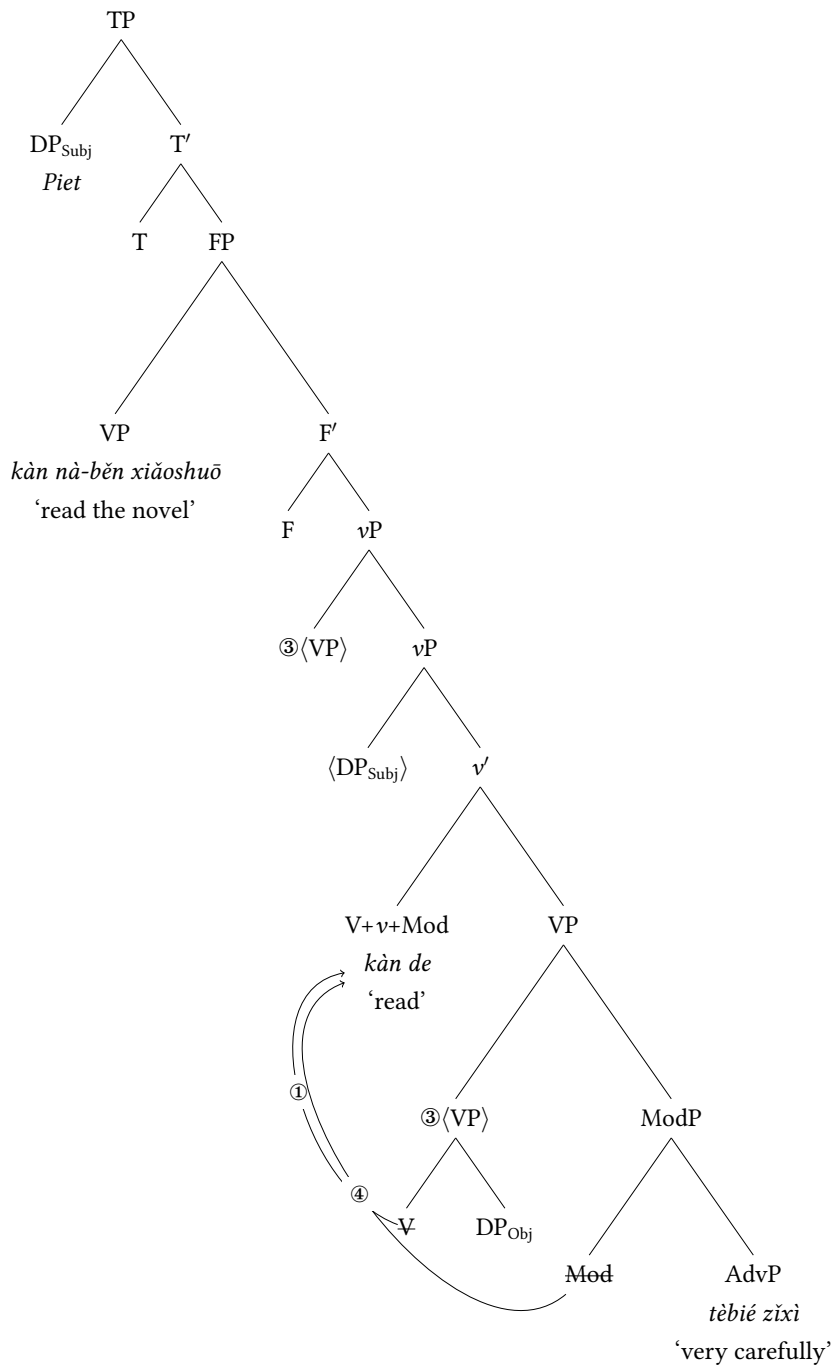
The third example is a low verb doubling configuration that contains *de* and the manner adjunct, as shown in (195).

- (195) Piet *kàn nà-běn xiǎoshuō* *kàn de tèbié zǐxì.*
 Piet read that-CL novel read DE particularly carefully
 'Piet read the novel very carefully.'

(=(154a))

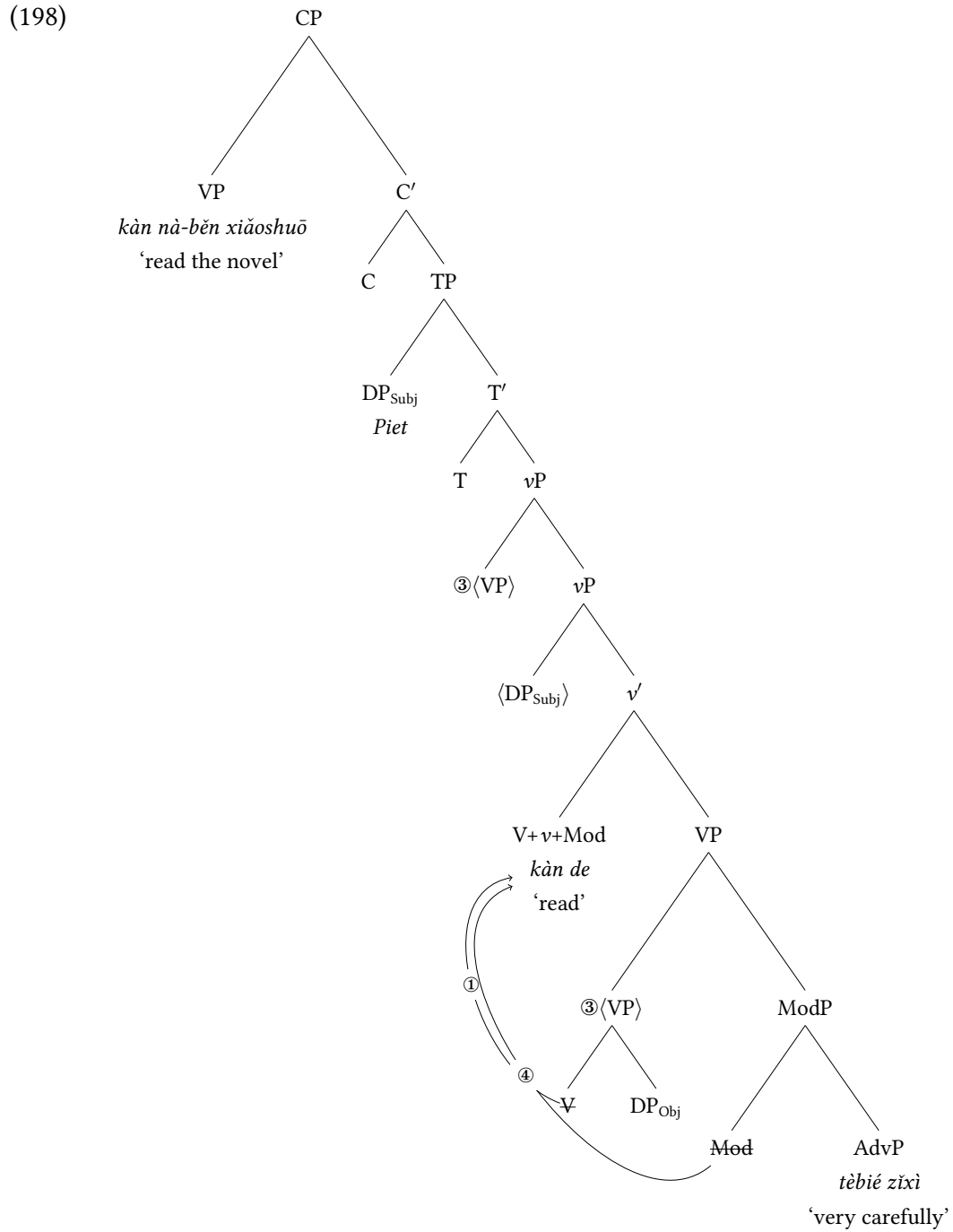
The post-syntactic structure of this verb doubling is illustrated in (196). Once the Head Movement (①) from V to *v* takes place, it is followed by Chian Reduction (③) that deletes the two VP copies left by verbal phrasal movement in narrow syntax. Then Vocabulary Insertion/Linearisation (not represented below) engages in this process, after which Local Dislocation (④) happens, resulting in the displacement of the particle *de* to the right side of the V+*v* complex.

(196)



The fourth examples is given in (197), which is the high verb doubling counterpart of (195). Its corresponding structure at PF is illustrated in (198).

- (197) Kàn nà-běn xiǎoshuō Piet kàn **de** tèbié zǐxì.
 read that-CL novel Piet read DE particularly carefully
 ‘As for reading the novel, Piet read it very carefully.’ (= (154b))



As with (192) and (194), the only difference between (196) and (198) is where the first occurrence of the doubled VP is pronounced: here the VP gets spelled out in Spec,CP rather than Spec,FP, which gives rise to the effect of high verb doubling.

5.3 Extending the PF algorithm

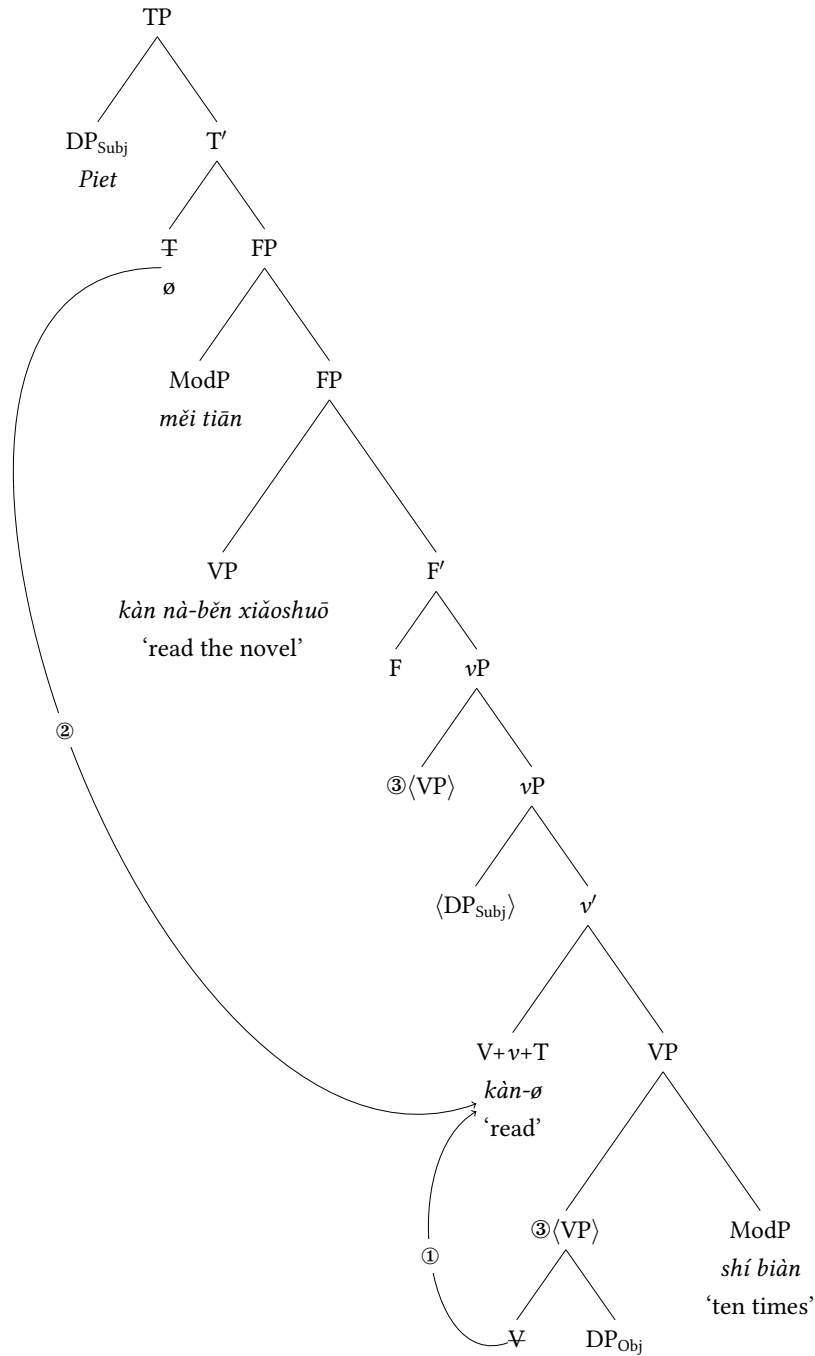
Before wrapping up the chapter, I would like to briefly show how my proposed post-syntactic account can be extended to two other configurations which are closely-related to canonical verb doubling. The first one is those verb doubling configurations which are constructed in the present tense and do not contain the TAM *le* postverbally. These examples are repeated in (199).

- (199) a. Piet *měi tiān* kàn nà-běn xiǎoshuō kàn shí biàn.
 Piet every day read that-CL novel read ten time
 ‘Piet reads the novel ten times every day.’
- b. Kàn nà-běn xiǎoshuō Piet *měi tiān* kàn shí biàn.
 read that-CL novel Piet every day read ten time
 ‘As for reading the novel, Piet reads it ten times every day.’ (= (172))

Unlike languages like English, Mandarin Chinese does not have grammaticalised tense markers (Li and Thompson 1981; Liu 2015). Therefore, it has to rely on overt temporal expressions (e.g., *měi tiān*) to distinguish one tense from another. This fact, however, does not rule out the possibility that Mandarin Chinese can have covert tense markers. To pursue a uniform analysis among frequency-type verb doubling configurations, I propose that Lowering still takes place when verb doubling configurations without overt TAMs like (199) are sent to PF. I posit that what undergoes Lowering in these cases is the null T head, which is the base position of the covert tense marker. Lowering of the T still targets the V+v complex, as illustrated in (200); but due to its covert essence (represented by ‘ $\emptyset$ ’), there will be no tense marker that appears on the V₂.¹⁴

¹⁴ For the sake of word order, here I simply regard the overt temporal expression *měi tiān* as an adjunct adjoined to the FP. There is no serious theoretical consideration that this choice hinges on.

(200)



The second configuration is the non-doubling counterpart of manner-type verb doubling, as given in (201a). In contrast to (201b), which is a standard example of manner-type low verb doubling, its non-doubling counterpart not only lacks the V_2 but also the particle *de*. Interestingly, there is no evident difference between these two in terms of interpretation. In other words, these two constructions can be used interchangeably in most scenarios (recall, e.g., (20) in section 2.1).

- (201) a. Piet kàn nà-běn xiǎoshuō tèbié zǐxì.
 Piet read that-CL novel particularly carefully
 ‘Piet read the novel very carefully.’
 b. Piet kàn nà-běn xiǎoshuō kàn de tèbié zǐxì.
 Piet read that-CL novel read DE particularly carefully
 ‘Piet read the novel very carefully.’ (= (195))

The question here is why these two structures look so different but can convey the same meaning. I argue that the difference between these two does not really lie in syntax but actually at PF. I propose that the non-doubling counterpart (201a), as with low verb doubling, involves verbal phrasal movement to Spec,FP, but differs in not spelling out its V₂ at PF. That is to say, this non-doubling configuration is in fact a string-vacuous, verbal phrasal movement.

There are at least two syntactic arguments that favour my proposal. First, as shown in (202a), the particle *de* is unable to appear on the verb *kàn* in the non-doubling counterpart. This situation is similar to what we have seen about the position of particles in section 5.1.1: all the particles in verb doubling can only appear on the V₂ instead of the V₁ (repeated in (202b)).

- (202) a. *Piet kàn **de** nà-běn xiǎoshuō tèbié zǐxì.
 Piet read DE that-CL novel particularly carefully
 ‘Piet read the novel very carefully.’
 b. *Piet kàn **de** nà-běn xiǎoshuō kàn tèbié zǐxì.
 Piet read DE that-CL novel read particularly carefully
 Int.: ‘Piet read the novel very carefully.’ (= (156a))

The fact that the non-doubling counterpart (202a) displays gets explained right away if we adopt the idea that the verbal phrase here undergoes phrasal movement as well. That is, the only verb found in (202a) is equivalent to the V₁ seen in (202b). Since the particle is only able to appear on the verb copy in the clausal spine (recall section 5.1.1), it is expected that the particle cannot be attached to the verb in the non-doubling counterpart.

Second, these two configurations behave identically in terms of their position relative to adverbs. (203b) shows that in a low verb doubling configuration, the fronted VP can actually occur in different places relative to the same frequency adverb *zǒngshì* ‘always’. Assuming that the frequency adverb is fixed to one position in the clausal spine, I attribute the flexible position of the fronted VP to the idea that there are multiple positions in the clausal-medial domain where the verbal phrase can move to (recall section 2.4; also see relevant discussion in chapter 6). As given in (203a), the non-doubling counterpart also shows similar flexibility with respect to its position relative to the same frequency adverb. This fact, once again, follows the proposed idea that there

is a string-vacuous, verbal phrasal movement in this non-doubling configuration.

- (203) a. Piet (**zǒngshì**) kàn nà-běn xiǎoshuō (**zǒngshì**) tèbié zǐxì.
 Piet always read that-CL novel always particularly carefully
 ‘Piet always reads the novel very carefully.’
- b. Piet (**zǒngshì**) kàn nà-běn xiǎoshuō (**zǒngshì**) kàn de tèbié zǐxì.
 Piet always read that-CL novel always read DE particularly carefully
 ‘Piet always reads the novel very carefully.’

If my proposal is on the right track, then the crucial question becomes where the V_2 is in the non-doubling counterpart. To put it differently, what happens to the lower verbal copy in this configuration? I propose that the order and the number of post-syntactic operations involved in this non-doubling construction is different from its low verb doubling counterpart, which is given in (204): in this case, Chain Reduction is the first operation that takes place at PF, which leads to the result that the entire lower verbal copy gets deleted. Suppose that Head Movement still takes place in this scenario: when this operation begins to look for its target, i.e., the lower V head, it no longer exists at that point due to the previous deletion caused by Chain Reduction. The failure of implementing the second operation results in the further failure of Local Dislocation, as the particle *de* is unable to find the $V+v$ complex it requires.

- (204) *Order of post-syntactic operations underlying (201a)*
 ... >> Chain Reduction >> (Head Movement) >> (Local Dislocation) >> ...

Given that the first application of Chain Reduction leads to a sequence of bleeding environments for further operations, it is also plausible to assume that neither Head Movement nor Local Dislocation needs to take place in this case (indicated by brackets in the schema above). In other words, what really matters to these non-doubling constructions is to apply Chain Reduction as early as possible at PF. One question that arises immediately is how we ensure that these post-syntactic operations can take place in the correct order, if their relative order is allowed to be flexible to some degree. I will leave this for future study.

To sum up, in this section I have developed a method of realising verb doubling effects at PF, which consists of a series of post-syntactic operations that take place in order. My proposed analysis not only captures both similarities and differences found between the two groups of particles, but also offers an account of how verb doubling effects vary from one to another. In the next chapter, I will turn my attention back to the syntactic properties of verb doubling, and focus on how its locality behaves in cross-clausal contexts.

Chapter 6

Improper Verb Doubling

The goal of this chapter is twofold: on the one hand, I present novel data from long-distance verb doubling configurations in Mandarin Chinese, showing that their locality profiles vary in three different embedding contexts; on the other hand, I use these varying locality profiles of verb doubling to offer support for the *Williams Cycle* (Williams 1974, 2003, 2011, 2013), a locality constraint which posits a general connection between the height of movement and its locality.¹

In the previous chapters, I have proposed that verb doubling configurations involve verbal phrasal movement, in which the base-position is partly realised overtly (cf. Lai 2021, 2024; Meadows and Yan 2023, 2025). As in familiar languages like English, this movement can land in a high position Spec,CP in the clausal spine (205a), above where subjects typically land, namely, Spec,TP. Furthermore, such movement can also land in a low position Spec,FP in the clausal spine, below the surface subject position (205b) but outside of the thematic domain.

(205) *Two landing sites for verbal phrasal movement with doubling in Mandarin Chinese*

- a. [CP [VP **kàn** nà-běn xiǎoshuō] [TP Piet [vP **kàn** de hěn kuài]]].
read that-CL novel Piet read DE very quickly
'As for reading the novel, Piet reads it quickly.' (high verb doubling)
- b. [CP [TP Piet [FP [VP **kàn** nà-běn xiǎoshuō] [vP **kàn** de hěn kuài]]]].
Piet read that-CL novel read DE very quickly
'Piet reads the novel quickly.' (low verb doubling)

Situations like this, where the same type of movement can land in multiple positions, are crucial testing grounds for the Williams Cycle, a locality principle which I summarised as in (206).

¹ The main content of this chapter is adapted from Yan and Meadows (2025), where I made major contributions to the data collection, and equal contributions to the theoretical proposal with my co-author. This chapter includes several changes, including my own comments on the Level Embedding approach proposed in Yan and Meadows 2025 in section 6.3, the closing remarks in section 6.5, etc.

(206) *The Williams Cycle*

For certain types of movement, there is a principled connection between landing and locality such that: the higher a movement operation lands in the clausal functional sequence, the more unbounded it is.

To investigate the influence of height on locality, I examine the interaction between the choice of final landing site of long-distance VP-movement, and three kinds of embedding predicate, respectively labelled ‘Group 1/2/3’ as given in (207). I provide independent evidence that these three classes of embedding predicate take complement clauses of three degrees of structural richness (cf. He 2020, 2024; Huang 2022, among others).² Group 1 complements are bigger than Group 2 complements, which are in turn bigger than Group 3. If it is correct, one may expect that VP-fronting to the lower position to have a more constrained locality profile than its higher counterpart.

(207) *Three groups of embedding predicate and complement clause in Mandarin Chinese*

- a. Wassily rènwéi_{Group 1} [_{CP} Piet kàn nà-běn xiǎoshuō kàn de hěn kuài].
Wassily think Piet read that-CL novel read DE very quickly
‘Wassily thinks that Piet reads the novel quickly.’
(Group 1 predicate + CP complement)
- b. Wassily qiǎngpò_{Group 2} Piet [_{TP} kàn nà-běn xiǎoshuō kàn de hěn kuài].
Wassily force Piet read that-CL novel read DE very quickly
‘Wassily forces Piet to read the novel quickly.’
(Group 2 predicate + TP complement)
- c. Wassily chángshì_{Group 3} [_{FP} kàn nà-běn xiǎoshuō kàn de hěn kuài].
Wassily try read that-CL novel read DE very quickly
‘Wassily tries to read the novel quickly.’
(Group 3 predicate + FP complement)

The basic pattern of long-distance VP-movement is schematised in (208). VP-movement to the clause-initial, pre-subject position is relatively unconstrained, compared to the highly restricted VP-movement to the clause-medial, post-subject position, a phenomenon that I term *improper verb doubling*.³ The possible landing sites for long-distance VP-movement are contingent on the size

² I use CP and TP as umbrella terms for clause initial and medial fields of functional projections, in line with the spirit of Ramchand and Svenonius (2014). Clarifying the ultimate identity of these functional projections is not my main concern. TP, in particular, is suggested to unpack into several projections in Section 6.4, concerning the precise size of Group 2 complements and the landing site of low object shift.

³ Part of the patterns have already been documented by Lai (2021, 2024) and Huang (2022). Yan and Meadows (2025) is the first to identify the full paradigm of long-distance verb doubling, including the previously unattested data concerned with what are defined as Group 2 predicates here.

of clausal complements containing the base position. Descriptively, it looks like VP-movement to the clause-medial position cannot cross Group 1 or Group 2 clause boundaries. This is exactly what the Williams Cycle leads us to expect if VP-fronting can in principle land below TP.

(208)	<i>Schematic long-distance VP-movement</i>	<u>Landing sites</u>
a.	(VP) Subj (*VP) Pred. _{Group 1} [CP ...VP ...]	✓ Subj ✗
b.	(VP) Subj (*VP) Pred. _{Group 2} [TP ...VP ...]	✓ Subj ✗
c.	(VP) Subj (VP) Pred. _{Group 3} [FP ...VP ...]	✓ Subj ✓

The Williams Cycle (as given in (206)) is a descriptive constraint on movement which needs to follow from general properties governing the narrow syntax. I adopt a version of the *Level Embedding* approach from Yan and Meadows (2025), which is initially pioneered by Williams (2003, 2011, 2013) and more recently developed by Poole (2022) and Meadows (2024). The central innovation of Level Embedding, *Parallel Derivation* (209a), is rethinking the timing of main and embedded clause structure building. The result of (209a) is that the bigger a clause is, the later in the derivation it is embedded - with embedding achieved through *Late Merge* (209b). The *Strict Cycle Condition* (209c) interacts with this novel timing of structure building and embedding to restrict long-distance movement. Long-distance movement can only be triggered by features introduced at stage of the derivation when embedding takes place, or later.

- (209) *Three key ingredients of Level Embedding*
- a. **PARALLEL DERIVATION:** The same component of the clausal functional sequence (*fseq*) (V, v, F, T, C, etc.) is externally merged in every workspace at the same time.
 - b. **LATE MERGE:** Clausal embedding involves external Late Merge.
 - c. **STRICT CYCLE CONDITION:** The output of prior cycles of rule application cannot be altered by rules in subsequent cycles. Every functional projection in the clausal *fseq* delineates a new cycle.

An analysis of improper verb doubling in terms of A/ $\bar{A}$ -distinction is conceivable: low verb doubling could be archetypal A-movement (see Travis and Massam 2021), which is not able to skip over intervening VPs. Setting aside what such an approach would need to say about Group 3 predicates, its key predictive difference with Level Embedding is that the latter expects low nominal movement, analogous to low verb doubling, to be similarly restricted. This expectation is borne out, by close attention to long-distance *object shift* (OS) (Ernst and Wang 1995; Shyu 1995; Paul 2002b, 2005; Chen 2023, 2025, a.o.). This nominal movement can likewise land in a pre-subject or post-subject position (210), but targeting Spec,TopP in the clause-medial domain -

a position slightly higher than the Spec,FP landing site of low verb doubling.

(210) *Two landing sites for object shift in Mandarin Chinese (within a rich clause structure)*

- a. [CP [DP **Nà-běn xiǎoshuō**] [TP Piet [VoiceP kàn ___ de hěn kuài]]].
 that-CL novel Piet read DE very quickly
 ‘THAT NOVEL, Piet reads quickly.’ (high object shift)
- b. [CP [TP Piet [TopP [DP **Nà-běn xiǎoshuō**] [VoiceP kàn ___ de hěn kuài]]]].
 Piet that-CL novel read DE very quickly
 ‘Piet reads THAT NOVEL quickly.’ (low object shift)

What emerges from comparing verb doubling and object shift are three locality profiles for embedding contexts (211). Group 1 complements let high VD/OS escape, but trap low VD/OS. Group 2 complements let high VD/OS and low OS escape, but trap low VD. Group 3 are the least restrictive, allowing high and low VD/OS to escape. We cannot treat the boundedness of low OS as a reflex of archetypal A-movement: low OS can move over intervening objects of Group 2 predicates, which shows the only and the most crucial difference from the locality profile of low VD. What we are seeing is precisely what the Level Embedding/Williams Cycle expect: the bigger the complement clause is, the higher we have to aim to land in the clausal *fseq* in order to escape the complement.

(211) *Three locality profiles for embedding contexts*

Locality profile	Pred. _{Group 1} + [CP ...]	Pred. _{Group 2} + [TP ...]	Pred. _{Group 3} + [FP ...]
Transparent to:	High VD/OS	High VD/OS, Low OS	High VD/OS, Low VD/OS
Opaque to:	Low VD/OS	Low VD	Ø

This chapter is organised as follows. Section 6.1 presents the paradigm of long-distance VP-fronting in different embedding contexts. Section 6.2 argues for the idea that embedded clausal complements vary in size. Section 6.3 sketches out the Williams Cycle in the first place, and then explores the Level Embedding approach in greater detail. Section 6.4 extends the Level Embedding account to restrictions on long-distance object shift. Section 6.5 closes the chapter by exhibiting two remaining puzzles regarding verb doubling’s interaction with other movement-related phenomena.

6.1 Long-distance verb doubling

In this section I show that different groups of embedding predicates give rise to different patterns of long-distance VP-fronting. Building on Huang (2022)'s classification of clausal complementation types in Mandarin Chinese, I identify three kinds of embedding contexts, demonstrated by three groups of predicates (212).

(212) *Three groups of embedding predicates in Mandarin Chinese*

Group 1: *fāxiàn* 'find out', *fǒurèn* 'deny', *juéde* 'feel', *rènwéi* 'think', *zhīdào* 'know'...

Group 2: *bī(pò)* 'force', *qiǎngpò* 'force', *quàn* 'urge, persuade', *yǐnyòu* 'lure', *zhǐdǎo* 'instruct'...

Group 3: *chángshì* 'try', *dǎsuan* 'intend', *jìhuà* 'plan', *juédìng* 'decide', *kāishǐ* 'begin'...

I start the discussion with Group 1 embedding predicates, represented by *rènwéi* 'think' in (213). The baseline configuration (213a) reveals that *rènwéi* can take a complement clause containing verb doubling.⁴ However, once the VP undergoes long-distance movement, there arises a position-contrast between (213b) and (213c). Though VP-fronting to the matrix-initial position is still possible (213c), the same phrasal movement to the matrix-medial position (213b) is prohibited. Such a configuration exemplifies what I term *improper verb doubling*.

- (213) a. Wassily rènwéi [Piet **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily think Piet read that-CL novel read DE very quickly
 'Wassily thinks that Piet reads the novel quickly.' ('think' + embedded VD ✓)
- b. *Wassily **kàn nà-běn xiǎoshuō** rènwéi [Piet kàn de hěn kuài].
 Wassily read that-CL novel think Piet read DE very quickly
 Int.: 'Wassily thinks that Piet reads the novel quickly.' ('think' + matrix low VD ✗)
- c. **Kàn nà-běn xiǎoshuō** Wassily rènwéi [Piet kàn de hěn kuài].
 read that-CL novel Wassily think Piet read DE very quickly
 'As for reading the novel, Wassily thinks that Piet reads it quickly.'
 ('think' + matrix high VD ✓)

The distinction between proper and improper verb doubling is also identified in the Group 2

⁴ Note that what is contained here is an instance of low VD. It is also possible to construct high VD in this Group 1 embedded clause, namely, the VP *kàn nà-běn xiǎoshuō* 'read the novel' moves to the embedded left periphery preceding the embedded subject *Piet* (and following the matrix predicate *rènwéi*). The same configuration cannot be easily constructed in Group 2 and 3 embedding contexts, since there is no overt subject in those embedded clauses. As will become clearer soon, I attribute such a difference to the size of different complement clauses.

embedding contexts, as illustrated by *qiǎngpò* ‘force’ in (214). Unlike their Group 1 counterparts seen above, Group 2 predicates take a surface object and a clausal complement simultaneously. As with (213a), the baseline configuration (214a) shows that verb doubling can be constructed within the complement clause taken by a Group 2 predicate. The same contrast regarding long-distance VP-fronting appears again between (214b) and (214c). That is, the cross-clausal VP-fronting is only able to target the clause-initial position (214c) rather than the clause-medial one (214b).⁵

- (214) a. Wassily qiǎngpò Piet [**kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily force Piet read that-CL novel read DE very quickly
 ‘Wassily forces Piet to read the novel quickly.’ (‘force’ + embedded VD ✓)
- b. *Wassily **kàn nà-běn xiǎoshuō** qiǎngpò Piet [kàn de hěn kuài].
 Wassily read that-CL novel force Piet read DE very quickly
 Int.: ‘Wassily forces Piet to read the novel quickly.’ (‘force’ + matrix low VD ✗)
- c. **Kàn nà-běn xiǎoshuō** Wassily qiǎngpò Piet [kàn de hěn kuài].
 read that-CL novel Wassily force Piet read DE very quickly
 ‘As for reading the novel, Wassily forces Piet to read it quickly.’
 (‘force’ + matrix high VD ✓)

We see a different locality profile of long-distance VP-fronting when we look at Group 3 embedding predicates, represented by *chángshì* ‘try’ in (215). The baseline configuration (215a) shows that, similar to its Group 1 and 2 counterparts, it is possible to construct verb doubling within the embedded clause taken by a Group 3 predicate.⁶ (215b) and (215c) further demonstrate that long-distance VP-fronting can land either after or before the matrix subject (i.e., *Wassily*) in a Group 3 embedding context (cf. (213b) and (214b)), presenting a scenario in which the cross-clausal verb doubling is no longer confined to the matrix left periphery (cf. (213c) and (214c)).

⁵ Since the Group 2 representative I choose is commonly regarded as an object control verb, one may wonder whether the same patterns of long-distance verb doubling still hold in the context of ECM verbs, such as *xiāngxìn* ‘believe’ or *qídài* ‘expect’. According to the data I collect, the result shows no difference from their object control counterparts. There are two potential concerns regarding ECM verbs, though. One is whether these alleged ‘ECM’ verbs in Mandarin Chinese really behave identically to those in English, since there is recent literature (e.g., Ussery et al. 2016; Paul 2021a) arguing against the existence of Mandarin ECM verbs. The other is that these ECM verbs pose a serious challenge to the Williams Cycle, namely that the raising of the argument constitutes a violation of this locality principle. As both of them are topics beyond the scope of this chapter, I confine Group 2 predicates to object control verbs only throughout the following discussion.

⁶ Some native speakers report to me that the meaning of (215a) sounds pragmatically awkward to them. However, presented with the scenario below, the acceptance of (215a) improves significantly: Wassily used to read the novel in a very slow manner. Now he wants to change his way of reading, trying to read the novel as quickly as possible, or at least more quickly than before.

- (215) a. Wassily chángshì [**kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily try read that-CL novel read DE very quickly
 ‘Wassily tries to read the novel quickly.’ (‘try’ + embedded VD ✓)
- b. Wassily **kàn nà-běn xiǎoshuō** chángshì [kàn de hěn kuài].
 Wassily read that-CL novel try read DE very quickly
 ‘Wassily tries to read the novel quickly.’ (‘try’ + matrix low VD ✓)
- c. **Kàn nà-běn xiǎoshuō** Wassily chángshì [kàn de hěn kuài].
 read that-CL novel Wassily try read DE very quickly
 ‘As for reading the novel, Wassily tries to read it quickly.’
 (‘try’ + matrix high VD ✓)

To summarise, I have shown that long-distance VP-fronting differs in targeting the two landing sites in various embedding contexts. Though it is always possible for the VP to move across the complement clause and target the matrix-initial position, it becomes more restricted when the VP targets the relatively lower, medial position in the matrix clausal spine, giving rise to the proper/improper verb doubling contrast. The overall paradigm I have gone through is generalised in (216).

(216) *Full paradigm of verb doubling in different embedding contexts*

Embedding predicate group	Matrix high VD	Matrix low VD	Embedded VD
<i>Group 1</i> : ‘think’ + [...]	✓	✗	✓
<i>Group 2</i> : ‘force’ + [...]	✓	✗	✓
<i>Group 3</i> : ‘try’ + [...]	✓	✓	✓

Viewed in this way, the embedding predicate groups that I have identified in the first place can also be divided only into two: one group (including Group 1 and 2) exhibits improper verb doubling, whereas another group (i.e., Group 3 only) does not show the same pattern. So it is worth questioning why and how I recognise three instead of two predicate groups. As will be clearer in the upcoming section 6.2, I argue that whether improper verb doubling is attested reflects the size of clausal complement. The fact that we have seen in this section indicates that both Group 1 and 2 predicates take bigger complements than their Group 3 counterparts. It is the empirical findings I use to diagnose clause size in section 6.2 that lead us to the distinction of three predicate groups as presented here.

6.2 Three sizes of clausal complement

To capture the locality differences presented above, I rely on the idea that clausal complements vary in size. In the literature of Mandarin syntax, clause-size variation has been proposed and discussed in, for instance, Xue and McFetridge (1996, 1998), Grano (2012, 2015, 2017), and Huang (2022), among many others. Along the same lines, I propose that Group 1 predicates take CP-sized complements, Group 2 TP-sized ones, and Group 3 FP-sized ones. The proposed size difference of clausal complements taken by different groups of embedding predicates is schematised in (217). In what follows I offer two suggestive empirical arguments in this direction.

(217) *Embedding predicates and their corresponding clausal complements*

- a. Pred._{Group 1} + [CP [MP^{epis} [TP [MP^{deon} [FP [vP ...]]]]]]]
- b. Pred._{Group 2} + [TP [MP^{deon} [FP [vP ...]]]]
- c. Pred._{Group 3} + [FP [vP ...]]

6.2.1 Object shift in embedding contexts

The first argument comes from *object shift* (OS), a well-known displacement phenomenon that is widely discussed in the Mandarin literature (e.g., Ernst and Wang 1995; Shyu 1995; Paul 2002b, 2005; Chen 2023, 2025). I will show that Group 1 clausal complements can provide more positions for object shift than their Group 2 and 3 counterparts, which I argue reflects reduced clause structure.

My discussion begins with the complement clauses taken by Group 1 predicates. As shown in (218), the object *nà-běn xiǎoshuō* ‘the novel’, which is canonically understood to be selected by the verb *kàn* ‘read’, can be displaced to the left of the verb and show up either before the embedded subject *Piet* (218a) or after it (218b). The displaced object receives a topicalised interpretation in both positions (indicated by the words in capitals).

- (218) a. Wassily rènwéi [Piet **nà-běn xiǎoshuō** *kàn de hěn kuài*].
 Wassily think Piet that-CL novel read DE very quickly
 ‘Wassily thinks that Piet reads THAT NOVEL quickly.’
- b. Wassily rènwéi [**nà-běn xiǎoshuō** Piet *kàn de hěn kuài*].
 Wassily think that-CL novel Piet read DE very quickly
 ‘Wassily thinks that, THAT NOVEL, Piet reads quickly.’

Chen (2023, 2025), following Paul (2002b, 2005), proposes that in Mandarin Chinese, projec-

tions in the domain below TP (= IP in her usage) parallel those in the articulated left periphery (Rizzi 1997). According to Chen, the shifted object is actually the consequences of DP movement that targets either TP-internal Spec,TopP (as in (218a)) or TP-external Spec,TopP (as in (218b)). As schematised in (219), that both kinds of topicalisation are available in Group 1 complement clauses follows my proposal that Group 1 predicates take full-fledged, CP-sized clausal complements.

- (219) *Object shift in the Group 1 embedding context*
 Pred._{Group 1} + [_{CP} (OS) [_{TP} Subj (OS) ...]]

Object shift is also attested in complement clauses taken by Group 2 predicates, as exemplified in (220). As with its counterpart in the Group 1 embedding context, the shifted object receives a topicalised interpretation, suggesting that the displacement of the object is triggered by the same mechanism of topicalisation in such a case as well.

- (220) Wassily qiǎngpò Piet [**nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily force Piet that-CL novel read DE very quickly
 ‘Wassily forces Piet to read THAT NOVEL quickly.’

The difference of the Group 2 embedding context is that there is no overt embedded subject in the construction, leading to the puzzle whether the shifted object lands within the TP- or the CP-domain.⁷ Putting aside this problem for a while (which I will return to very soon), the fact that Group 2 complements can host object shift indicates that they are at least TPs, as schematised in (221).

- (221) *Object shift in the Group 2 embedding context*
 Pred._{Group 2} + [_{TP} OS ...]

Unlike what we see in the previous two situations, object shift is not acceptable in Group 3 complement clauses, as shown in (222).

⁷ Object shift cannot target the position between Group 2 predicate and its surface object, as given in ((i)).

- (i) *Wassily qiǎngpò nà-běn xiǎoshuō Piet [kàn de hěn kuài].
 Wassily force that-CL novel Piet read DE very quickly
 Int: ‘Wassily forces Piet to read THAT NOVEL quickly.’

The ungrammatical result is actually what we predict, since Group 2 predicates are all object control verbs, and there is no available position to host object shift before the object.

- (222) *Wassily chángshì [**nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily try that-CL novel read DE very quickly
 Int.: ‘Wassily tries to read THAT NOVEL quickly.’

The ungrammaticality of (222) implies that complement clauses taken by Group 3 predicates are smaller than their Group 1 and 2 counterparts. Comparing with the case of Group 2 complements seen in (220), I conclude that Group 3 clausal complements are not as big as TP. Recall that embedded verb doubling is possible within the Group 3 context (e.g., (215a)), and that low verb doubling is proposed to land in Spec,FP. I postulate that Group 3 complement clauses are FP-sized, and that FP is lower than TP in the clausal spine. That no object shift is attested in Group 3 complements is illustrated in (223).

- (223) *No object shift in the Group 3 embedding context*
 Pred._{Group 3} + [FP (*OS) ...]

Variation in clause size seems to provide a direct account for the distribution of object shift within different embedding contexts. Further evidence for my account comes from the distribution of adverbs and their interaction with the shifted object. Let us first have a look at another pair of object shift cases within the Group 1 embedding context that co-occur with different adverbs respectively, as given in (224).

- (224) a. Wassily rènwéi [_{CP} [_{TP} Piet **nà-běn xiǎoshuō** yòu kàn-le sān biàn]].
 Wassily think Piet that-CL novel again read-PRF three time
 ‘Wassily thinks that Piet again read THAT NOVEL three times.’
 b. Wassily rènwéi [_{CP} **nà-běn xiǎoshuō**, xìngkuī [_{TP} Piet kàn-le sān biàn]].
 Wassily think that-CL novel fortunately Piet read-PRF three time
 ‘Wassily thinks that, THAT NOVEL, fortunately Piet read three times.’

(224a) presents a configuration where the TP-internal object shift is higher than *yòu* ‘again’, a VP-level adverb in the sense of Paul (2017) (cf. Ernst 2014). According to Paul, such adverbs refer to as well as modify the internal structure of an event: in (224a) *yòu* is applied to the event of ‘reading that novel three times’, thus the whole event gets repeated and happens again. The positions of these so-called ‘VP-level’ adverbs are generally assumed to be relatively low in the *fseq* (Cinque 1999). Here I simply presume that *yòu* marks the edge of the event/verbal domain which is for sure lower than TP.⁸

⁸ As for the exact position of *yòu*, Liu (2021, 2024) proposes that it can be base-generated with an unvalued Asp feature in [Spec, vP], which would, for the sake of specifier-head agreement, further move to the specifier position of

(224b) exemplifies an instance of TP-external object shift that precedes not only the embedded subject *Piet* but also a sentence-level adverb *xìngkuī* ‘fortunately’. The label ‘sentence-level’, as defined by Paul (2017), actually consists of a number of sub-categories (e.g., as proposed in Ernst 2014). Though most sentence-level adverbs are liable to occur both to the right and left of the subject, a few of them are confined to the clause-initial position. Paul (2017) claims that *kěxī* ‘unfortunately’ is the only candidate for the latter group, which is confined to clause-initial and excluded from clause-medial position. However, I find that the latter position for *kěxī* is marginally acceptable. *Xìngkuī* ‘fortunately’ shares the same distribution with *kěxī* according to my data: it is definitely preferred in the clause-initial position. This observation is crucial because it would be very likely for any constituent to the left of *xìngkuī* (as well as *kěxī*) to be automatically assumed to occur in the clausal left periphery, just like the shifted object in (224b).⁹ Based on (224), I schematise the distribution of object shift relative to distinct types of adverbs in the Group 1 embedding context as follows:

- (225) *Object shift in the Group 1 embedding context (relative to adverbs)*
 Pred-Group 1 + [CP (OS) Adv_{S-Level} [TP Subj (OS) Adv_{VP-Level} ...]]

Proceeding to the Group 2 embedding context, we first see that the sentence-level adverb *xìngkuī* is not allowed to appear independently in the Group 2 embedding context, as shown in (226).

- (226) *Wassily qiǎngpò Piet [TP *xìngkuī* kàn-le **nà-běn xiǎoshuō** sān biàn].
 Wassily force Piet fortunately read-PRF that-CL novel three time

The ungrammatical result follows naturally my proposal that Group 2 clausal complements are generally TP-sized. Since the left periphery is totally absent within the Group 2 embedding context, there exists no position in the clausal spine for sentence-level adverbs like *xìngkuī* to be hosted properly. Furthermore, in this case, no object shift is predicted to target a position that is even higher than the sentence-level adverbs. This is borne out by (227).

- (227) *Wassily qiǎngpò Piet [TP **nà-běn xiǎoshuō** *xìngkuī* kàn-le sān biàn].
 Wassily force Piet that-CL novel fortunately read-PRF three time

the closet AspP, a position he takes to be lower than TP but higher than vP. Liu’s proposal seems compatible with my assumption. However, since his analysis lays the foundation on the ambiguous interpretations of *yòu*, which do not concern us here, I maintain agnostic towards its precise position.

⁹ Note that the shifted object can also occur to the right of *xìngkuī* in (224b), which follows Rizzi (1997)’s proposal that there are two topic positions in the left periphery. By contrast, it is impossible for object shift to appear at the right side of *yòu* in (224a), which is compatible with our assumption that *yòu* stays at the edge of verbal domain lower than TP. The same fact also implies that the position of *yòu* is too low to be included in the TP-domain.

By comparison, the shifted object is still able to co-occur with the VP-level adverb *yòu*, as given in (228). This is because object shift can target the position that is within the TP but higher than the entire event/verbal domain including the position of *yòu*.

- (228) Wassily qiǎngpò Piet [_{TP} **nà-běn xiǎoshuō** *yòu* kàn-le sān biàn].
 Wassily force Piet that-CL novel again read-PRF three time
 ‘Wassily forced Piet to again read THAT NOVEL three times.’

It is worth pointing out that my account does not empirically rule out the possibility that the shifted object in (228) in effect lands in a TP-external position that is lower than *xìngkuī* (see fn. 9), which is why I give a question marker to the corresponding space in Table (234). If that was the actual situation, one needs to come up with some different assumption in order to explain why the independent appearance of *xìngkuī* is impossible in (226). This possibility can be ruled out by assuming that the TP-external shifted object may act as an intervenor between the surface object and its controlled PRO in the embedded clause. At the moment I simply conclude that Group 2 complement clauses are TP-sized as represented in (229).

- (229) *Object shift in the Group 2 embedding context (relative to adverbs)*
 Pred._{Group 2} + [_{TP} **OS** Adv_{VP-Level} ...]

Moving on to Group 3 complement clauses, it is not a surprise that object shift is unable to co-occur with either the VP-level adverb *zài* ‘again’ (230a) or the sentence-level one *xìngkuī* (230b), since this specific kind of DP-movement is prohibited in the Group 3 embedding context at the outset (as seen in (222)), leading to the hypothesis that Group 3 complements are FP-sized only (as represented in (223)). Such an idea is reinforced by the fact that the sentence-level adverb is also unable to show up on its own in the Group 3 embedding context (231).

- (230) a. *Wassily chángshì [_{FP} **nà-běn xiǎoshuō** *zài* kàn sān biàn].
 Wassily try that-CL novel again read three time
 ‘Wassily tries to again read THAT NOVEL three times.’
 b. *Wassily chángshì [_{FP} **nà-běn xiǎoshuō** *xìngkuī* kàn sān biàn].
 Wassily try that-CL novel fortunately read three time
- (231) *Wassily chángshì [_{FP} *xìngkuī* kàn **nà-běn xiǎoshuō** sān biàn].
 Wassily try fortunately read that-CL novel three time

Interestingly, as exemplified in (232), the VP-level adverb is able to co-occur with the embedded verb doubling, contrary to the configuration involving embedded object shift as in (230a).

- (232) Wassily chángshì [_{FP} *zài kàn nà-běn xiǎoshuō* kàn sān biàn].
 Wassily try again read that-CL novel read three time
 ‘Wassily tries to again read THAT NOVEL three times.’

Empirically, the contrast between (232) and (230a) indicates clearly that low verb doubling and TP-internal object shift target different positions in the clausal spine: low verb doubling can appear below the VP-level adverb, whereas object shift (in all possible cases) has to occur above the same adverb. Theory-wise, such a contrast indicates that FP is lower than TP in the clausal spine: Spec,FP is targeted by low verb doubling rather than TP-internal object shift, as schematised in (233). I will rely on this idea while extending the Level Embedding approach to long-distance object shift in section 6.4.

- (233) *Verb doubling but no object shift in the Group 3 embedding context (relative to adverbs)*
 Pred._{Group 3} + [_{FP} (*OS) Adv_{VP-Level} **VD** ...]

To recap my argument, I have shown that the distribution of embedded object shift varies in different embedding contexts, and argued that it is the clause-size variation that determines the result of displacing the object. The key patterns of object shift as well as the proposed clause sizes are summarised in (234).

- (234) *Object shift in different embedding contexts*

Embedding predicate group	OS above TP _{Emb.}	OS below TP _{Emb.}	Clause size
<i>Group 1</i> : ‘think’ + [...]	✓	✓	CP-sized
<i>Group 2</i> : ‘force’ + [...]	?	✓	TP-sized
<i>Group 3</i> : ‘try’ + [...]	✗	✗	FP-sized

6.2.2 Modal hierarchy in embedding contexts

The second argument for variation in clause size focuses on the distribution of various modals in complement clauses. In what follows I will show that Group 1 clausal complements display a full range of modals within the embedding context, whereas their Group 2 and 3 counterparts are only able to exhibit a reduced range of, or even a complete absence of, the same modals. I argue such a contrast again reflects the reduced clause structure of Group 2 and 3 complements.

One characteristic of modals in Mandarin Chinese is that they display rigid word orders when

they co-occur (Lin 2011, 2012; Lai and Li 2023, a.o.).¹⁰ For instance, as shown in (235a), the epistemic modal *yīnggāi* ‘should’ must precede another epistemic modal *kěnéng* ‘may’ when they appear together. When both epistemic modals respectively co-occur with the future modal *huì*, either of them precedes the future one, as given in (235b) and (235c). If these three appear together in the same configuration, such as (235d), their relative positions to each other remain the same as what we have seen above, that is, the epistemic ‘may’ is preceded by the epistemic ‘should’ and followed by the future ‘will’.

- (235) a. Piet **yīnggāi**^{epis} **kěnéng**^{epis} líkāi lúndūn le.
 Piet should may leave London SFP
 ‘It should be the case that Piet may have left London.’ (should^{epis} >> may^{epis})
- b. Piet **yīnggāi**^{epis} **huì**^{future} líkāi lúndūn.
 Piet should will leave London
 ‘It should be the case that Piet will leave London.’ (should^{epis} >> will^{future})
- c. Piet **kěnéng**^{epis} **huì**^{future} líkāi lúndūn.
 Piet may will leave London
 ‘It may be the case that Piet will leave London.’ (may^{epis} >> will^{future})
- d. Piet **yīnggāi**^{epis} **kěnéng**^{epis} **huì**^{future} líkāi lúndūn.
 Piet should may will leave london
 ‘It should be the case that it is likely that Piet will leave London.’
 (should^{epis} >> may^{epis} >> will^{future})

Another rigid sequence of word order can be found in cases involving deontic modals relative to epistemic ones. As given in (236a), the deontic modal *bìxū* ‘must’ can appear before another deontic modal *děi* ‘have to’.¹¹ When either of them co-occurs with an epistemic modal, e.g., *kěnéng* ‘may’, the latter has to be positioned before the former, as seen respectively in (236b) and (236c). When these two deontic modals show up with the epistemic one at the same time, they are both preceded by the epistemic modal, with the internal order between them unchanged, as illustrated in (236d).

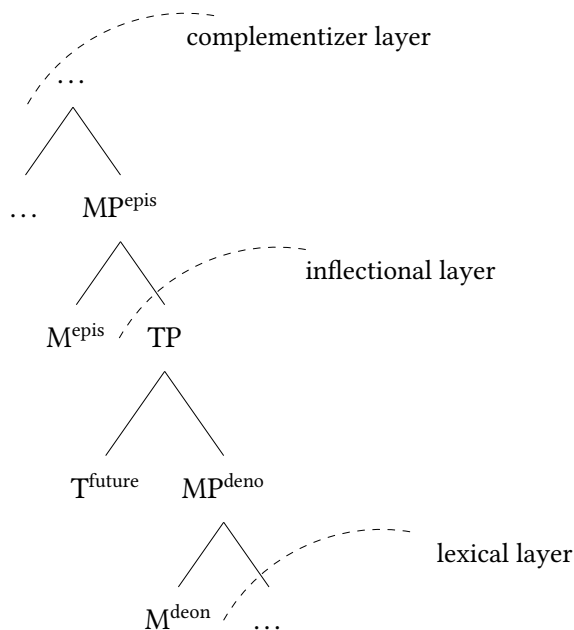
¹⁰ Throughout the discussion, I remain agnostic towards the actual category of modals, since in Mandarin it is not immediately obvious whether a modal is verbal or adverbial. What matters to us is that every modal is licensed by a unique functional projection, and the presence/absence of each modal is contingent on the existence of its corresponding functional projection in the clausal *fseq*.

¹¹ There are two points worth mentioning here. First, the meaning difference between *bìxū* and *děi* is actually quite trivial. The different translations that I give to these two deontic modals only serve the purpose of tearing them apart. Second, though I agree that *bìxū* preceding *děi* is the preferred order in general, I find the reverse word order acceptable as well. However, the flexible word order between deontic modals, if it holds, would not undermine the point I want to raise here, which is, epistemic modals always precede deontic modals.

- (236) a. Piet **bìxū**^{deon} **děi**^{deon} líkāi lúndūn.
 Piet must have-to leave London
 ‘Piet must have to leave London.’ (must^{deon} >> have to^{deon})
- b. %Piet **kěnéng**^{epis} **bìxū**^{deon} líkāi lúndūn le.
 Piet may must leave London SFP
 ‘It may be the case that Piet must leave London.’ (may^{deon} >> must^{deon})
- c. Piet **kěnéng**^{epis} **děi**^{deon} líkāi lúndūn le.
 Piet may have-to leave London SFP
 ‘It may be the case that Piet has to leave London.’ (may^{epis} >> have to^{deon})
- d. Piet **kěnéng**^{epis} **bìxū**^{deon} **děi**^{deon} líkāi lúndūn le.
 Piet may must have-to leave London SFP
 ‘It may be the case that Piet must have to leave London.’
 (may^{epis} >> must^{deon} >> have to^{deon})

Tsai (2015) claims that the rigid word order among modals in Mandarin, as seen above, can be aligned with the syntactic configuration advocated in the cartographic fashion (Rizzi 1997; Cinque 1999). Assuming that the future modal *huì* acts as the head of TP, he proposes that epistemic modals, which are structurally higher than the future, are situated in the complementizer layer outside the TP. As for deontic modals, Tsai posits that they are lower than the future modal, but still located in the same inflectional layer with the future. The modal hierarchy in the sense of Tsai (2015) is simply illustrated in (237) (cf. Lin 2012):

(237) *Modal hierarchy in Mandarin Chinese (in the sense of Tsai 2015)*



With this hierarchy in mind, let us start looking at the distribution of modals in different embedding contexts, with a focus on epistemic and deontic modals. As illustrated in (238), the modals which I am concerned with are all compatible with embedded verb doubling in Group 1 complement clauses. (238a) and (238b) respectively show the configurations involving the epistemic *yīnggāi* and *kěnéng*. (238c) and (238d) are cases where the deontic *bìxū* and *děi* occur in the embedded clauses. Note that, to better accommodate to the meaning of deontics, the postverbal adverbial in the latter two examples is slightly different from its counterpart in the former two, conveying a comparative state that has yet to achieve (i.e., *gèng kuài yīxiē* ‘more quickly’). The empirical picture presented in (238) is in line with my proposal regarding the clause size: since Group 1 predicates are proposed to take CP-sized complement clauses, it follows that all the modals in principle are able to occur in such configurations.¹²

- (238) a. Wassily *rènwéi* [_{CP} Piet *yīnggāi*^{epis} **kàn nà-běn xiǎoshuō** *kàn de hěn kuài*].
 Wassily think Piet should read that-CL novel read DE very quickly
 ‘Wassily thinks it should be the case that Piet reads the novel quickly.’
 (should^{epis} ✓)
- b. Wassily *rènwéi* [_{CP} Piet *kěnéng*^{epis} **kàn nà-běn xiǎoshuō** *kàn de hěn kuài*].
 Wassily think Piet may read that-CL novel read DE very quickly
 ‘Wassily thinks it may be the case that Piet reads the novel quickly.’ (may^{epis} ✓)
- c. Wassily *rènwéi* [_{CP} Piet *bìxū*^{deon} **kàn nà-běn xiǎoshuō** *kàn de gèng kuài yīxiē*].
 Wassily think Piet must read that-CL novel read DE more quickly
 a-bit
 ‘Wassily thinks that Piet must read the novel more quickly.’ (must^{deon} ✓)
- d. Wassily *rènwéi* [_{CP} Piet *děi*^{deon} **kàn nà-běn xiǎoshuō** *kàn de gèng kuài yīxiē*].
 Wassily think Piet will read that-CL novel read DE more quickly
 a-bit
 ‘Wassily thinks that Piet has to read the novel more quickly.’ (have to^{deon} ✓)

The difference with respect to the modal hierarchy starts appearing in Group 2 clausal complements. As shown in (239a) and (239b) separately, neither the epistemic *yīnggāi* nor the epistemic *kěnéng* is allowed to exist in the Group 2 embedding context, contrary to their status in the Group 1 context (cf. (238a) and (238b)). The deontic *bìxū* and *děi*, however, are still able to show up

¹² In the relevant literature, both Lin (2011, 2012) and Zhang (2019) argue that epistemic modals (and future modals) may occur in finite complements. Such an idea is theoretically compatible with my proposal, as CP-sized complements are generally considered finite. However, I do not have too much to comment on the relation between finiteness and clause size in this paper. I refer readers to the references cited above as well as recent work by Huang (2022).

in Group 2 complement clauses, as given in (239c) and (239d). Under my proposal that Group 2 complements are TP-sized, such a contrastive result is in fact what we would expect. Recall that in the modal hierarchy (237), epistemic modals are higher than TP and assumed to be excluded from the inflectional layer, whereas deontic modals are argued to be lower than TP and inside the inflectional layer. Thus, the contrast between epistemics and deontics within the TP-sized, Group 2 embedding context fits neatly with the division of layers in the modal hierarchy.

- (239) a. *Wassily qiǎngpò Piet [_{TP} *yīnggāi*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily force Piet should read that-CL novel read DE very quickly
(should^{epis} ✗)
- b. *Wassily qiǎngpò Piet [_{TP} *kěnéng*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
Wassily force Piet may read that-CL novel read DE very quickly
(may^{epis} ✗)
- c. Wassily qiǎngpò Piet [_{TP} *bìxū*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài
Wassily force Piet must read that-CL novel read DE more quickly
yīxiē].
a-bit
‘Wassily forced Piet to have to read the novel more quickly.’ (must^{deon} ✓)
- d. %Wassily qiǎngpò Piet [_{TP} *děi*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài
Wassily force Piet must read that-CL novel read DE more quickly
yīxiē].
a-bit
‘Wassily forced Piet to have to read the novel more quickly.’ (have to^{deon} ✓)

The modal hierarchy continues to ‘vanish’ within the Group 3 embedding context. As shown in (240), all the epistemic and deontic modals that I am concerned with are unable to occur in any complement clause taken by Group 3 predicates. By comparison, if the modals are removed from the embedded clauses, all of them become grammatical, exhibiting embedded verb doubling as we have already seen in section 6.1 (e.g., (215a)). The results in (240) suggest that Group 3 complement clauses, which I propose are only FP-sized, might be structurally too small to host any modals that we discuss here.

- (240) a. *Wassily chángshì [_{FP} *yīnggāi*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily try may read that-CL novel read DE very quickly
 (should^{epis} ✗)
- b. *Wassily chángshì [_{FP} *kěnéng*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily try may read that-CL novel read DE very quickly
 (may^{epis} ✗)
- c. *Wassily chángshì [_{FP} *bìxū*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài
 Wassily try must read that-CL novel read DE more quickly
 yīxiē].
 a-bit
 (must^{deon} ✗)
- d. *Wassily chángshì [_{FP} *děi*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
 Wassily try must read that-CL novel read DE more quickly a-bit
 (have to^{deon} ✗)

The patterns regarding the distribution of modals in clausal complements are summarised in (241). With the modal hierarchy (237) taken into account, I have argued that the distributional patterns can receive a direct explanation from my proposal. That is, whether a modal shows up in a complement clause depends on the size of the clause it enters.

(241) *Modal hierarchy in different embedding contexts*

Embedding predicate group	Should ^{epis}	May ^{epis}	Must ^{deon}	Have to ^{deon}	Clause size
<i>Group 1</i> : ‘think’ + [...]	✓	✓	✓	✓	CP-sized
<i>Group 2</i> : ‘force’ + [...]	✗	✗	✓	✓	TP-sized
<i>Group 3</i> : ‘try’ + [...]	✗	✗	✗	✗	FP-sized

6.2.3 Note on types of clausal complementation

At the beginning of Sect. 6.1, I mentioned that the predicate groups I identify are built on the clause types proposed by Huang (2022). In the recent literature, He (2020, 2024) and Huang (2022), among many others, propose that the notion of finiteness can be viewed as part of the *Implicational Complementation Hierarchy* (ICH) proposed by Wurmbrand and Lohninger (2023). Along the ICH-based typology, Huang (2022:22-24) recognises three types of verbal complements in Mandarin Chinese, characterising Type I as ‘the most finite’, Type II as ‘somewhat infinitival’, and Type III as ‘the most infinitival’ (see a similar proposal in He 2020, 2024).

There are overlaps between Huang’s clause types and my predicate groups. To be precise, Huang’s Type I is my Group 1, and his Type III falls into my Group 3. The actual difference lies

in his Type II versus my Group 2: only those Type II predicates which are able to take a surface object plus a clausal complement (i.e., object control verbs) are defined as my Group 2 predicates, and the remaining Type II ones all enter my Group 3. The correspondence between these two ways of classifying predicates/clauses is illustrated in (242).

(242) *Correspondence between our predicate groups and Huang's clause types*

Predicate group proposed in here	Clause type proposed in Huang (2022)
Group 1	Type I
Group 2	Type II
Group 3	Type II & III

The question that arises immediately is why Group 3 corresponds to both Type II and III. I argue that this mismatch comes as a natural result of classifying predicates and clauses based on different criteria. Recall that my evidence to distinguish predicate groups as well as their corresponding embedding context is primarily those patterns that can be directly explained by varying clause size. The distinction of clause types, however, is originally motivated by the meaning of their embedding predicates (see Wurmbrand and Lohninger 2023 for details). The two different criteria together lead us to expect there should be predicates that syntactically behave identically to other Group 3 and semantically select clausal complements of Type II. In what follows I use *dǎsuàn* ‘plan’ as the representative to illustrate the property of this specific group of predicates.

As discussed by both He (2020, 2024) and Huang (2022), one defining characteristic of differentiating clause types is their temporal independence (see a summarised overview in He 2024:280). Type I complements can flexibly have independent time adverbs in each of the clauses, as shown in (243). Though Type II complements also permit time adverbs different from those in the matrix, the former must be constrained by the latter in terms of their temporal relation, as seen in (244) and (245). Type III complements are the most temporally dependent, which forbid the co-occurrence of different time adverbs at all, as given in (246). In this respect, it is reasonable to label *dǎsuàn* as Type II (as with *qiǎngpò*) rather than Type III (as with *chángshì*).

- (243) a. **Zuótiān** Wassily rènwéi [Piet **jīntiān** huì líkāi lúndūn].
yesterday Wassily think Piet today will leave London
‘Yesterday Wassily thought that Piet will leave London today.’
- b. **Zuótiān** Wassily rènwéi [Piet **shàng-xīngqī** líkāi-le lúndūn].
yesterday Wassily think Piet last-week leave-PFV London
‘Yesterday Wassily thought that Piet left London last week.’

- (244) a. **Zuótiān** Wassily qiǎngpò Piet [**jīntiān** líkai lúndūn].
yesterday Wassily force Piet today leave London
'Yesterday Wassily forced Piet to leave London today.'
- b. ***Zuótiān** Wassily qiǎngpò Piet [**shàng-xīngqí** líkai lúndūn].
yesterday Wassily force Piet last-week leave London
'*Yesterday Wassily forced Piet to leave London last week.'
- (245) a. **Zuótiān** Wassily dǎsuàn [**jīntiān** líkai lúndūn].
yesterday Wassily plan today leave London
'Yesterday Wassily planned to leave London today.'
- b. ***Zuótiān** Wassily dǎsuàn [**shàng-xīngqí** líkai lúndūn].
yesterday Wassily plan last-week leave London
'*Yesterday Wassily planned to leave London last week.'
- (246) a. ***Zuótiān** Wassily chángshì [**jīntiān** líkai lúndūn].
yesterday Wassily try today leave London
'*Yesterday Wassily tried to leave London today.'
- b. ***Zuótiān** Wassily chángshì [**shàng-xīngqí** líkai lúndūn].
yesterday Wassily try last-week leave London
'*Yesterday Wassily tried to leave London last week.'

The picture becomes different when checking all the patterns we have seen in this section on *dǎsuàn*. It turns out that *dǎsuàn* shows all the same patterns as *chángshì* (a Group 3 predicate in my sense) instead of *qiǎngpò* (my Group 2 predicate): both kinds of long-distance verb doubling are attested (247) (cf. (215)); the judgement of embedded object shift is relatively degraded (248) (cf. (222); and modal hierarchy is totally absent in its clausal complements (249) (cf. (240)).

- (247) a. Wassily dǎsuàn [**kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
Wassily plan read that-CL novel read DE more quickly a-bit
'Wassily plans to read the novel more quickly.' ('plan' + embedded VD ✓)
- b. Wassily **kàn nà-běn xiǎoshuō** dǎsuàn [kàn de gèng kuài yīxiē].
Wassily read that-CL novel plan read DE more quickly a-bit
'Wassily plans to read the novel more quickly.' ('plan' + matrix low VD ✓)
- c. **Kàn nà-běn xiǎoshuō** Wassily dǎsuàn [kàn de gèng kuài yīxiē].
read that-CL novel Wassily plan read DE more quickly a-bit
'As for reading the novel, Wassily plans to read it more quickly.'
('plan' + matrix high VD ✓)

- (248) ??Wassily dǎsuàn [**nà-běn xiǎoshuō** kàn de gèng kuài yīxiē / sān biàn].
 Wassily plan that-CL novel read DE more quickly a-bit three time
 Int.: ‘Wassily plans to read THAT NOVEL more quickly / three times.’
- (249) a. *Wassily dǎsuàn [*yīnggāi*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily plan may read that-CL novel read DE very quickly
 (should^{epis} ✗)
- b. *Wassily dǎsuàn [*kěnéng*^{epis} **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily plan may read that-CL novel read DE very quickly
 (may^{epis} ✗)
- c. *Wassily dǎsuàn [*bìxū*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
 Wassily plan must read that-CL novel read DE more quickly a-bit
 (must^{deon} ✗)
- d. *Wassily dǎsuàn [*děi*^{deon} **kàn nà-běn xiǎoshuō** kàn de gèng kuài yīxiē].
 Wassily plan must read that-CL novel read DE more quickly a-bit
 (have to^{deon} ✗)

The comparisons made above clearly demonstrate that the predicate group in my sense and the clause type in Huang (2022)’s sense are not the same concept due to their different criteria of classification. Therefore it does not constitute a huge surprise that my predicate groups and Huang’s clause types do not exhibit one-on-one correspondence. Nevertheless, we still face the dilemma that Type II clauses seem to correspond to two different sizes, namely, TP taken by Group 2 predicates and FP taken by Group 3. Here I simply suggest two potential accounts. The first one is that, since neither Huang’s classification nor mine is exhaustive, there might be more predicate groups/clause types in Mandarin Chinese than the tripartite analysis presented in either work. That is to say, verbs like *dǎsuàn* might constitute a group on its own independently of other Group 3 predicates because of the distinct semantic requirements on their complements. Another possible explanation is that semantic distinctions like temporal independence might not be a good diagnostic for tearing apart infinitives, since tense in infinitives is always constrained to some extent by the embedding predicates. I leave both possibilities for future research, and stick to my tripartite classification of predicates in the rest of the chapter.

6.2.4 Improper verb doubling as a Height-Locality effect

This section has so far outlined what Keine (2016, 2019, 2020) calls a *Height-Locality effect* as generalised in (250): VP-movement to a lower position in the clause is possible out of smaller range of complement clauses than VP-movement to the clausal periphery. Mandarin Chinese

is therefore part of an increasingly diverse range of languages in which such effects have been observed in movement dependencies.¹³

(250) *Height-Locality effect*

Locality of a syntactic dependency appears to be conditioned by the position of its trigger (or landing site) in the clausal functional sequence. (based on Keine 2020:11-15, *et seq.*)

In English this most obviously concerns restrictions on raising-to-subject, which cannot proceed out of finite clausal complements (e.g., (251a) vs. (251b)). This contrasts with the behaviour of, e.g., *wh*-movement, which can escape finite complements (251c). What this contrast seems to track is the difference in landing site: *wh*-movement and raising-to-subject target different linear and structural positions (Spec,CP vs. Spec,TP). In other words, *wh*-movement lands higher in the clausal spine than raising, and it is less bounded than raising.

(251) *Height-Locality effects in English raising-to-subject*

- a. Piet seems to like fish and chips.
- b. *Piet seems likes fish and chips.
- c. Who does it seem likes fish and chips?

In the growing cross-linguistic picture of Height-Locality effects, various movement types beyond raising exhibit clause-boundedness without the full range of archetypal A-movement properties, as given in (252). To my knowledge, Height-Locality effects with movement have up to now only been observed with the movement of nominals. The constraints on VP-movement in Mandarin Chinese show us that VP-movement is not exceptional in this regard. I go on to show that Height-Locality effects with nominal movement also exist in Mandarin, with reference to long-distance object shift in section 6.4.

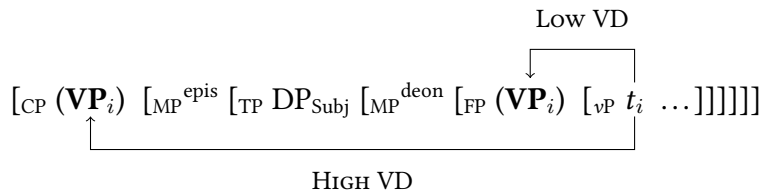
(252) *Height-Locality effects beyond raising*

Scrambling in Hindi	(Keine 2016, 2020)
<i>Wh</i> -movement out of V-final vs. V2 complements in German	(Keine 2016, 2020)
Particle movement in Hungarian	(Egressy 2023)
<i>Wh</i> -movement in Georgian	(Bondarenko 2024)
Movement forming relative clauses in Swahili	(Meadows 2024)

¹³ I set aside the issue of how this manifests with cases discussed extensively by sub-extraction and remnant movement (e.g., Abels 2007, 2009; Neeleman and van de Koot 2010; Egressy 2023), as well domains beyond movement where these effects may also hold, such as agreement (Keine 2016, 2019, 2020) and case-assignment (Poole 2022).

With this cross-linguistic context clear, let me round out this section. I have motivated an analysis of long-distance verb doubling in which a key variable is the size of the embedded clausal complement. I have presented straightforwardly syntactic arguments, from nominal movement within embedded clauses and the distribution of modals, that complements come in three sizes. What matters most going forward is that low verb doubling lands lower in the clausal spine than the top of Group 1/2 complements. In the process I have committed myself to the maximal clause structure in (253). In the next section I will flesh out an Williams-Cycle-based treatment of improper verb doubling.

(253) *Clause structure and verb doubling in Mandarin Chinese*



6.3 Improper verb doubling and the Williams Cycle

This section demonstrates how improper verb doubling can be accounted for by the Williams Cycle (Williams 1974, 2003, 2011, 2013), a locality constraint positing a quite general connection between the height of movement and the size of clausal complement it can escape. I first present the analysis proposed in Yan and Meadows (2025): section 6.3.1 fleshes out a Level Embedding derivation of the Williams Cycle; section 6.3.2 explicates how that derives the restrictions on verb doubling; and section 6.3.3 verifies a couple of predictions concerning multiple layers of clausal embedding. I then focus on several problems that arise from the current analysis: section 6.3.4 discusses three challenges to the Level Embedding approach, and proposes a constraint based on the Grimshawan extended projection (Grimshaw 1991, 2000, 2005) to ameliorate all of them; section 6.3.5 offers a solution to a non-intervention puzzle relevant to low verb doubling out of Group 3 complements.

6.3.1 A Level Embedding approach

In Yan and Meadows (2025), we define the Williams Cycle as a descriptive constraint on movement of the form in (254). More or less descriptively restrictive versions of this constraint are special cases of the Williams Cycle. They all involve being more precise about the kinds of movement that are subject to this general Height-Locality constraint.

(254) *The Williams Cycle*

For certain types of movement, there is a principled connection between landing and locality such that: the higher a movement operation lands in the clausal functional sequence, the more unbounded it is.

There are a range of different theoretical models which derive some version of the Williams Cycle, all of which could explain improper verb doubling (see Abels 2007, 2009, Müller 2014a,b, Keine 2016, 2019, 2020, among others). In Yan and Meadows (2025), we focus on the *Level Embedding* approach originally proposed by Williams (2003, 2011, 2013), and further developed by Poole (2022) and Meadows (2024). The major theoretical innovation of Level Embedding concerns the timing of main and embedded structure-building. The standard assumption is that an embedded clause is built before merging into the matrix clause, e.g., as an argument of the VP, and then go to finish off constructing the matrix clause. Level Embedding instead maintains that main and embedded clauses are built in parallel, in the sense of (255). Following Meadows (2024), we term this central assumption *Parallel Derivation*. This is concretely illustrated in (256). Whilst still in separate workspaces (notated by comma in subsequent diagrams), it is not possible to finish building one clause before another. Only embedding a clause can prevent it from being built to completion. This is the major theoretical ‘price’ to pay for use of Level Embedding architecture.¹⁴

(255) *Parallel Derivation*

The same component of the clausal functional sequence (*fseq*) (V, *v*, F, T, C, etc.) is externally merged in every workspace at the same time.

(256) *Illustration of Parallel Derivation*

- a. [CP Piet said [CP that he will make fish and chips]]
- b. [VP say ____], [VP make fish and chips] (Merge V)
- c. [vP Piet [VP say ____]], [vP he [VP make fish and chips]] (Merge *v*)
- d. [TP Piet_i [vP t_i [VP said ____]]], [TP he_i will [vP t_i [VP make fish and chips]]] (Merge T)
- e. [CP [TP Piet_i [vP t_i [VP said ____]]]], [CP that [TP he_i will [vP t_i [VP make fish and chips]]]] (Merge C)

In combination with Parallel Derivation, the Level Embedding approach makes use of two

¹⁴ To implement other theoretical models, one has to first have an understanding of the novel theoretical concepts that each of them comes up with (e.g., Keine (2016, 2019, 2020)’s *Horizons*), and then think about to what extent they would like to accept them. By comparison, the theoretical ‘price’ that the Level Embedding approach requires us to pay for here seems much lower, which I consider as an advantage of choosing this implementation over the others.

devices which used in contemporary generative syntax to greater or less extents. The first device is *Late Merge* (Lebeaux 1988, 1998), for use in clausal embedding, as defined in (257). Something equivalent to Late Merge is inescapable given a commitment to Parallel Derivation. Only the very smallest clauses can be embedded in the same cycle in which they are built. As illustrated in (258), if we want to embed a CP anywhere other than at the CP-level, we will have to wait around until both main and embedded clauses are CP-sized, and then Late Merge the CP into, e.g., an argument position of the VP (indicated by the underscore).

- (257) *Late Merge*
 Clausal embedding involves external Late Merge.

- (258) *Illustration of Embedding with Late Merge*
- a. $[_{CP} [_{TP} \text{Piet}_i [_{VP} t_i [_{VP} \text{said } ____]]]], [_{CP} \text{that } [_{TP} \text{he}_i \text{ will } [_{VP} t_i [_{VP} \text{make fish and chips}]]]]$
(Build to CP)
 - b. $[_{CP} [_{TP} \text{Piet}_i [_{VP} t_i [_{VP} \text{said } [_{CP} \text{that } [_{TP} \text{he}_i \text{ will } [_{VP} t_i [_{VP} \text{make fish and chips}]]]]]]]]]]$
(Late Merge CP)

The second device is a version of the *Strict Cycle Condition* (SCC) summarised in (259) (cf. Müller 2023:12, (13)). Some version of the SCC is instantiated in the *Extension Condition* (Chomsky 1995), and *Featural Cyclicity* (Richards 1999). This version of the SCC employs much more cycles than traditionally assumed. For every projection in the clausal functional sequence, there is cycle of Merge/Agree associated with it. Once a new clausal F is merged into the spine, a new cycle starts and material introduced in prior cycles cannot be altered. The SCC essentially forbids Late applications of Merge, though we stipulate that Late External Merge is unaffected in Yan and Meadows (2025) (see relevant discussion in section 6.3.4).

- (259) *Strict Cycle Condition*
 The output of prior cycles of rule application cannot be altered by rules in subsequent cycles. Every functional projection in the clausal *fseq* delineates a new cycle.

Long-distance movement is restricted by the interaction between Parallel Derivation and the SCC. We take for granted that movement cannot leave a clause until it is embedded. By Parallel Derivation, the bigger a clause is, the later in the derivation it will be embedded. This waiting around for embedding is where the Strict Cycle comes in. Once the complement is embedded, subsequent long-distance movement is only possible to positions introduced in the same cycle in which embedding takes place. Any movement triggers introduced lower/earlier in the matrix

clause (e.g., on T, F, etc.) have been forced to wait too long, as schematised in (260).¹⁵

(260) *Illustration of the Strict Cycle Condition*

- a. [VP] (Parallel Derivation)
- b. [CP [TP [VP]]], [CP [TP [VP]]] (Parallel Derivation)
- c. [CP [TP [VP [CP [TP [VP]]]]]] (Embedding with Late Merge)
- d. [CP ✓ [TP ✗ [VP ✗ [CP t [TP [VP]]]]]] (SCC-compliant subsequent movement)

Combining a Strict Cycle Condition with Parallel Derivation and Late Merge derives a particularly strong version of the Williams Cycle. This is the *Generalised Ban on Improper Movement* (GBOIM) given in (261) (Williams 2003; Poole 2022). According the GBOIM, all movements must land at the same height or higher in the clausal *fseq*.

(261) *Generalised Ban on Improper Movement (GBOIM)*

Movement to Spec,XP cannot proceed from Spec,YP or across YP, where Y is higher than X in the clausal functional sequence (*fseq*).

(Poole 2022:348, based on Williams 2003)

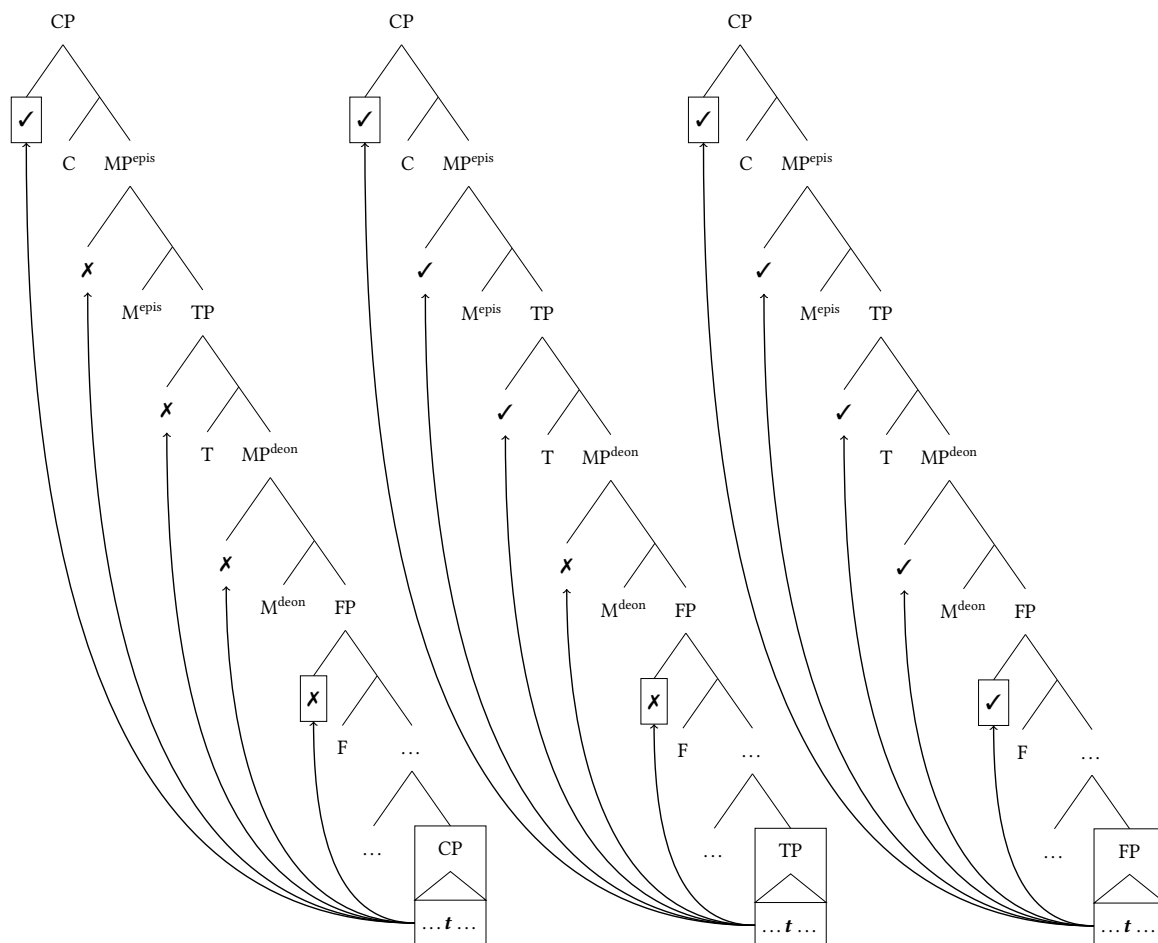
The three trees below display representations of movement constrained by the GBOIM out of CP-sized (262), TP-sized (263) and FP-sized (264) complements. Recall that these are three complement clause sizes motivated in section 6.2. Movement out of a CP complement can land no lower than Spec,CP in the clausal *fseq* (262). This means that VP-movement is not possible to the post-subject position identified as Spec,FP. The same situation holds for movement from TP complements, where movement can land no lower than Spec,TP (263). Only in the case of FP complement, from which movement can land no lower than Spec,FP, is VP-movement to that post-subject position possible (264).

¹⁵ There is some potential tension between Parallel Derivation and *vP*-phasehood that calls for attention here: recall that in several previous sections, I adopted the assumption that *vP* is a phase. This idea, however, is incompatible with Parallel Derivation: if Parallel Derivation proceeds, e.g., to the CP-level, then movement through the edge of the *vP* (which is lower than CP in the *fseq*) would violate the SCC defined here. One way of loosening this tension is to assume that the Williams Cycle is a global constraint that concerns most the final derivational output, and that intermediate movement steps through phase edges can somehow be exempted from Parallel Derivation (as well as the SCC). I leave details of this thought for future research.

(262)

(263)

(264)



6.3.2 Deriving improper verb doubling

We have seen the key ingredients of the Level Embedding approach, and how they derive a locality restriction of which improper verb doubling is a special case. I now go through the relevant derivations in a more mechanistic fashion as what we have done in Yan and Meadows (2025).

6.3.2.1 Low VD out of Group 3 complements

Recall that Group 3 clauses are complements to predicates like *chángshì* ‘try’. I analysed these complements as smaller than TP, since they lack distinct subjects and any modals. Nevertheless these complements can display clause-internal verb doubling. I therefore treat such complements as maximally FP-sized, i.e., built to the level at which low VP-movement takes places. Recall that it is only out of these complements that cross-clausal low verb doubling is possible. This follows from our assumptions so far.

We first build the main and complement clause to FP-size (265a)-(265b), according to Parallel Derivation. At this stage, the Cycle starts by Merging F into the clausal spine, and then we embed the complement into the matrix clause (265c). There is still time in the F-cycle for movement triggered at this cycle to occur. There is a featural trigger on matrix F, which attracts embedded VP to matrix Spec,FP (265d). This movement is completely compliant with the Strict Cycle Condition, since we affect only the material introduced in the present cycle. The FP-sized matrix clauses continues growing to the full CP-size (265e).

- (265) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [_{VP} try ____], [_{VP} read the novel]
- b. *Step 2: Build up to the FP-level* (Parallel Derivation)
 [_{FP} F_[uV] [_{vP} Wassily try ____]], [_{FP} [_{vP} read the novel quickly]]
- c. *Step 3: Embed the FP in the other FP* (Embedding with Late Merge)
 [_{FP} F_[uV] [_{vP} Wassily try [_{FP} [_{vP} read the novel quickly]]]]
- d. *Step 4: Probe on matrix F triggers VP-movement* (Strict Cycle ✓)
 [_{FP} [_{VP} read the novel] F_[uV] [_{vP} Wassily try [_{FP} [_{vP} read DE quickly]]]]
- e. *Step 5: Build up to the CP-level in the matrix*
 [_{CP} [_{TP} Wassily [_{FP} [_{VP} read the novel] [_{vP} try [_{FP} [_{vP} read DE quickly]]]]]]

6.3.2.2 Low VD out of Group 1 and 2 complements

It is with other complement types that restrictions on cross-clausal verb doubling emerge. Remember that I analysed Group 1 complements as CP-sized, and Group 2 complements as TP-sized. Group 1 complements behave essentially like matrix clauses in their internal richness, whereas Group 2 complements stand somewhere between Group 1 and Group 3.

The crucial point is that both complements are built to higher in the clausal *fseq* than FP (as in (266a)-(266b) and (267a)-(267b)). It is this, combined with the Strict Cycle Condition, that produces a missed opportunity. Imagine that we build our main and to-be-embedded clauses to FP-level, where there is a featural trigger for VP-movement on matrix F. The problem is that unless we embed an FP complement, no such movement can take places. Suppose we wait until the clauses have grown to TP-sized for the Group 2 case (266c), or grown all the way to CP-sized for the Group 1 case (267c). In either situation we do Embedding with Late Merge, but movement to Spec,FP is forbidden by the Strict Cycle Condition (266d)/(267d). The derivation is thus forbidden by the competing requirements of Parallel Derivation and the Strict Cycle Condition.

- (266) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [_{VP} force Piet ____], [_{VP} read the novel]
- b. *Step 2: Build up to the TP-level* (Parallel Derivation)
 [_{TP} Wassily [_{FP} F_[uV] [_{VP} force Piet ____]]], [_{TP} [_{FP} [_{VP} read the novel quickly]]]
- c. *Step 3: Embed the TP in the other TP* (Embedding with Late Merge)
 [_{TP} Wassily [_{FP} F_[uV] [_{VP} force Piet [_{TP} [_{FP} [_{VP} read the novel quickly]]]]]]
- d. *Step 4: Probe on matrix F cannot trigger VP-movement any more* (Strict Cycle ✗)
 *[[_{TP} Wassily [_{FP} [_{VP} read the novel] F_[uV] [_{VP} force Piet [_{TP} [_{FP} [_{VP} read DE quickly]]]]]]]]
- (267) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [_{VP} think ____], [_{VP} read the novel]
- b. *Step 2: Build up to the CP-level* (Parallel Derivation)
 [_{CP} [_{TP} Wassily [_{FP} F_[uV] [_{VP} think ____]]]], [_{CP} [_{TP} Piet [_{FP} [_{VP} read the novel quickly]]]]
- c. *Step 3: Embed the CP in the other CP* (Embedding with Late Merge)
 [_{CP} [_{TP} Wassily [_{FP} F_[uV] [_{VP} think [_{CP} [_{TP} Piet [_{FP} [_{VP} read the novel quickly]]]]]]]]
- d. *Step 4: Probe on matrix F cannot trigger VP-movement any more* (Strict Cycle ✗)
 *[[_{CP} [_{TP} Wassily [_{FP} [_{VP} read the novel] F_[uV] [_{VP} think [_{CP} [_{TP} Piet [_{FP} [_{VP} read DE quickly]]]]]]]]]]

6.3.2.3 High VD

High verb doubling is unaffected by these competing constraints, precisely because it lands as high in the functional sequence as biggest complement type is built to - the C/CP-level. In other words, there will be no complement embedding that happens in a later cycle than when high verb doubling occurs. As a result, movement to Spec,CP out of any size of complement complies with the Strict Cycle Condition. We simply build our main and embedded clauses in parallel (268a)-(268b), then embed at the relevant cycle (FP, TP or CP) (268c). Within the same, or later cycle (if embedding FP or TP), movement of the embedded VP to Spec,CP is driven by a trigger on matrix C (268d)-(268e).

- (268) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [_{VP} try ____], [_{VP} read the novel]

- b. *Step 2: Build up to the FP-level (or TP/CP-level)* (Parallel Derivation)
 $[_{FP} [_{VP} \text{Wassily try } ___]], [_{FP} [_{VP} \text{read the novel quickly }]]$
- c. *Step 3: Embed the FP in the other FP* (Embedding with Late Merge)
 $[_{FP} [_{VP} \text{Wassily try } [_{FP} [_{VP} \text{read the novel quickly }]]]]$
- d. *Step 4: Build up the CP-level in the matrix*
 $[_{CP} C_{[uTop]} [_{TP} \text{Wassily } [_{FP} [_{VP} \text{try } [_{FP} [_{VP} \text{read the novel quickly }]]]]]]$
- e. *Step 5: Probe on matrix C triggers VP-movement* (Strict Cycle ✓)
 $[_{CP} [_{VP} \text{read the novel }] C_{[uTop]} [_{TP} \text{Wassily } [_{FP} [_{VP} \text{try } [_{FP} [_{VP} \text{read DE quickly }]]]]]]$

6.3.3 Prediction: Movement out of multiply-embedded clauses

In this section I show that VP-fronting over multiple levels of embedding conforms to the Williams Cycle that emerges from the Level Embedding approach.

6.3.3.1 Restricted verb doubling across multiple embeddings

The first type of multiple-embedding configuration is schematised in (269), where a non-Group 3 predicate is embedded under a Group 3. As already argued in section 6.2, Group 3 predicates take FP-sized complements, while non-Group 3 ones take clauses bigger than FP as their complement. Since the VP in the most embedded clause needs to move out of an environment bigger than FP, we predict that low verb doubling targeting Spec,FP is restricted compared to its high counterpart landing in Spec,CP of the matrix.

- (269) *Non-Group 3 predicate embedded under Group 3*
 $[_{CP} \dots [_{TP} [_{FP} \dots [_{VP} \text{Pred.}_{\text{Group } 3} [_{FP} \text{Pred.}_{\text{Group } -3} [_{>FP} \dots \mathbf{VP} \dots]]]]]]]]$

This prediction is first borne out by the configuration (270) where *xiāngxìn* ‘believe’, a Group 1 predicate, is embedded under *jùjué* ‘refuse’, a Group 3 predicate. Since *xiāngxìn* takes a CP-sized clausal complement, the fronted VP is prohibited from landing in positions lower than CP. Therefore low verb doubling across multiple levels of embedding is not permitted in (270a), but high verb doubling in the same configuration is attested in (270b).

- (270) *Group 1 embedded under Group 3: no low VD*
- a. *Wassily **kàn nà-běn xiǎoshuō** jùjué $[_{FP} \text{xiāngxìn } [_{CP} \text{Piet kàn de hěn kuài}]]$.
Wassily read that-CL novel refuse believe Piet read DE very quickly
Int.: ‘Wassily refuses to believe that Piet reads the novel quickly.’

- b. **Kàn nà-běn xiǎoshuō** Wassily jùjué [_{FP} xiāngxìn [_{CP} Piet kàn de hěn kuài]].
 read that-CL novel Wassily refuse believe Piet read DE very quickly
 ‘As for reading the novel, Wassily refuses to believe that Piet reads it quickly.’

We also correctly predict that the same contrast between low and high verb doubling exists in the configuration where a Group 2 predicate is embedded under a Group 3 one, as exemplified in (271). The Group 3 *chángshì* and the group 2 *qiǎngpò* respectively take FP- and TP-sized clausal complements. As the VP still has to move out of an embedded clause that is bigger than its low landing site, low verb doubling in (271a) is ruled out. By contrast, high verb doubling (271b) is licensed since the final landing site is higher than the TP complement containing the base-position.

(271) *Group 2 embedded under Group 3: no low VD*

- a. *Wassily **kàn nà-běn xiǎoshuō** chángshì [_{FP} qiǎngpò Piet [_{TP} kàn de hěn kuài]].
 Wassily read that-CL novel try force Piet read DE very quickly
 Int.: ‘Wassily tries to force Piet to read the novel quickly.’
- b. **Kàn nà-běn xiǎoshuō** Wassily chángshì [_{FP} qiǎngpò Piet [_{TP} kàn de hěn kuài]].
 read that-CL novel Wassily try force Piet read DE very quickly
 ‘As for reading the novel, Wassily tries to force Piet to read it quickly.’

6.3.3.2 No restrictions on multiple Group 3 embeddings

We have seen that improper verb doubling occurs in configurations where the bigger-sized complement clauses are more embedded than the smaller-sized ones. The second prediction I will make is if multiple-embedding configurations consist of Group 3 predicates only (as schematised in (272)), then neither low nor high verb doubling is restricted.

(272) *Group 3 predicate embedded under Group 3*

[_{CP} ... [_{TP} [_{FP} ... [_{VP} Pred._{Group 3} [_{FP} Pred._{Group 3} [_{FP} ... **VP** ...]]]]]]]

This prediction is corroborated empirically by (273), in which the Group 3 predicate *juédìng* embeds another Group 3 *chángshì*. Since both of them take FP-sized clausal complements, moving the VP from the most embedded clause to the matrix Spec,FP does not cross any higher projections in the clausal *fseq*. Thus low verb doubling in such a configuration is allowed, as shown in (273a).

High verb doubling is also unaffected as in (273b).

(273) *Group 3 embedded under Group 3: no restrictions*

- a. Wassily **kàn nà-běn xiǎoshuō** juédìng [_{FP} chángshì [_{FP} kàn de hěn kuài]].
 Wassily read that-CL novel decide try read DE very quickly
 ‘Wassily decides to try to read the novel quickly.’
- b. **Kàn nà-běn xiǎoshuō** Wassily juédìng [_{FP} chángshì [_{FP} kàn de hěn kuài]].
 read that-CL novel Wassily decide try read DE very quickly
 ‘As for reading the novel, Wassily decides to try to read it quickly.’

There is no constrain on how many Group 3 predicates can be stacked together. One further prediction is that both low and high verb doubling are still permitted even if the multiple-embedding configuration is made up of more than two Group 3 predicates. This is validated by the configuration in (274) where three Group 3 predicates are embedded consecutively one after another.

(274) *Triple Group 3 embedding: no restrictions*

- a. Wassily **kàn nà-běn xiǎoshuō** juédìng [_{FP} kāishǐ [_{FP} chángshì [_{FP} kàn de hěn kuài]]].
 Wassily read that-CL novel decide begin try read DE very quickly
 ‘Wassily decides to begin trying to read the novel quickly.’
- b. **Kàn nà-běn xiǎoshuō** Wassily juédìng [_{FP} kāishǐ [_{FP} chángshì [_{FP} kàn de hěn kuài]]].
 read that-CL novel Wassily decide begin try read DE very quickly
 ‘As for reading the novel, Wassily decides to begin trying to read it quickly.’

Having confirmed these basic predictions, I turn to three latent challenges posed to the Level Embedding approach in the next section.

6.3.4 Three challenges to the Level Embedding approach

The key ingredients of the Level Embedding approach, namely, Parallel Derivation, Late Merge, and the Strict Cycle Condition, are not unproblematic. In this section I discuss one challenge to each of these three ideas in turn, and propose a derivational constraint on all of them based on

Grimshaw (1991, 2000, 2005)’s theory of extended projection.¹⁶

6.3.4.1 Problems underlying the three key ingredients

Let me start off with Parallel Derivation. The way this idea defines ‘parallel’ not only requires two clauses to be built in separate workspace at the same time, but also to contain exactly the same component in each clausal spine. This leads to a prediction that we are unable to find a functional projection which exists in only one of the parallel clauses. For instance, Parallel Derivation cannot generate an asymmetrical derivation where only the main clause contains a NegP but the embedded one does not, as schematised in (275a), or the other way around, as schematised in (275b). But this is contrary to the fact, as given in (276): the negation mark *méiyǒu* can occur alone either in the matrix clause (276a) or in the embedded one (276b).

(275) *Impossible derivations predicted by Parallel Derivation*

- a. [CP [TP [NegP [vP [VP]]]]], [CP [TP [vP [VP]]]]
- b. [CP [TP [vP [VP]]]], [CP [TP [NegP [vP [VP]]]]]

- (276) a. Wassily **méiyǒu** rènwéi [Piet kàn nà-běn xiǎoshuō kàn de hěn kuài].
Wassily NEG think Piet read that-CL novel read DE very quickly
‘Wassily did not think that Piet read the novel quickly.’
- b. Wassily rènwéi [Piet **méiyǒu** kàn nà-běn xiǎoshuō kàn de hěn kuài].
Wassily think Piet NEG read that-CL novel read DE very quickly
‘Wassily thought that Piet did not read the novel quickly.’

Due to the existence of such examples, it looks like Parallel Derivation is too strict to rule out a lot of possible derivations. The simplest way of solving this problem is to loosen Parallel Derivation to the extent that the two parallel clauses only need to match up with each other in terms of the topmost projection, which is to say, it does not matter whether the component of each clausal spine also needs to be exactly the same.

Now we can move on to the second problem concerned with Late Merge, of which the relevant illustration is repeated in (277). It has been commonly accepted that the nature of Late Merge can be viewed as adjunction (see Privizentseva 2023 for relevant discussion). Recall that in section 3.2, I proposed that adjuncts can be freely-adjoined to maximal projections, in line with Adger (2003). If we stick to this proposal and assume that Late Merge, as adjunction, also applies for free, then it will become unclear how Late Merge is aware of where exactly the clausal embedding can take place. In theory, the embedded clause can be Late Merged to any maximal functional projection

¹⁶ I thank David Adger for the challenges and the proposed solution discussed here.

in the matrix clause. But why does Late Merge know the position of clausal embedding is the internal argument position of the matrix verb (indicated by the underscore)?¹⁷

(277) *Illustration of Embedding with Late Merge* (=258)

- a. $[_{CP} [_{TP} \text{Piet}_i [_{vP} t_i [_{VP} \text{said } ____]]]], [_{CP} \text{that } [_{TP} \text{he}_i \text{ will } [_{vP} t_i [_{VP} \text{make fish and chips }]]]]$
(Build to CP)
- b. $[_{CP} [_{TP} \text{Piet}_i [_{vP} t_i [_{VP} \text{said } [_{CP} \text{that } [_{TP} \text{he}_i \text{ will } [_{vP} t_i [_{VP} \text{make fish and chips }]]]]]]]]]]$
(Late Merge CP)

To address the problem above, the most straightforward method seems to replace Late Merge with another operation. One alternative is *Substitution* proposed by Poole (2022), which is analogous to generalised transformations in Transformational Grammar (Chomsky 1955, 1957), substitution in the early Minimalist Program (Chomsky 1995), and substitution in Tree Adjoining Grammar (Joshi et al. 1975). As illustrated below, the only change made to Parallel Derivation is that the matrix verb *said* merges with a CP node at the very beginning of the derivation (278a), which is later substituted by the embedded CP (278b).¹⁸ It can be said that Substitution is constrained by the same categorial label, which renders it different from Late Merge. Though the nature of Substitution is no clearer than Late Merge to me, at least it has the advantage of avoiding the potential free-adjoining problem that Late Merge contains.

(278) *Illustration of Embedding with Substitution*

- a. $[_{CP} [_{TP} \text{Piet}_i [_{vP} t_i [_{VP} \text{said } \mathbf{CP}]]]], [_{CP} \text{that } [_{TP} \text{he}_i \text{ will } [_{vP} t_i [_{VP} \text{make fish and chips }]]]]$
(Build to CP)
- b. $[_{CP} [_{TP} \text{Piet}_i [_{vP} t_i [_{VP} \text{said } [_{CP} \text{that } [_{TP} \text{he}_i \text{ will } [_{vP} t_i [_{VP} \text{make fish and chips }]]]]]]]]]]$
(Substitute CP)

The third problem I would like to discuss here is related to the Strict Cycle Condition. As already mentioned in section 6.3.1, there are various versions of the Strict Cycle Condition (e.g., Chomsky 1973, 1995, 2001, 2008; Müller 2017, 2023, among many others); the version given above is just specifically formulated to be compliant with the Williams Cycle. However, no matter which version of the Strict Cycle Condition is adopted, the clausal embedding operation (either Late Merge or Substitution) always seemingly conflicts with this derivational constraint, because the former will alter the output already established in the previous derivation, a clear violation of the

¹⁷ A more trivial puzzle here is related to the nature of this underscore, which I will address right below.

¹⁸ In this sense, the substituted CP node in the matrix clause is actually a placeholder for a CP-clause. See Poole 2022:383, footnote 48 for more details.

latter.¹⁹ Given that the clausal embedding operation is a necessary follow-up to Parallel Derivation but the Strict Cycle Condition is not, this suggests that the former may win out the conflict, and that the latter can be given up. However, if so, then there would be no way of constraining the long-distance movement of the verbal phrase, e.g., in (279) in a parallel fashion, rendering the William Cycle/GBOIM invalid. What the Strict Cycle Condition brings to us, eventually, is such a dilemma.

(279) *Unexplained derivation without the Strict Cycle Condition*

*[_{CP} [_{TP} Wassily [_{FP} [_{VP} read the novel] F_[+V] [_{VP} think [_{CP} [_{TP} Piet [_{FP} [_{VP} read DE quickly]]]]]]]]]]
 =(267d)

6.3.4.2 Proposal: Grimshawan extended projection

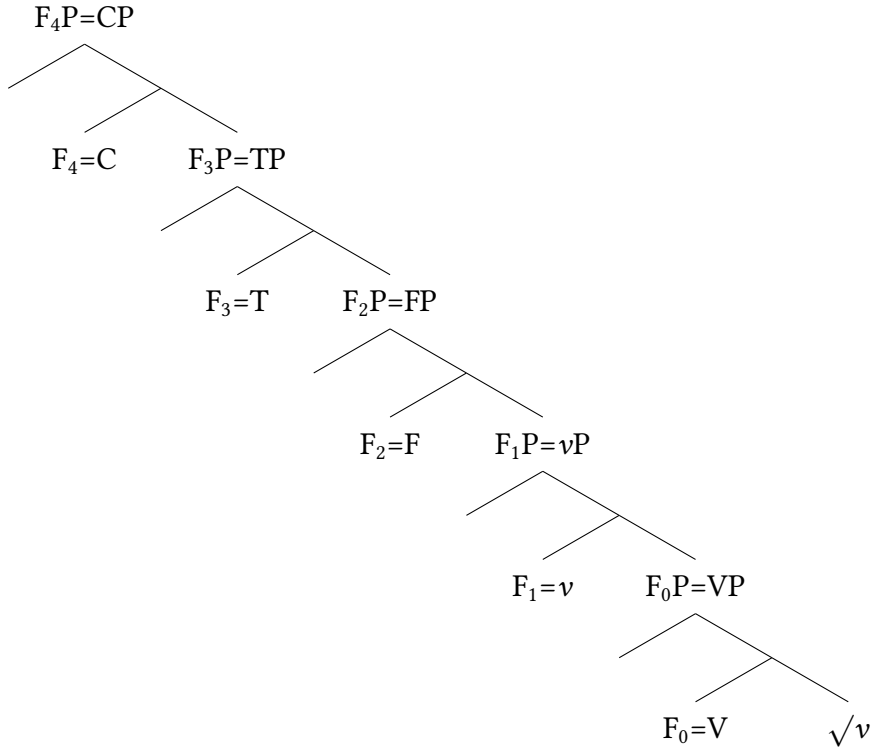
I have shown the three challenges to the Level Embedding approach in the previous section. I have sketched out separate solutions to the first two challenges, but not found a way to deal with the dilemma created by the Strict Cycle Condition. In what follows I first propose a constraint based on Grimshaw (1991, 2000, 2005)'s theory of extended projection to address the dilemma, and then extend my proposed idea to the other two problems.

Grimshaw proposes that higher functional projections in the clausal spine are extended projections of the lowest, lexical V. According to this theory, a single category feature is decomposed into a lexical feature ([V] or [N]) and a functional feature [F_n] (n ≥ 0). For instance, V is specified as a feature bundle [V, F₀], which means that it functions as the lowest verbal position. In contrast, *v* bears the same lexical feature [V] but differs in its functional one [F_n] (n > 0), indicating that this functional projection is placed somewhere higher along the verbal extended projection. A clausal spine which consists of V, *v*, F, T, C, thus, can be converted to a verbal extended projection in which each layer bears its own functional feature, as illustrated in (280).²⁰

¹⁹ In Yan and Meadows (2025), we have to stipulate that Late Merge is unaffected by the Strict Cycle Condition, which has been brought up in section 6.3.1. Poole (2022:384, footnote 49) discusses a little bit about how Embedding with Substitution obeys the Strict Cycle Condition, in accordance with the version of the Strict Cycle Condition (Müller 2017) he adopts.

²⁰ I put a verbal root at the bottom of (280) only to indicate that this is a verbal extended projection (cf. Borer 2005). Nothing else hinges on it.

(280) *Grimshawan verbal extended projection*



The number of the functional feature borne by each projection is not absolute. For instance, the functional feature of C can be $[F_{31}]$, and that of T can be $[F_{27}]$. No matter what exact the number is, the verbal extended projection will not be altered. What matters to the well-formedness of the verbal extended projection is the relative number of the functional feature: as long as the higher projection bears a functional feature whose number is larger than the lower one, the verbal extended projection is well established. The functional feature's number of the same projection can also vary from one language to another, which can be used as an account of linguistic parametrisation, an idea I will not further elaborate on here (see Grimshaw 2005 for relevant discussion).

With the Grimshawan extended projection in mind, let us reconsider the dilemma of the Strict Cycle Condition first. As discussed in the previous section, without the Strict Cycle Condition, it will be difficult to derive either the Williams Cycle or the GBOIM. What the Williams Cycle states (recall (254)) or what the GBOIM requires (recall (261)) can be simply generalised as follows: movement cannot take place from a higher position to a lower one. This simplified idea can be further translated in terms of the functional feature's number under the Grimshawan extended projection: movement cannot take place from a projection with a larger functional feature's number to one with a smaller number. A more formal formulation of this idea is given in (281), which I term *never-go-down constraint*.²¹

²¹ I attribute this constraint to Adger (2025), where an idea in the similar spirit is applied to account for how lineari-

(281) *Never-go-down constraint*

Every syntactic operation must not decrease the number of the functional feature the current derivational stage concerns.

Let me be more explicit about how this constraint works for the Williams Cycle/GBOIM and replace the Strict Cycle Condition. Imagine a derivational stage where movement takes place. What needs to be concerned in this case is the functional feature's number on the landing site of the moved object compared with the other number borne on its movement pathway. For the Williams Cycle/GBOIM, the derivational stage concerns particularly the comparison between the landing site and the launch site or the topmost functional projection moved across: the functional feature's number on the former cannot be smaller than that on the latter; otherwise, the derivation crashes. By the never-go-down constraint, the effect of the Strict Cycle Condition is still achievable, but without appealing to the Strict Cycle Condition proper. This is an ideal result, because that means, on the one hand, that we can give up the Strict Cycle Condition, and on the other hand, that Substitution can be relieved of the Strict Cycle Condition.

Even more pleasing is that the never-go-down constraint can restrict Substitution as well. If we adopt the idea that Substitution is also a separate syntactic operation, then when it applies in a parallel derivation, what is concerned becomes the functional feature's number on the substitute compared with that on the substitutor. For example, in the illustrated derivation (278) above, both the substitute and the substitutor are CPs, which means that the functional feature's number on both of them is the same. According to the never-go-down constraint, this will generate a legitimate derivation, because the concerned functional feature's number does not decrease. In this sense, Substitution is no longer simply restricted by the categorial label; it is actually restricted by a derivational constraint that also applies elsewhere.

Last but not least, let us have a look again at the asymmetrical derivation (275) that arises from Parallel Derivation. From a perspective of never-go-down constraint, the appearance of the asymmetrical derivation constitutes no problem: in each workspace, we just merge one functional projections with another. In this kind of situation, what the derivational stage concerns is the functional feature's number borne on the newly-merged object compared with the already established one. As long as the never-go-down constraint is obeyed, it does not matter how many times Internal Merge applies in the single workspace, before Parallel Derivation starts the clausal embedding. It can be said that my attempt to loosen Parallel Derivation is not unwarranted; it receives support from the never-go-down constraint.

To sum up, I have shown how the never-go-down constraint can replace the Strict Cycle Condition, and how it can be used to justify my proposed solutions to problems underlying Late Merge

sation works in mereological syntax. I also owe many thanks to David Adger for discussion on this idea.

and Parallel Derivation. The nature of this constraint makes use of monotonicity in syntax, an idea proposed by Graf (2020). In the rest of the chapter, I will still illustrate all the derivations in a parallel fashion as what I have done so far, mainly for the purpose of consistency. However, all my proposed solutions here continue to take effect, though they are not displayed explicitly.

6.3.5 The non-intervention problem

I conclude this section with a puzzle concerning the featural status of low VD which I have abstracted away from thus far.

Recall that in chapter 2, I proposed that low verb doubling is triggered by a categorial feature $[uV]$ on F, which is often used for obligatory/EPP-like VP-movement (e.g., Massam and Smallwood 1997; Coon 2010; Collins 2017; Travis and Massam 2021; van Urk 2024). Without further assumptions, a $[uV]$ probe will encounter the most local (and therefore preferred) $[iV]$ goal that is part of the embedding predicate. By default then, our featural commitment leads to low verb doubling being clause-bound. Empirically this is problematic for Group 3 clausal complements, which seem to be transparent for low verb doubling. It is also problematic for central contention of the analysis just sketched out above, since the feature-induced minimality rids us of the need to appeal to the Williams Cycle/Level Embedding to explain clause-boundedness. I refer to this puzzle as *the non-intervention problem*.

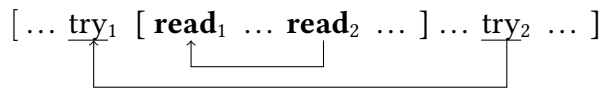
It is necessary to address how it is possible in principle for $[uV]$ to ‘ignore’ intervening embedding material. An intuitive solution is that Group 3 predicates are actually functional restructuring verbs (Cinque 2004b; Wurmbrand 2004), which lack the categorial feature $[iV]$, and thus do not count as intervenors. If this was the case, this would require us to reanalyse Group 3 complements as mono-clausal structures, and it would permit a straightforward featural intervention account of low verb doubling’s clause-boundedness without needing the Williams Cycle. Nevertheless, there is a crucial problem with this solution: if Group 3 predicates were indeed functional verbs (rather than lexical ones), we predict that Group 3 predicates can never display verb doubling effects. This is not borne out by (282), in which the matrix, Group 3 predicate *chángshi* and the embedded one *kàn* are both doubled and co-occur.

- (282) a. Piet chángshì [**kàn** nà-běn xiǎoshuō **kàn** de gèng tóurù yīxiē] chángshì-le
 Piet try read that-CL novel read DE more attentively a.bit try-PFV
 hěnduō-cì.
 many-time
 ‘Piet tried many times to read the novel more attentively.’

- b. Piet chángshì [**kàn** nà-běn xiǎoshuō **kàn** bàn-gè-zhōngtóu] chángshì-le
 Piet try read that-CL novel read half-CL-hour try-PFV
 hěnduō-cì.
 many-time
 ‘Piet tried many times to read the novel for half an hour.’

In both examples, the first occurrence of the matrix predicate *chángshì* takes a clausal complement, which contains a verb doubling erected on the embedded predicate *kàn*. Moreover, the second occurrence of *chángshì* is followed by an adverbial, resulting in another verb doubling erected on the matrix *chángshì*. Since both doubling constructions are supposed to involve the same mechanism, i.e., VP-movement, this creates a derivation in which the VP-movement chain related to the embedded *kàn* is nested within the other one that concerns the matrix *chángshì*, as illustrated in (283) (for the sake of clarity, only the doubled verbs are included in the representation). Thus, I refer to this kind of configuration as *nested verb doubling*, whose existence clearly indicates that Group 3 predicates cannot be analysed as functional restructuring verbs.

(283) *Representation of nested verb doubling in (282)*



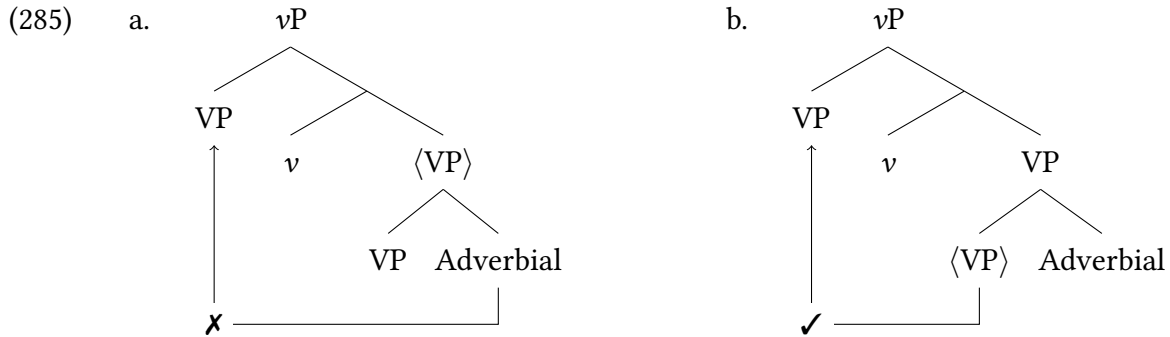
The conclusion made above requires us to adopt an alternative idea, that is, no embedding predicate by default constitutes a true intervenor for $[uV]$ on F. The non-intervention problem, then, can be understood as follows: though the most local goal for Agree, movement of the matrix VPs in this configuration is independently ruled out (I will account for this very soon below); after this failure, Agree works in such a way that it can apply again, leaving the embedded VP the closest remaining goal.²² The intended derivation is sketched below in (284), which illustrates a stage of the derivation at the FP-cycle, after an FP-sized complement has been embedded.

- (284) a. $[_{FP} F [_{uV} [_{vP} VP [_{FP} [_{vP} VP]]]]]$ (Agree with matrix VP ✓)
 $\begin{array}{c} \text{O} \quad \text{O} \\ \text{AGREE} \end{array}$
- b. $[_{FP} F [_{uV} [_{vP} VP [_{FP} [_{vP} VP]]]]]$ (Attract matrix VP to Spec,FP ✗)
 $\begin{array}{c} \uparrow \\ \text{MOVE} \end{array}$

²² The derivation here could appeal to either *Multiple Agree* (Hiraiwa 2001, 2005) or *Interaction and Satisfaction Agree* (Deal 2015, 2024).

- c.
$$\begin{array}{c} [\text{FP } \underset{\circ}{\text{F}} [\text{uV}] [\text{vP } \text{VP} [\text{FP } [\text{vP } \text{VP}]]]] \\ \text{-----} \text{AGREE } \checkmark \end{array}$$
 (Agree with embedded VP ✓)
- d.
$$\begin{array}{c} [\text{FP } \text{F} [\text{uV}] [\text{vP } \text{VP} [\text{FP } [\text{vP } \text{VP}]]]] \\ \text{-----} \text{MOVE } \checkmark \end{array}$$
 (Attract embedded VP to Spec,FP ✓)

The non-intervention problem is reminiscent of the stranding problem that I discussed in section 4.3. As a reminder, the two contrastive derivations are illustrated again in (285) (repeated from (133)): the closet goal here is the topmost VP, but its movement to the edge of vP is ruled out by Abels (2003)' anti-locality constraint (repeated in (286)). Therefore, the proper goal for the feature-triggered movement becomes the second highest VP within the entire verbal structure, which does not encounter the same anti-locality issue due to the existence of the adverbial.

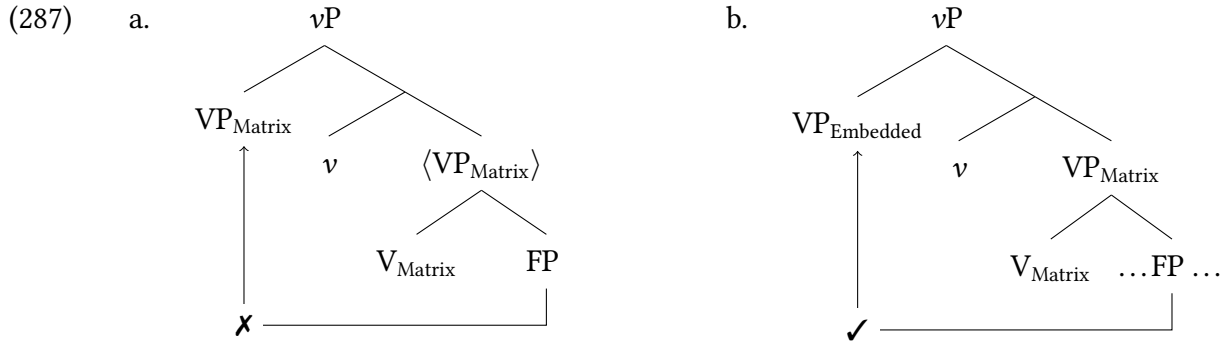


(286) *Comp-to-Spec Anti-locality Constraint*

Movement of the complement of a head X must cross a head other than X.

(Abels 2003:12-14, *et seq.*)

Considering the close similarities between these two puzzles, I propose that the solution to the non-intervention problem can also be attributed to the anti-locality constraint on complement-to-specifier movement: as schematised in (287a), all unmodified embedding predicates, by virtue of their truncated verbal structures, do not allow the matrix VP to move to the clause-internal edge (i.e., Spec,vP). This constraint is evidenced by the ungrammatical nested configurations in (288) (cf. (282)). [uV] on matrix F, thus, can only subsequently find a suitable goal in FP-sized complements, out of which movement will not violate the anti-locality constraint, as represented in (287b). These thoughts combined together explain why matrix VPs do not generally count as intervenors.



- (288) a. *Piet chángshì [**kàn** nà-běn xiǎoshuō **kàn** de gèng tóurù yīxiē]
 Piet try read that-CL novel read DE more attentively a.bit
chángshì-le.
 try-PFV
- b. *Piet chángshì [**kàn** nà-běn xiǎoshuō **kàn** bàn-gè-zhōngtóu] chángshì-le.
 Piet try read that-CL novel read half-CL-hour try-PFV

Now that I have addressed this lacuna in our analysis, I wrap up this section. I have outlined a version of Level Embedding approach, a theoretical architecture which derives a strict version of the Williams Cycle. I have shown how this derives improper verb doubling effects and makes correct predictions about movement out of multiply-embedded clauses. I have also discussed several problems underlying the Level Embedding approach and proposed a general derivational constraint to ameliorate them. In the next section I turn my attention to long-distance object shift, which will be used as evidence to further support the idea that the Williams Cycle in Height-Locality effects can extend across movement of different categories and featural triggers.

6.4 Clause-bound movement of nominals: object shift

In section 6.2.1, we have seen that object shift does not behave identically in various embedding contexts. I have used the distinct distribution of object shift as a crucial argument to favour my proposal that embedding predicates take clausal complements with different sizes. Unlike long-distance VP-fronting discussed throughout the paper, long-distance object shift has received considerable attention in the literature, such as Ernst and Wang (1995), Shyu (1995), Paul (2002b, 2005), Chen (2023, 2025), among many others. The purpose of the current section is to show how the current approach can be extended to those cases.

6.4.1 Outlining the basic paradigm

I first present the full paradigm of object shift in different embedding contexts in what follows. I construct the examples in exactly the same environments as long-distance verb doubling in section 6.1. This ensures that the distribution of long-distance object shift is only affected by the size of complement clause from which the nominal movement starts.

Within Group 1 complement clauses, we have already known that object shift can target two positions due to their CP-sized status. One of them follows the embedded subject, and another precedes the same subject, as respectively given again in (289a) and (289b). Similar to the way of distinguishing verb doubling, I dub the former situation *low object shift* and the latter *high object shift* from this point onwards. When the object moves across the full clause-boundary, low and high object shift show the same dichotomy as their verb doubling counterparts: shifting the object to the lower site of the matrix is banned (289c), whereas landing in the higher site is permitted (289d). I treat (289c) as an instance of *improper object shift*.

- (289) a. Wassily rènwéi [Piet **nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily think Piet that-CL novel read DE very quickly
 ‘Wassily thinks that Piet reads THAT NOVEL quickly.’
 (‘think’ + embedded low OS ✓ (=218a))
- b. Wassily rènwéi [**nà-běn xiǎoshuō** Piet kàn de hěn kuài].
 Wassily think that-CL novel Piet read DE very quickly
 ‘Wassily thinks that, THAT NOVEL, Piet reads quickly.’
 (‘think’ + embedded high OS ✓ (=218b))
- c. *Wassily **nà-běn xiǎoshuō** rènwéi [Piet kàn de hěn kuài].
 Wassily that-CL novel think Piet read DE very quickly
 Int.: ‘Wassily thinks that Piet reads THAT NOVEL quickly.’
 (‘think’ + matrix low OS ✗)
- d. **Nà-běn xiǎoshuō** Wassily rènwéi [Piet kàn de hěn kuài].
 that-CL novel Wassily think Piet read DE very quickly
 ‘As for THAT NOVEL, Wassily thinks Piet reads it quickly.’
 (‘think’ + matrix high OS ✓)

We have also seen that embedded object shift is possible in the Group 2 embedding context (290a), and argued that Group 2 clausal complements are TP-sized due to the absence of overt embedded subject and sentence-level adverb (recall section 6.2.1). What is intriguing here is that the potential occurrence of improper object shift is not attested. As shown in (290b), long-distance object shift is allowed in the matrix clause-medial position, unlike its low verb doubling counterpart in the

same environment (214b). Moving the object to the matrix clause-initial position is permitted as usual.

- (290) a. Wassily qiǎngpò Piet [**nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily force Piet that-CL novel read DE very quickly
 ‘Wassily forces Piet to read THAT NOVEL quickly.’
 (‘force’ + embedded OS ✓ (=220))
- b. Wassily **nà-běn xiǎoshuō** qiǎngpò Piet [kàn de hěn kuài].
 Wassily that-CL novel force Piet read DE very quickly
 ‘Wassily forces Piet to read THAT NOVEL quickly.’
 (‘force’ + matrix low OS ✓)
- c. **Nà-běn xiǎoshuō** Wassily qiǎngpò Piet [kàn de hěn kuài].
 that-CL novel Wassily force Piet read DE very quickly
 ‘As for THAT NOVEL, Wassily forces Piet to read it quickly.’
 (‘force’ + matrix high OS ✓)

Proceeding to object shift in the Group 3 embedding context, we have been aware that embedded object shift in such an environment is not possible (as in (291a)) because of its most reduced, FP-sized status. However, no restrictions are imposed on the movement of object across Group 3 complement clauses. As given in (291b) and (291c), the shifted object can flexibly target its landing site either below or above the surface subject.

- (291) a. *Wassily chángshì [**nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily try read that-CL novel DE very quickly
 Int.: ‘Wassily tries to read THAT NOVEL quickly.’ (‘try’ + embedded OS ✗ (=222))
- b. Wassily **nà-běn xiǎoshuō** chángshì [kàn de hěn kuài].
 Wassily that-CL novel try read DE very quickly
 ‘Wassily tries to read THAT NOVEL quickly.’ (‘try’ + matrix low OS ✓)
- c. **Nà-běn xiǎoshuō** Wassily chángshì [kàn de hěn kuài].
 that-CL novel Wassily try read DE very quickly
 ‘As for THAT NOVEL, Wassily tries to read it quickly.’ (‘try’ + matrix high OS ✓)

The full paradigm of object shift in three embedding contexts is summarised in (292). The variations in embedded object shift have already been explained by the size of different clausal complements in section 6.2.1. The key problem is the different distributions of long-distance object shift in the matrix clause-medial position. Object shift to the matrix left periphery is always clause-unbounded, as with matrix high verb doubling (cf. (216)). For the sake of space I will not discuss

these cases in what follows, assuming that the Level Embedding approach to long-distance high verb doubling outlined in section 6.3.2 fully applies to long-distance high object shift as well.

(292) *Full paradigm of object shift in different embedding contexts*

Embedding predicate group	Matrix high OS	Matrix low OS	Embedded OS
Group 1: ‘think’ + [CP ...]	✓	✗	✓
Group 2: ‘force’ + [TP ...]	✓	✓	✓
Group 3: ‘try’ + [FP ...]	✓	✓	✗

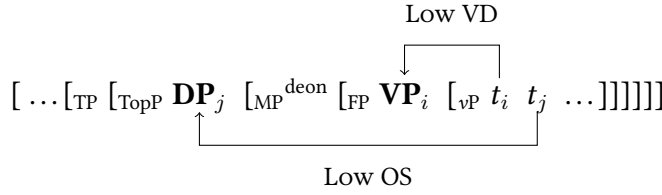
6.4.2 The landing site of low object shift vs. low verb doubling

Before presenting my analyses of long-distance low object shift, I need to highlight an important difference between low object shift and low verb doubling concerning their landing sites after movement. Recall that at the end of section 6.2.1, I reached the conclusion that low verb doubling lands into a position lower than low object shift in the clausal spine. This is motivated by the empirical observation that within Group 3 complement clauses, verb doubling is possible while object shift is not. The contrast is illustrated again in (293).

- (293) a. Wassily chángshì [FP **kàn nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily try read that-CL novel read DE very quickly
 ‘Wassily tries to read the novel quickly.’ (Embedded VD in Group 3 ✓ (=215a))
- b. *Wassily chángshì [FP **nà-běn xiǎoshuō** kàn de hěn kuài].
 Wassily try that-CL novel read DE very quickly
 Int.: ‘Wassily tries to read THAT NOVEL quickly.’
 (Embedded OS in Group 3 ✗ (=222/291a))

That (293a) is well-established lays the foundation of my proposal that Group 3 complement clauses are FP-sized, since verb doubling can target the clause-medial position Spec,FP and occur in such embedded configurations. Then the ungrammatical (293b) gets explained right away once we adopt the assumption that object shift generally targets a position higher than the clause size of Group 3 complements; otherwise, we would predict no difference between (293a) and (293b). Following Paul (2002b, 2005) and Chen (2023, 2025), I assume that there is a TopP in the TP-domain, and that low object shift is triggered by [*u*Top] on this Top head and moves to its specifier position. A more articulated clause-medial structure is illustrated in (294), with both low verb doubling and low object shift indicated (cf. (253)).

(294) *Clause-medial structure and low object shift & verb doubling in Mandarin Chinese*



The different landing sites of low object shift and low verb doubling lead us to expect that their movement behaviour is not totally identical. As pointed out in the previous section, in the Group 2 embedding context, low object shift does not manifest improper movement (e.g., (290b)), in contrast to low verb doubling that does behave improperly (e.g., (214b)). In the following section I show how the Level Embedding approach can capture this contrast.

6.4.3 Deriving long-distance low object shift

In what follows I use the Level Embedding approach to derive proper/improper object shift attested in the clause-medial position.

6.4.3.1 Low OS out of Group 1 complements

I begin with the only situation where improper object shift is observed, namely low object shift out of Group 1 complements, as exemplified in (289b). According to Parallel Derivation, we start building up the main and embedded clause simultaneously from VP (295a) to CP (295b). Once reaching the CP-cycle, we embed the CP complement into the matrix CP via Late Merge (295c). However, what is forbidden in the CP-cycle is to move the object from the embedded CP complement to the matrix Spec,TopP, since such a movement violates the Strict Cycle Condition (295d), thereby giving rise to improper object shift.

- (295) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [_{VP} think ____], [_{VP} read the novel]
- b. *Step 2: Build up to the CP-level* (Parallel Derivation)
 [_{CP} [_{TP} Wassily [_{TopP} Top_[uTop] [_{vP} think ____]]]], [_{CP} [_{TP} Piet [_{TopP} [_{vP} read the novel quickly]]]]
- c. *Step 3: Embed the CP in the other CP* (Embedding with Late Merge)
 [_{CP} [_{TP} Wassily [_{TopP} [_{vP} think [_{CP} [_{TP} Piet [_{TopP} [_{vP} read the novel quickly]]]]]]]]

- d. *Step 4: Probe on matrix Top cannot trigger DP-movement anymore* (Strict Cycle ✗)
 *_{[CP [TP Wassily [_{TopP} [_{DP} the novel] Top_[≠Top] [_{VP} think [_{CP} [TP Piet [_{TopP} [_{VP} read DE quickly]]]]]]]]}

6.4.3.2 Low OS out of Group 2 complements

The derivation of long-distance low object shift out of Group 2 complements also starts with building up both clauses to the VP-level in parallel (296a). Parallel Derivation keeps going to the TopP-level (296b). In this cycle, the TopP-sized complement clause is embedded (using Late Merge) into the congruent matrix clause (296c), and the feature on the matrix Top head triggers the nominal movement to its specifier (296d). Once object shift is done, the matrix clause continues to grow and ends up at the CP-level (296e).

- (296) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
_{[VP force Piet ____], [_{VP} read the novel]}
- b. *Step 2: Build up to the TopP-level* (Parallel Derivation)
{[{TopP} Top_[uTop] [_{VP} Wassily force Piet ____]], [_{TopP} [_{VP} read the novel quickly]]}
- c. *Step 3: Embed the TopP in the other TopP* (Embedding with Late Merge)
{[{TopP} Top_[uTop] [_{VP} Wassily force Piet [_{TopP} [_{VP} read the novel quickly]]]]}
- d. *Step 4: Probe on matrix Top triggers DP-movement* (Strict Cycle ✓)
{[{TopP} [_{DP} the novel] Top_[≠Top] [_{VP} Wassily force Piet [_{TopP} [_{VP} read DE quickly]]]]}
- e. *Step 5: Build up to the CP-level in the matrix*
{[{CP} [TP Wasily [_{TopP} [_{DP} the novel] [_{VP} force Piet [_{TopP} [_{VP} read DE quickly]]]]]]}

The derivation above shows that the clause size of Group 2 complements can actually be smaller than TP (i.e., TopP) because of the Strict Cycle Condition. As mentioned in footnote 2, I use TP (and also CP) as an umbrella term throughout the paper. That is to say, what I call TP in principle can split into several functional projections, e.g., the articulated clause-medial structure given in (294). Adopting this split-TP assumption would not undermine what I have said previously about improper verb doubling in the Group 2 embedding context (recall section 6.3.2): as long as the size of Group 2 complement clauses is bigger than FP in the clausal *fseq*, low verb doubling is not permitted to land in the matrix Spec,FP. Abstracting away from the precise size of Group 2 complements, I will stick to the label TP till the end of the paper.

6.4.3.3 Low OS out of Group 3 complements

Low object shift out of Group 3 complements does not significantly differ from its Group 2 counterpart. Again, we build up two clause structures in parallel from VP (297a) to FP (297b). Clause embedding with Late Merge happens in this FP-cycle (297c), followed by the matrix clause being built up to the Top-level (297d). The feature on the matrix Top triggers low object shift at this level without violating the Strict Cycle Condition (297e). In the end the matrix clause grows up and stops at the CP-level (297f).

- (297) a. *Step 1: Build up to the VP-level* (Parallel Derivation)
 [_{VP} try ____], [_{VP} read the novel]
- b. *Step 2: Build up to the FP-level* (Parallel Derivation)
 [_{FP} [_{VP} Wassily try ____]], [_{FP} [_{VP} read the novel quickly]]
- c. *Step 3: Embed the FP in the other FP* (Embedding with Late Merge)
 [_{FP} [_{VP} Wassily try [_{FP} [_{VP} read the novel quickly]]]]
- d. *Step 4: Build up to the TopP-level in the matrix*
 [_{TopP} Top[_{uTop}] [_{FP} [_{VP} Wassily try [_{FP} [_{VP} read the novel quickly]]]]]]
- e. *Step 5: Probe on matrix Top triggers DP-movement* (Strict Cycle ✓)
 [_{TopP} [_{DP} the novel] Top[_{uTop}] [_{FP} [_{VP} Wassily try [_{FP} [_{VP} read DE quickly]]]]]]
- f. *Step 6: Build up to the CP-level in the matrix*
 [_{CP} [_{TP} Wassily [_{TopP} [_{DP} the novel] [_{FP} [_{VP} try [_{FP} [_{VP} read DE quickly]]]]]]]]

6.4.4 Three locality profiles for embedding contexts

By comparing the behaviour of cross-clausal verb doubling and cross-clausal object shift, we can construct locality profiles for our three embedding contexts - repeated from the introduction in (298). Complements to Group 1 predicates are opaque to the most kinds of movements (low verb doubling and low object shift). At the other end of the scale, complements to Group 3 predicates are transparent to everything. Complements to Group 2 predicates occupy an intermediate status in being opaque to just low verb doubling.²³

²³ There are of course other types of movement which could be used to enrich this picture further. For more details of these movements, see relevant discussion in section 2.4.

(298)	Locality profile	Pred. _{Group 1} + [CP ...]	Pred. _{Group 2} + [TP ...]	Pred. _{Group 3} + [FP ...]
	Transparent to:	High VD/OS	High VD/OS, Low OS	High VD/OS, Low VD/OS
	Opaque to:	Low VD/OS	Low VD	Ø

Showing that two categorially distinct kinds of movement strengthens the case for treating these Height-Locality effects as instances of the Williams Cycle. Looking at each movement in isolation, it is possible in principle to imagine probe-specific restrictions which do not concern their height in the clausal spine. However, the more we find different types of movement within a language that display *prima facie* Height-Locality effects, the probability of a merely coincidental connection to position decreases. The Level Embedding approach to the Williams Cycle expects Height-Locality effects to hold of movements with different featural triggers.

6.5 Closing remarks

In this final section, I close the whole chapter by showing two puzzles which illustrate how verb doubling, as an instance of verbal phrasal movement, interacts with other movement-related phenomena in Mandarin.

The first one is a follow-up to the nested verb doubling configuration illustrated in section 6.3.5. The representation of nested verb doubling is repeated below.

- (299) *Representation of nested verb doubling*
- [... try₁ [read₁ ... read₂ ...] ... try₂ ...]
-

Throughout this chapter we have seen that VPs can move out of embedded clauses and bring about long-distance verb doubling. In a configuration which exhibits double verb doubling effects (such as (299) above), it is still feasible for the VP in the embedded clause to undergo long-distance movement to the matrix clause, and create what I term *crossing verb doubling*. As exemplified in (300), here both the matrix and the embedded predicates display doubling effects, as with nested verb doubling (recall (282)). The key difference between them lies in the position of the VP *kàn nà-běn xiǎoshuō*: in (300), this VP occurs out of the embedded complement clause, taking up a position between the subject *Piet* and the first occurrence of the matrix predicate *chángshì*.

- (300) a. Piet [**kàn** nà-běn xiǎoshuō] [chángshì [**kàn** de gèng tóurù yīxiē]
 Piet read that-CL novel try read DE more attentively a.bit
chángshì-le hěnduō-cì].
 try-PFV many-time
 ‘Piet tried many times to read the novel more attentively.’
- b. Piet [**kàn** nà-běn xiǎoshuō] [chángshì [**kàn** bàn-gè-zhōngtóu] chángshì-le
 Piet read that-CL novel try read half-CL-hour try-PFV
 hěnduō-cì].
 many-time
 ‘Piet tried many times to read the novel for half an hour.’

Comparing to long-distance low verb doubling, it seems reasonable to entertain this displacement as another instance of long-distance VP-movement across the embedded clause. This analysis gives us the crossing representation in (301): that is, the movement chain of the embedded VP crosses that of the matrix VP, unlike the nested one in (299).

- (301) *Representation of crossing verb doubling*
- [... [**read**₁ ...] [try₁ [... **read**₂ ...] ... try₂ ...]]
-

The distribution of the displaced embedded VP may leave us an impression that its landing site in the matrix clause is the canonical position that hosts low verb doubling in root clauses. Nevertheless, this line of thinking cannot hold here: for the reason of cyclicity, let us assume that crossing verb doubling derives from nested verb doubling, which is to say, the embedded VP-fronting (as in (301)) takes place after the completion of the double verb doubling configuration (as in (299)). Since the matrix predicate forms verb doubling in some earlier derivation, this indicates that the matrix VP (i.e., the matrix predicate itself and its clausal complement) has already occupied the low landing site in the matrix clause. Then for the embedded VP, there would be no available landing site in the matrix clause at the point when it moves out of the embedded one, because it is supposed to target the same landing site as the matrix VP does.

How do we account for this issue? I propose that, since the clause-medial domain below the subject actually consists of multiple positions (e.g., Spec,TopP for object shift and Spec,FP for low verb doubling), crossing verb doubling can be understood as a configuration where the matrix and the embedded VPs move to two distinct positions rather than compete for the same landing site in the matrix clause. However, where exactly the embedded VP further moves to remains unclear to me at the moment.

The second puzzle relates verb doubling to other kinds of overt movements in Mandarin Chinese. For the sake of convenience, here I take object shift as an instance. (302) and (303) respectively show two baseline examples of object shift and verb doubling. So far we have only seen cases where each movement type occurs on its own. The question I pursue is whether verb doubling and object shift are able to co-occur. In other words, is it possible to subextract the object out of the fronted VP?

- (302) a. Piet [*Níngméng Shù*] zǒngshì **chàng** de hěn dàshēng.
 Piet lemon tree always sing DE very loudly
 ‘Piet always sings *Lemon Tree* very loudly.’
- b. [*Níngméng Shù*] Piet zǒngshì **chàng** de hěn dàshēng.
 lemon tree Piet always sing DE very loudly
 ‘*Lemon Tree*, Piet always sings very loudly.’
- (303) a. Piet [**chàng** *Níngméng Shù*] zǒngshì **chàng** de hěn dàshēng.
 Piet sing lemon tree always sing DE very loudly
 ‘Piet always sings *Lemon Tree* very loudly.’
- b. [**Chàng** *Níngméng Shù*] Piet zǒngshì **chàng** de hěn dàshēng.
 sing lemon tree Piet always sing DE very loudly
 ‘As for singing *Lemon Tree*, Piet always sings it very loudly.’

At the first glance, subextraction out of verb doubling appears impossible, as given in (304). In (304a), VP-fronting targets the clause-medial position, followed by a local object shift. This does not result in a grammatical configuration. In (304b), the fronted VP still lands into the clause-medial position, but the object is shifted to the higher, pre-subject position. The construction, nevertheless, is still ungrammatical. In (304c), both VP-fronting and object shift enter the left periphery, but they still fail to yield a grammatical sentence. All the results in (304) seem to fall under what Müller (2010, 2011) characterises as *freezing effects*, namely, a moved item may become a barrier for extraction: in these cases, one can assume that the fronted VP loses its transparency for extraction after phrasal movement, which explains why further object shift is not permitted.

- (304) *Subextraction out of verb doubling with freezing effects*
- a. *Piet *Níngméng Shù* **chàng** zǒngshì **chàng** de hěn dàshēng.
 Piet lemon tree sing always sing DE very loudly
 Int.: ‘Piet always sings *LEMON TREE* very loudly.’

- b. ??*Níngméng Shù* Piet **chàng** zǒngshì **chàng** de hěn dàshēng.
 lemon tree Piet sing always sing DE very loudly
 Int.: ‘As for *Lemon Tree*, Piet always sings it very loudly.’
- c. **Níngméng Shù* **chàng** Piet zǒngshì **chàng** de hěn dàshēng.
 lemon tree sing Piet always sing DE very loudly
 Int.: ‘As for singing *Lemon Tree*, Piet always sings it very loudly.’

However, this is not the entire picture of subextraction out of verb doubling. (305) shows that, once another constituent, e.g., a PP *zài jiā-lǐ* ‘at home’, is adjoined to the fronted VP, all the ungrammatical instances of subextraction seen above turn to be grammatical. It looks like the originally ‘frozen’ barrier (i.e., the fronted VP) is ‘melted’ (i.e., transparent for extraction) by the occurrence of some extra constituent. According to Müller (2010, 2011), this opposite phenomenon is referred to as a *melting effect*.

(305) *Subextraction out of verb doubling with melting effects*

- a. Piet *Níngméng Shù* zài jiā-lǐ **chàng** zǒngshì **chàng** de hěn dàshēng.
 Piet lemon tree LOC home-inside sing always sing DE very loudly
 ‘Piet always sings *LEMON TREE* very loudly at home.’
- b. *Níngméng Shù* Piet zài jiā-lǐ **chàng** zǒngshì **chàng** de hěn dàshēng.
 lemon tree Piet LOC home-inside sing always sing DE very loudly
 ‘As for *Lemon Tree*, Piet always sings it very loudly at home.’
- c. *Níngméng Shù* zài jiā-lǐ **chàng** Piet zǒngshì **chàng** de hěn dàshēng.
 lemon tree LOC home-inside sing Piet always sing DE very loudly
 ‘As for singing *Lemon Tree* at home, Piet always sings it very loudly.’

The contrast between (304) and (305) is robust, but an account of it based on freezing/melting effects seems a bit fragile. This is because a lot of empirical facts pose challenges to the freezing effects, such as quantifier floating in West Ulster English (McCloskey 2000), possessor extraction in colloquial English (Davis 2020, 2021), etc. Even so, I will still stick to these labels as they capture the contrast in a pretty straightforward way. Once again, I do not have any specific solutions to explaining these patterns at the moment. How verb doubling interacts with different types of movement calls for more future investigation.

Chapter 7

Conclusion

The primary goal of this thesis was to develop a comprehensive account of deriving verb doubling in Mandarin Chinese. Though similar doubling effects are found across a variety of languages, Mandarin verb doubling displays (at least) three language-specific properties that merit particular attention as well as detailed investigation. First, the first occurrence of the doubled verb plus the object that follows it are not confined to a fixed position before the subject, in contrast to the majority of languages that display verb doubling effects. Instead, this verb-object unit is also possible to occur in a clause-medial position after the subject. Second, the postverbal adverbials are always involved in the formation of verb doubling. More precisely speaking, it is always the same case that the verb doubling configurations in question have an adverbial occur in their postverbal field. Third, particles that follow the second occurrence of the doubled verb vary from one configuration to another, depending on the type of the adverbial showing up postverbally. To understand and account for all these properties, I laid out the majority of my proposal through chapter 2-5.

Chapter 3 presented the core of my account. I proposed that deriving verb doubling involves verbal phrasal movement to some higher position in the clausal spine. Since the fronted verbal phrase can occur at two different positions around the subject, I further proposed that there are two separate landing sites which can be independently targeted by the feature-triggered verbal phrasal movement. One crucial question was that where exactly these two positions are in the clausal spine. I postulated that there is a specific functional projection FP between TP and vP which is responsible for hosting the fronted verbal phrase in the clause-medial domain, and that the higher position before the subject is situated in the left periphery. I argued that my proposed two-landing-site analysis not only has more theoretical advantages over other plausible alternative accounts (e.g., the fronting-plus-topicalisation one), but also displays compatibility with our current understanding of the clausal architecture in Mandarin Chinese, namely that there is an articulated clause-medial domain below TP that parallels the left periphery (à la Belletti 2004).

Chapter 3 investigated various aspects of the postverbal adverbials. I started off by classifying adverbials into six types on the basis of their meaning, in line with prior literature (e.g., Li and Thompson 1981; Tai 1999; Cheng 2017). Then by examining their co-occurrence conditions with a variety of verbs, I argued that these six types of postverbal adverbials can be further classified into two sub-groups, i.e., adverbial adjuncts which have no difficulty occurring postverbally in any doubling configuration, and adverbial arguments whose occurrence is conditioned by the specific choice of verbs. I contended that this adjunct/argument dichotomy among adverbials needs to be reflected in the way they are introduced to the verbal projection. Therefore, I proposed that adverbial adjuncts can be freely-adjoined to the maximal verbal projection (e.g., VP or ApplP), following Ernst (2002), whereas adverbial arguments are selected by some higher verbal head (such as the applicative one) via features and base-generated in its specifier (i.e., Spec,ApplP). I showed that the dichotomy correctly predicts a range of possible and impossible co-occurrences of multiple adverbials in the postverbal field, but fails to account for the ban on the co-occurrence of two adjuncts in the same field. To address the latter, I offered a non-structural, PF-based account, adopting the idea of distinctness proposed by Richards (2010). The most important takeaway from this chapter was the proposed structural relationship between the adverbials and the verbal projections: through the lens of my proposal, the fronted verbal phrases involved in verb doubling always come out of an articulated verbal structure consisting of one or more adverbials (as part of it). This idea is completely different from other highly-relevant analyses, e.g., the one proposed by Lai (2021, 2024), which regards adverbials as specifier of some low functional projection in the Cinquean Hierarchy (Cinque 1999), and makes a clear separation between the lexical VP at the bottom and other low functional projections.

Against the backdrop set up by the previous two chapters, chapter 4 conducted a more in-depth examination on the distribution of verbal dependents within verb doubling configurations. The main claim I made here is that verbal phrases which are larger than simple VPs can also be the target of movement involved in verb doubling. The motivation behind this claim was from doubling configurations built upon ditransitive verbs, especially those involving double object constructions: in those cases, what undergoes movement can be characterised as either a V-IO-DO string, or just a string of V plus the DO. By contrast, the same alternation was not found in cases where prepositional dative constructions are contained. I proposed that this contrast can receive a straightforward account as long as we accept that (i) ditransitives have asymmetric verbal structures, namely, either an ApplP or a VP, and that (ii) phrasal movement involved in verb doubling is not confined to VPs only; instead, bigger structures like ApplPs can undergo fronting to some higher position as well. This, however, is contrary to Lai (2021, 2024)'s proposal, in which he insists that VPs are the only suitable target of phrasal movement. Further support in favour of my proposal came from the prediction about the distribution of multiple adverbials within verb

doubling constructions. I showed that there is a systematic way of distributing adverbials across an articulated verbal structure, as a result of allowing larger verbal phrases to undergo fronting. At the end of the chapter, I narrowed down to two analytical issues, and had some brief discussions on each of them. One was concerned with the fact that in each verb doubling configuration, there is always an adverbial stranded in the postverbal field. Borrowing insights from Lai (2021, 2024), I adopted Abels (2003)'s version of anti-locality to address this stranding problem. The other issue was related to several unique characteristics of verbal compounds, which I argued can be attributed to the properties of dummy objects (in the sense of Cheng and Sybesma 1998) taken by the verbs, and do not affect the transitive verbal nature of these compounds.

Chapter 5 turned to the question about how to pronounce the same verb twice at PF, thereby realising the verb doubling effects. Here I placed a particular focus on the realisation of the second verb, and proposed that this process consists of a series of operations that only take place in order post-syntactically (see also Embick and Noyer 2001; Embick 2007; Hein 2017, 2018, 2020, among others). I began with comparing two particles (i.e., *le* and *de*) which are involved in different verb doubling configurations. The result of the comparison showed that, though these two particles do share some similarities, they are fundamentally two distinct elements. Given this fact, I argued that it is impossible for these particles to originate from the same source in the clausal spine, which is to say, they must come from different base positions. Therefore, I postulated that tense and aspect markers like *le* are base-generated in the head of AspP, while markers for descriptive expressions like *de* are base-generated in the head of ModP, which itself is a freely-adjoined adjunct and internally takes adverbials as its complement. The distinction between particles motivated me to develop a PF-based account of realising verb doubling effects, comprising a variety of post-syntactic operations which can be freely combined to deal with elements from different sources. After applying these operations to canonical verb doubling effects, I further demonstrated how this PF-toolkit can be used to account for other closely-related configurations.

Since the preceding four chapters already offered a complete account that concerns all the idiosyncrasies Mandarin verb doubling displays, chapter 6 turned its attention to a brand-new topic, namely, the locality of long-distance verb doubling. Building upon recent literature on Mandarin clause structures (e.g., He 2020, 2024; Huang 2022, among many), I first identified three groups of embedding predicates (i.e., Group 1/2/3 predicates) and their corresponding embedding contexts where long-distance verb doubling can be tested. Then I showed that long-distance verb doubling does not display a uniform behaviour in these different embedding contexts: though long-distance high verb doubling that targets the matrix left periphery is always permissible in each context, its low counterpart that targets the matrix clause-medial position is more restricted and only allowed in Group 3 contexts. In Group 1 and 2 embedding contexts, long-distance low verb doubling is completely prohibited, a phenomenon which I referred to as improper verb dou-

bling. I attributed the occurrence of improper verb doubling to the size of embedded clauses from which long-distance VP-fronting sets off, and argued that different groups of embedding predicates take clausal complements with different sizes. Relying on two language-specific diagnostics, I showed that Group 1 predicates take the largest, CP-sized complements, Group 2 more reduced, TP-sized ones, and Group 3 the smallest, FP-sized ones. With the variations in clause size in mind, I proposed that the full paradigm of long-distance verb doubling can be properly accounted for by the Williams Cycle, that is, there is a general connection between the height of movement and the size of clausal complement it can escape. I presented the Level Embedding approach proposed in Yan and Meadows (2025) to implementing this locality principle as well as accounting for the patterns in question. I recognised that there are several issues arising from this specific theoretical implementation, and therefore proposed a derivational constraint based on the Grimshawan extended projection (Grimshaw 1991, 2000, 2005). I also noticed that there is a potential problem of intervention in Group 3 contexts, of which I made use of Abels (2003)'s anti-locality constraint on phrasal movement again to offer an account. At the end of the chapter, I examined another kind of movement phenomenon in the three embedding contexts, namely, long-distance object shift. Though not totally identical to cross-clausal verb doubling, the similar varying locality profiles of cross-clausal object shift once again confirmed the intuition behind the Williams Cycle: the higher a movement operation lands in the clausal functional sequence, the more unbounded it is.

In a word, this thesis provides us with a detailed analysis of a wide range of aspects of Mandarin verb doubling. I have no doubt that there are a lot of phenomena related to verb doubling still awaiting explanations. I hope that this work will pave the way for inquiries into them.

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